18.100A: Complete Lecture Notes

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1 Lecture 1

Sets, Set Operations, and Mathematical Induction

For this class, we will be using the book Introduction to Real Analysis, Volume I by Jiří Lebl **[L]**. I will use ■ to end proofs of examples, and □ to end proofs of theorems.

Basic Set Theory

Remark 1. *There are two main goals of this class:*

1. *Gain experience with proofs.*
2. *Prove statements about real numbers, functions, and limits.*

Sets

A set is a collection of objects called elements or members of that set. The empty set (denoted $\emptyset$) is the set with no elements. There are a few symbols that are super helpful to know as a shorthand, and will be used throughout the course. Let S be a set. Then

- $a \in S$ means that "a is an element in S."
- $a \notin S$ means that "a is not an element in S."
- $\forall$ means "for all."
- $:=$ means "define."
- $\exists$ means "there exists."
- $\exists!$ means "there exists a unique."
- $\implies$ means "implies."
- $\iff$ means "if and only if."

Definition 2 (Set Relations)
We want to relate different sets, and thus we get the following notation/definitions:

1. A set A is a subset of B, $A \subset B$, if every element of A is in B. Given $A \subset B$, if $a \in A \implies a \in B$.
2. Two sets A and B are equal, $A = B$, if $A \subset B$ and $B \subset A$.
3. A set A is a proper subset of B, $A \subsetneq B$ if $A \subset B$ and $A \neq B$.

One way we can describe a set is using "set building notation". We write

$$\{x \in A \mid P(x)\} \quad \text{or} \quad \{x \mid P(x)\}$$

to mean "all $x \in A$ that satisfies property $P(x)$". One example of this would be $\{x \mid x \text{ is an even number}\}$. There are a few key sets that we will use throughout this class:

1. The set of natural numbers: $\mathbb{N} = \{1, 2, 3, 4, \dots\}$.
2. The set of integers: $\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}$.
3. The set of rational numbers: $\mathbb{Q} = \{\frac{m}{n} \mid m, n \in \mathbb{Z} \text{ and } n \neq 0\}$.
4. The set of real numbers: $\mathbb{R}$.

It follows that

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

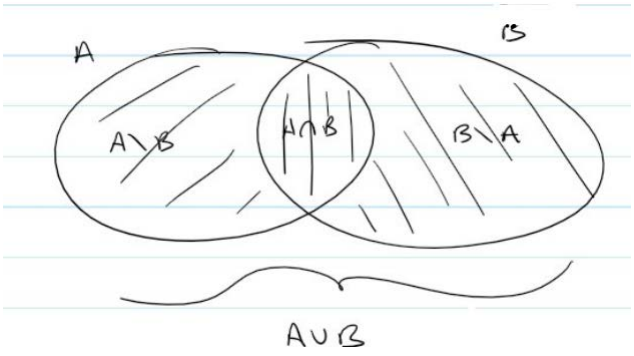
The fourth item on this list brings us to an important question, and the first goal of our course:

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Problem 3
How do we describe $\mathbb{R}$?

We will answer this question in Lectures 3 and 4. In the meantime, let's continue our study of sets and proof methods. Given sets A and B, we have the following definitions:

1. The union of A and B is the set $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.
2. The intersection of A and B is the set $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.
3. The set difference of A and B is the set $A \setminus B = \{x \in A \mid x \notin B\}$.
4. The complement of A is the set $A^c = \{x \mid x \notin A\}$.
5. A and B are disjoint if $A \cap B = \emptyset$.



Theorem 4 (De Morgan's Laws)
If A, B, C are sets then

1. $(B \cup C)^c = B^c \cap C^c$,
2. $(B \cap C)^c = B^c \cup C^c$,
3. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$,
4. and $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

We will prove the first statement to give an example of how such a proof would go, but the rest will be left to you.
Proof: Let B, C be sets. We must prove that

$$(B \cup C)^c \subset B^c \cap C^c \quad \text{and} \quad B^c \cap C^c \subset (B \cup C)^c.$$

If $x \in (B \cup C)^c \implies x \notin B \cup C \implies x \notin B$ and $x \notin C$. Hence, $x \in B^c$ and $x \in C^c \implies x \in B^c \cap C^c$. Thus, $(B \cup C)^c \subset B^c \cap C^c$.

If $x \in B^c \cap C^c$ then $x \in B^c$ and $x \in C^c \implies x \notin B$ and $x \notin C$. Hence, $x \notin B \cup C \implies x \in (B \cup C)^c$. Thus, $B^c \cap C^c \subset (B \cup C)^c$. □

Mathematical Induction

We will now talk about some of the biggest proof methods there are. Firstly, note that $\mathbb{N} = \{1, 2, 3, \dots\}$ has an ordering (as $1 < 2 < 3 < \dots$).

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Axiom 5 (Well-ordering property)
The well-ordering property of $\mathbb{N}$ states that if $S \subset \mathbb{N}$ then there exists an $x \in S$ such that $x \leq y$ for all $y \in S$. In other words, there is always a smallest element.

Note that this is an axiom, and thus we have to assume this without proof.

Theorem 6 (Induction)
This concept was invented by Pascal in 1665. Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. Assume that

1. (Base case) $P(1)$ is true and
2. (Inductive step) if $P(m)$ is true then $P(m+1)$ is true.

Then, $P(n)$ is true for all $n \in \mathbb{N}$.

Proof: Let $S = \{n \in \mathbb{N} \mid P(n) \text{ is not true}\}$. We wish to show that $S = \emptyset$. We will prove this by contradiction.

Remark 7. *When we prove something by contradiction, we assume the conclusion we want is false, and then show that we will reach a false statement. Rules of logic thus imply that the initial statement must be false. Thus in this case, we will assume $S \neq \emptyset$ and derive a false statement.*

Suppose that $S \neq \emptyset$. Then, by the well-ordering property of $\mathbb{N}$, S has a least element $m \in S$. Since $P(1)$ is true, $m \neq 1$, i.e. $m > 1$. Since m is a least element, $m - 1 \notin S \implies P(m-1)$ is true. This implies that $P(m)$ is true $\implies m \notin S$ by assumption. But then $m \in S$ and $m \notin S$. This is a contradiction. Thus $S = \emptyset$ and hence $P(n)$ is true for all $n \in \mathbb{N}$. □

Let's see an example of induction in action.

Theorem 8
For all $c \neq 1$ in the real numbers, and for all $n \in \mathbb{N}$,

$$1 + c + c^2 + \cdots + c^n = \frac{1 - c^{n+1}}{1 - c}.$$

Proof: We will prove this by induction. First, we prove the base case ($n = 1$). The left hand side of the equation is $1 + c$ for $n = 1$. The right hand side is $\frac{1-c^2}{1-c} = \frac{(1-c)(1+c)}{1-c} = 1 + c$. Hence, the base case has been shown.

Assume that the equation is true for $k \in \mathbb{N}$, in other words

$$1 + c + c^2 + \cdots + c^k = \frac{1 - c^{k+1}}{1 - c}.$$

Thus,

$$\begin{aligned} \implies 1 + c + c^2 + \cdots + c^k + c^{k+1} &= (1 + c + c^2 + \cdots + c^k) + c^{k+1} \\ &= \frac{1 - c^{k+1}}{1 - c} + c^{k+1} \\ &= \frac{1 - c^{k+1} + c^{k+1}(1 - c)}{(1 - c)} \\ &= \frac{1 - c^{(k+1)+1}}{1 - c}. \end{aligned}$$

Therefore, our proof is complete. □

Let's do another example:

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Theorem 9

For all $c \geq -1$, $(1+c)^n \geq 1+nc$ for all $n \in \mathbb{N}$.

Proof: We prove this through induction. In the base case, we have: $(1+c)^1 = 1+1\cdot c$. For the inductive step, suppose that

$$(1+c)^m \geq 1+mc.$$

Then,

$$(1+c)^{m+1} = (1+c)^m \cdot (1+c).$$

By assumption,

$$\begin{aligned} &\geq (1+mc)\cdot(1+c) \\ &= 1+(m+1)c+mc^2 \\ &\geq 1+(m+1)c. \end{aligned}$$

By induction, our proof is complete. □

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2 Lecture 2

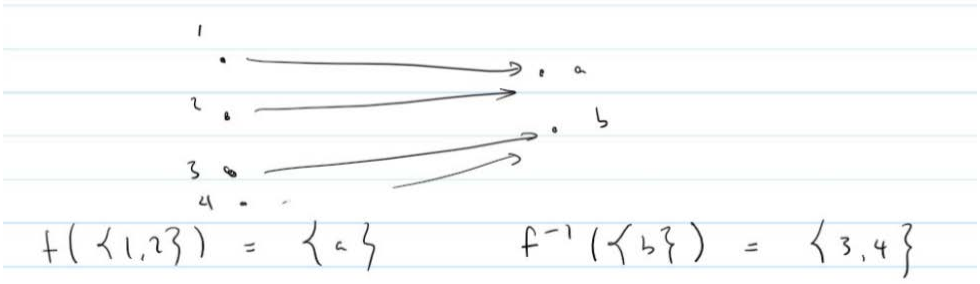
Cantor's Theory of Cardinality (Size)

Functions

If A and B are sets, a function $f : A \to B$ is a mapping that assigns each $x \in A$ to a unique element in B denoted $f(x)$. Let $f : A \to B$. Then

1. If $C \subset A$, we define $f(C) := \{y \in B \mid y \in f(x) \text{ for some } x \in C\}$.
2. If $D \subset B$, we define $f^{-1}(D) := \{x \in A \mid f(x) \in D\}$.

As an example, consider the following mapping $f : \{1, 2, 3, 4\} \to \{a, b\}$:



We can categorize functions in 3 important ways. Let $f : A \to B$.

1. f is injective or one-to-one (1-1) if $f(x_1) = f(x_2) \implies x_1 = x_2$.
2. f is surjective or onto if $f(A) = B$.
3. f is bijective if it is 1-1 and onto.

If a function $f : A \to B$ is bijective, then $f^{-1} : B \to A$ is the function which assigns each $y \in B$ to the unique $x \in A$ such that $f(x) = y$. Note that $f(f^{-1}(x)) = x$.

Cardinality

Question 10. *When do two sets have the same **size**?*

Cantor answered this question in the 1800s, stating that two sets have the same size when you can pair each element in one set with a unique element in the other.

Definition 11 (Cardinality)
We state that two sets A and B have the same cardinality if there exists a bijection $f : A \to B$.

With this new concept comes some new notation:

1. $|A| = |B|$ if A and B have the same cardinality.
2. $|A| = n$ if $|A| = |\{1, \ldots, n\}|$. If this is the case we say A is finite.
3. $|A| \leq |B|$ if there exists an injection $f : A \to B$.
4. $|A| < |B|$ if $|A| \leq |B|$ but $|A| \neq |B|$.

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Theorem 12 (Cantor-Schröder-Bernstein)
If $|A| \leq |B|$ and $|B| \leq |A|$ then $|A| = |B|$.

If $|A| = |\mathbb{N}|$, then A is countably infinite. If A is finite or countably infinite, we say A is countable. Otherwise, we say A is uncountable.

Example 13
There are a few key theorems that we can prove with this new concept:

1. $|\{2n \mid n \in \mathbb{N}\}| = |\mathbb{N}|$.
2. $|\{2n-1 \mid n \in \mathbb{N}\}| = |\mathbb{N}|$.
3. $|\{x \in \mathbb{Q} \mid x > 0\}| = |\mathbb{N}|$.

The first two statements can be summarized by Feynman: "There are twice as many numbers as numbers."
Proof:

1. Define the function $f : \mathbb{N} \to \{2n \mid n \in \mathbb{N}\}$ as $f(n) = 2n$. Then, f is 1-1– if $f(n) = f(m)$ then $2n = 2m \implies n = m$. Furthermore, f is also onto, as if $m \in \{2n \mid n \in \mathbb{N}\}$ then $\exists n \in \mathbb{N}$ such that $m = 2n = f(n)$.
2. The second statement can be proven similarly.
3. This is left as an exercise to the reader in Assignment 1.

■

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3 Lecture 3

Cantor's Remarkable Theorem and the Rationals' Lack of the Least Upper Bound Property

Question 14. *Is anything bigger than* $\mathbb{N}$*?*

If A is a set then $\mathcal{P}(A) = \{B \mid B \subset A\}$. Here are a few examples:

1. $A = \emptyset$ then $\mathcal{P}(A) = \{\emptyset\}$.
2. $A = \{1\}$, then $\mathcal{P}(A) = \{\emptyset, \{1\}\}$.
3. $A = \{1, 2\}$, then $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

In general, if $|A| = n$ then $|\mathcal{P}(A)| = 2^n$. This is why we call $\mathcal{P}(A)$ the power set of A.

Theorem 15 (Cantor)
If A is a set, then $|A| < |\mathcal{P}(A)|$.

Remark 16. *Therefore,*

$$\mathbb{N} < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < \dots.$$

Hence, there are an infinite number of infinite sets.

Proof: Define the function $f : A \to \mathcal{P}(A)$ by $f(x) = \{x\}$. Then, f is 1-1– as if $\{x\} = \{y\} \implies x = y$. Thus, $|A| \leq |\mathcal{P}(A)|$. To finish the proof now all we need to show is that $|A| \neq |\mathcal{P}(A)|$. We will do so through contradiction. Suppose that $|A| = |\mathcal{P}(A)|$. Then, there exists a surjection $g : A \to \mathcal{P}(A)$. Let

$$B := \{x \in A \mid x \notin g(x)\} \in \mathcal{P}(A).$$

Since g is surjective, there exists a $b \in A$ such that $g(b) = B$. There are two cases:

1. $b \in B$. If this is the case, then $b \notin g(b) = B \implies b \notin B$.
2. $b \notin B$. If this is the case, then $b \notin g(b) = B \implies b \in B$.

In either case we obtain a contradiction. Thus, g is not surjective $\implies |A| \neq |\mathcal{P}(A)|$. □

Remark 17. *This is another proof method: casework. If the conclusion for every case is true, then the conclusion must be true.*

Corollary 18
For all $n \in \mathbb{N} \cup \{0\}$, $n < 2^n$.

Remark 19. *This can also be shown by induction, see Assignment 1.*

Real Numbers

Remark 20. *In a sense, to be made precise, the set of real numbers is the unique set with all of the algebraic and ordering properties of the rational numbers, but none of the holes.*

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Problem 21

Now let's try to precisely describe $\mathbb{R}$.

We will start by stating what our end result will be, and then we will derive it:

Theorem 22 (Real Numbers ($\mathbb{R}$))

There exists a unique **ordered field** containing $\mathbb{Q}$ with the **least upper bound property**. We denote this field by $\mathbb{R}$.

Ordered Sets & Fields

Definition 23 (Ordered set)

An ordered set is a set S with a relation $<$ called an "ordering" such that

1. $\forall x, y \in S$ either $x < y$, $y < x$, or $x = y$.
2. If $x < y$ and $y < z$ then $x < z$.

Here are a few examples and one non-example:

- $\mathbb{Z}$ is an ordered set, with the relation that $m > n \iff m - n \in \mathbb{N}$.
- $\mathbb{Q}$ is an ordered set, with the relation that $p > q \iff \exists m, n \in \mathbb{N}$ such that $p - q = \frac{m}{n}$.
- $\mathbb{Q} \times \mathbb{Q}$ is an ordered set with the relation $(q, r) > (s, t) \iff q > s$ or $q = s$ and $r > t$.
- Consider the set $\mathcal{P}(\mathbb{N})$. Let $A, B \in \mathcal{P}(\mathbb{N})$ and let $A \prec B$ if $A \subset B$. This is **NOT** an ordered set– it doesn't satisfy the first property of an ordered set.

Definition 24 (Bounded Above/Below)

Let S be an ordered set and let $E \subset S$. Then,

1. If there exists a $b \in S$ such that $x \leq b$ for all $x \in E$, then E is bounded above and b is an vocab of E.
2. If $\exists c \in S$ such that $x \geq c$ for all $x \in E$, then E is bounded below and c is a lower bound of E.

From here, there are some very important definitions in real analysis. We say that b_0 is the **least upper bound**, or the supremum of E if

A) b_0 is an upper bound for E and

B) if b is an upper bound for E then $b_0 \leq b$.

We denote this as $b_0 = \sup E$. Similarly, we say that c_0 is the **greatest lower bound**, or the infinimum of E if

A) c_0 is a lower bound for E and

B) if c is a lower bound for E then $c < c_0$.

We denote this as $c_0 = \inf E$.

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Example 25

Here are a few examples of infimums and supremums:

- $S = \mathbb{Z}$ and $E = \{-2, -1, 0, 1, 2\}$. Then, $\inf E = -2$ and $\sup E = 2$.
- But, note that the supremum nor the infimum need to be in E. Consider the sets $S = \mathbb{Q}$ and

$$E = \{q \in \mathbb{Q} \mid 0 \leq q < 1\}.$$

  Then, $\inf E = 0 \in E$, but $\sup E = 1 \notin E$.
- Furthermore, neither the supremum nor the infimum need exist. Consider the sets $S = \mathbb{Z}$ and $E = \mathbb{N}$. Then, $\inf E = 1$, but $\sup E$ does not exist as there is not an integer greater than all natural numbers.

Definition 26 (Least Upper Bound Property)

An ordered set S has the least upper bound property if every $E \subset S$ which is nonempty and bounded above has a supremum in S.

One example of such a set is

$$-\mathbb{N} = \{-1, -2. - 3. \ldots\}.$$

Then, $E \subset S$ is bounded above if and only if $-E \subset \mathbb{N}$ is bounded below. By the well-ordering principle, $-E$ has a least element $x \in -E$, and thus $-x = \sup E$.

We will now show that $\mathbb{Q}$ does not have the least upper bound property.

Theorem 27

If $x \in \mathbb{Q}$ and

$$x = \sup\{q \in \mathbb{Q} \mid q > 0, q^2 < 2\}$$

then $x > 0$ and $x^2 = 2$.

Proof: Let E equal the set on the right hand side, and suppose $x \in \mathbb{Q}$ such that $x = \sup E$. Then, since $1 \in E$ and x is an upper bound for E, $1 \leq x \implies x > 0$.

We now prove that $x^2 \geq 2$. Suppose that $x^2 < 2$. Define $h = \min\left\{\frac{1}{2}, \frac{2-x^2}{2(2x+1)}\right\} < 1$. Then, if $x^2 < 2$ then $h > 0$. We now prove that $x + h \in E$. Indeed,

$$\begin{aligned}(x+h)^2 &= x^2 + 2xh + h^2 \\ &< x^2 + h(2x+1)\end{aligned}$$

as $h < 1$. Hence

$$\begin{aligned}(x+h)^2 &\leq x^2 + (2 - x^2) \cdot \frac{2x+1}{2(2x+1)} \\ &= x^2 + \frac{2-x^2}{2} \\ &< 2 + \frac{2-2}{2} \\ &= 2.\end{aligned}$$

Therefore, $x + h \in E$ and $x + h > x \implies x$ is not an upper bound for E. Therefore, $x \neq \sup E$ which is a contradiction. Hence, $x^2 \geq 2$.

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We now prove that $x^2 \leq 2$. Suppose $x^2 > 2$. Let $h = \frac{x^2-2}{2x}$. Hence, if $x^2 > 2$ then $h > 0$ and $x - h > 0$. We will show that $x - h$ is an upper bound for E. We have

$$\begin{aligned}(x-h)^2 &= x^2 - 2xh + h^2 \\ &= x^2 - (x^2 - 2) + h^2 \\ &= 2 + h^2 \\ &> 2.\end{aligned}$$

Let $q \in E$. Then, $q^2 < 2 < (x-h)^2 \implies (x-h)^2 - q^2 > 0$. Hence,

$$((x-h)+q)((x-h)+q) > 0 \implies (x-h) - q > 0.$$

Thus, for all $q \in E$, $q < x - h < x \implies x \neq \sup E$. This is a contradiction. Therefore, $x^2 = 2$. □

Theorem 28

The set $E = \{q \in \mathbb{Q} \mid q > 0 \text{ and } q^2 < 2\}$ does not have a supremum in $\mathbb{Q}$.

Proof: Suppose there exists an $x \in \mathbb{Q}$ such that $x = \sup E$. Then, by our previous theorem, $x^2 = 2$. In particular, note that $x > 1$ as otherwise $x \leq 1 \implies 2 = x^2 < 1^2$. Thus, $\exists m, n \in \mathbb{N}$ such that $m > n$ and $x = \frac{m}{n}$. Therefore, $\exists n \in \mathbb{N}$ such that $nx \in \mathbb{N}$. Let

$$S = \{k \in \mathbb{N} \mid kx \in \mathbb{N}\}.$$

Then, $S \neq \emptyset$ since $n \in S$. By the well-ordering property of $\mathbb{N}$, S has a least element $k_0 \in S$. Let $k_1 = k_0 x - k_0 \in \mathbb{Z}$. Then, $k_1 = k_0(x-1) > 0$ since $k_0 \in \mathbb{N}$ and $x > 1$. Therefore, $k_1 \in \mathbb{N}$. Now $x^2 = 2 \implies x < 2$, as otherwise $x^2 > 4 > 2$. Thus, $k_1 = k_0(x-1) < k_0(2-1) = k_0$. So, $k_1 \in \mathbb{N}$ and $k_1 < k_0 \implies k_1 \notin S$ as k_0 is the least element of S. But,

$$xk_1 = k_0 x^2 - xk_0 = 2k_0 - xk_0 = k_0 - k_1 \in \mathbb{N} \implies k_1 \in S.$$

This is a contradiction. Thus, $\nexists x \in \mathbb{Q}$ such that $x = \sup E$. □

$\mathbb{Q}$ is an example of a field, which we will start to discuss in the next lecture.

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4 Lecture 4

The Characterization of the Real Numbers

Question 29. *Last time we stated that $\mathbb{Q}$ was an example of a field, but what is a* ***field****?*

Definition 30 (Field)
A set F is a field if it has two operations: addition $(+)$ and multiplication $(\cdot)$ with the following properties.

A1) If $x, y \in F$ then $x + y \in F$.

A2) *(Commutativity)* $\forall x, y \in F$, $x + y = y + x$.

A3) *(Associativity)* $\forall x, y, z \in F$, $(x + y) + z = x + (y + z)$.

A4) $\exists$ an element $0 \in F$ such that $0 + x = x = x + 0$.

A5) $\forall x \in F$, $\exists y \in F$ such that $x + y = 0$. We denote $y = -x$.

M1) If $x, y \in F$, then $x \cdot y \in F$.

M2) *(Commutativity)* $\forall x, y \in F$, $x \cdot y = y \cdot x$.

M3) *(Associativity)* $\forall x, y, z \in F$, $(x \cdot y) \cdot z = x \cdot (y \cdot z)$.

M4) $\exists 1 \in F$ such that $1 \cdot x = x = x \cdot 1$ for all $x \in F$.

M5) $\forall x \in F \setminus \{0\}$, $\exists x^{-1}$ such that $x \cdot x^{-1} = 1$.

D) *(Distributativity)* $\forall x, y, z \in F$, $(x + y)z = xz + yz$.

These may seem like trivial properties, but consider the following non-example: $\mathbb{Z}$. $\mathbb{Z}$ fails M5)– multiplicative inverses do not exist in the integers.

Example 31
Here are two examples of fields:

1. $\mathbb{Z}_2 = \{0, 1\}$ where $1 + 1 = 0$.
2. $\mathbb{Z}_3 = \{0, 1, 2\}$ with $c := a + b \pmod 3$. In other words,

$$2 + 1 = 3 = 0 \pmod 3 \quad \text{and} \quad 2 \cdot 2 = 4 = 3 + 1 = 1 \pmod 3.$$

Simple properties follow from the properties of a field!

Theorem 32
If $x \in F$ where F is a field, $0x = 0$.

Proof: If $x \in F$, then

$$0 = 0 \cdot x - 0 \cdot x = (0 + 0) \cdot x - 0 \cdot x = 0 \cdot x + 0 \cdot x - 0 \cdot x = 0 \cdot x.$$

□

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Definition 33 (Ordered field)

A field F is an ordered field if F is also an ordered set with ordering $<$ and

i) $\forall x, y, z \in F$, $x < y \implies x + z < y + z$.

ii) If $x > 0$ and $y > 0$ then $xy > 0$.

If $x > 0$ we say x is **positive**, and if $x \geq 0$ we say x is **non-negative**.

Example 34

$\mathbb{Q}$ is an ordered field.

A non-example would be $\mathbb{Z}_2$. For instance, consider $0 < 1$. If $0 < 1$, then $1 + 0 < 1 + 1 = 0 \implies 1 < 0$ which is a contradiction. If $1 < 0$, then $0 = 1 + 1 < 0 + 1 \implies 0 < 1$ which is a contradiction. Hence, $\mathbb{Z}_2$ is not an ordered field.

Using the definition of an ordered field, one can easily prove all of the usual facts about inequalities.

Theorem 35

If $x > 0$, then $-x < 0$ (and vice versa).

Proof: If $x \in F$ and $x > 0$, then by i),

$$-x + x > -x \implies 0 > -x.$$

□

One can see Proposition 1.1.8 **[L]** for a list of other simple inequality facts.

Theorem 36

Let $x, y \in F$ where F is an ordered field. If $x > 0$ and $y < 0$ or $x < 0$ and $y > 0$, then $xy < 0$.

Proof: Suppose $x > 0$ and $y < 0$. Then, $x > 0$ and $-y > 0$. Hence, $-xy = x(-y) > 0$. Thus, $xy < 0$. If $x < 0$ and $y > 0$, then $-x > 0$ and $y > 0 \implies -xy = (-x)y > 0 \implies xy < 0$. □

Question 37. *Is there a greatest lower bound property?*

For an ordered field F, if F has the least upper bound property then F has a greatest lower bound property.

Theorem 38

Let F be an ordered field with the least upper bound property. If $A \subset F$ is nonempty and bounded below, then $\inf A$ exists in F.

Proof: Suppose $A \subset F$ is nonempty and bounded below, i.e. $\exists a \in F$ such that $\forall x \in A$, $a \leq x$. Let $B = \{-x \mid x \in A\}$. Then, $\forall x \in A$, $-x \leq -a \implies -a$ is an upper bound for B. Since F has the least upper bound property, $\exists c \in F$ such that $c = \sup B$. Then, $\forall x \in A$, $-x \leq c \implies \forall x \in A$, $-c \leq x$. Hence, $-c$ is a lower bound for A. We have also shown that if a is a lower bound for A, then $-a$ is an upper bound for B. Therefore, $c \leq -a$ since $c = \sup B \implies a \leq -c$. Hence, $-c$ is the greatest lower bound for A. □

Real Numbers

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Theorem 39

There exists a "unique" ordered field, labeled $\mathbb{R}$, such that $\mathbb{Q} \subset \mathbb{R}$ and $\mathbb{R}$ has the least upper bound property.

One can construct $\mathbb{R}$ using Dedekind cuts or as equivalence classes of Cauchy sequences. (We will define Cauchy sequences later in the course.)

Theorem 40

$\exists! r \in \mathbb{R}$ such that $r > 0$ and $r^2 = 2$. In other words, $\sqrt{2} \in \mathbb{R}$ but $\sqrt{2} \notin \mathbb{Q}$.

Proof: Let $\tilde{E} = \{x \in \mathbb{R} \mid x > 0 \text{ and } x^2 < 2\}$. Then, since $\tilde{E}$ is bounded above (by 2 for instance), we have that $r := \sup \tilde{E}$ exists in $\mathbb{R}$. Then, one can show that $r > 0$ and $r^2 = 2$. This is left as an exercise.

We now prove uniqueness. Suppose that there is a $\tilde{r} > 0$ with $\tilde{r}^2 = 2$. Then, since $(r + \tilde{r}) > 0$,

$$0 = r^2 - \tilde{r}^2 = (r + \tilde{r})(r - \tilde{r}) \implies r - \tilde{r} = 0 \implies r = \tilde{r}.$$

□

Remark 41. *In Assignment 2 Exercise 7, you will show that* $\sqrt[3]{2} \in \mathbb{R}$.

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5 Lecture 5

The Archimedian Property, Density of the Rationals, and Absolute Value

For all $x, y \in \mathbb{R}$ and $x < y$, there exists an $r \in \mathbb{R}$ such that $x < r < y$ (take $r = \frac{x+y}{2}$).

Question 42. *Can we find $r \in \mathbb{Q}$ such that $x < r < y$?*

Theorem 43

The answer is yes!

i) (Archimedian Property) If $x, y \in \mathbb{R}$ and $x > 0$, then $\exists n \in \mathbb{N}$ such that $nx > y$.

ii) (Density of $\mathbb{Q}$) If $x, y \in \mathbb{R}$ and $x < y$ then $\exists r \in \mathbb{Q}$ such that $x < r < y$.

Proof:

i) Suppose that $x, y \in \mathbb{R}$ and $x > 0$. Then we wish to show that $\exists n \in \mathbb{N}$ such that $n > \frac{y}{x}$. Suppose this is not the case. Then, $\forall n \in \mathbb{N}$, $n \leq \frac{y}{x}$. In other words, $\mathbb{N}$ is bounded above by $\frac{y}{x}$. Hence, $\exists a = \sup \mathbb{N} \in \mathbb{R}$. Since a is the least upper bound for $\mathbb{N}$, $a - 1$ cannot be an upper bound for $\mathbb{N}$. Hence, $\exists m \in \mathbb{N}$ such that

$$a - 1 < m \implies a < m + 1 \in \mathbb{N}.$$

However, this is a contradiction, because then a is not an upper bound for $\mathbb{N}$. Therefore, $\exists n \in \mathbb{N}$ such that $n \geq \frac{y}{x}$.

ii) Suppose $x, y \in \mathbb{R}$ and $x < y$. Then, there are three cases:

- $0 \leq x < y$,
- $x < 0 < y$, and
- $x < y \leq 0$.

For the second case, take $r = 0 \in \mathbb{Q}$. So, assume that $0 \leq x < y$. Then, by the Archimedian Property, $\exists n \in \mathbb{N}$ such that $n(y - x) > 1$. Again by the Archimedean property, $\exists l \in \mathbb{N}$ such that $l > nx$. Thus, consider the set

$$S = \{k \in \mathbb{N} \mid k > nx\}.$$

By the well-ordering property of $\mathbb{N}$, S has a least element, $m \in S \implies nx < m \implies x < \frac{m}{n} \in \mathbb{Q}$.

Since $m - 1 \notin S$, $m - 1 \leq nx \implies m \leq nx + 1 < ny$. Hence, $\frac{m}{n} < y$. Therefore,

$$x < \frac{m}{n} < y.$$

If instead we have $x < y \leq 0$, then $0 \leq -y < -x \implies \exists \tilde{r} \in \mathbb{Q}$ such that

$$-y < \tilde{r} < x \implies x < -\tilde{r} < y$$

by the previous case.

□

Theorem 44

$1 = \sup\left\{1 - \frac{1}{n} \mid n \in \mathbb{N}\right\}$.

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Proof: If $n \in \mathbb{N}$, then $1 - \frac{1}{n} < 1 \implies 1$ is an upper bound of this set. Suppose that x is an upper bound for the set $\{1 - 1/n \mid n \in \mathbb{N}\}$. We now prove that $x \geq 1$. For the sake of contradiction, assume that $x < 1$. By the Archimedean property, there exists an $n \in \mathbb{N}$ such that $1 < n(1 - x)$. Therefore, $\exists n \in \mathbb{N}$ such that $x < 1 - 1/n$. Hence, x is not an upper bound for the set $\{1 - 1/n \mid n \in \mathbb{N}\}$ if $x < 1$. Thus, if x is an upper bound, $x \geq 1$. Therefore,

$$\sup\left\{1 - \frac{1}{n} \mid n \in \mathbb{N}\right\} = 1.$$

□

We now begin proving some theorems about supremums and infinimums which will make them easier to use.

Theorem 45

Suppose that $S \subset \mathbb{R}$ is nonempty and bounded above. Then, $x = \sup S$ if and only if

1. x is an upper bound for S.
2. for all $\epsilon > 0$, $\exists y \in S$ such that $x - \epsilon < y \leq x$.

Proof: This is left as an exercise in Assignment 3. □

Notation 46

For $x \in \mathbb{R}$ and $A \subset \mathbb{R}$, define

$$x + A := \{x + a \mid a \in A\}$$
$$xA := \{xa \mid a \in A\}.$$

Theorem 47

Using this new notation, we have the following theorems:

1. If $x \in \mathbb{R}$ and A is bounded above, then $x + A$ is bounded above and
$$\sup(x + A) = x + \sup A.$$
2. If $x > 0$ and A is bounded above then xA is bounded above and
$$\sup(xA) = x \sup A.$$

Proof:

1. Suppose that $x \in \mathbb{R}$ and A is bounded above. Therefore, $\sup A \in \mathbb{R}$ by the least upper bound property of $\mathbb{R}$. Then, $\forall a \in A$, $a \leq \sup A$. Hence,
$$\forall a \in A, \quad x + a \leq x + \sup A.$$
Hence, $x + \sup A$ is an upper bound for $x + A$. Let $\epsilon > 0$. Then, $\exists y \in A$ such that
$$\sup A - \epsilon < y \leq \sup A \implies (x + \sup A) - \epsilon < y + x \leq x + \sup A.$$
Therefore, by our previous theorem, $x + \sup A = \sup(x + A)$.
2. Suppose that $x > 0$ and A is bounded above. Thus, $\sup A \in \mathbb{R}$. Then, $\forall a \in A$, $a \leq \sup A$ and thus

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$xa \leq x \sup A$. Hence, $x \sup A$ is an upper bound of xA. Let $\epsilon > 0$. Then $\exists y \in A$ such that

$$\sup A - \frac{\epsilon}{x} < y \leq \sup A \implies x \sup A - \epsilon < xy \leq x \sup A.$$

Therefore, by the previous theorem, $\sup(xA) = x \sup A$.

□

Theorem 48

Let $A, B \subset \mathbb{R}$ such that $\forall x \in A, \forall y \in B$, $x \leq y$. Then, $\sup A \leq \inf B$.

Proof: The proof of this is left to the reader. □

Absolute Value

Definition 49

If $x \in \mathbb{R}$ we define

$$|x| := \begin{cases} x, & x \geq 0 \\ -x, & x \leq 0 \end{cases}.$$

Theorem 50

We can prove a bunch of theorems about the absolute value function that we usually take for granted:

1) $|x| \geq 0$ and $|x| = 0 \iff x = 0$.
2) $\forall x \in \mathbb{R}$, $|-x| = |x|$.
3) $\forall x, y \in \mathbb{R}$, $|xy| = |x||y|$.
4) $|x^2| = x^2 = |x|^2$.
5) If $x, y \in \mathbb{R}$, then $|x| \leq y \iff -y \leq x \leq y$.
6) $\forall x \in \mathbb{R}$, $x \leq |x|$.

Proof:

1) If $x \geq 0$ then $|x| = x \geq 0$. If $x \leq 0$, then $-x \geq 0 \implies |x| = -x \geq 0$. Thus, $|x| \geq 0$. Now suppose $x = 0$. Then, $|x| = x = 0$. For the other direction, suppose $|x| = 0$. Then, if $x \geq 0 \implies x = |x| = 0$. If $x \leq 0$, then $-x = |x| = 0$. Therefore, $x = 0 \iff |x| = 0$.
2) If $x \geq 0$ then $-x \leq 0$. Thus, $|x| = x = -(-x) = |-x|$. If $x \leq 0$ then $-x \geq 0$ and thus $|-x| = |-(-x)| = |x|$.
3) If $x \geq 0$ and $y \geq 0$, then $xy \geq 0$ and $|xy| = xy = |x||y|$. If $x \leq 0$ and $y \leq 0$, then

$$xy \leq 0 \implies |xy| = -xy = (-x)y = |x||y|.$$

4) Take $x = y$ in 3). Then, $|x^2| = |x|^2$. Since $x^2 \geq 0$, it follows that $|x^2| = x^2$.
5) Suppose $|x| \leq y$. If $x \geq 0$, then $-y \leq 0 \leq x = |x| \leq y$. Therefore, $-y \leq x \leq y$. If $x \leq 0$, then $-x \geq 0$ and $|-x| \leq y$. Hence, $-y \leq -x \leq y \implies -y \leq x \leq y$.
6) Take $y = |x|$ in 5).

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□

On its own, these properties of the absolute values may not seem all that useful, but in the next lecture we will prove the *extremely* important Triangle Inequality.

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6 Lecture 6

The Uncountabality of the Real Numbers

Theorem 51 (Triangle Inequality)
$\forall x, y \in \mathbb{R}$,

$$|x + y| \leq |x| + |y|.$$

Proof: Let $x, y \in \mathbb{R}$. Then, $x + y \leq |x| + |y|$ and

$$(-x) + (-y) \leq |-x| + |-y| = |x| + |y|.$$

Therefore, $-(|x| + |y|) \leq x + y \leq |x| + |y|$. Hence,

$$|x + y| \leq |x| + |y|$$

by our previous theorem. □

Remark 52. *We may denote the Triangle Inequality with $\triangle$-inequality as a shorthand.*

Question 53. *As we showed in Assignment 1, we know that $\mathbb{Q}$ is countable. Is the set of real numbers countable?*

Recall 54
Recall that a set A is countable if A is either finite or $|A| = |\mathbb{N}|$.

We can think of $\mathbb{Q}$ as decimal expansions. In other words, we can think of a rational number x as being in the form

$$x = 10^k d_k + \cdots + 10 d_1 + d_0 + 10^{-1} d_{-1} + \cdots + 10^{-M} d_{-M}$$

with $d_i \in \{0, 1, 2, 3, \ldots, 9\}$. We may write

$$x = d_k d_{k-1} \ldots d_1 d_0 \bullet d_{-1} \ldots d_{-M}$$

where $\bullet$ is the decimal point. The same can be said about real numbers if we allow for infinite decimal expansions.

Definition 55
Let $x \in (0, 1]$ and let $d_{-j} \in \{0, 1, \ldots, 9\}$. We say that x is **represented** by the digits $\{d_{-j} \mid j \in \mathbb{N}\}$, i.e. $x = 0 \bullet d_{-1} d_{-2} \ldots$, if

$$x = \sup\{10^{-1} d_{-1} + 10^{-2} d_{-2} + \cdots + 10^{-n} d_{-n} \mid n \in \mathbb{N}\}.$$

Here is an example: $.2500 = \sup\{2 \cdot 10^{-1}, 2 \cdot 10^{-1} + 5 \cdot 10^{-2}, 2 \cdot 10^{-1} + 5 \cdot 10^{-2} + 0 \cdot 10^{-3}, \ldots\}$. Notice here that after a while the previous set becomes $\frac{1}{4}$ repeating. Hence, we have $.2500 = \sup\left\{\frac{1}{5}, \frac{1}{4}\right\} = \frac{1}{4}$.

Theorem 56
For every $x \in (0, 1]$, there exists a unique sequence of digits d_{-j} such that $x = 0 \bullet d_{-1} d_{-2} \ldots$ and

$$0 \bullet d_{-1} d_{-2} \ldots d_{-n} < x \leq 0 \bullet d_{-1} d_{-2} \ldots d_{-n} + 10^{-n}.$$

Furthermore, for every set of digits $\{d_{-j} \mid j \in \mathbb{N}\}$, there exists a unique $x \in [0, 1]$ such that $x = 0 \bullet d_{-1} \ldots$.

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Notice however that the representative of $\frac{1}{2}$ is $0.4999\ldots$.

Theorem 57 (Cantor)
(0,1] is uncountable.

Proof: We will prove this through contradiction. Suppose that $(0,1]$ is countable. Therefore, there exists a bijection $x : \mathbb{N} \to (0,1]$. We now construct a $y \in (0,1]$ such that y is not in the range of x. We write

$$x(n) = 0 \bullet d_{-1}^{(n)} d_{-2}^{(n)} \ldots.$$

These are not exponents! This is the set of digits for a given $n \in \mathbb{N}$. In other words, x takes in a natural number n and maps it to the sequence of digits $\left\{ d_{-j}^{(n)} \mid n \in \mathbb{N} \right\}$. Let

$$e_{-j} = \begin{cases} 1, & d_{-j}^{(j)} \neq 1 \\ 2, & d_{-j}^{(j)} = 1 \end{cases}.$$

Let $y = 0 \bullet e_{-1} e_{-2} \ldots$. Then, $\forall n \in \mathbb{N}$,

$$0 \bullet e_{-1} e_{-2} \ldots e_{-n} \leq y 0 \bullet e_{-1} \ldots e_{-n} + 10^{-n}$$

since all e_{-j}s are positive. Thus, $0 \bullet e_{-1} \ldots$ is the unique decimal expansion of y. However, for all $n \in \mathbb{N}$, $d_{-n}^{(n)} \neq e_{-n}$. Therefore, $\forall n$, $x(n) \neq y$. This is a contradiction, and thus $(0,1]$ is uncountable. □

So $(0,1] \subset \mathbb{R}$ is uncountable!

Corollary 58
The set of real numbers, $\mathbb{R}$, is uncountable.

Sequences and Series

Remark 59. *Analysis is the study of limits.*

Sequences and Limits

Definition 60 (Sequence of Reals)
A sequence of real numbers is a function $x : \mathbb{N} \to \mathbb{R}$. We denote $x(n) = x_n$ and we denote the sequence by $\{x_n\}_{n=1}^{\infty}$, $\{x_n\}$, or $x_1, x_2, \ldots$.

Definition 61
A sequence $\{x_n\}$ is bounded if $\exists B \geq 0$ such that $\forall n$, $|x_n| \leq B$.

One example of a bounded sequence is $x_n = \frac{1}{n}$, since $\left|\frac{1}{n}\right| \leq 1$ for all $n \in \mathbb{N}$. However, $x_n = n$ is not bounded.

Remark 62. *A sequence is different from a set!*

For example,

$$-1, 1, -1, 1, \cdots = \{(-1)^n\}_{n=1}^{\infty},$$

while

$$\{(-1)^n \mid n \in \mathbb{N}\} = \{-1, 1\}.$$

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Definition 63 (Sequence Convergence of Reals)
A sequence $\{x_n\}$ converges to $x \in \mathbb{R}$ if $\forall \epsilon > 0$, $\exists M \in \mathbb{N}$ such that $\forall n \geq M$,

$$|x_n - x| < \epsilon.$$

A sequence that converges is said to be **convergent**, and otherwise is said to be **divergent**. We can also define divergence as the negation of convergent.

Negation 64 (Not Convergent)
The sequence $\{x_n\}$ is not convergent, or divergent if $\exists \epsilon_0 > 0$ such that $\forall M \in \mathbb{N}$, $\exists n \geq M$ so that

$$|x_n - x| \geq \epsilon_0.$$

We now prove two theorems:

Theorem 65
If $\{x_n\}$ converges for x and y, then $x = y$. In other words, limits of convergent sequences of real numbers are unique.

Theorem 66
Let $x, y \in \mathbb{R}$. If $\forall \epsilon > 0$, $|x - y| < \epsilon$, then $x = y$.

Proof: We first prove the second theorem. Suppose that $x \neq y$. Then, $|x - y| > 0$. Hence, choosing $\epsilon = \frac{|x-y|}{2}$, we have

$$|x - y| \leq \frac{|x - y|}{2} \implies \frac{|x - y|}{2} < 0$$

which is a contradiction. □

Using this we prove the former theorem. Suppose x_n converges to x and to y. We will show that for all $\epsilon > 0$, $|x - y| < \epsilon$. Firstly, given $x_n \to x$, for $\epsilon > 0$ there exists an $N_1 \in \mathbb{N}$ such that $\forall n \geq N_1$,

$$|x_n - x| < \frac{\epsilon}{2}.$$

Then, given $x_n \to y$, for $\epsilon > 0$ there exists an $N_2 \in \mathbb{N}$ such that $\forall n \geq N_2$,

$$|x_n - y| < \frac{\epsilon}{2}.$$

Let $N = \max\{N_1, N_2\}$. Then, for all $\epsilon > 0$ there exists an $N = \max\{N_1, N_2\}$ such that for all $n \geq N$,

$$|x - y| \leq |x - x_n| + |x_n - y| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

by the Triangle Inequality. Hence, for all $\epsilon > 0$, $|x - y| < \epsilon$. Therefore, $x = y$. □

Notation 67
We write $x = \lim_{n \to \infty} x_n$ or $x_n \to x$.

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Example 68

Given the sequence $x_n = c \ \forall n$, $\lim_{n\to\infty} x_n = c$.

Proof: Let $\epsilon > 0$ and $M = 1$. Thus, for all $n \geq 1$,

$$|x_n - c| = |c - c| = 0 < \epsilon.$$

■

Example 69

$\lim_{n\to\infty} \frac{1}{n} = 0$.

Proof: Let $\epsilon > 0$. Choose $M \in \mathbb{N}$ such that $M^{-1} > \epsilon^{-1}$. Hence, for all $n \geq M$, $\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{M} \leq \epsilon$. ■

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7 Lecture 7

Convergent Sequences of Real Numbers

We will do another example of limits that converge:

Example 70
$\lim_{n\to\infty} \frac{1}{n^2+2n+100} = 0$.

Proof: Let $\epsilon > 0$ and choose $M \in \mathbb{N}$ such that $M > \frac{\epsilon^{-1}}{2}$. Then, $\forall n \geq M$,

$$\left|\frac{1}{n^2+2m+100} - 0\right| \leq \frac{1}{n^2+2m+100} \leq \frac{1}{2n} \leq \frac{1}{2M} < \epsilon.$$

■

The fact that we can go from a complicated rational function to one that works for our purposes (namely to prove the sequence converges to 0) is *awesome*.

Example 71
Consider the sequence $x_n = (-1)^n$. This sequence is divergent.

Proof: Let $x \in \mathbb{R}$. We claim $\lim_{n\to\infty}(-1)^n \neq x$. To prove this, we simply need find *an* epsilon that stops the sequence from converging. For instance, consider $\epsilon_0 = \frac{1}{2}$. Then, for $M \in \mathbb{N}$,

$$1 = |(-1)^M - (-1)^{M+1}| \leq |(-1)^M - x| + |(-1)^{M+1} - x|.$$

Thus, either $|(-1)^M - x| \geq \frac{1}{2}$ or $|(-1)^{M+1} - x| \geq \frac{1}{2}$. In either case, this shows that the limit cannot converge to x. ■

Theorem 72
If $\{x_n\}$ is convergent, then $\{x_n\}$ is bounded.

Before we start the proof, let's first talk about the idea of the proof. Let $\epsilon = 1$ such that $|x_n - x| < 1$ for all $n \leq M$ for some $M \in \mathbb{N}$. Then, there are finitely many elements not in the interval $(x-1, x+1)$. We use this to our advantage.

Proof: Suppose that $\lim_{n\to\infty} x_n = x$. Thus, there exists an $M \in \mathbb{N}$ such that $|x_n - x| < 1$ for all $n \geq M$. Let

$$B = \max\{|x_1|, |x_2|, \ldots, |x_{M-1}|, |x| + 1\}.$$

If $n < M$, then $|x_n| \leq B$ by construction. If $n \geq M$, then

$$|x_n| \leq |x_n - x| + |x| < 1 + |x| \leq B.$$

□

Definition 73 (Monotone)
A sequence $\{x_n\}$ is monotone increasing if $\forall n \in \mathbb{N}$, $x_n \leq x_{n+1}$. A sequence $\{x_n\}$ is monotone decreasing if $\forall n \in \mathbb{N}$, $x_n \geq x_{n+1}$. If $\{x_n\}$ is either monotone increasing or monotone decreasing, we say $\{x_n\}$ is monotone or **monotonic.**

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Example 74

For example, $x_n = \frac{1}{n}$ is monotone, $y_n = -\frac{1}{n}$ is monotone increasing, and $(-1)^n$ is neither.

Theorem 75

Let $\{x_n\}$ be a monotone increasing sequence. Then, $\{x_n\}$ is convergent if and only if $\{x_n\}$ is bounded. Moreover, $\lim_{n\to\infty} x_n = \sup\{x_n \mid n \in \mathbb{N}\}$.

Proof: Firstly, we know that if $\{x_n\}$ is convergent then it is bounded by the previous theorem, Now assume that $\{x_n\}$ is bounded. Then, $x := \sup\{x_n \mid n \in \mathbb{N}\}$ exists in $\mathbb{R}$ by the lowest upper bound property of $\mathbb{R}$. We now prove that

$$\lim_{n\to\infty} x_n = x.$$

Let $\epsilon > 0$. Then, $\exists M_0 \in \mathbb{N}$ such that

$$x - \epsilon < x_{M_0} < x$$

since x is the supremum of the set. Let $M = M_0$. Then, $\forall n \geq M$, we have

$$x - \epsilon < x_{M_0} = x_M \leq x_n \leq x < x + \epsilon.$$

Therefore, $|x_n| < |x + \epsilon|$. Therefore, $x_n \to x$. □

Theorem 76

Let $\{x_n\}$ be a monotone decreasing function. Then, $\{x_n\}$ is convergent if and only if $\{x_n\}$ is bounded. Moreover,

$$\lim_{n\to\infty} x_n = \inf\{x_n \mid n \in \mathbb{N}\}$$

The proof of this is similar to the previous theorem and is thus omitted.

Definition 77 (Subsequence)

Informally, a subsequence is a sequence with entries coming from another given sequence. In other words, let $\{x_n\}$ be a sequence and let $\{n_k\}$ be a strictly increasing sequence of natural numbers. Then the sequence

$$\{x_{n_k}\}_{k=1}^{\infty}$$

is called a subsequence of $\{x_n\}$.

Consider the sequence $\{x_n\} = n$ – in other words, the sequence $1, 2, 3, 4, \ldots$. Then, the following are subsequences of x_n :

$$1, 3, 5, 7, 9, 11, \ldots$$
$$2, 4, 6, 8, 10, \ldots$$
$$2, 3, 5, 7, 11, 13, \ldots.$$

The first two are described by $x_{n_k} = x_{2k}$ and $x_{n_k} = x_{2k-1}$ respectively.

Question 78. *How would we describe the third?*

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Continuing to let $\{x_n = n\}_n$, the following are *not* subsequences:

$$1, 1, 1, 1, 1, 1, \dots$$
$$1, 1, 3, 3, 5, 5, \dots.$$

Now consider the sequence $\{(-1)^n\}$. Then we have the subsequences

$$x_{n_k} = x_{2k-1} \to -1, -1, -1, \dots$$
$$y_{n_k} = x_{2k} \to 1, 1, 1, \dots.$$

Theorem 79
If $\{x_n\}$ converges to x, then any subsequence of x_n will converge to x.

Proof: Suppose $\lim_{n\to\infty} x_n = x$. Let $\epsilon > 0$. Then, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$,

$$|x_n - x| < \epsilon.$$

Choose $M = M_0$. If $k \geq M$, then $n_k \geq k \geq M = M_0$. Hence, for all $\epsilon > 0$ there exists an M such that for all $n_k > M$,

$$|x_{n-k} - x| < \epsilon.$$

□

Remark 80. *Notice that this also implies that the sequence $\{(-1)^n\}_n$ is divergent.*

Notation 81 (DNC)
We can denote the statement "a sequence does not converge"/"a sequence is divergent" as "the sequence DNC".

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8 Lecture 8

The Squeeze Theorem and Operations Involving Convergent Sequences

Facts About Limits

Theorem 82 (Squeeze Theorem)
Let $\{a_n\}$, $\{b_n\}$, and $\{x_n\}$ be sequences such that $\forall n \in \mathbb{N}$,

$$a_n \leq x_n \leq b_k.$$

Suppose that $\{a_n\}$ and $\{b_n\}$ converge and

$$\lim_{n\to\infty} a_n = x = \lim_{n\to\infty} b_n.$$

Therefore, $\{x\}$ converges and $\lim_{n\to\infty} x_n = x$.

Remark 83. *We sometimes abbreviate the Squeeze Theorem to ST.*

Proof: Let $\epsilon > 0$. Since $\lim_{n\to\infty} a_n = x$, there exists an $M_0 \in \mathbb{N}$ such that for all $n \geq M_0$,

$$|a_n - x| < \epsilon \implies x - \epsilon < a_n.$$

Since $\lim_{n\to\infty} b_n = x$, $\exists M_1 \in \mathbb{N}$ such that $\forall n \geq M_1$,

$$|b_n - x| < \epsilon \implies b_n < x + \epsilon.$$

Choose $M = \max\{M_0, M_1\}$. Then, if $n \geq M$, then

$$x - \epsilon < a_n \leq x_n \leq b_n < x - \epsilon \implies |x_n - x| < \epsilon.$$

Therefore, $\{x_n\}$ is convergent and $\lim_{n\to\infty} x_n = x$. □

Theorem 84
Another way to check that a sequence $x_n \to x$, is stated below:

$$\lim_{n\to\infty} x_n = x \iff \lim_{n\to\infty} |x_n - x| = 0.$$

Hence, we can consider a sequence like the following:

Example 85
Show that

$$\lim_{n\to\infty} \frac{n^2}{n^2+n+1} = 1.$$

Proof: We have

$$\left|\frac{n^2}{n^2+n+1} - 1\right| = \left|\frac{-n-1}{n^2+n+1}\right| = \frac{n+1}{n^2+n+1} \leq \frac{n+1}{n^2+n} = \frac{1}{n}.$$

Thus,

$$0 \leq \left|\frac{n^2}{n^2+n+1} - 1\right| \leq \frac{1}{n} \to 0 \implies \lim_{n\to\infty} \left|\frac{n^2}{n^2+n+1} - 1\right| = 0$$

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by the Squeeze Theorem. ■

Question 86. *How do limits interact with ordering?*

Theorem 87

Let $\{x_n\}$ and $\{y_n\}$ be sequences of real numbers. Then,

1. if $\{x_n\}$ and $\{y_n\}$ are convergent sequences and $\forall n \in \mathbb{N}\ x_n \leq y_n$, then $\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n$.
2. if $\{x_n\}$ is a convergent sequence and $\forall n \in \mathbb{N}\ x \leq x_n \leq b_n$ then $a \leq \lim_{n\to\infty} x_n \leq b$.

Proof:

1. Let $x = \lim_{n\to\infty} x_n$ and $y = \lim_{n\to\infty} y_n$. Suppose for the sake of contradiction that $y < x$. Then, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$
$$|y_n - y| < \frac{x-y}{2}$$
And $\exists M_1 \in \mathbb{N}$ such that for all $n \geq M_1$,
$$|x_n - x| < \frac{x-y}{2}.$$
Then, if $M = M_0 + M_1 \geq \max\{M_0, M_1\}$,
$$y_M < \frac{x-y}{2} + y = \frac{x+y}{2} = x - \frac{x-y}{2} + x < x_M.$$
However, this would imply that $y_M < x_M$ which contradicts $\forall n \in \mathbb{N} x_n \leq y_n$.
2. Apply part 1 to proof part 2, by considering $y_n = a \leq x_n \leq b = z_n$ for all $n \in \mathbb{N}$.

□

Question 88. *How do limits interact with algebraic operations?*

Theorem 89

Suppose $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$. Then,

1. $\{x_y + y_n\}_n$ is convergent and $\lim_{n\to\infty}(x_n + y_n) = x + y$.
2. $\forall c \in \mathbb{R}$, $\{cx_n\}_n$ is convergent and $\lim_{n\to\infty} cx_n = cx$.
3. $\{x_n \cdot y_n\}$ is convergent and $\lim_{n\to\infty} x_n y_n = xy$.
4. If $\forall n \in \mathbb{N}$, $y_n \neq 0$ and $y \neq 0$, then $\{x_n/y_n\}_n$ is convergent and
$$\lim_{n\to\infty} \frac{x_n}{y_n} = \frac{x}{y}.$$

Proof:

1. Let $\epsilon > 0$. Then, since $x_n \to x$, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$, $|x_n - x| < \frac{\epsilon}{2}$. Since $y_n \to y$, $\exists M_1 \in \mathbb{N}$ such that $\forall n \geq M_1$, $|y_n - y| < \frac{\epsilon}{2}$. Hence, letting $M = \max\{M_0, M_1\}$, we get for all $n \geq M$,
$$|x_n + y_n - (x+y)| \leq |x_n - x| + |y_n - y| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

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2. Let $\epsilon > 0$. Since $x_n \to x$, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$, $|x_n - x| < \frac{\epsilon}{|c|+1}$. Let $M = M_0$. Then, $\forall n \geq M$,

$$|cx_n - cx| = |c||x_n - x| \leq \frac{|c|}{|c|+1} \cdot \epsilon < \epsilon$$

since $\frac{|c|}{|c|+1} < 1$.

3. Since $y_n \to y$, $\{y_n\}$ is bounded. In other words, $\exists B \geq 0$ such that $\forall n \in \mathbb{N}$, $|y_n| \leq B$. Then,

$$\begin{aligned} |x_n y_n - xy| &= |(x_n - x)y_n + (y_n - y)x| \\ &\leq |x_n - x||y_n + |x||y_n - y| \\ &\leq B|x_n - x| + |x||y_n - y|. \end{aligned}$$

Therefore, $0 \leq |x_n y_n - xy| \leq B|x_n - x| + |x||y_n - y|$. Since $B|x_n - x| + |x||y_n - y| \to 0$, by the Squeeze Theorem $\lim_{n\to\infty} |x_n y_n - xy| = 0$.

4. We prove $\frac{1}{y_n} \to \frac{1}{y}$. We first prove $\exists b > 0$ such that $\forall n \in \mathbb{N}$, $|y_n| \geq b$. Since $y_n \to y$ and $y \neq 0$, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$,

$$|y_n - y| < \frac{|y|}{2}.$$

By the Triangle Inequality, $\forall n \geq M_0$,

$$|y| \leq |y_n - y| + |y_n| \leq \frac{|y|}{2} + |y_n| \implies |y_n| \geq \frac{|y|}{2}.$$

Let $b = \min\left\{|y_1|, \ldots, |y_{M_0-1}|, \frac{|y|}{2}\right\}$. Then, $\forall n \in \mathbb{N}$, $|y_n| \geq b$. Therefore,

$$0 \leq \left|\frac{1}{y_n} - \frac{1}{y}\right| = \frac{|y_n - y|}{|y_n||y|} \leq \frac{1}{b|y|}|y_n - y|.$$

By the Squeeze Theorem, $\lim_{n\to\infty} \left|\frac{1}{y_n} - \frac{1}{y}\right| = 0$. Therefore, $\lim_{n\to\infty} \frac{1}{y_n} = \frac{1}{y}$. Furthermore, by the proof before this (3.), it follows that $\lim_{n\to\infty} \left(x_n \cdot \frac{1}{y_n}\right) = \frac{x}{y}$.

□

Remark 90. *By induction, one can prove that*

$$\lim_{n\to\infty} (x_n)^k = x^k.$$

Theorem 91

If $\{x_n\}$ is a convergent sequence such that $\forall n \in \mathbb{N}$, $x_n \geq 0$, then $\{\sqrt{x_n}\}$ is convergent and

$$\lim_{n\to\infty} \sqrt{x_n} = \sqrt{\lim_{n\to\infty} x_n}.$$

Proof: Let $x = \lim_{n\to\infty} x_n$.

Case 1: $x = 0$. Let $\epsilon > 0$. Then, since $x_n \to 0$, there exists an $M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$, $x_n = |x_n - 0| < \epsilon^2$. Choose $M = M_0$. Then, $\forall n \geq M$,

$$|\sqrt{x_n} - \sqrt{0}| = \sqrt{x_n} < \sqrt{\epsilon^2} = \epsilon.$$

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Case 2: $x > 0$. We have $\forall n \in \mathbb{N}$,

$$\begin{aligned}|\sqrt{x_n} - \sqrt{x}| &= \left|\frac{\sqrt{x_n} - \sqrt{x}}{\sqrt{x_n} + \sqrt{x}} \cdot (\sqrt{x_n} + \sqrt{x})\right| \\ &= \frac{1}{\sqrt{x_n} + \sqrt{x}}|x_n - x| \\ &\leq \frac{1}{\sqrt{x}}|x_n - x|.\end{aligned}$$

Hence,

$$0 \leq |\sqrt{x_n} - \sqrt{x}| \leq \frac{1}{\sqrt{x}}|x_n - x|$$

$\forall n \in \mathbb{N}$. Hence, by the Squeeze Theorem,

$$\lim_{n\to\infty} |\sqrt{x_n} - \sqrt{x}| = 0.$$

□

Remark 92. *Why must we do casework in the above proof?*

Theorem 93
If $\{x_n\}$ is convergent and $\lim_{n\to\infty} x_n = x$, then $\{|x_n|\}$ is convergent and $\lim_{n\to\infty} |x_n| = |x|$.

Proof: Firstly, note that $\forall x \in \mathbb{R}$, $\sqrt{x^2} = |x|$. Then,

$$\lim_{n\to\infty} |x_n| = \lim_{n\to\infty} \sqrt{x_n^2} = \sqrt{x^2} = |x|$$

by the previous theorem. □

Theorem 94
If $c \in (0, 1)$, then $\lim_{n\to\infty} c^n = 0$. If $c > 1$, then $\{c^n\}$ is unbounded.

Proof: If $0 < c < 1$, we claim that $\forall n \in \mathbb{N}$, $0 < c^{n+1} < c^n < 1$. We can prove this through induction. Firstly, notice that $0 < c^2 < c < 1$ since $c > 0$ and $c < 1$. Now assume that $0 < c^{m+1} < c^m$. Then, multiply by $c > 0$ to obtain

$$0 < c^{m+1} \cdot c = c^{(m+1)+1} < c^m \cdot c = c^{(m+1)}.$$

By induction, our claim holds. Thus, $\{c^n\}$ is a monotone decreasing sequence and is bounded below. Thus, $\{c^n\}$ is convergent. Let $L = \lim_{n\to\infty} c^n$. We will prove that $L = 0$. Let $\epsilon > 0$. Then, $\exists M \in \mathbb{N}$ such that $\forall n \geq M$, $|c^n - L| < (1 - c)\frac{\epsilon}{2}$. Therefore,

$$\begin{aligned}(1 - c)|L| = |L - cL| &= |L - c^{M+1} + c^{M+1} - cL| \\ &\leq |L - c^{M+1}| + c|c^M - L| \\ &< (1 - c)\frac{\epsilon}{2} + c(1 - c)\frac{\epsilon}{2} < (1 - c)\epsilon.\end{aligned}$$

Therefore, $\forall \epsilon > 0$, $|L| < \epsilon \implies L = 0$.

Now let $c > 1$. We have to show that $\forall B \geq 0$, $\exists n \in \mathbb{N}$ such that $c^n > B$. Let $B \geq 0$. Choose $n \in \mathbb{N}$ such that $n > \frac{B}{c-1}$. Then,

$$c^n = (1 + (1 - c))^n \geq 1 + n(c - 1) \geq n(c - 1) > B.$$

To see why this center inequality is true, see the last theorem shown in Lecture 1. □

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9 Lecture 9

Limsup, Liminf, and the Bolzano-Weierstrass Theorem

Theorem 95 (Some Special Sequences)
What follows are some special sequences to have in our toolbox.

1. If $p > 0$, then $\lim_{n\to\infty} n^{-p} = 0$.
2. If $p > 0$ then $p^{\frac{1}{n}} = 1$.
3. $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$.

Proof:

1. Let $\epsilon > 0$. Then, choose $M > (1/\epsilon)^{1/p}$. Hence, if $n \geq M$,
$$\left|\frac{1}{n^p} - 0\right| = \frac{1}{|n^p|} \leq \frac{1}{M^p} < \epsilon.$$
2. Suppose $p > 1$. Then, $p^{1/n} - 1 > 0$ which may be proven by induction. Furthermore, we have
$$\begin{aligned} p &= (1 + (p^{1/n} - 1))^n \\ &\geq 1 + n(p^{1/n} - 1). \end{aligned}$$
Therefore, $0 < p^{1/n} - 1 \leq \frac{p-1}{n}$. Hence, we may apply the Squeeze Theorem, obtaining $\lim_{n\to\infty} |p^{1/n} - 1| = 0$. If $p < 1$, then
$$\lim_{n\to\infty} p^{1/n} = \lim_{n\to\infty} \frac{1}{(1/p)^{1/n}} = \frac{1}{1} = 1.$$
Furthermore, if $p = 1$ then it is clear that $\lim_{n\to\infty} p^{1/n} = 1$. Hence, in all cases, the limit is 1.
3. Let $x_n = n^{1/n} - 1 \geq 0$. We want to show that $\lim_{n\to\infty} x_n = 0$, as this will imply the end result. Notice that
$$n = (1 + x_n)^n = \sum_{j=0}^{n} \binom{n}{j} x_n^j \geq \binom{n}{2} x_n^2 = \frac{n!}{2(n-2)!} \cdot x_n^2 = \frac{n(n-1)}{2} \cdot x_n^2.$$
Thus, for $n > 1$,
$$0 \leq x_n \leq \sqrt{\frac{2}{n-1}} \implies x_n \to 0.$$

□

Limsup/Liminf

Question 96. *Does a bounded sequence have a convergent subsequence?*

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Definition 97 (Limsup/Liminf)

Let $\{x_n\}$ be a bounded sequence. We define, if the limits exist,

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} (\sup\{x_k \mid k \geq n\})$$
$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} (\inf\{x_k \mid k \geq n\}).$$

These are called the limit superior and limit inferior respectively.

We will now show that these limits always exist.

Theorem 98

Let $\{x_n\}$ be a bounded sequence, and let

$$a_n = \sup\{x_k \mid k \geq n\}$$
$$b_n = \inf\{x_k \mid k \geq n\}.$$

Then,

1. $\{a_n\}$ is monotone decreasing and bounded, and $\{b_n\}$ is monotone increasing and bounded.
2. $\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n$.

Proof

1. Since, $\forall \in \mathbb{N}$,
$$\{x_k \mid k \geq n+1\} \subseteq \{x_k \mid k \geq n\},$$
we have that $a_{n+1} = \sup\{x_k \mid k \geq n+1\} \leq \sup\{x_k \mid k \geq n\} = a_n$.
Similarly, $\forall n \in \mathbb{N}$, $b_{n+1} \geq b_n$. Given $\{x_n\}$ is a bounded sequence, $\exists B \geq 0$ such that $\forall n \in \mathbb{N}$,
$$-B \leq x_n \leq B.$$
Therefore, $\forall n \in \mathbb{N}$,
$$-B \leq b_n \leq a_n \leq B$$
which implies both sequences are bounded.
2. By the above equation, $\forall n \in \mathbb{N}$, $b_n \leq a_n \implies \liminf_{n\to\infty} x_n = \lim_{n\to\infty} b_n \leq \lim_{n\to\infty} a_n = \limsup_{n\to\infty} x_n$.

□

Let's consider a few examples.

Example 99

Let $x_n = (-1)^n$. Calculate the lim inf and lim sup of this sequence.

Proof: Notice that $\{(-1)^k \mid l \geq n\} = \{-1, 1\}$. Thus, the supremum of these sets is always 1 and the infimum is always -1. Therefore,

$$\limsup_{n\to\infty} x_n = 1 \quad \text{and} \quad \liminf_{n\to\infty} x_n = -1.$$

■

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Example 100

Let $x_n = \frac{1}{n}$. Calculate the lim inf and lim sup of this sequence.

Proof: We may do this directly:

$$\sup\{1/k \mid k \geq n\} = \frac{1}{n} \to 0 \implies \limsup_{n\to\infty} x_n = 0.$$
$$\inf\{1/k \mid k \geq n\} = 0 \to 0 \implies \liminf_{n\to\infty} x_n = 0.$$

■

The limit inferior and the limit superior allow us to answer the question posed at the beginning of this section.

Theorem 101

Let $\{x_n\}$ be a bounded sequence. Then, there exists subsequences $\{x_{n_k}\}$ and $\{x_{m_k}\}$ such that

$$\lim_{k\to\infty} x_{n_k} = \limsup_{n\to\infty} x_n$$
$$\lim_{k\to\infty} x_{m_k} = \limsup_{n\to\infty} x_n.$$

Proof: Let $a_n = \sup\{x_k \mid k \geq n\}$. Then, $\exists n_1 \in \mathbb{N}$ such that $a_1 - 1 < x_{n_1} \leq a_1$. Now, $\exists n_2 > n_1$ such that

$$a_{n_1+1} - \frac{1}{2} < x_{n_2} \leq a_{n_1+1}$$

since

$$a_{n+1} = \sup\{x_k \mid k \geq n_1 + 1\}.$$

Similarly, $\exists n_3 > n_2$ such that

$$a_{n_2+1} - \frac{1}{3} < x_{n_3} \leq a_{n_2+1}.$$

Continuing in this way, we obtain a sequence of integers $n_1 < n_2 < n_3 < \dots$ such that

$$a_{n_k+1} - \frac{1}{k+1} < x_{n_k} \leq a_{n_k+1}.$$

Given $\lim_{k\to\infty} a_{n_k+1} = \limsup_{n\to\infty} x_n$, by the Squeeze Theorem,

$$\lim_{k\to\infty} x_{n_k} = \limsup_{n\to\infty} x_n.$$

The direction for the lim inf works out the same way so that portion of the proof is left to the reader. □

Theorem 102 (Bolzano-Weierstrass)

Every bounded sequence has a convergent subsequence.

Remark 103. *We may abbreviate the Bolzano-Weierstrass theorem to B-W.*

Proof: This follows immediately from the previous theorem, but is so important that it itself is a theorem. □

Notation 104

When it is clear, we may have the following notational shorthand: $\liminf_{n\to\infty} x_n := \liminf x_n$, and $\limsup_{n\to\infty} x_n := \limsup x_n$.

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Theorem 105

Let $\{x_n\}$ be a bounded sequence. Then, $\{x_n\}$ converges if and only if $\liminf x_n = \limsup x_n$.

Proof ($\Longleftarrow$) Suppose $\liminf x_n = \limsup x_n$. Then, $\forall n \in \mathbb{N}$,

$$\inf\{x_k \mid k \geq n\} \leq x_n \leq \sup\{x_k \mid k \geq n\}.$$

By the Squeeze Theorem, since $\lim_{k\to\infty} \inf\{x_k \mid k \geq n\} = \lim_{k\to\infty} \sup\{x_k \mid k \geq n\}$ by assumption, we have

$$\lim_{n\to\infty} x_n = \liminf x_n = \limsup x_n.$$

Therefore, x_n converges.

($\Longrightarrow$) Let $x = \lim_{n\to\infty} x_n$. Therefore, every subsequence of $\{x_n\}$ converges to x, so $\liminf x_n = x$ and $\limsup x_n = x$ by a theorem we proved in Lecture 7. Hence, $\liminf x_n = \limsup x_n$. □

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10 Lecture 10

The Completeness of the Real Numbers and Basic Properties of Infinite Series

Cauchy Sequences

Definition 106

Cauchy A sequence $\{x_n\}$ is Cauchy if $\forall \epsilon > 0$ $\exists M \in \mathbb{N}$ such that for all $n, k \geq M$,

$$|x_n - x_k| < \epsilon.$$

Example 107

Show the sequence $x_n = \frac{1}{n}$ is Cauchy.

Proof: Let $\epsilon > 0$ and choose $M \in \mathbb{N}$ such that $\frac{1}{M} < \frac{\epsilon}{2}$. Then, if $n, k \geq M$, then

$$\left|\frac{1}{n} - \frac{1}{k}\right| \leq \frac{1}{n} + \frac{1}{k} \leq \frac{2}{M} < \epsilon.$$

■

Negation 108 (Not Cauchy)

By the negation of the definition, a sequence $\{x_n\}$ is not Cauchy if $\exists \epsilon_0 > 0$ such that for all $M \in \mathbb{N}$, $\exists n, k \geq M$ such that $|x_n - x_k| \geq \epsilon_0$.

Example 109

Show the sequence $x_n = (-1)^n$ is not Cauchy.

Proof: Choose $\epsilon = 1$ and let $M \in \mathbb{N}$. Choose $n = M$ and $k = M + 1$. Then,

$$|(-1)^n - (-1)^k| = 2 \geq 1.$$

■

Theorem 110

If $\{x_n\}$ is Cauchy, then $\{x_n\}$ is bounded.

Proof: If $\{x_n\}$ is Cauchy then $\exists M \in \mathbb{N}$ such that for all $n, k \geq M$,

$$|x_n - x_k| < 1.$$

Then, for all $n \geq M$, $|x_n - x_M| < 1$. Hence,

$$|x_n| \leq |x_n - x_M| + |x_M| < |x_M| + 1.$$

Let $B = |x_1| + \cdots + |x_M| + 1$. Then, for all $n \in \mathbb{N}$, $|x_n| \leq B$. □

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Theorem 111
If $\{x_n\}$ is Cauchy and a subsequence $\{x_{n_k}\}$ converges, then $\{x_n\}$ converges.

Proof: Suppose that $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$ such that $\lim_{k\to\infty} x_{n_k} = x$. We claim that $x_n \to x$. Let $\epsilon > 0$. Since $x_{n_k} \to x$, there exists $M_0 \in \mathbb{N}$ such that $\forall k \geq M_0$,

$$|x_{n_k} - x| < \frac{\epsilon}{2}.$$

Since $\{x_n\}$ is Cauchy, there exists an $M_1 \in \mathbb{N}$ such that for all $n \geq M_1$ and $m \geq M_1$,

$$|x_n - x_m| < \frac{\epsilon}{2}.$$

Choose $M = M_0 + M_1$. If $n \geq M$, then $n_M \geq M \geq M_0$ and $n_{\geq} M_1$. Therefore,

$$|x_n - x| \leq |x_n - x_{n_M}| + |x_{n_M} - x| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

□

Theorem 112
A sequence of real numbers $\{x_n\}$ is Cauchy if and only if $\{x_n\}$ is convergent.

Proof: ($\implies$) If $\{x_n\}$ is Cauchy, then $\{x_n\}$ is bounded. Therefore, $\{x_n\}$ has a convergent subsequence by Bolzano-Weierstrass. By the previous theorem, we thus have that $\{x_n\}$ is convergent.

($\impliedby$) Suppose that $\{x_n\}$ is convergent and $x = \lim_{n\to\infty} x_n$. Let $\epsilon > 0$. Since $x_n \to x$, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$,

$$|x_n - x| < \frac{\epsilon}{2}.$$

Choose $M = M_0$. Then, if $n, k \geq M$,

$$|x_n - x_k| \leq |x_n - x| + |x_k - x| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Therefore, $\{x_n\}$ is Cauchy. □

Series

Remark 113. *Series were the original motivation for analysis.*

Definition 114
Given $\{x_n\}$, the symbol $\sum_{n=1}^{\infty} x_n$ or $\sum x_n$ is the series associated to $\{x_n\}$. We say $\sum x_n$ converges if the sequence

$$\left\{ s_m = \sum_{n=1}^{m} x_n \right\}_{m=1}^{\infty}$$

converges. We call the terms of $\{s_m\}$ the partial sums. If $\lim_{m\to\infty} s_m = s$, we write $s = \sum x_n$ and treat $\sum x_n$ as a number.

Remark 115. *A series need not start at $n = 1$.*

Example 116
$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges.

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Proof: We may do show this directly by consider the partial sums:

$$\begin{aligned} s_m = \sum_{n=1}^{m} \frac{1}{n(n+1)} &= \sum_{n=1}^{m} \frac{1}{n} - \frac{1}{n+1} \\ &= \left(1 + \frac{1}{2} + \cdots + \frac{1}{m}\right) - \left(\frac{1}{2} + \ldots \frac{1}{m} + \frac{1}{m+1}\right) \\ &= 1 - \frac{1}{m+1}. \end{aligned}$$

Thus, $s_m = 1 - \frac{1}{m+1} \to 1$. Hence, the partial sums converge and thus the series converges. ■

Theorem 117

If $|r| < 1$ then $\sum_{n=0}^{\infty} r^n$ converges and

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Proof: We have $\forall m \in \mathbb{N}$,

$$s_m = \sum_{n=0}^{m} r^n = \frac{1 - r^{m+1}}{1-r}$$

by induction. Since $|r| < 1$, $\lim_{m\to\infty} |r|^{m+1} = 0$. Therefore,

$$\lim_{m\to\infty} s_m = \frac{1-0}{1-r} = \frac{1}{1-r}.$$

□

Remark 118. *Series of the form $\sum_{n=0}^{\infty} \alpha(r)^n$ for $\alpha \in \mathbb{R}$ and $r \in \mathbb{R}$ are called geometric series.*

Theorem 119

Let $\{x_n\}$ be a sequence and let $M \in \mathbb{N}$. Then, $\sum_{n=1}^{\infty} x_n$ converges if and only if $\sum_{n=M}^{\infty} x_n$ converges.

Proof: The partial sums satisfy, for all $m \in \mathbb{N}$,

$$\sum_{n=1}^{m} x_n = \sum_{n=M}^{m} x_n + \sum_{n=1}^{M} x_n.$$

□

Definition 120

$\sum x_n$ is Cauchy if the sequence of partial sums is Cauchy.

Theorem 121

$\sum x_n$ is Cauchy $\iff$ $\sum x_n$ is convergent.

Proof: This follows by the analogous theorem for regular sequences of real numbers proven earlier. □

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Theorem 122

$\sum x_n$ is Cauchy if and only if $\forall \epsilon > 0$, $\exists M \in \mathbb{N}$ such that for all $m \geq M$ and $\ell > m$,

$$\left| \sum_{n=m+1}^{\ell} x_n \right| < \epsilon.$$

Proof: ($\implies$) Suppose $\sum x_n$ is Cauchy. Let $\epsilon > 0$. Then, $\exists M_0 \in \mathbb{N}$ such that $\forall m, \ell \geq M_0$,

$$|s_m - s_\ell| < \epsilon.$$

Choose $M = M_0$. Then, if $m \geq M$ and $\ell > m$, then

$$\left| \sum_{n=m+1}^{\ell} x_n \right| = |s_\ell - s_m| < \epsilon.$$

The other direction is left as an exercise. □

Theorem 123

If $\sum x_n$ converges then $\lim_{n\to\infty} x_n = 0$.

Proof: Suppose $\sum x_n$ converges. Then, $\sum x_n$ is Cauchy. Let $\epsilon > 0$. Since $\sum x_n$ is Cauchy, $\exists M_0 \in \mathbb{N}$ such taht for all $\ell > m \geq M_0$,

$$\left| \sum_{n=m+1}^{\ell} x_n \right| < \epsilon.$$

Choose $M = M_0 + 1$. Then, if $m \geq M \implies m - 1 \geq M_0$. Therefore,

$$|x_m| = \left| \sum_{n=m}^{m} x_n \right| < \epsilon$$

by taking $\ell = m$. □

Theorem 124

If $|r| \geq 1$, then $\sum_{n=0}^{\infty} r^n$ diverges.

Proof: If $|r| \geq 1$, then $\lim_{m\to\infty} r^m \neq 0$. Therefore, $\sum_{n=0}^{\infty} r^n$ diverges, as if this wasn't the case then $\lim_{m\to\infty} r^m = 0$ by the previous theorem which is a contradiction. □

Corollary 125

The series $\sum_{n=0}^{\infty} \alpha(r)^n$ converges if and only if $|r| < 1$.

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11 Lecture 11

Absolute Convergence and the Comparison Test for Series

Recall 126
Last time we showed that if $\sum x_n$ converges then $\lim_{n\to\infty} x_n = 0$.

Question 127. *Is the converse true? Does* $\lim_{n\to\infty} x_n = 0 \implies \sum x_n$ *converges?*

Theorem 128
The series $\sum_{n=1}^{\infty} \frac{1}{n}$ does not converge.

Proof: We will show that there exists a subsequence of $s_m = \sum_{n=1}^{m} \frac{1}{n}$ which is unbounded, which will imply the series diverges. Consider, for $\ell \in \mathbb{N}$,

$$s_{2^\ell} = \sum_{n=1}^{2^\ell} \frac{1}{n}.$$

Then,

$$\begin{aligned}
s_{2^\ell} &= 1 + \left(\frac{1}{2}\right) + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \ldots \left(\frac{1}{2^{\ell-1}+1} + \cdots + \frac{1}{2^\ell}\right) \\
&= 1 + \sum_{\lambda=1}^{\ell} \sum_{n=2^{\lambda-1}+1}^{2^\lambda} \frac{1}{n} \\
&\geq 1 + \sum_{\lambda=1}^{\ell} \sum_{n=2^{\lambda-1}+1}^{2^\lambda} \frac{1}{2^\lambda} \\
&= 1 + \sum_{\lambda=1}^{\ell} \frac{1}{2^\lambda}(2^\lambda - (2^{\lambda-1}+1) + 1) \\
&= 1 + \sum_{\lambda=1}^{\ell} \frac{2^{\lambda-1}}{2^\lambda} \\
&= 1 + \frac{\ell}{2}.
\end{aligned}$$

Thus, $\{s_{2^\ell}\}_{\ell=1}^{\infty}$ is unbounded which implies $\{s_{2^\ell}\}$ does not converge. □

Remark 129. *The series* $\sum \frac{1}{n}$ *is called the* ***harmonic series.***

Theorem 130
Let $\alpha \in \mathbb{R}$ and $\sum x_n$ and $\sum y_n$ be convergent series. Then the series $\sum(\alpha x_n + y_n)$ converges and

$$\sum(\alpha x_n + y_n) = \alpha \sum x_n + \sum y_n.$$

Proof: The partial sums satisfy

$$\sum_{n=1}^{m}(\alpha x_n + y_n) = \alpha \sum_{n=1}^{m} x_n + \sum_{n=1}^{m} y_n.$$

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By linear properties of limits, it follows that

$$\lim_{m\to\infty}\sum_{n=1}^{m}(\alpha x_n + y_n) = \alpha\sum x_n + \sum y_n.$$

□

Series with non-negative terms are easier to work with than general series as then $\{s_n\}$ is a monotone sequence.

Theorem 131
If $\forall n \in \mathbb{N}\ x_n \geq 0$, then $\sum x_n$ converges if and only if $\{s_m\}$ is bounded.

Proof: If $x_n \geq 0$ for all $n \in \mathbb{N}$ then

$$s_{m+1} = \sum_{n=1}^{m+1} x_n = \sum_{n=1}^{m} x_n + x_{m+1} = s_m + x_{m+1} \geq s_m$$

Thus, $\{s_m\}$ is a monotone increasing sequence. Therefore, $\{s_m\}$ converges if and only if $\{s_m\}$ is bounded. □

Definition 132
$\sum x_n$ converges absolutely if $\sum |x_n|$ converges.

Theorem 133
If $\sum x_n$ converges absolutely then $\sum x_n$ converges.

Proof: Suppose $\sum |x_n|$ converges. We will then show that $\sum x_n$ is Cauchy.

Claim: $\forall m \geq 2$, $|\sum_{n=1}^{m} x_n| \leq \sum_{n=1}^{m} |x_n|$. We prove this claim by induction. For $m = 2$, this states that $|x_1 + x_2| \leq |x_1| + |x_2|$, which follows by the Triangle Inequality. Suppose for all $\left|\sum_{n=1}^{\ell} x_n\right| \leq \sum_{n=1}^{\ell} |x_n|$. Then,

$$\left|\sum_{n=1}^{\ell+1} x_n\right| \leq \left|\sum_{n=1}^{\ell} x_n\right| + |x_{\ell+1}| \leq \sum_{n=1}^{\ell} |x_n| + |x_{\ell+1}| = \sum_{n=1}^{\ell+1} |x_n|.$$

We now prove that $\sum x_n$ is Cauchy. Let $\epsilon > 0$. Since $\sum |x_n|$ converges, $\sum |x_n|$ is Cauchy. Therefore, there exists an $M_0 \in \mathbb{N}$ such that for all $\ell > m \geq M_0$,

$$\sum_{n=m+1}^{\ell} |x_n| < \epsilon.$$

Choose $M = M_0$. Then, for all $\ell > m \geq M$,

$$\left|\sum_{n=m+1}^{\ell} x_n\right| \leq \sum_{n=m+1}^{\ell} |x_n| < \epsilon.$$

Hence, $\sum x_n$ is Cauchy, and thus converges. □

Remark 134. *We will see that* $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ *is convergent but not absolutely convergent.*

Notice that it is immediately clear that this series is not absolutely convergent as $\sum \left|\frac{(-1)^n}{n}\right| = \sum \frac{1}{n}$ (the harmonic series), which doesn't converge.

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Convergence tests

Theorem 135 (Comparison Test)
Suppose for all $n \in \mathbb{N}$ $0 \leq x_n \leq y_n$. Then,

1. if $\sum y_n$ converges, then $\sum x_n$ converges.
2. if $\sum x_n$ diverges, then $\sum y_n$ diverges.

Proof:

1. If $\sum y_n$ converges, then $\{\sum_{n=1}^{m} y_n\}_{m=1}^{\infty}$ is bounded. In other words, there exists a $B \geq 0$ such that for all $m \in \mathbb{N}$,
$$\sum_{n=1}^{m} y_n \leq B.$$
Thus, for all $m \in \mathbb{N}$, $\sum_{n=1}^{m} x_n \leq \sum_{n=1}^{m} y_n \leq B$. Therefore, the partial sums of $\{x_n\}$ are bounded, which implies $\sum x_n$ converges.
2. If $\sum x_n$ diverges, then $\{\sum_{n=1}^{m} x_n\}_{m=1}^{\infty}$ is unbounded. We now prove that
$$\left\{\sum_{n=1}^{m} y_n\right\}_{m=1}^{\infty}$$
is also unbounded. Let $B \geq 0$. Then, $\exists m \in \mathbb{N}$ such that
$$\sum_{n=1}^{m} x_n \geq B.$$
Therefore, $\sum_{n=1}^{m} y_n \geq \sum_{n=1}^{m} x_n \geq B$. Thus, $\{\sum_{n=1}^{m} y_n\}_{m=1}^{\infty}$ is unbounded, which implies $\sum y_n$ diverges.

□

Remark 136. *We will see that geometric series and the Comparison Test imply everything!*

Theorem 137
For $p \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

Proof: ($\Longrightarrow$) We prove this direction through contradiction. Suppose $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges and $p \leq 1$. Then, $\frac{1}{n^p} \geq \frac{1}{n}$, and $\sum \frac{1}{n}$ diverges. Therefore, by the Comparison Test, $\sum \frac{1}{n^p}$ also diverges. Hence, if $\sum \frac{1}{n^p}$ converges, then $p > 1$.

($\Longleftarrow$) Suppose $p > 1$. We first prove that a subsequence of the partial series is bounded.

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Claim 1: $\forall k \in \mathbb{N}$, $s_{2^k} \leq 1 + \frac{1}{1-2^{-(p-1)}}$. Proof:

$$\begin{aligned}
s_{2^k} &= 1 + \sum_{\ell=1}^{k} \sum_{n=2^{\ell-1}+1}^{2^\ell} \frac{1}{n^p} \\
&\leq 1 + \sum_{\ell=1}^{k} \sum_{n=2^{\ell-1}+1}^{2^\ell} \frac{1}{(2^{\ell-1}+1)^p} \\
&\leq 1 + \sum_{\ell=1}^{k} 2^{-p(\ell-1)}(2^\ell - (2^{\ell-1}+1) + 1) \\
&= 1 + \sum_{\ell=1}^{k} 2^{-(p-1)(\ell-1)} \\
&= 1 + \sum_{\ell=0}^{k-1} 2^{-(p-1)\ell} \\
&\leq 1 + \sum_{\ell=0}^{\infty} 2^{-(p-1)\ell} \\
&= 1 + \frac{1}{1-2^{-(p-1)}}
\end{aligned}$$

using the fact that $p - 1 > 0$, and using properties of geometric series. Thus, Claim 1 is proven.

Claim 2: $\{s_m = \sum_{n=1}^{m} \frac{1}{n^p}\}$ is bounded. Proof: Let $m \in \mathbb{N}$. Since $2^m > m$, we have that

$$s_m = \sum_{n=1}^{m} \frac{1}{n^p} \leq \sum_{n=1}^{2^m} n^{-p} \leq 1 + \frac{1}{1-2^{-(p-1)}}.$$

Hence, the partial sums are bounded, which implies $\{s_m\}$ converges. □

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12 Lecture 12

The Ratio, Root, and Alternating Series Tests

We continue our study of convergence tests.

Theorem 138 (Ratio test)
Suppose $x_n \neq 0$ for all n and

$$L = \lim_{n\to\infty} \frac{|x_{n+1}|}{|x_n|}$$

exists. Then,

1. if $L < 1$ then $\sum x_n$ converges absolutely.
2. if $L > 1$ then $\sum x_n$ diverges.

Proof: We will first prove the second part of this theorem.

2) Suppose $L > 1$ and $\alpha \in (1, L)$. Then, there exists $M_0 \in \mathbb{N}$ such that for all $N \geq M_0$, $\frac{|x_{n+1}|}{|x_n|} \geq \alpha \geq 1$. Thus, for all $n \geq M_0$,

$$|x_{n+1}| \geq |x_n| \implies \lim_{n\to\infty} |x_n| \neq 0.$$

Therefore, $\sum x_n$ diverges.

1) Now suppose that $L < 1$. Let $\alpha \in (L, 1)$. Then, there exists $M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$, $\frac{|x_{n+1}|}{|x_n|} < \alpha$. Therefore,$\forall n \geq M_0$, $|x_{n+1}| \leq \alpha|x_n|$. In other words, for all $n \geq M_0$,

$$|x_n| \leq \alpha|x_{n-1}| \leq \alpha^2|x_{n-2}| \leq \cdots \leq \alpha^{n-M_0}|x_{M_0}|.$$

Let $m \in \mathbb{N}$. Then,

$$\begin{aligned}
\sum_{n=1}^{m} |x_n| &= \sum_{n=1}^{M_0-1} |x_n| + \sum_{n=M_0}^{m} |x_n| \\
&\leq \sum_{n=1}^{M_0-1} |x_n| + |x_{M_0}| \sum_{n=M_0}^{m} \alpha^{n-M_0} \\
&\leq \sum_{n=1}^{M_0-1} |x_n| + |x_{M_0}| \sum_{\ell=0}^{\infty} \alpha^{\ell} \\
&= \sum_{n=1}^{M_0-1} |x_n| + \frac{|x_{M_0}|}{1-\alpha}.
\end{aligned}$$

Therefore, $\{\sum_{n=1}^{m} |x_n|\}_{m=1}^{\infty}$ is bounded, and thus $\sum |x_n|$ converges. Hence, x_n is absolutely convergent.

□

Let's consider two examples where we can use the Ratio test.

Example 139
Show the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2+1}$ converges absolutely.

Proof: Notice

$$\left|\frac{(-1)^n}{n^2+1}\right| \leq \frac{1}{n^2+1} < \frac{1}{n^2},$$

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and hence

$$\lim_{n\to\infty}\left|\frac{\frac{(-1)^{n+1}}{(n+1)^2+1}}{\frac{(-1)^n}{n^2+1}}\right| < \lim_{n\to\infty}\frac{n^2}{(n+1)^2} = 1.$$

■

Example 140

Show that $\forall x \in \mathbb{R}$, $\sum_{n=0}^{\infty}\frac{x^n}{n!}$ converges absolutely.

Proof: This immediately follows from the Ratio test, noting that

$$\lim_{n\to\infty}\frac{|x|^{n+1}}{(n+1)!}\cdot\frac{n!}{|x|^n} = \lim_{n\to\infty}\frac{|x|}{n+1} = 0.$$

■

Remark 141. *As seen above, the Ratio test can be really helpful to use when we have a $(-1)^n$ or a factorial in the argument. Also note that if $L = 1$ then we the test doesn't apply.*

Theorem 142 (Root test)

Let $\sum x_n$ be a series and suppose that

$$L = \lim_{n\to\infty}|x_n|^{1/n}$$

exists. Then,

1. if $L < 1$ then $\sum x_n$ converges absolutely.
2. if $L > 1$ then $\sum x_n$ diverges.

Proof:

1. Suppose $L < 1$. Let $L < r < 1$. Then, since $|x_n|^{1/n} \to L$, $\exists M \in \mathbb{N}$ such that $\forall n \geq M$, $|x_n|^{1/n} < r$. Therefore, for all $n \geq M$, $|x_n| \leq r^n$. Thus, for all $m \in \mathbb{N}$,

$$\begin{aligned}\sum_{n=1}^{m}|x_n| &= \sum_{n=1}^{M-1}|x_n| + \sum_{n=M}^{m}|x_n| \\ &\leq \sum_{n=1}^{M-1}|x_n| + \sum_{n=M}^{m}r^n \\ &\leq \sum_{n=1}^{M-1}|x_n| + \sum_{n=M}^{\infty}r^n \\ &= \sum_{n=1}^{M-1}|x_n| + \frac{r^M}{1-r}.\end{aligned}$$

   Thus, $\{\sum_{n=1}^{m}|x_n|\}_{m=1}^{\infty}$ is bounded, and thus $\sum|x_n|$ converges.

2. Suppose $L > 1$. Then, since $|x_n|^{1/n} \to L > 1$, there exists an $M \in \mathbb{N}$ such that for all $n \geq M$, $|x_n|^{1/n} > 1$. In other words, for all $n \geq M$, $|x_n| > 1$. Therefore, $\lim_{n\to\infty} x_n \neq 0$, and thus $\sum x_n$ diverges.

□

Remark 143. *Again, note that if $L = 1$ then the test doesn't apply.*

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Theorem 144 (Alternating Series test)
Let $\{x_n\}$ be a monotone decreasing sequence such that $x_n \to 0$. Then, $\sum(-1)^n x_n$ converges.

Proof: Let $s_m = \sum_{n=1}^{m}(-1)^n x_n$. Then,

$$\begin{aligned} s_{2k} &= \sum_{n=1}^{2k}(-1)^n x_n \\ &= (x_2 - x_1) + (x_4 - x_3) + \cdots + (x_{2k} - x_{2k-1}) \\ &\geq (x_2 - x_1) + \cdots + (x_{2k} - x_{2k-1}) + (x_{2k+2} - x_{2k+1}) \\ &= s_{2(k+1)} \end{aligned}$$

as $\{x_n\}$ is a monotone decreasing sequence. Thus, $\{s_{2k}\}_{k=1}^{\infty}$ is monotone decreasing. Furthermore,

$$s_{2k} = -x_1 + (x_2 - x_3) + (x_4 - x_5) + \cdots + (x_{2k-2} - x_{2k-1}) + x_{2k} \geq -x_1.$$

In other words, $\{s_{2k}\}$ is a bounded below monotone decreasing sequence. Thus, $\{s_{2k}\}_{k=1}^{\infty}$ converges. Let $s = \lim_{k\to\infty} s_{2k}$. We now prove $\{s_m\}_{m=1}^{\infty}$ converges to s.

Let $\epsilon > 0$. Since $s_{2k} \to s$, $\exists M_0 \in \mathbb{N}$ such that for all $k \geq M_0$,

$$|s_{2k} - s| < \frac{\epsilon}{2}.$$

Since $x_n \to 0$, $\exists M_1 \in \mathbb{N}$ such that $\forall n \geq M_1$,

$$|x_n| < \frac{\epsilon}{2}.$$

Choose $M = \max\{2M_+0 + 1, M_1\}$. Suppose $m \geq M$. If m is even, then $\frac{m}{2} \geq M_0 + 1/2 \geq M_0$. Therefore,

$$|s_m - s| = |s_{2\cdot\frac{m}{2}} - s| < \frac{\epsilon}{2} < \epsilon.$$

If m is odd, let $k = \frac{m-1}{2}$ so $m = 2k + 1$. Then, $m \geq M \implies k \geq M_0$ and $m \geq M_1$. Then,

$$\begin{aligned} |s_m - s| &= |s_{m-1} + x_m - s| \\ &\leq |s_{2k} - s + x_m| \\ &\leq |s_{2k} - s| + |x_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Thus, $s_m \to s$, and thus $\sum(-1)^n x_n$ converges. □

Corollary 145
We already showed that $\sum \frac{(-1)^n}{n}$ does not absolutely converge. However, $\sum \frac{(-1)^n}{n}$ converges.

Proof: This follows immediately from the Alternating Series test.

Theorem 146
Suppose $\sum x_n$ converges absolutely and $\sum x_n = x$. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijective function. Then, $\sum x_{\sigma(n)}$ is absolutely convergent and $\sum x_{\sigma(n)} = x$. In other words, absolute convergence implies if we rearrange the sequence the new series will still converge to the same value of the original series.

Proof: We first show $\sum |x_{\sigma(n)}|$ converges, which is equivalent to showing the partial sums $\sum_{n=1}^{m} |x_{\sigma(n)}|$ is bounded.

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Since $\sum x_n$ converges, $\exists B \geq 0$ such that for all $\ell \in \mathbb{N}$,

$$\sum_{n=1}^{\ell} |x_n| \leq B.$$

Let $m \in \mathbb{N}$. Then, $\sigma(\{1, \ldots, m\})$ is a finite subset of $\mathbb{N}$. Thus, there exists an $\ell \in \mathbb{N}$ such that

$$\sigma(\{1, \ldots, m\}) \subset \{1, \ldots, \ell\}.$$

Thus,

$$\sum_{n=1}^{m} |x_{\sigma(n)}| = \sum_{n \in \sigma(\{1,\ldots,m\})} |x_n| \leq \sum_{n=1}^{\ell} |x_n| \leq B.$$

Therefore, $\sum |x_{\sigma(n)}|$ converges. Let $x = \sum_{n=1}^{\infty} x_n$, and let $\epsilon > 0$. Then, $\exists M_0 \in \mathbb{N}$ such that $\forall m \geq M_0$,

$$\left| \sum_{n=1}^{m} x_n - x \right| < \frac{\epsilon}{2}.$$

Since $\sum |x_n|$ converges, $\exists M_1 \in \mathbb{N}$ such that for all $\ell > m \geq M_1$,

$$\sum_{n=m+1}^{\ell} |x_n| < \frac{\epsilon}{2}.$$

Let $M_2 = \max\{M_0, M_1\}$. Then, $\forall \ell > m \geq M_2$,

$$\left| \sum_{n=1}^{m} x_n - x \right| < \frac{\epsilon}{2} \quad \text{and} \quad \sum_{n=m+1}^{\ell} |x_n| < \frac{\epsilon}{2}.$$

Since $\sigma^{-1}(\{1, \ldots, M_2\})$ is a finite set, $\exists M_3 \in \mathbb{N}$ such that

$$\{1, \ldots, M_2\} \subset \sigma(\{1, \ldots, M_3\}).$$

Choose $M = M_3$. Thus, if $m' \geq M$,

$$\begin{aligned}
\left| \sum_{n'=1}^{m'} x_{\sigma(n')} - x \right| &= \left| \sum_{n \in \sigma(\{1,\ldots,m'\})} x_n - x \right| \\
&= \left| \sum_{n=1}^{M} x_n - x + \sum_{n \in \sigma(\{1,\ldots,m'\}) \setminus \{1,\ldots,M\}} x_n \right| \\
&\leq \left| \sum_{n=1}^{M} x_n - x \right| + \sum_{n=M+1}^{\max \sigma(\{1,\ldots,m'\})} |x_n| \\
&\leq \left| \sum_{n=1}^{M} x_n - x \right| + \sum_{n=M+1}^{\ell} |x_n| \\
&< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.
\end{aligned}$$

□

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13 Lecture 13

Limits of Functions

Continuous Functions

Remark 147. *Continuous functions are those functions where tolerable changes to outputs accompany sufficiently small differences of inputs.*

Limits of Functions

Definition 148 (Cluster Point)
Let $S \subset \mathbb{R}$. $x \in \mathbb{R}$ is a cluster point of S if $\forall \delta > 0$, $(x-\delta, x+\delta) \cap S \setminus \{x\} \neq \emptyset$.

Let's look at some examples.

1. $S = \{1/n \mid n \in \mathbb{N}\}$. Here, 0 is a clusterpoint of S.
2. $S = (0,1)$. The set of cluster points of S is $[0,1]$.
3. $S = \mathbb{Q}$. The set of cluster points of S is $\mathbb{R}$.
4. $S = \{0\}$. There are no cluster points of S.
5. $S = \mathbb{Z}$. There are no cluster points of S.

Theorem 149
Let $S \subset \mathbb{R}$. Then, x is a cluster point of S if and only if there exists a sequence $\{x_n\}$ of elements in $S \setminus \{x\}$ such that $x_n \to x$.

Definition 150 (Function Convergence)
Let $S \subset \mathbb{R}$, let c be a cluster point of S, and $f : S \to \mathbb{R}$. We say that $f(x)$ converges to $L \in \mathbb{R}$ at c if $\forall \epsilon > 0$ $\exists \delta > 0$ such that if $x \in S$ and $0 < |x - c| < \delta$, then $|f(x) - L| < \epsilon$.

Notation 151
Notationally, we may write $f(x) \to L$ as $x \to c$, or $\lim_{x \to c} f(x) = L$.

Theorem 152
Let c be a cluster point of $S \subset \mathbb{R}$, and let $f : S \to \mathbb{R}$. If $f(x) \to L_1$ and $f(x) \to L_2$ as $x \to c$, then $L_1 = L_2$.

Proof: We will show $\forall \epsilon > 0$, $|L_1 - L_2| < \epsilon$. Let $\epsilon > 0$. Then, since $f(x) \to L_1$ and $f(x) \to L_2$, $\exists \delta_1$ such that if $x \in S$ and $0 < |x - c| < \delta_1$ then

$$|f(x) - L_1| < \epsilon/2$$

and $\exists \delta_2 > 0$ such that if $x \in S$ and $0 < |x - c| < \delta_2$, then

$$|f(x) - L_2| < \epsilon/2.$$

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Let $\delta = \min\{\delta_1, \delta_2\} > 0$. Then, since c is a cluster point of S, $\exists x_0 \in S$ such that

$$0 < |x_0 - c| < \delta \implies |L_1 - L_2| = |L_1 - f(x_0) + f(x_0) + L_2| \leq |L_1 - f(x_0)| + |f(x_0) - L_2| < \epsilon.$$

□

Let's see some examples of limits of functions.

Example 153
Let $f(x) = ax + b$. Then, for all $c \in \mathbb{R}$, $\lim_{x \to c} f(x) = ac + b$.

Proof: Let $\epsilon > 0$. Choose $\delta = \frac{\epsilon}{1+|a|}$. Then, if $x \in \mathbb{R}$ and $0 < |x - c| < \delta$, then

$$\begin{aligned} |f(x) - (ac + b)| &= |a(x - c)| \\ &= |a||x - c| \\ &< |a|\delta \\ &= \frac{|a|}{1 + |a|}\epsilon < \epsilon. \end{aligned}$$

■

Example 154
Let $f(x) = \sqrt{x}$. Then, $\forall c > 0$, $\lim_{x \to c} f(x) = \sqrt{c}$.

Proof: Let $\epsilon > 0$. Choose $\delta = \epsilon\sqrt{c}$. Then, if $x > 0$ and $0 < |x - c| < \delta$, then

$$\begin{aligned} |f(x) - \sqrt{c}| &= |\sqrt{x} - \sqrt{c}| \\ &= \left| \frac{(\sqrt{x} - \sqrt{c})(\sqrt{x} + \sqrt{c})}{\sqrt{x} + \sqrt{c}} \right| \\ &= \frac{|x - c|}{\sqrt{x} + \sqrt{c}} \\ &\leq \frac{|x - c|}{\sqrt{c}} \\ &< \frac{\delta}{\sqrt{c}} = \epsilon. \end{aligned}$$

■

Example 155
Let $f(x) = \begin{cases} 1 & x \neq 0 \\ 2 & x = 0 \end{cases}$. Then, $\lim_{x \to 0} f(x) = 1$. Notably, $\lim_{x \to 0} f(x) \neq f(0)$!

Proof: Let $\epsilon > 0$ and choose $\delta = 1$. Then, if $0 < |x - 0| < 1$ then $x \neq 0 \implies$

$$|f(x) - 1| = |1 - 1| = 0 < \epsilon.$$

■

Question 156. *How do limits of functions relate to limits of sequences?*

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Theorem 157

Let $S \subset \mathbb{R}$, c a cluster point of S, and let $f : S \to \mathbb{R}$. Then, the following are equivalent:

1. $\lim_{x \to c} f(x) = L$ and
2. for every sequence $\{x_n\}$ in $S \setminus \{c\}$ such that $x_n \to c$, we have $f(x_n) = L$.

Proof: (1. $\implies$ 2.): Suppose $\lim_{x \to c} f(x) = L$. Let $\{x_n\}$ be a sequence in $S \setminus \{c\}$ such that $x_n \to c$. We want to show that $f(x_n) \to L$. Let $\epsilon > 0$. Given $\lim_{x \to c} f(x_n) = L$, $\exists \delta > 0$ such that if $x \in S$ and $0 < |x - c| < \delta$ then $|f(x) - L| < \epsilon$. Since $x_n \to c$, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$, $0 < |x_n - c| < \delta$.

Choose $M = M_0$. Then, $\forall n \geq M$, if $0 < |x_n - c| < \delta$ then $|f(x_n) - L| < \epsilon$. Thus, $f(x_n) \to L$.

(2. $\impliedby$ 1.): Suppose 2. holds, and assume for the sake of contradiction that 1) is false. Then, $\exists \epsilon_0 > 0$ such that $\forall \delta > 0$, $\exists x \in S$ such that

$$0 < |x - c| < \delta \quad \text{and} \quad |f(x) - L| \geq \epsilon_0.$$

Then, $\forall n \in \mathbb{N}$, $\exists x_n \in S$ such that $0 < |x_n - c| < \frac{1}{n}$ and $|f(x_n) - L| \geq \epsilon_0$. By the Squeeze Theorem applied to

$$0 < |x_n - c| < \frac{1}{n},$$

$x_n \to c$. Then, by 2.,

$$0 = \lim_{n \to \infty} |f(x_n) - L| \geq \epsilon_0$$

which is a contradiction. $\square$

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14 Lecture 14

Limits of Functions in Terms of Sequences and Continuity

Theorem 158
For all $c \in \mathbb{R}$, $\lim_{x\to c} x^2 = c^2$.

Proof: Let $\{x_n\}$ be a sequence in $\mathbb{R} \setminus \{c\}$ such that $x_n \to c$. Then, $x_n^2 \to c^2$ by a theorem shown in Lecture 8. Thus,

$$\lim_{x\to c} x^2 = c^2.$$

□

Theorem 159
We show that

1. $\lim_{x\to 0} \sin(1/x)$ does not exist, and
2. $\lim_{x\to 0} x\sin(1/x) = 0$.

Proof:

1. Let $x_n = \frac{2}{(2n-1)\pi}$. Then, $x_n \neq 0$, and $x_n \to 0$. But,

$$\sin(1/x_n) = \sin\left(\frac{(2n-1)\pi}{2}\right) = (-1)^{n+1}$$

for all n. However, this sequence does not converge (i.e. the limit does not exist).

2. Suppose $x_n \neq 0$ and $x_n \to 0$. Then,

$$0 \leq |x_n \sin(1/x_n)| = |x_n||\sin(1/x_n)| \leq |x_n|.$$

By the Squeeze Theorem, $\lim_{n\to\infty} |x_n \sin(1/x_n)| = 0$. □

We can use the 'sequential limit' characterization to prove analogs of previous theorems for limits of sequences.

Theorem 160
Let $S \subset \mathbb{R}$, c a cluster point of S, and $f, g : S \to \mathbb{R}$. Suppose $\forall x \in S$, $f(x) \leq g(x)$ and $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ exist. Then,

$$\lim_{x\to c} f(x) \leq \lim_{x\to c} g(x).$$

Proof: Let $L_1 = \lim_{x\to c} f(x)$ and $L_2 = \lim_{x\to c} g(x)$. Let $\{x_n\}$ be a sequence in $S \setminus \{c\}$ such that $x_n \to c$. Then, $\forall n \in \mathbb{N}$, $f(x_n) \leq g(x_n)$. Therefore,

$$L_1 = \lim_{n\to\infty} f(x_n) \leq \lim_{n\to\infty} g(x_n) = L_2.$$

□

Similarly, we have analogs of the Squeeze Theorem, limits of algebraic operations, and limits of absolute values. You may read the end of Section 3.1.3 **[L]** for this.

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Definition 161

Let $S \subset \mathbb{R}$ and suppose c is a cluster point of $S \cap (-\infty, c)$. Then, we say $f(x)$ converges to L as $x \to c^-$ if $\forall \epsilon > 0\ \exists \delta > 0$ such that if $x \in S$ and $c - \delta < x < c$ then $|f(x) - L| < \epsilon$.

Notation 162

This is denoted $L = \lim_{x \to c^-} f(x)$.

Definition 163

Similarly, let $S \subset \mathbb{R}$ and suppose c is a cluster point of $S \cap (c, \infty)$. Then, we say $f(x)$ converges to L as $x \to c^+$ if $\forall \epsilon > 0\ \exists \delta > 0$ such that if $x \in S$ and $c < x < c + \delta$ then $|f(x) - L| < \epsilon$.

Notation 164

This is denoted $L = \lim_{x \to c^+} f(x)$.

Example 165

Let $f(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$. Then,

$$\lim_{x \to 0^-} f(x) = 0 \quad \text{and} \quad \lim_{x \to 0^+} f(x) = 1,$$

even though $f(0)$ is undefined.

Theorem 166

Let $S \subset \mathbb{R}$ and let c be a cluster point of $S \cap (-\infty, c)$ and $S \cap (c, \infty)$. Then, c is a cluster point of S. Moreover,

$$\lim_{x \to c} f(x) = L \iff \lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L.$$

Continuous Functions

As we have seen, limits do not care about $f(x)$ when $x = c$. Continuity is a condition that connects $\lim_{x \to c} f(x)$ with $f(c)$.

Definition 167 (Continuous Functions)

Let $S \subset \mathbb{R}$ and let $c \in S$. We say f is continuous at c if $\forall \epsilon > 0\ \exists \delta > 0$ such that if $x \in S$ and $|x - c| < \delta$ then $|f(x) - f(c)| < \epsilon$ We say f is continuous on U for $U \subset S$ if f is continuous at every point in U.

Example 168

$f(x) = ax + b$ is continuous on $\mathbb{R}$.

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Proof: Let $\epsilon > 0$ and choose $\delta = \frac{\epsilon}{1+|a|}$. If $|x - c| < \delta$, then

$$\begin{aligned} |f(x) - f(c)| &= |ax + b - (ac + b)| \\ &= |a||x - c| \\ &< |a|\delta \\ &= \frac{|a|}{1 + |a|}\epsilon < \epsilon. \end{aligned}$$

□

Example 169

Show that $f(x) = \begin{cases} 1 & x \neq 0 \\ 2 & x = 0 \end{cases}$ is *not* continuous at $c = 0$.

First we write the negation of the definition of continuity.

Negation 170 (Not Continuous)

f is not continuous at c if $\exists \epsilon_0$ such that for all $\delta > 0$, $\exists x \in S$ such that $|x - c| < \delta$ and $|f(x) - f(c)| \geq \epsilon_0$.

Proof: Choose $\epsilon_0 = 1$ and let $\delta > 0$. Then, $x = \frac{\delta}{2}$ satisfies $|x - 0| < \delta$ and

$$|f(x) - f(0)| = |2 - 1| \geq 1 = \epsilon_0.$$

■

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15 Lecture 15

The Continuity of Sine and Cosine and the Many Discontinuities of Dirichlet's Function

Theorem 171

Let $S \subset \mathbb{R}$, $c \in S$, and $f : S \to \mathbb{R}$. Then,

1. if c is not a cluster point of f, then f is continuous at c.
2. if c is a cluster point of S, then f is continuous at c if and only if $\lim_{x\to c} f(x) = f(c)$.
3. f is continuous at c if and only if for every sequence $\{x_n\}$ of elements of S such that $x_n \to c$, we have $f(x_n) \to f(c)$.

Proof:

1. Let $\epsilon > 0$. Since c is not a cluster point of S, $\exists \delta_0 > 0$ such that $(c - \delta_0, c + \delta_0) \cap S = \{c\}$. Choose $\delta = \delta_0$. If $x \in S$ and $|x - c| < \delta \implies x = c \implies |f(x) - f(c)| = 0 < \epsilon$. Therefore, f is continuous at c.
2. This part of the theorem is left as an exercise (or read the short proof in the book).
3. ($\implies$) Suppose f is continuous at c. Let $\{x_n\}$ be a sequence such that $x_n \to c$. Let $\epsilon > 0$. Since f is continuous at c, $\exists \delta > 0$ such that if $x \in S$ and $|x - c| < \delta$ then $|f(x) - f(c)| < \epsilon$. Since $x_n \to c$, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0$, $|x_n - c| < \delta$. Choose $M = M_0$. Then, $\forall n \geq M$,

$$|x_n - c| < \delta \implies |f(x_n) - f(c)| < \epsilon.$$

Thus, $f(x_n) \to f(c)$.

($\impliedby$) Suppose that for every sequence $\{x_n\}$ of elements of S such that $x_n \to c$, we have that $f(x_n) \to f(c)$. We will work towards a contradiction. Suppose $f(x)$ is not continuous at c. Then, $\exists \epsilon_0$ such that $\forall \delta > 0$ $\exists x \in S$ such that $|x - c| < \delta$ and $|f(x) - f(c)| \geq \epsilon_0$.

Thus, $\forall n \in \mathbb{N}$, $\exists x_n \in S$ such that $|x_n - c| < \frac{1}{n}$ and

$$|f(x_n) - f(c)| \geq \epsilon_0.$$

Thus, by the Squeeze Theorem, $|x_n - c| \to 0 \implies x_n \to c$. Therefore,

$$0 = \lim_{n\to\infty} |f(x_n) - f(c)| \geq \epsilon_0$$

which is a contradiction.

□

Theorem 172

The functions $f(x) = \sin x$ and $g(x) = \cos x$ are continuous functions on $\mathbb{R}$.

Proof: From their definitions in terms of the unit circle, we have that $\sin^2(x) + \cos^2(x) = 1$. Also note the following:

1. $\forall x \in \mathbb{R}$, $|\sin x| \leq 1$ and $|\cos x| \leq 1$
2. $\forall x \in \mathbb{R}$, $|\sin x| \leq |x|$

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3. The angle formulae:

$$\sin(a+b) = \cos(a)\sin(b) + \sin(a)\cos(b) \quad \text{and} \quad \sin(a) - \sin(b) = 2\sin\left(\frac{a-b}{2}\right)\cos\left(\frac{a+b}{2}\right).$$

We now show that $\sin x$ is continuous on $\mathbb{R}$. Let $\epsilon > 0$. Choose $\delta = \epsilon$. Then, if $|x-c| < \delta$, then

$$|\sin x - \sin c| = 2\left|\sin\frac{x-c}{2}\cos\frac{x+c}{2}\right| \leq 2\left|\sin\frac{x-c}{2}\right| \leq 2\frac{|x-c|}{2} = |x-c| < \delta = \epsilon.$$

Therefore, $\sin x$ is continuous on $\mathbb{R}$. We now show that $\cos x$ is continuous. Recall that $\forall x \in \mathbb{R}$, $\cos x = \sin(x+\pi/2)$. Let $c \in \mathbb{R}$ and let $\{x_n\}$ be a sequence such that $x_n \to c$. Then, $x_n + \pi/2 \to c + \pi/2$. Since $\sin x$ is continuous on $\mathbb{R}$,

$$\lim_{n\to\infty} \cos x_n = \lim_{n\to\infty} \sin\left(x_n + \frac{\pi}{2}\right) = \sin\left(c + \frac{\pi}{2}\right) = \cos c.$$

Therefore, $\cos x$ is continuous on $\mathbb{R}$. □

Theorem 173

Let f be a polynomial, in other words let f be of the form

$$f(x) = a_d x^d + \cdots + a_1 x + a_0.$$

Then, f is continuous on all of $\mathbb{R}$.

Proof: Let $c \in \mathbb{R}$ and let $\{x_n\}$ be a sequence such that $x_n \to c$. Then, by the limit theorem for sequences,

$$\begin{aligned}
\lim_{n\to\infty} f(x_n) &= \lim_{n\to\infty} (a_d x^d + \cdots + a_1 x + a_0) \\
&= a_d(\lim_{n\to\infty} x_n)^d + \cdots + a_1(\lim_{n\to\infty} x_n) + a_0 \\
&= a_d c^d + \cdots + a_1 c + a_0 \\
&= f(c).
\end{aligned}$$

Thus, f is continuous at c for all $c \in \mathbb{R}$. □

Theorem 174

If $f : S \to \mathbb{R}$, $g : S \to \mathbb{R}$ are continuous at $c \in S$, then

1. $f + g$ is continuous at c,
2. $f \cdot g$ is continuous at c,
3. and if $\forall x \in S$ $g(x) \neq 0$, then $\frac{f}{g}$ is continuous at c.

Proof: These proofs are left to the reader. □

Theorem 175

Let $A, B \subset \mathbb{R}$, $f : B \to \mathbb{R}$, $g : A \to B$. Then, if g is continuous at c and f is continuous at $g(c)$, then $f \circ g$ is continuous at c.

Proof: Suppose $x_n \to c$. Then, $g(x_n) \to g(c)$, and thus

$$f(g(x_n)) \to f(g(c)).$$

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□

Example 176

These theorems allow us to say that some functions are continuous without a huge $\epsilon - \delta$ proof:

i) $\frac{1}{x^2}$ is continuous on $(0, \infty)$. This follows as $g(x) = x^2$ is continuous on $(0, \infty)$ and thus $\frac{1}{g(x)} = 1/x^2$ is continuous on $(0, \infty)$.

ii) $(\cos \frac{1}{x^2})^2$ is continuous on $(0, \infty)$. This follows as $\cos x$ is continuous on $\mathbb{R}$, and thus $g(x) = \cos(1/x^2)$ is continuous on $(0, \infty)$. Furthermore, since $f(x) = x^2$ is continuous on $\mathbb{R}$, $(f \circ g)(x) = (\cos 1/x^2)^2$ is continuous on $(0, \infty)$.

Question 177. *Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Does there exists a point $c \in \mathbb{R}$ such that f is continuous at c?*

Theorem 178

The function

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

is not continuous on all of $\mathbb{R}$. This function is called the **Dirichlet function**.

Proof: We have two cases: $c \in \mathbb{Q}$ or $c \notin \mathbb{Q}$.

1. $c \in \mathbb{Q}$. For each $n \in \mathbb{N}$, $\exists x_n \notin \mathbb{Q}$ such that $c < x_n < c + 1/n$, and thus $x_n \to c$ but $f(x_n) = 1$ for all n so
$$1 = \lim_{n \to \infty} f(x_n) \neq f(c) = 0.$$

2. $c \notin \mathbb{Q}$. Similarly, for each $n \in \mathbb{N}$, $\exists x_n \in \mathbb{Q}$ such that $c < x_n < c + 1/n$, and thus $x_n \to c$ but $f(x_n) = 0$ for all n so
$$0 = \lim_{n \to \infty} f(x_n) \neq f(c) = 1.$$

□

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16 Lecture 16

The Min/Max Theorem and Bolzano's Intermediate Value Theorem

As we will see in today's lecture, continuous functions are well behaved on closed intervals of the form $[a, b]$, with $f([a, b]) = [e, f]$ for some $e, f \in \mathbb{R}$.

Definition 179 (Bounded Functions)
A function $f : S \to \mathbb{R}$ is bounded if $\exists B \geq 0$ such that for all $x \in S$,

$$|f(x)| \leq B.$$

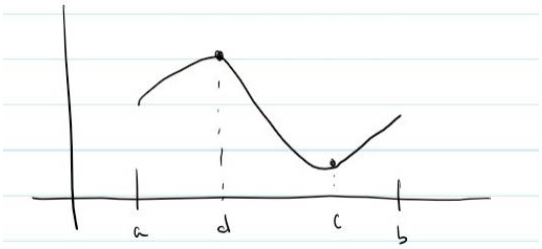
Theorem 180
If $f : [a, b] \to \mathbb{R}$ is continuous then f is bounded.

Proof: Suppose for the sake of contradiction that $f : [a, b] \to \mathbb{R}$ is continuous and f is unbounded. Then, $\forall n \in \mathbb{N}$, $\exists x_n \in [a, b]$ such that $|f(x_n)| \geq n$. By the Bolzano-Weierstrass theorem, $\exists$ a subsequence $\{x_{n_k}\}_k$ of $\{x_n\}_n$ and an $x \in \mathbb{R}$ such that $x_{n_k} \to x$. Since $a \leq x_{n_k} \leq b$ for all k, $a \leq x \leq b$. Given f is continuous at x by assumption,

$$f(x) = \lim_{k\to\infty} f(x_{n_k}) \implies |f(x)| = \lim_{k\to\infty} |f(x_{n_k})|.$$

Therefore, $\{|f(x_{n_k})|\}$ is bounded, and thus $\{n_k\}$ is bounded since $n_k \leq |f(x_{n_k})|$. But by the definition of a subsequence, we must have $k \leq n_k$ for all k, contradicting the boundedness of $\{n_k\}$. □

Definition 181 (Absolute Minimum/Maximum)
Let $f : S \to \mathbb{R}$. Then, f achieves an absolute minimum at c if $\forall x \in S$, $f(x) \geq f(c)$. Similarly, f achieves an absolute maximum at d if $\forall x \in S$, $f(x) \leq f(d)$.



Theorem 182 (Min-Max Theorem)
Let $f : [a, b] \to \mathbb{R}$. If f is continuous, then f achieves an absolute maximum and absolute minimum.

Remark 183. *Note that this is also called the Extreme Value Theorem or EVT for short, though to stay consistent with the Lebl's book I will be calling it the Min-Max theorem.*

Proof: We will prove this for the absolute maximum. If f is continuous, then f is bounded by the previous theorem. Thus, the set

$$E = \{f(x) \mid x \in [a, b]\}$$

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is bounded above. Let $L = \sup E$. Then,

1. L is an upper bound for E, i.e.
$$\forall x \in [a,b],\ f(x) \leq L.$$

2. There exists a sequence $\{f(x_n)\}_n$ with $x_n \in [a,b]$ such that $f(x_n) \to L$.

By the Bolzano-Weierstrass theorem, there exists a subsequence $\{x_{n_k}\}_k$ of $\{x_n\}$ and $d \in [a,b]$ such that $x_{n_k} \to d$ as $k \to \infty$. Hence,
$$f(d) = \lim_{k\to\infty} f(x_{n_k}) = \lim_{n\to\infty} f(x_n) = L$$
by the continuity of f. Thus, f achieves an absolute maximum at d.

We leave the absolute minimum proof to the reader. □

Remark 184. *As students of mathematics, we also care about the necessity of the hypotheses!*

For example, what if $f : [a,b] \to \mathbb{R}$ is not continuous? Does the Min-Max theorem apply? The answer is **no**. Consider
$$f(x) = \begin{cases} \frac{1}{2} & x = 0, 1 \\ x & x \in (0,1) \end{cases}.$$
Here, f neither achieves an absolute maximum nor an absolute minimum on $[0,1]$.

What if $f : S \to \mathbb{R}$ and S is not closed and bounded? Does the Min-Max theorem apply? Again, the answer is **no**. Consider $f(x) = \frac{1}{x} - \frac{1}{1-x}$ on $S = (0,1)$. Even though f is continuous on S, f neither achieves an absolute minimum nor an absolute maximum.

So far we have shown that if $f : [a,b] \to \mathbb{R}$ is continuous, then $f([a,b]) \subset [f(c), f(d)]$ where f achieves an absolute minimum at c and an absolute maximum at d.

Question 185. *Does f achieve all values in $f(c)$ and $f(d)$?*

The answer is **yes**, by Bolzano's Intermediate Value Theorem as we will show.

Theorem 186
Let $f : [a,b] \to \mathbb{R}$. If $f(a) < 0$ and $f(b) > 0$, then $\exists c \in (a,b)$ such that $f(c) = 0$.

Proof: We prove this using a bisection method. Let $a_1 = a$ and $b_1 = b$, and define a_2, b_2 as follows: If $f((a_1 + b_1)/2) \geq 0$, define $a_2 = a_1$, $b_2 = \frac{a_1+b_1}{2}$. If $f((a_1 + b_1)/2) < 0$, define $a_2 = \frac{a_1+b_1}{2}$ and $b_2 = b_1$. In general, if we know a_n, b_n, we choose a_{n+1} and b_{n+1} as follows: If $f((a_n + b_n)/2) \geq 0$, define $a_{n+1} = a_n$, $b_2 n + 1 = \frac{a_n+b_n}{2}$. If $f((a_n + b_n)/2) < 0$, define $a_{n+1} = \frac{a_n+b_n}{2}$ and $b_{n+1} = b_n$. Thus, we have:

1. $\forall n \in \mathbb{N}$, $a \leq a_n \leq a_{n+1} \leq b_{n+1} \leq b_n \leq b$.
2. $\forall n \in \mathbb{N}$, $b_{n+1} - a_{n+1} = \frac{b_n - a_n}{2}$.
3. $\forall n \in \mathbb{N}$, $f(a_n) < 0$ and $f(b_n) \geq 0$.

By 1., $\{a_n\}$ and $\{b_n\}$ are monotone increasing and monotone decreasing respectively, both of which are bounded. Thus, $\exists c, d \in [a,b]$ such that $a_n \to c$ and $b_n \to d$. By 2.,
$$b_n - a_n = \frac{b_{n-1} - a_{n-1}}{2} = \frac{1}{4}(b_{n-2} - a_{n-2}) = \cdots = \frac{1}{2^{n-1}}(b-a).$$
Thus,
$$d - c = \lim_{n\to\infty}(b_n - a_n) = \lim_{n\to\infty} \frac{1}{2^{n-1}}(b-a) = 0 \implies d = c.$$

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Therefore, $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = c$. By 3., $f(c) = \lim_{n\to\infty} f(a_n) \leq 0$ and $f(c) = \lim_{n\to\infty} f(b_n) \geq 0$. Therefore, $f(c) = 0$. □

Theorem 187 (Bolzano IVT)
Let $f : [a, b] \to \mathbb{R}$ be continuous. If $f(a) < f(b)$, and $y \in (f(a), f(b))$, $\exists c \in (a, b)$ such that $f(c) = y$. If $f(b) < f(a)$ and $y \in (f(b), f(a))$, $\exists c \in (a, b)$ such that $f(c) = y$.

Remark 188. *This is known as the Intermediate Value Theorem or IVT for short.*

Proof: Suppose $f(a) < f(b)$. Let $y \in (f(a), f(b))$. Define $g(x) = f(x) - y$. Then, $g : [a, b] \to \mathbb{R}$ is continuous, $g(a) = f(a) - y < 0$ and $g(b) = f(b) - y > 0$. Therefore, by the previous theorem, $\exists c \in (a, b)$ such that $g(c) = 0$. Therefore, $\exists c \in (a, b)$ such that $g(c) = f(c) - y = 0 \implies f(c) = y$.

The other direction is analogous. □

Theorem 189
Let $f : [a, b] \to \mathbb{R}$ be continuous. let $c \in [a, b]$ be where f achieves an absolute minimum and $d \in [a, b]$ be where f achieves an absolute maximum. Then,

$$f([a, b]) = [f(c), f(d)].$$

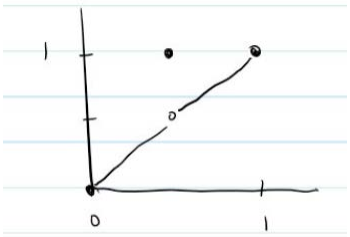
In other words, every value between the absolute minimum value and the absolute maximum value is achieved.

Proof: We know that $f([a, b]) \subseteq [f(c), f(d)]$. Hence, we prove the other direction. By the IVT applied to $f : [c, d] \to \mathbb{R}$,

$$[f(c), f(d)] \subseteq f([c, d]) \subseteq f([a, b]).$$

Therefore, $f([a, b]) = [f(c), f(d)]$. □

Of course, Bolzano IVT is false if we assume f is not continuous (as can be seen by the following diagram):



Theorem 190
The polynomial $f(x) = x^{2021} + x^{2020} + 9.03x + 1$ has at least one real root.

Proof: Notice that $f(0) = 1 > 0$ and $f(-1) = -1 + 1 - 9.03 + 1 = -8.03 < 0$. Thus, by IVT, $\exists c \in (-1, 0)$ such that $f(c) = 0$. □

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17 Lecture 17

Uniform Continuity and the Definition of the Derivative

Uniform Continuity

Recall 191

Recall the definition of continuity: $f : S \to \mathbb{R}$ is continuous on S if $\forall c \in S$ and $\forall \epsilon > 0$, $\exists \delta = \delta(\epsilon, c) > 0$ such that $\forall x \in S$, $|x - c| < \delta \implies |f(x) - f(c)| < \epsilon$.

Here, $\delta(\epsilon, c)$ denotes the fact that δ **can** depend on ϵ and c.

Example 192

Consider the function $f(x) = \frac{1}{x}$. f is continuous on $(0, 1)$.

Proof: Let $\epsilon > 0$. Choose $\delta = \min\left\{\frac{c}{2}, \frac{c^2}{2}\epsilon\right\}$. Suppose $|x - c| < \delta$. Then, $|x - c| < \frac{c}{2} \implies |x| > c - |x - c| > \frac{c}{2}$. Thus, $\frac{1}{|x|} < \frac{2}{c}$. Therefore,

$$\begin{aligned}\left|\frac{1}{x} - \frac{1}{c}\right| &= \frac{|x - c|}{|xc|} \\ &< \frac{\delta}{|x||c|} \\ &< \frac{2}{c^2}\delta \\ &\leq \frac{2}{c^2}\frac{c^2\epsilon}{2} = \epsilon.\end{aligned}$$

■

As shown in the previous example. δ depended on **both** ϵ and c.

Definition 193 (Uniformly Continuous)

Let $f : S \to \mathbb{R}$. Then, f is uniformly continuous on S if $\forall \epsilon > 0$, $\exists \delta = \delta(\epsilon) > 0$ such that $\forall x, c \in S$,

$$|x - c| < \delta \implies |f(x) - f(c)| < \epsilon.$$

Remark 194. *Thus, in the definition of uniform continuity, δ only depends on ϵ!*

Example 195

The function $f(x) = x^2$ is uniformly continuous on $[0, 1]$.

Proof: Let $\epsilon > 0$. Choose $\delta = \frac{\epsilon}{2}$. Then, if $x, c \in [0, 1]$ then $|x - c| < \delta$ implies that

$$|x^2 - c^2| = |x + c||x - c| \leq 2|x - c| < 2\delta = \epsilon.$$

■

However, there are of course continuous functions that are not uniformly continuous. For example, we will show that $f(x) = \frac{1}{x}$ is not uniformly continuous on (0,1), but first we consider the negation of the definition.

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Negation 196 (Not Uniformly Continuous)
Let $f : S \to \mathbb{R}$. Then, f is not uniformly continuous on S if $\exists \epsilon_0 > 0$, $\forall \delta > 0$ such that $\exists x, c \in S$ with

$$|x - c| < \delta \quad \text{and} \quad |f(x) - f(c)| \geq \epsilon_0.$$

Proof: Choose $\epsilon_0 = 2$ (in fact, any $\epsilon_0 > 0$ will show that $\frac{1}{x}$ is not uniformly continuous on $(0, 1)$). Then, let $\delta > 0$. Choose $c = \min\left\{\delta, \frac{1}{2}\right\}$ and $x = \frac{c}{2}$. Then, $|x - c| = \frac{c}{2} \leq \frac{\delta}{2} < \delta$ and

$$\left|\frac{1}{x} - \frac{1}{c}\right| = \left|\frac{2}{c} - \frac{1}{c}\right| = \frac{1}{c} \geq \frac{1}{\frac{1}{2}} = 2.$$

□

Theorem 197
Let $f : [a, b] \to \mathbb{R}$. Then, f is continuous if and only if f is uniformly continuous.

Proof: ($\Longleftarrow$) This direction is left as an exercise to the reader.

($\Longrightarrow$) Suppose f is continuous and assume for the sake of contradiction that f is *not* uniformly continuous. Then, $\exists \epsilon_0 > 0$ such that for all $n \in \mathbb{N}$, $\exists x_n, c_n \in [a, b]$ such that

$$|x_n - c_n| < \frac{1}{n} \quad \text{and} \quad |f(x_n) - f(c_n)| > \epsilon_0.$$

By Bolzano-Weierstrass, $\exists$ a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ and $x \in [a, b]$ such that $\lim_{k\to\infty} x_{n_k} = x$. Similarly, by Bolzano-Weierstrass, $\exists$ a subsequence $\{c_{n_k}\}$ of $\{c_n\}$ and $c \in [a, b]$ such that $\lim_{k\to\infty} c_{n_k} = c$. Note that subsequence $\{x_{n_{k_j}}\}$ of $\{x_{n_k}\}$ satisfies $\lim_{j\to\infty} x_{n_{k_j}} = x$.

Then,

$$|x - c| = \lim_{j\to\infty} |x_{n_{k_j}} - c_{n_{k_j}}| \leq \lim_{j\to\infty} \frac{1}{n_{k_j}} - 0.$$

Thus, $x = c$. But, since f is continuous at c,

$$0 = |f(c) - f(c)| = \lim_{j\to\infty} |f(x_{n_{k_j}}) - f(c_{n_{k_j}})| \geq \epsilon_0.$$

This is a contradiction. □

Derivative

Definition 198
Let I be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that f is differentiable at c if the limit

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c}$$

exists.

Notation 199
If f is differentiable at c, we write

$$f'(c) := \lim_{x\to c} \frac{f(x) - f(c)}{x - c}.$$

Furthermore, if f is differentiable at every $c \in I$, we write f' or $\frac{\mathrm{d}f}{\mathrm{d}x}$ for the function $f'(x)$.

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Example 200

Consider the function $f(x) = ax + b$. Then, for all $c \in \mathbb{R}$, $f'(c) = a$.

Proof: This follows as

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} = \lim_{x\to c} \frac{ax + b - (ac + b)}{x - c} = a \lim_{x\to c} \frac{x - c}{x - c} = \lim_{x\to c} a = a.$$

■

Example 201 (The Power Rule)

For all $n \in \mathbb{N}$, if $f(x) = \alpha x^n$, then for all $c \in \mathbb{R}$,

$$f'(c) = \alpha n c^{n-1}.$$

Proof: We note that for all $n \in \mathbb{N}$,

$$(x - c) \sum_{j=0}^{n-1} x^{n-1-j} c^j = \sum_{j=0}^{n-1} x^{n-j} c^j - \sum_{j=0}^{n-1} x^{n-1-j} c^{j+1}.$$

Letting $\ell = j + 1$, we obtain

$$\begin{aligned}(x - c) \sum_{j=0}^{n-1} x^{n-1-j} c^j &= \sum_{j=0}^{n-1} x^{n-j} c^j - \sum_{\ell=1}^{n} x^{n-\ell} c^\ell \\ &= x^{n-0} c^0 - x^{n-n} c^n \\ &= x^n - c^n.\end{aligned}$$

Therefore,

$$\lim_{x\to c} \frac{\alpha x^n - \alpha c^n}{x - c} = \alpha \lim_{x\to c} \sum_{j=0}^{n-1} x^{n-1-j} c^j = \alpha \sum_{j=0}^{n-1} c^{n-1-j} c^j = \alpha n c^{n-1}.$$

■

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18 Lecture 18

Weierstrass's Example of a Continuous and Nowhere Differentiable Function

Theorem 202
If $f : I \to \mathbb{R}$ is differentiable at $c \in I$, then f is continuous at c.

Proof: Since every point of I is a cluster point of I, f is continuous at $c \in I \iff \lim_{x\to c} f(x) = f(c)$. Now,

$$\begin{aligned}\lim_{x\to c} f(x) &= \lim_{x\to c}(f(x) - f(c) + f(c)) \\ &= \lim_{x\to c}\left((x-c)\frac{f(x)-f(c)}{x-c} + f(c)\right) \\ &= 0 \cdot f'(c) + f(c) = f(c).\end{aligned}$$

□

Question 203. *Is the converse true? Does f being continuous imply that f is differentiable?*

The answer, is **no**.

Example 204
Let $f(x) = |x|$. Then, f is not differentiable at 0.

Proof: We find a sequence $x_n \to 0$ such that

$$\lim_{n\to\infty} \frac{f(x_n) - f(0)}{x_n - 0} \text{ does not exist.}$$

Let $x_n = \frac{(-1)^n}{n}$. Then, $\lim_{n\to\infty} x_n = 0$. However,

$$\frac{f(x_n) - f(0)}{x_n - 0} = \frac{|(-1)^n/n|}{(-1)^n/n} = (-1)^n,$$

and $\lim_{n\to\infty}(-1)^n$ does not exist. ■

Question 205. *If $f : \mathbb{R} \to \mathbb{R}$ is continuous, then does there exist a $c \in \mathbb{R}$ such that f is differentiable at c?*

The answer is again **no**! This was shown by Weierstrass, aka the Godfather.
The basic idea is to build a continuous function that is a sum of highly oscillating functions.

Remark 206. *Note that we number the upcoming theorems so we may reference them a bit later in this lecture.*

Theorem 207 (Theorem I)
We will show the following

1. $\forall x, y \in \mathbb{R}$, $|\cos x - \cos y| \le |x - y|$.
2. Let $c \in \mathbb{R}$. Then, for all $K \in \mathbb{N}$, $\exists y \in (c + \pi/K, c + 3\pi/K)$ such that
$$|\cos(Kc) - \cos(Ky)| \ge 1.$$

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Proof:

1. In the proof of continuity of $\sin x$, we showed that $\forall x, y \in \mathbb{R}$, $|\sin x - \sin y| \leq |x - y|$. Thus,

$$|\cos x - \cos y| = |\sin(x + \pi/2) - \sin(y + \pi/2)| \leq |x - y|.$$

2. The function $f(x) = \cos(Kx)$ is a $\frac{2\pi}{K}$-periodic function. In particular, $([-1, 1] \setminus \cos(Kc)) \subset f(c + \pi/K, c + 3\pi/K)$.

   If $\cos Kc \geq 0$, then we choose y such that $\cos(Ky) = -1$. If $\cos(Kc) < 0$, then we choose y such that $\cos(Ky) = 1$. This completes the proof

□

Theorem 208 (Theorem II)
For all $a, b, c \in \mathbb{R}$,

$$|a + b + c| \geq |a| - |b| - |c|.$$

Proof: We apply the Triangle Inequality twice:

$$|a| = |a + b + b + (-b) + (-c)| \leq |a + b + b| + |b + c| \leq |a + b + c| + |b| + |c|.$$

□

Theorem 209 (Theorem III)
We will show the following:

1. $\forall x \in \mathbb{R}$, $\sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k}$ is absolutely convergent.
2. The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k}$ is bounded and continuous.

Proof:

1. $\forall k$, $\left|\frac{\cos(160^k x)}{4^k}\right| \leq 4^{-k}$. Hence, by the Comparison Test,

$$\sum_{k=0}^{\infty} \left|\frac{\cos(160^k x)}{4^k}\right| \quad \text{converges.}$$

2. For all $x \in \mathbb{R}$, $|f(x)| \leq \sum_{k=0}^{\infty} \frac{|\cos(160^k x)|}{4^k} \leq \sum 4^{-k} = \frac{4}{3}$. Therefore, f is bounded.

   We now show that f is continuous over $\mathbb{R}$. Suppose $c \in \mathbb{R}$ and $x_n \to c$. Note that $\{|f(x_n) - f(c)|\}_n$ is bounded, and thus

$$\lim_{n\to\infty} |f(x_n) - f(c)| = 0 \iff \limsup_{n\to\infty} |f(x_n) - f(c)| = 0.$$

   We claim that for all $\epsilon > 0$, $\limsup_{n\to\infty} |f(x_n) - f(c)| \leq \epsilon$. Let $\epsilon > 0$. Choose M_0 such that $\sum_{k=M_0+1}^{\infty} 4^{-k} < \frac{\epsilon}{2}$.

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Then,

$$\begin{aligned}\limsup_{n\to\infty}|f(x_n)-f(c)| &= \limsup_n \left|\sum_{k=0}^{M_0}\frac{\cos(160^k x_n)}{4^k}-\frac{\cos(160^k c)}{4^k}+\sum_{k=M_0+1}^{\infty}\frac{\cos(160^k x_n)}{4^k}-\frac{\cos(160^k c)}{4^k}\right| \\ &\le \limsup_n \sum_{k=0}^{M_0} 4^{-k}|\cos(160^k x_n)-\cos(160^k c)|+\sum_{k=M_0+1}^{\infty}4^{-k}(|\cos(160^k x_n)|+|\cos(160^k c)|) \\ &\le \limsup_n \left(\sum_{k=0}^{M_0}40^k\right)|x_n-c|+\epsilon=\epsilon.\end{aligned}$$

□

Theorem 210 (Weierstrass)
The function $f(x)=\sum_{k=0}^{\infty}\frac{\cos(160^k x)}{4^k}$ is nowhere differentiable.

Proof: Let $c\in\mathbb{R}$. We will construct a sequence $x_n\to c$ such that $\left\{\frac{f(x_n)-f(c)}{x_n-c}\right\}_n$ is unbounded. By Theorem I 2), $\forall n\in\mathbb{N}$ there exists an x_n such that **a)** $\frac{\pi}{160^n}<x_n-c<\frac{3\pi}{160^n}$ and **b)** $|\cos(160^n c)-\cos(160^n x_n)|\ge 1$.

By **a)**, $x_n\neq 0\forall n$ and $|x_n-c|\le\frac{3\pi}{160^n}\to 0$. Let $f_k(x)=\frac{\cos(160^k x)}{4^k}$ so $f(x)=\sum f_k(x)$. Let $n\in\mathbb{N}$. Thus, denote

$$\begin{aligned}f(c)-f(x_n) &= f_n(c)-f_n(x_n)+\sum_{k=0}^{n-1}(f_k(c)-f_k(x_n))+\sum_{k=n}^{\infty}(f_k(c)-f_k(x_n)) \\ &:= a_n+b_n+c_n.\end{aligned}$$

Therefore, by Theorem II,

$$|f(c)-f(x_n)|\ge|a_n|-|b_n|-|c_n|.$$

By **b)**, $|a_n|=4^{-n}|\cos(160^k x_n)-\cos(160^k c)|\ge 4^{-n}$. Furthermore, we have

$$|b_n|\le\sum_{k=0}^{n-1}4^{-k}|\cos(160^k c)-\cos(160^k x_n)|\le\sum_{k=0}^{n-1}4^{-k}\cdot 160^k|x_n-c|\le\frac{3\pi}{160^n}\sum_{k=0}^{n-1}40^k=\frac{3\pi}{160^n}\cdot\frac{40^n-1}{39}\le\frac{4^{-n+1}}{13}.$$

Finally, we have

$$|c_n|\le\sum_{k=n+1}^{\infty}4^{-k}(|\cos(160^k c)|+|\cos(160^k x_n)|)\le 2\sum_{k=n+1}^{\infty}4^{-k}=2\cdot 4^{-n-1}\cdot\frac{4}{3}=4^{-n}\frac{2}{3}.$$

Therefore, by the above inequalities, we have

$$|f(c)-f(x_n)|\ge 4^{-n}\left(1-\frac{4}{13}-\frac{2}{3}\right)=4^{-n}\cdot\frac{1}{39}.$$

Therefore,

$$\frac{|f(c)-f(x_n)|}{|c-x_n|}\ge\frac{160^n}{3\pi}\cdot 4^{-n}\cdot\frac{1}{39}=\frac{40^n}{117\pi}.$$

Thus, $\left\{\frac{f(x_n)-f(c)}{x_n-c}\right\}_n$ is unbounded. □

Remark 211. *In other words, this proof by Weierstrass shows that there exists a continuous function that is nowhere differentiable!*

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19 Lecture 19

Differentiation Rules, Rolle's Theorem, and the Mean Value Theorem

Theorem 212

Let $f : I \to \mathbb{R}$, $g : I \to \mathbb{R}$ be differentiable at $c \in I$. Then,

1. (Linearity) $\forall \alpha \in \mathbb{R}$, $(\alpha f + g)'(c) = \alpha f'(c) + g'(c)$.
2. (Product rule) $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$.
3. (Quotient rule) If $g(x) \neq 0$ for all $x \in I$, then

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Proof:

1. We can compute this directly:

$$\lim_{x\to c} \frac{(\alpha f + g)(x) - (\alpha f + g)(c)}{x - c} = \lim_{x\to c} \alpha \frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c} = \alpha f'(x) + g'(x).$$

2. We first write

$$\frac{f(x)g(x) - f(c)g(c)}{x - c} = \frac{f(x) - f(c)}{x - c} \cdot g(x) + f(c) \cdot \frac{g(x) - g(c)}{x - c}$$

and use the fact that $\lim_{x\to c} g(x) = f(c)$.

3. The quotient rule is left as an exercise to the reader.

□

Theorem 213 (Chain Rule)

Let I_1, I_2 be two intervals, $g : I_1 \to I_2$ be differentiable at $c \in I_1$, and $f : I_2 \to \mathbb{R}$ differentiable at $g(c)$. Then, $f \circ g : I_1 \to \mathbb{R}$ is differentiable at c and

$$(f \circ g)'(c) = f'(g(c))g'(c).$$

Proof: Let $h(x) = f(g(x))$ and $d = g(c)$. We want to prove that $h'(c) = f'(d)g'(c)$. Define the following

$$u(y) = \begin{cases} \frac{f(y)-f(d)}{y-x} & y \neq d \\ f'(d) & y = d \end{cases} \quad \text{and} \quad v(y) = \begin{cases} \frac{g(x)-g(c)}{x-c} & x \neq c \\ g'(d) & x = c \end{cases}.$$

Then,

$$\lim_{y\to d} u(y) = \lim_{y\to d} \frac{f(y) - f(d)}{y - d} = f'(d) = u(d).$$

Similarly,

$$\lim_{x\to c} v(x) = \lim_{x\to c} \frac{g(x) - g(c)}{x - c} = g'(c) = v(c).$$

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In other words, u is continuous at d and v is continuous at c. Now,

$$\begin{aligned} f(y) - f(d) &= u(y)(y-d) \\ g(x) - g(c) &= v(x)(x-c) \\ \implies h(x) - h(c) &= f(g(x)) - f(d) \\ &= u(g(x))(g(x) - g(c)) \\ &= u(g(x))v(x)(x-c). \end{aligned}$$

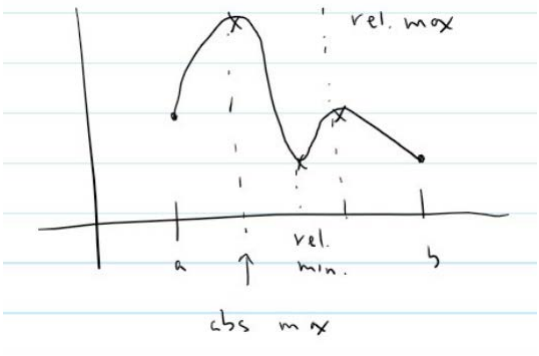
Therefore,

$$\begin{aligned} \lim_{x\to c} \frac{h(x)-h(c)}{x-c} &= \lim_{x\to c} u(g(x))v(x) \\ &= u(g(c))v(c) \\ &= f'(g(c))g'(c). \end{aligned}$$

□

Mean Value Theorem

Definition 214 (Relative Maximum/Minimum)
Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$. Then, f has a relative maximum at $c \in S$ if $\exists \delta > 0$ such that for all $x \in S$, $|x - c| < \delta \implies f(x) \leq f(c)$. The definition for relative maximum is analogous.



Theorem 215
If $f : [a, b] \to \mathbb{R}$ has a relative max or min at $c \in (a, b)$ and f is differentiable at c, then

$$f'(c) = 0.$$

Proof: If f has a relative maximum at $c \in (a, b)$ then $\exists \delta > 0$ such that $(c-\delta, c+\delta) \subset (a, b)$ and $\forall x \in (c-\delta, c+\delta)$, $f(x) \leq f(c)$. Let

$$x_n = c - \frac{\delta}{2n} \in (c - \delta, c).$$

Then, $x_n \to c$ so

$$f'(c) = \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} \geq 0.$$

Now let

$$y_n = c + \frac{\delta}{2n} \in (c, c + \delta).$$

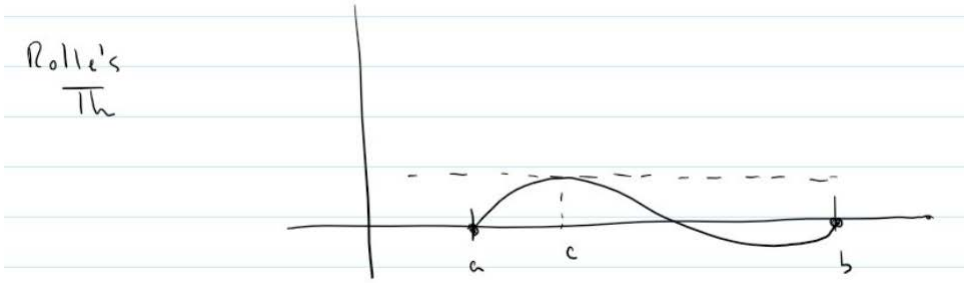
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Then, $y_n \to c$ so

$$f'(c) = \lim_{n\to\infty} \frac{f(y_n) - f(c)}{y_n - c} \leq 0.$$

Therefore, $f'(c) = 0$. The proof for relative minimum is similar and thus left to the reader. □

Theorem 216 (Rolle)
Let $f : [a, b] \to \mathbb{R}$ be continuous and differentiable on (a, b). If $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$.



Remark 217. *Are the hypotheses all necessary? This is left to the reader to figure out.*

Proof: Let $K = f(a) = f(b)$. Since f is continuous on $[a, b]$, $\exists c_1, c_2 \in [a, b]$ such that f achieves an absolute maximum at c_1 and absolute minimum at c_2. If $f(c_1) > K \implies c_1 \in (a, b)$. Therefore, $f'(c_1) = 0$ by the previous theorem. Similarly, if $f(c_2) < K$, then $c_2 \in (a, b) \implies f'(c_2) = 0$. If

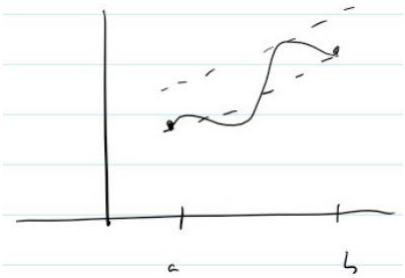
$$f(c_1) \leq K \leq f(c_2) \implies f(x) = K \ \forall x \in [a, b] \implies f'(c) - 0 \ \text{ for any } \ c \in (a, b).$$

□

Theorem 218 (Mean Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ be continuous, and let f be differentiable on (a, b). Then, $\exists c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

Remark 219. *The Mean Value Theorem is sometimes denoted MVT.*



Proof: Define $g : [a, b] \to \mathbb{R}$ with

$$g(x) = f(x) - f(b) + \frac{f(b) - f(a)}{b - a}(b - x).$$

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Then, $g(a) = g(b) = 0$. Thus, by Rolle's theorem, $\exists c \in (a, b)$ with $g'(c) = 0$, and hence

$$0 = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

□

We now look at some useful applications of the MVT.

Theorem 220
If $f : I \to \mathbb{R}$ is differentiable and $f'(x) = 0$ for all $x \in I$, then f is constant.

Proof: Let $a, b \in I$ with $a < b$. Then, f is continuous on $[a, b]$ and differentiable on (a, b). Therefore, $\exists c \in (a, b)$ such that $f(b) - f(a) = (b - a)f'(c) = 0$. Hence, $f(b) = f(a)$ for all $a, b \in I$ such that $a < b$. □

Theorem 221
Let $f : I \to \mathbb{R}$ be differentiable. Then,

1. f is increasing if and only if $f'(x) \geq 0$ for all $x \in I$ and
2. f is decreasing if and only if $f'(x) \leq 0$ for all $x \in I$.

Proof:

1. ($\Longleftarrow$) Suppose $f'(x) \geq 0$ for all $x \in I$. Let $a, b \in I$ with $a < b$. Then, by MVT, $\exists c \in (a, b)$ such that

$$f(b) - f(a) = (b - a)f'(c) \geq 0 \implies f(a) \leq f(b).$$

($\Longrightarrow$) Suppose f is increasing. Let $c \in I$ and let $\{x_n\}$ be a sequence in I such that $x_n \to c$ such that $\forall n$, $x_n < c$. Then, for all n, $f(x_n) - f(c) \leq 0 \implies \frac{f(x_n)-f(c)}{x_n-c} \geq 0$. Therefore,

$$f'(c) = \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} \geq 0.$$

Now let $\{x_n\}$ be a sequence in I such that $x_n \to c$ such that $\forall n$, $x_n > c$. Then, for all n, $f(x_n) - f(c) \geq 0 \implies \frac{f(x_n)-f(c)}{x_n-c} \geq 0$. Therefore,

$$f'(c) = \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} \geq 0.$$

Hence, in either case, $f'(c) \geq 0$.

2. Notice that f is decreasing if and only if $-f$ is increasing, and apply 1. to $-f$.

□

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20 Lecture 20

Taylor's Theorem and the Definition of Riemann Sums

Taylor's Theorem

Remark 222. *Taylor's theorem is essentially the Mean Value Theorem for higher order derivatives.*

Definition 223 (n-times Differentiable)
We say $f : I \to \mathbb{R}$ is n-times differentiable on $J \subset I$ if $f', f'', \ldots, f^{(n)}$ exist at every point in J.

Notation 224
We denote the n-th derivative of f as $f^{(n)}$ (as used above).

Theorem 225 (Taylor)
Suppose $f : [a, b] \to \mathbb{R}$ is continuous and has n continuous derivatives on $[a, b]$ such that $f^{(n+1)}$ exists on (a, b). Given $x_0, x \in [a, b]$, there exists a $c \in (x_0, x)$ such that

$$f(x) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}.$$

Denote the large sum as $P_n(x)$ and the last term with $R_n(x)$.

Definition 226
$P_n(x)$ is the n-th order Taylor polynomial for f at x_0. $R_n(x)$ is the n-th order remainder term.

We will essentially apply the Mean Value Theorem $n + 1$ times to prove Taylor's theorem.

Proof: Let $x, x_0 \in [a, b]$. If $x = x_0$ then any c will satisfy the theorem. So, suppose $x \neq x_0$. Let $M_{x,x_0} = \frac{f(x) - P_n(x)}{(x - x_0)^{n+1}}$. Hence,

$$f(x) = P_n(x) + M_{x,x_0}(x - x_0)^{n+1}.$$

Now, for $0 \leq k \leq n$,

$$f^k(x_0) = P_n^{(k)}(x_0).$$

Let $g(s) = f(s) - P_n(s) - M_{x,x_0}(s - x_0)^{n+1}$ (notably, $n + 1$-times differentiable. Then,

$$\begin{aligned}
g(x_0) &= f(x_0) - P_n(x_0) - M_{x,x_0}(x_0 - x_0)^{n+1} = 0 \\
g'(x_0) &= f'(x_0) - P_n'(x_0) - M_{x,x_0}(n+1)(x_0 - x_0)^n = 0 \\
&\vdots \\
g^{(n)}(x_0) &= f^{(n)}(x_0) - P_n^{(n)}(x_0) - M_{x,x_0}(n+1)!(x_0 - x_0) = 0.
\end{aligned}$$

Now, notice that $g(x) = 0$ and $g(x_0) = 0$. By the MVT, there exists an $x_1 \in (x_0, x)$ such that $g'(x_1) = 0$. Thus, $g'(x_0) = 0$ and $g'(x_1) = 0$. Therefore, $\exists x_2 \in (x_0, x_1)$ such that $g''(x_2) = 0$. Continuing, we analogously find x_n

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between x_0 and x_{n-1} such that $g^{(n)}(x_n) = 0$. Then, finally, $g^{(n)}(x_0) = 0$ and $g^{(n)}(x_n) = 0$ implies $\exists c \in (x_0, x_n)$ (and thus between x_0 and x) such that

$$g^{(n+1)}(c) = 0.$$

We may compute

$$\frac{\mathrm{d}^{n+1}}{\mathrm{d}s^{(n+1)}} M_{x,x_0}(s - x_0)^{n+1} = M_{x,x_0}(n+1)!.$$

Furthermore, $P_n^{(n+1)}(c) = 0$ since P_n is a polynomial of degree n. Thereforee,

$$0 = g^{(n+1)}(c) = f^{(n+1)}(c) - M_{x,x_0}(n+1)!,$$

which implies $M_{x,x_0} = \frac{f^{(n+1)!}(c)}{(n+1)!}$ and thus

$$f(x) = P_n(x) + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}.$$

□

Theorem 227 (Second Derivative Test)
Suppose $f : (a, b) \to \mathbb{R}$ has two continuous derivatives. If $x_0 \in (a, b)$ such that $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a strict relative minimum at x_0.

Proof: Since f'' is continuous at x_0 and

$$\lim_{c \to x_0} f''(c) = f''(x_0) > 0,$$

we have that $\exists \delta > 0$ such that for all $c \in (x_0 - \delta, x_0 + \delta)$, $f''(c) > 0$. Let $x \in (x_0 - \delta, x_0 + \delta)$ (as you will show in your homework). Then, by Taylor's theorem, $\exists c$ between x and x_0 (hence $c \in (x_0 - \delta, x_0 + \delta)$) such that

$$f(x) = f(x_0) + \frac{f''(c)}{2}(x - x_0)^2 \geq f(x_0),$$

with $f(x) > f(x_0)$ if $x \neq x_0$. □

The Riemann Integral

Remark 228. *Riemann integration is the first rigorous theory of 'area' that agrees with experience (areas of rectangles, triangles, circles), and it is the inverse of differentiation. However, it is not a complete theory of area (see Lebesgue integration).*

The Riemann Integral

Definition 229
We define the set

$$C([a, b]) := \{f : [a, b] \to \mathbb{R} \mid f \text{ is continuous}\}.$$

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Definition 230 (Partition)

A partition $\underline{x}$ of $[a,b]$ is a finite set

$$\underline{x} = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

The norm of $\underline{x}$, denoted $\|\underline{x}\|$, is the number

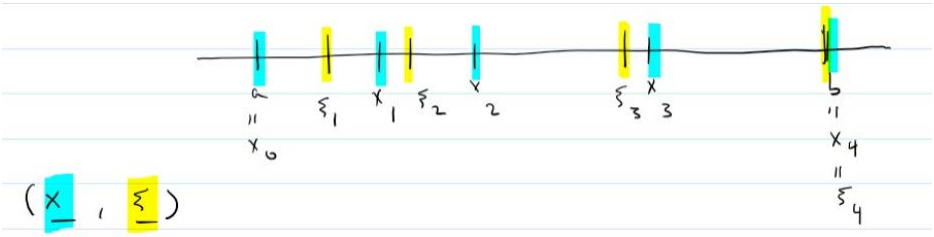
$$\|\underline{x}\| := \max\{x_1 - x_0, x_2 - x_1, \ldots, x_n - x_{n-1}\}.$$

Definition 231 (Tag)

If $\underline{x}$ is a partition, a tag of $\underline{x}$ is a finite set $\underline{\xi} = \{\xi_1, \ldots, \xi_n\}$ such that

$$a = x_0 \leq \xi_1 \leq x_1 \leq \xi_2 \leq x_2 \leq \cdots \leq x_{n-1} \leq \xi_n \leq x_n = b.$$

The pair $(\underline{x}, \underline{\xi})$ is referred to as a tagged partition.



Example 232

Consider the tagged partition $(\underline{x}, \underline{\xi}) = (\{1, 3/2, 2, 3\}, \{5/4, 7/4, 5/2\}$. Then,

$$\|\underline{x}\| = \max\{3/2 - 1, 2 - 3/2, 3 - 2\} = 1.$$

Definition 233 (Riemann sum)

The Riemann sum of f corresponding to $(\underline{x}, \underline{\xi})$ is the number

$$S_f(\underline{x}, \underline{\xi}) := \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}).$$

Let's try to understand how to interpret this using a picture. For $f \in C([a,b])$ positive, $S_f(\underline{x}, \underline{\xi})$ is an approximate area under the graph of f. As $\|\underline{x}\| \to 0$, we *should* expect these approximate areas to converge to a number A, which we **interpret** as the area under the curve f on the interval $[a,b]$.

Question 234. *Do these approximate sums actually converge?*

We will answer this question and more during the next few lectures.

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21 Lecture 21

The Riemann Integral of a Continuous Function

Theorem 235 (Riemann Integral)
Let $f \in C([a,b])$. Then, there exists a unique number denoted $\int_a^b f(x)\,\mathrm{d}x \in \mathbb{R}$ with the following property: for all sequences of tagged partitions $\{(\underline{x}^r, \underline{\xi}^r)\}$ such that $\|\underline{x}^r\| \to 0$, we have

$$\lim_{r\to\infty} S_f(\underline{x}^r, \underline{\xi}^r) = \int_a^b f(x)\,\mathrm{d}x.$$

Remark 236. *Uniqueness follows immediately from uniqueness of limits of sequences of real numbers. All we need to prove is existence of* $\int_a^b f(x)\,\mathrm{d}x$.

Before giving the proof of the theorem, we first prove some useful facts. Note that we number the next few theorems to use them in our proof of the above theorem.

Definition 237 (Modulus of Continuity)
For $f \in C([a,b])$, $\eta > 0$, we define the modulus of continuous

$$w_f(\eta) = \sup\{|f(x) - f(y)| \mid |x-y| \le \eta\}.$$

Remark 238. *Note that we number the following theorems to reference later.*

Theorem 239 (Theorem I)
For all $f \in C([a,b])$, $\lim_{\eta\to 0} w_f(\eta) = 0$. In other words, for all $\epsilon > 0$, there exists a $\delta > 0$ such that $\forall \eta < \delta$, $w_f(\eta) < \epsilon$.

Proof: Let $\epsilon > 0$. Since $f \in C([a,b])$, f is uniformly continuous on $[a,b]$ (as continuity on a bounded interval is equivalent to uniform continuity on the same interval). Thus, $\exists \delta_0 > 0$ such that if $|x-y| < \delta_9$, then $|f(x)-f(y)| < \epsilon/2$. Choose $\delta = \delta_0$ and let $\eta < \delta$. Then, if $|x-y| \le \eta < \delta = \delta_0$, then

$$|f(x) - f(y)| < \frac{\epsilon}{2}.$$

Therefore, $\epsilon/2$ is an upper bound for $\{|f(x)-f(y)| \mid |x-y| \le \eta\}$. Hence,

$$w_f(\eta) \le \epsilon/2 < \epsilon.$$

□

Theorem 240 (Theorem II)
If $(\underline{x}, \underline{\xi})$ and $(\underline{x}', \underline{\xi}')$ are tagged partitions of $[a,b]$ such that $\underline{x} \subset \underline{x}'$, then if $f \in C([a,b])$ then

$$|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| \le w_f(\|\underline{x}\|)(b-a).$$

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Definition 241 (Refinement)
If $(\underline{x}, \underline{\xi})$ and $(\underline{x}', \underline{\xi}')$ are tagged partitions of $[a, b]$ such that $\underline{x} \subset \underline{x}'$, we say $\underline{x}'$ is a refinement of $\underline{x}$.

Remark 242. *Refinements of $\underline{x}$ are obtained by adding more partition points.*

Proof: For $k = 1, \ldots, n$, let

$$\underline{y}(k) = \{x_{k-1} = x'_\ell, x'_{\ell+1}, \ldots, x'_m = x_k\}$$
$$\underline{\eta}(k) = \{\xi'_{\ell+1}, \xi'_{\ell+2}, \ldots, \xi'_m\}.$$

Then,

$$\begin{aligned} |f(\xi_k)(x_k - x_{k-1}) - S_f(\underline{y}(k), \underline{\eta}(k))| &= \left| f(\xi_k) - \sum_{j=\ell+1}^{m} f(\xi'_j)(x'_j - x'_{j-1}) \right| \\ &= \left| \sum_{j=\ell+1}^{m} (f(\xi_k) - f(\xi'_j))(x'_j - x'_{j-1}) \right| \end{aligned}$$

since $\sum_{j=1}^{m} x'_j - x'_{j-1} = x_m - x'_\ell = x_k - x_{k-1}$. Hence,

$$\begin{aligned} |f(\xi_k)(x_k - x_{k-1}) - S_f(\underline{y}(k), \underline{\eta}(k))| &\leq \sum_{j=\ell+1}^{m} |f(\xi_k) - f(\xi'_j)|(x'_j - x'_{j-1}) \\ &\leq \sum_{j=\ell+1}^{m} w_f(|x_k - x_{k-1}|)(x'_j - x'_{j-1}) \\ &\leq w_f(\|\underline{x}\|)(x_k - x_{k-1}). \end{aligned}$$

Thus,

$$\begin{aligned} |S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| &= \left| \sum_{k=1}^{m} (f(\xi_k)(x_k - x_{k-1}) - S_f(\underline{y}(k), \underline{\eta}(k))) \right| \\ &\leq \sum_{k=1}^{m} |f(\xi_k)(x_k - x_{k-1}) - S_f(\underline{y}(k), \underline{\eta}(k))| \\ &\leq w_f(\|\underline{x}\|) \sum_{k=1}^{n} x_k - x_1 = w_f(\|\underline{x}\|)(b - a). \end{aligned}$$

□

Theorem 243 (Theorem III)
If $(\underline{x}, \underline{\xi})$ and $(\underline{x}', \underline{\xi}')$ are any two tagged partitions of $[a, b]$ and $f \in C([a, b])$, then

$$|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| \leq (w_f(\|\underline{x}) + w_f(\|\underline{x}'\|))(b - a).$$

Proof: Let $\underline{x}'' = \underline{x} \cup \underline{x}'$ (i.e. a common refinement), and $\underline{\xi}''$ be a tag of $\underline{x}''$. Then, by Theorem II,

$$\begin{aligned} |S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| &\leq |S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}'', \underline{\xi}'')| - |S_f(\underline{x}', \underline{\xi}') - S_f(\underline{x}'', \underline{\xi}'')| \\ &\leq w_f(\|\underline{x}\|)(b - a) + w_f(\|\underline{x}'\|)(b - a). \end{aligned}$$

□

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We now have the theorems necessary to prove the big theorem at the beginning of this section.

Proof: Let $\{\underline{y}(r), \underline{\zeta}(r)\}_r$ be a sequence of tagged partitions with $\|\underline{y}(r)\| \to 0$ as $r \to \infty$.

Claim 1: $\{S_f(\underline{y}(r), \underline{\zeta}(r)\}_r$ is a Cauchy sequence. Proof: Let $\epsilon > 0$. By Theorem I, $\exists \delta > 0$ such that $\forall \eta < \delta$,

$$w_f(\eta) < \frac{\epsilon}{2(b-a)}.$$

Since $\|\underline{y}(r)\| \to 0$, $\exists M_0 \in \mathbb{N}$ such that $\forall r \geq M_0$,

$$\|\underline{y}(r)\| < \delta.$$

Choose $M = M_0$. Then, if $r, s \geq M = M_0$,

$$|S_f(\underline{y}(r), \underline{\zeta}(r)) - S_f(\underline{y}(s), \underline{\zeta}(r))| \leq (w(\|\underline{y}(r)\| + w(\|\underline{y}(s)\|))(b-a)$$

by Theorem III. Hence, by the above inequalities, it follows that

$$|S_f(\underline{y}(r), \underline{\zeta}(r)) - S_f(\underline{y}(s), \underline{\zeta}(r))| < \left(\frac{\epsilon}{2(b-a)} + \frac{\epsilon}{2(b-a)}\right)(b-a) = \epsilon.$$

This proves Claim 1. Let $L = \lim_{r\to\infty} S_f(\underline{y}(r), \underline{\zeta}(r))$, which exists as Cauchy sequences of real numbers are convergent sequences.

Claim 2: Let $\{(\underline{x}(r), \underline{\xi}(r))\}_r$ be **any** sequence of partitions with $\|\underline{x}(r)\| \to 0$. Then,

$$\lim_{r\to\infty} S_f(\underline{x}(r), \underline{\xi}(r)) = L.$$

With $(\underline{y}(r), \underline{\zeta}(r))$ as before, we have by the Triangle Inequality and Theorem III that

$$\begin{aligned} |S_f(\underline{x}(r), \underline{\xi}(r)) - L| &\leq |S_f(\underline{x}(r), \underline{\xi}(r)) - S_f(\underline{y}(r), \underline{\zeta}(r))| + |S_f(\underline{y}(r), \underline{\zeta}(r)) - L| \\ &\leq (w_f(\|\underline{x}(r)\|) + w_f(\|\underline{y}(r)\|))(b-a) + |S_f(\underline{y}(r), \underline{\zeta}(r)) - L| \to 0 \end{aligned}$$

as $r \to \infty$ by Theorem I and by the definition of L. Thus, by the Squeeze Theorem,

$$|S_f(\underline{x}(r), \underline{\xi}(r)) - L| \to 0 \quad \text{as} \quad r \to 0.$$

□

Properties of the Riemann Integral

Notation 244

We will often abbreviate $\int_a^b f(x)\,\mathrm{d}x$ to $\int_a^b f$.

Theorem 245 (Linearity)

If $f, g \in C([a,b])$ and $\alpha \in \mathbb{R}$, then

$$\int_a^b (\alpha f + g) = \alpha \int_a^b f + \int_a^b g.$$

Proof: Let $\{(\underline{x}(r), \underline{\xi}(r))\}_r$ be a sequence of tagged partitions such that $\|\underline{x}(r)\| \to 0$. Then,

$$S_{\alpha f+g}(\underline{x}(r), \underline{\xi}(r)) = \alpha S_f(\underline{x}(r), \underline{\xi}(r)) + S_g(\underline{x}(r), \underline{\xi}(r)).$$

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Therefore,

$$\begin{aligned}\int_a^b (\alpha f + g) &= \lim_{r\to\infty} S_{\alpha f+g}(\underline{x}(r), \underline{\xi}(r)) \\ &= \lim_{r\to\infty} (\alpha S_f(\underline{x}(r), \underline{\xi}(r)) + S_g(\underline{x}(r), \underline{\xi}(r)) \\ &= \alpha \int_a^b f + \int_a^b g.\end{aligned}$$

□

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22 Lecture 22

The Fundamental Theorem of Calculus, Integration by Parts, and Change of Variable Formula

Theorem 246 (Additivity)
If $f \in C([a,b])$ and $a < c < b$, then

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof: Let $\{(\underline{y}(r), \underline{\zeta}(r))\}_r$ and $\{(\underline{z}(r), \underline{\eta}(r))\}_r$ be tagged partitions of $[a,c]$ and $[c,b]$ respectively such that $\|\underline{y}(r)\| \to 0$ and $\|\underline{z}(r)\| \to 0$. Define

$$\underline{x}(r) = \underline{y}(r) \cup \underline{z}(r)$$
$$\underline{\xi}(r) = \underline{\zeta}(r) \cup \underline{\eta}(r),$$

a sequence of tagged partitions of $[a,b]$. Then,

$$\|\underline{x}(r)\| \leq \|\underline{y}(r)\| + \|\underline{z}(r)\| \to 0.$$

Thus,

$$\begin{aligned}\int_a^b f &= \lim_{t\to\infty} S_f(\underline{x}(r), \underline{\xi}(r)) \\ &= \lim_{r\to\infty} (S_f(\underline{y}(r), \underline{\zeta}(r)) + S_f(\underline{z}(r), \underline{\eta}(r))) \\ &= \int_a^c f + \int_c^b f.\end{aligned}$$

□

Theorem 247
Let $f \in C([a,b])$, and

$$m_f = \inf\{f(x) \mid x \in [a,b]\} \in \mathbb{R}$$
$$M_f = \sup\{f(x) \mid x \in [a,b]\} \in \mathbb{R}.$$

Then,

$$m_f(b-a) \leq \int_a^b f \leq M_f(b-a).$$

Proof: Let $\{(\underline{x}(r), \underline{\xi}(r))\}_r$ be a sequence of tagged partitions with $\|\underline{x}(r)\| \to 0$. Then,

$$S_f(\underline{x}(r), \underline{\xi}(r)) = \sum_{k=1}^n f(\xi_k(r))(x_k(r) - x_{k-1}(r)) \geq m_f \sum_{k=1}^n (x_k(r) - x_{k-1}(r)) = m_f(b-a).$$

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Similarly,

$$S_f(\underline{x}(r), \underline{\xi}(r)) = \sum_{k=1}^{n} f(\xi_k(r))(x_k(r) - x_{k-1}(r)) \leq M_f \sum_{k=1}^{n} (x_k(r) - x_{k-1}(r)) = M_f(b-a).$$

Therefore, for all r,

$$m_f(b-a) \leq S_f(\underline{x}(r), \underline{\xi}(r)) \leq M_f(b-a) \implies m_f(b-a) \leq \int_a^b f \leq M_f(b-a).$$

□

Theorem 248

Suppose $f \in C([a,b])$ and $g \in C([a,b])$.

1. If $\forall x \in [a,b]$ $f(x) \leq g(x)$, then
$$\int_a^b f \leq \int_a^b g.$$
2. (Triangle Inequality for integrals): $|\int_a^b f| \leq \int_a^b |f|$.

Proof:

1. Let $\{(\underline{x}(r), \underline{\xi}(r)\}_r$ be a sequence of tagged partitions such that $\|\underline{x}(r)\| \to 0$. Then, for all $r \in \mathbb{N}$,
$$\begin{aligned} S_f(\underline{x}(r), \underline{\xi}(r)) &= \sum_{j=1}^{n} f(\xi_j(r))(x_j(r) - x_{j-1}(r)) \\ &\leq \sum_{j=1}^{n} g(\xi_j(r))(x_j(r) - x_{j-1}(r)) \\ &= S_g(\underline{x}(r), \underline{\xi}(r)). \end{aligned}$$
Then, letting $r \to \infty$, we get that
$$\int_a^b f \leq \int_a^b g.$$
2. Notice that $\pm f(x) \leq |f(x)|$ for all x, and thus
$$\pm \int_a^b f \leq \int_a^b |f| \implies -\int_a^b f \leq \int_a^b f \leq \int_a^b |f|.$$
Therefore, $|\int_a^b f| \leq \int_a^b |f|$.

□

Remark 249. *There are some conventions that are worth noting:*

1. $\int_a^a f := 0$. *This is consistent with our definitions and theorems thus far as* $\lim_{b \to a} |\int_a^b f| = 0$.
2. $\int_a^b f = -\int_b^a f$.

Fundamental Theorem of Calculus

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Theorem 250 (Fundamental Theorem of Calculus)
Suppose $f \in C([a,b])$.

1. If $F : [a,b] \to \mathbb{R}$ is differentiable and $F' = f$, then
$$\int_a^b f = F(b) - F(a).$$
2. The function $G(x) := \int_a^x f$ is differentiable on $[a,b]$ and
$$\begin{cases} G' = f \\ G(a) = 0 \end{cases}.$$

Remark 251. *We sometimes abbreviate the Fundamental Theorem of Calculus to FTC.*

Proof:

1. Let $\{(\underline{x}(r))\}_r$ be a sequence of partitions with $\|\underline{x}\| \to 0$. Then, by the Mean Value Theorem, $\forall r \forall j$, there exists a $\xi_j(r) \in [x_{j-1}(r), x_j(r)]$ such that
$$F(x_j(r)) - F(x_{j-1}(r)) = F'(\xi_j(r))(x_j(r) - x_{j-1}(r)) = f(\xi_j(r))(x_j(r) - x_{j-1}(r)).$$
Thus,
$$\begin{aligned} \int_a^b f &= \lim_{r\to\infty} \sum_{j=1}^{n(r)} f(\xi_j(r))(x_j(r) - x_{j-1}(r)) \\ &= \lim_{r\to\infty} \sum_{j=1}^{n(r)} F(x_j(r)) - F(x_{j-1}(r)) \\ &= \lim_{r\to\infty} (F(b) - F(a)) = F(b) - F(a). \end{aligned}$$
2. Let $c \in [a,b]$. We wish to show that
$$\lim_{x\to c} \frac{\int_a^x f - \int_a^c f}{x - c} = f(c).$$
Let $\epsilon > 0$. Then, since f is continuous at c, $\exists \delta_0 > 0$ such that
$$|t - c| < \delta_0 \implies |f(t) - f(c)| < \epsilon/2.$$
Choose $\delta = \delta_0$. Suppose $0 < x - c < \delta$. If $t \in [c, x]$, then
$$|t - c| \le |x - c| < \delta = \delta_0.$$

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Thus,

$$\begin{aligned}\left|\frac{1}{x-c}\int_c^x f(t)\,\mathrm{d}t - f(c)\right| &= \left|\frac{1}{x-c}\int_c^t f(t)\,\mathrm{d}t - \frac{1}{x-c}\int_x^c f(c)\,\mathrm{d}t\right| \\ &= \frac{1}{x-c}\left|\int_c^x (f(t)-f(c))\,\mathrm{d}t\right| \\ &\leq \frac{1}{x-c}\int_c^x |f(t)-f(c)|\,\mathrm{d}t \\ &\leq \frac{1}{x-c}\int_c^x \epsilon/2\,\mathrm{d}t \\ &= \frac{1}{x-c}\cdot\frac{\epsilon}{2}(x-c) = \frac{\epsilon}{2}.\end{aligned}$$

A similar argument holds for $0 < c - x < \delta$. Thus,

$$0 < |x-c| < \delta \implies \left|\frac{\int_a^x f - \int_a^c f}{x-c} - f(c)\right| \leq \frac{\epsilon}{2} < \epsilon.$$

□

Theorem 252 (Integration by Parts)
Suppose $f, g \in C([a,b])$ and $f', g' \in C([a,b])$. Then,

$$\int_a^b f'g = (f(b)g(b) - f(a)g(a)) - int_a^b fg'.$$

Proof: We have

$$(fg)' = f'g + fg'.$$

Therefore, by the Fundamental Theorem of Calculus,

$$f(b)g(b) - f(a)g(a) = \int_a^b f'g + \int_a^b fg'.$$

□

Remark 253. *We sometimes abbreviate Integration By Parts as IBP.*

Lemma 254 (Riemann-Lebesgue)
Suppose $f \in C([-\pi,\pi])$ and $f' \in C([-\pi,\pi])$ with f 2π-periodic with $f(-\pi) = f(\pi)$. For $n \in \mathbb{N} \cup \{0\}$, let

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,\mathrm{d}x$$
$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,\mathrm{d}x.$$

Then,

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = 0.$$

Definition 255 (Fourier Coefficients)
The a_n, b_n defined in the above lemma are referred to as the Fourier coefficients of f.

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Proof: Using IBP, we have

$$\begin{aligned}
|b_n| &= \frac{1}{\pi}\left|\int_{-\pi}^{\pi} f(x)\cos(nx)\,\mathrm{d}x\right| \\
&= \frac{1}{\pi}\left|\int_{-\pi}^{\pi} (\frac{1}{n}\sin(nx))' f(x)\,\mathrm{d}x\right| \\
&= \left|\frac{1}{n}(f(\pi)\sin(n\pi) - f(-pi)\sin(n(-\pi))) - \frac{1}{n}\int_{-\pi}^{\pi}\sin(nx)f'(x)\,\mathrm{d}x\right|.
\end{aligned}$$

Notice that $\sin(n\pi) = \sin(n(-\pi)) = 0$ for all $n \in \mathbb{N}$. Hence,

$$\begin{aligned}
|b_n| &\le \frac{1}{n}\int_{-\pi}^{\pi} |\sin(nx)||f'(x)|\,\mathrm{d}x \\
&\le \frac{1}{n}\int_{-\pi}^{\pi} |f'| \to 0.
\end{aligned}$$

By the Squeeze Theorem, $|b_n| \to 0$. A similar arguments works for a_n. □

Theorem 256 (Change of Variables)
Let $\varphi : [a,b] \to [c,d]$ be continuously differentiable with $\varphi' > 0$ on $[a,b]$, $\varphi(a) = c$, and $\varphi(b) = d$. Then,

$$\int_c^d f(u)\,\mathrm{d}u = \int_a^b f(\varphi(x))\varphi'(x)\,\mathrm{d}x.$$

Proof: Let $F : [a,b] \to \mathbb{R}$ such that $F' = f$. Then,

$$F(\varphi(x))' = f(\varphi(x)).$$

Hence, by the FTC,

$$\begin{aligned}
\int_a^b f(\varphi(x))\varphi'(x)\,\mathrm{d}x &= \int_a^b F(\varphi(x))'\,\mathrm{d}x \\
&= F(\varphi(b)) - F(\varphi(a)) \\
&= F(d) - F(c).
\end{aligned}$$

Furthermore, by the FTC,

$$\int_c^d f(u)\,\mathrm{d}u = \int_c^d F(u)'\,\mathrm{d}u = F(d) - F(c).$$

□

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23 Lecture 23

Pointwise and Uniform Convergence of Sequences of Functions

Sequences of Function

Power Series

Remark 257. *Power series motivate the general discussion of sequences of functions.*

Definition 258 (Power series)
A power series about x_0 is a series of the form

$$\sum_{m=0}^{\infty} a_m(x-x_0)^m.$$

Theorem 259
Suppose

$$R = \lim_{m\to\infty} |a_m|^{1/n}$$

exists, and let

$$p = \begin{cases} \frac{1}{R} & R > 0 \\ \infty & R = 0. \end{cases}$$

Then, $\sum a_m(x-x_0)^m$ converges absolutely if $|x-x_0| < p$ and diverges if $|x-x_0| > p$.

Definition 260 (Radius of Convergence)
In the above theorem, we define p to be the radius of convergence.

Proof: We have

$$\lim_{n\to\infty} |a_m(x-x_0)^m|^{1/m} = R|x-x_0|,$$

and the theorem follows by the Root test. □

Suppose $\sum a_m(x-x_0)^m$ is a power series with radius of convergence p. Furthermore, define $f : (x_0-p, x_0+p) \to \mathbb{R}$ such that

$$f(x) := \sum_{m=0}^{\infty} a_m(x-x_0)^m.$$

Then, f is a limit of a sequence of functions

$$f(x) = \lim_{n\to\infty} f_n(x),$$

for $x \in (x_0-p, x_0+p)$ and where

$$f_n(x) = \sum_{m=0}^{n} a_m(x-x_0)^m.$$

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Example 261

For example, we have

$$f(x) = \frac{1}{1-x} = \sum_{m=0}^{\infty} x^m.$$

Question 262. *This concept begs a number of questions:*

1. *Is f continuous?*
2. *Is f differentiable, and does $f' = \lim_{n\to\infty} f_n'$?*
3. *If 1. is true, does*

$$\int_a^b f = \lim_{n\to\infty} \int_a^b f_n?$$

These questions will be the key motivator for the last section of this course.

Pointwise and Uniform Convergence

We now consider a setting far more general than power series.

Definition 263 (Pointwise Convergence)

For $n \in \mathbb{N}$, let $f_n : S \to \mathbb{R}$. Let $f : S \to \mathbb{R}$. We say that $\{f_n\}$ converges pointwise to f if for all $x \in S$,

$$\lim_{n\to\infty} f_n(x) = f(x).$$

Let's consider some examples.

1. Let $f_n(x) = x^n$ on [0,1]. Then,

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 0 & x \in [0,1) \\ 1 & x = 1 \end{cases}.$$

   Thus, $\{f_n\}$ converges to the above pointwise function. Hence, notice that a sequence of continuous functions may not converge pointwise to a continuous function!

2. Let $f_n(x) = \sum_{m=0}^{n} x^m$ for $x \in (-1,1)$. Then,

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \sum_{m=0}^{n} x^m = \frac{1}{1-x}.$$

   Hence, pointwise, this sequence converges to its power series (see the above example).

3. Let $f_n(x) : [0,1] \to \mathbb{R}$ be defined by

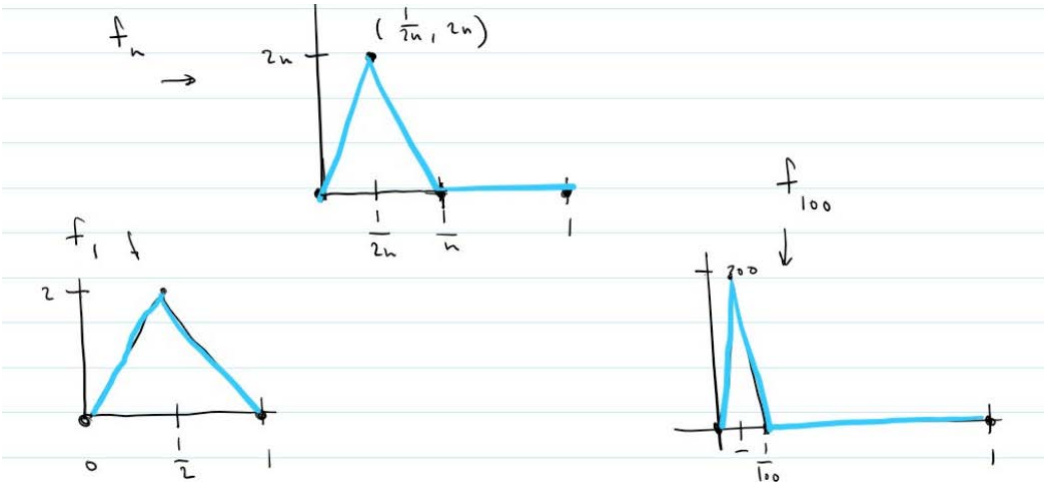
$$f_n(x) = \begin{cases} 4n^2 x & x \in \left[0, \frac{1}{2n}\right] \\ 4n - 4n^2 x & x \in \left[\frac{1}{2n}, \frac{1}{n}\right] \\ 0 & x \in \left[\frac{1}{n}, 1\right] \end{cases}.$$

   We can picture this sequence (on the next page)

   Then, $\lim_{n\to\infty} f_n(0) = \lim_{n\to\infty} 0 = 0$. Let $x \in (0,1]$. Let $N \in \mathbb{N}$ such that $\frac{1}{N} < x$. Then, for all $n \geq N$,

$$f_n(x) = 0.$$

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Therefore,

$$\{f_n(x)\} = f_1(x), \ldots, f_{N-1}(x), 0, 0, 0, \ldots.$$

Hence, $\lim_{n\to\infty} f_n(x) = 0$ for all $x \in [0,1]$. Thus, $\{f_n\}$ converges pointwise to $f(x) = 0$ on $[0,1]$.

Definition 264 (Uniform Convergence)
For $n \in \mathbb{N}$, let $f_n : S \to \mathbb{R}$, and let $f : S \to \mathbb{R}$. Then, we say f_n converges to f uniformly or **converges uniformly to** f if $\forall \epsilon > 0$ $\exists M \in \mathbb{N}$ such that for all $n \geq M$ $\forall x \in S$,

$$|f_n(x) - f(x)| < \epsilon$$

Theorem 265
If $f_n : S \to \mathbb{R}$, $f : S \to \mathbb{R}$, and $f_n \to f$ uniformly, then $f_n \to f$ pointwise.

Proof: Let $c \in S$ and let $\epsilon > 0$. Then, $f_n \to f$ uniformly implies that there exists $M_0 \in \mathbb{N}$ such that for all $n \geq M, \forall x \in S$, $|f_n(x) - f(x)| < \epsilon$. Choose $M = M_0$. Then, $\forall n \geq M$,

$$|f_n(c) - f(c)| < \epsilon.$$

Thus, $\lim_{n\to\infty} f_n(c) = f(c)$ for all $c \in S$, and therefore $f_n \to f$ pointwise. □

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24 Lecture 24

Uniform Convergence, the Weierstrass M-Test, and Interchanging Limits

Theorem 266

Let $f_n(x) = x^n$, and let $f(x) = \begin{cases} 0 & x \in [0,1) \\ 1 & x = 1 \end{cases}$.

1. $\forall 0 < b < 1$, $f_n \to f$ uniformly on $[0, b]$.
2. f_n does not converge to f uniformly on $[0, 1]$.

Proof:

1. Let $\epsilon > 0$. Since $b \in (0,1)$, $b^n \to 0$. Therefore, $\exists M_0 \in \mathbb{N}$ such that for all $n \geq M_0$, $b^n < \epsilon$. Choose $M = M_0$. Then, $\forall n \geq M, \forall x \in [0, b]$,
$$|f_n(x) - f(x)| = |f_n(x)| = x^n \leq b^n < \epsilon.$$
Thus, $f_n \to f$ uniformly on $[0, b]$.

Before proving the other part, we first note the following negation:

Negation 267 (Not Uniformly Convergent)

$f_n : S \to \mathbb{R}$ does not converge to $f : S \to \mathbb{R}$ uniformly if $\exists \epsilon_0 > 0$ such that $\forall M \in \mathbb{N}$, $\exists n \geq M$ and $\exists x \in S$ with $|f_n(x) - f(x)| \geq \epsilon_0$.

2. Hence, for our example, choose $\epsilon_0 = \frac{1}{4}$. Let $M \in \mathbb{N}$ and choose $x = \left(\frac{1}{2}\right)^{\frac{1}{M}} \in (0,1)$. Thus,
$$|f_M(x) - f(x)| = f_M(x) = \frac{1}{2} > \epsilon_0.$$

□

Theorem 268 (Weierstrass M-test)

let $f_j : S \to \mathbb{R}$ and suppose $\exists M_j > 0$ such that

a) $\forall x \in S$, $|f_j(x)| \leq M_j$.

b) $\sum_{j=1}^{\infty} M_j$ converges.

Then,

1. $\forall x \in S$, $\sum_{j=1}^{\infty} f_j(x)$ converges absolutely.
2. Let $f(x) = \sum_{j=1}^{\infty} f_j(x)$ for $x \in S$. Then,
$$\sum_{j=1}^{n} f_j \to f \quad \text{uniformly on} \quad S.$$

Proof:

1. The first part follows from a), b), and the Comparison Test.

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2. Let $\epsilon > 0$. Since $\sum M_j$ converges, $\exists N_0 \in \mathbb{N}$ such that $\forall n \geq N_0$,

$$\sum_{j=n+1}^{\infty} M_j = \left| \sum_{j=1}^{\infty} M_j - \sum_{j=1}^{n} M_j \right| < \epsilon.$$

Choose $N = N_0$. Then, for all $n \geq N$ and $\forall x \in S$,

$$\begin{aligned} \left| f(x) - \sum_{j=1}^{n} f_j(x) \right| &= \left| \sum_{j=n+1}^{\infty} f_j(x) \right| \\ &\leq \sum_{j=n+1}^{\infty} |f_j(x)| \\ &\leq \sum_{j=n+1}^{\infty} M_j < \epsilon. \end{aligned}$$

Thus, $\sum_{j=1}^{n} f_j \to f$ uniformly on S.

□

Interchange of Limits

Remark 269. *In general, limits cannot be interchanged.*

Example 270

For instance, consider the following example:

$$\lim_{n\to\infty} \lim_{k\to\infty} \frac{n/k}{n/k+1} = \lim_{n\to\infty} \frac{0}{0+1} = 0$$
$$\lim_{k\to\infty} \lim_{n\to\infty} \frac{n/k}{n/k+1} = \lim_{k\to\infty} 1 = 1.$$

Question 271. *Hence, we ask three questions about interchanging limits:*

1. *If $f_n : S \to \mathbb{R}$, f_n continuous and $f_n \to f$ pointwise or uniform, then is f continuous?*
2. *If $f_n : [a,b] \to \mathbb{R}$, f_n differentiable, and $f_n \to f$ with $f_n' \to g$, then is f differentiable and does $g = f'$?*
3. *If $f_n : [a,b] \to \mathbb{R}$, with f_n and f continuous such that $f_n \to f$, then does*

$$\int_a^b f_n = \int_a^b f?$$

The answer to the above questions are all **yes**, if the convergence is uniform.

Question 272. *What if the convergence is only pointwise?*

If convergence is only pointwise, the answer to the above questions are all **no**, which we will show next time.

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25 Lecture 25

Power Series and the Weierstrass Approximation Theorem

Last time, we asked three questions about interchanging limits:

Question 273. *Hence, we ask three questions about interchanging limits:*

1. *If $f_n : S \to \mathbb{R}$, f_n continuous and $f_n \to f$ pointwise or uniform, then is f continuous?*
2. *If $f_n : [a,b] \to \mathbb{R}$, f_n differentiable, and $f_n \to f$ with $f_n' \to g$, then is f differentiable and does $g = f'$?*
3. *If $f_n : [a,b] \to \mathbb{R}$, with f_n and f continuous such that $f_n \to f$, then does*

$$\int_a^b f_n = \int_a^b f?$$

The answer to the above questions are all **no**, if the convergence is pointwise as seen by the following counterexamples:

1. Let $f_n(x) = x^n$ on $[0,1]$ is continuous $\forall n$. As we noted earlier, $f_n(x) \to f(x) = \begin{cases} 0 & x \in [0,1) \\ 1 & x = 1 \end{cases}$. Notice that f is not continuous.
2. Let $f_n(x) = \frac{x^{n+1}}{n+1}$ on $[0,1]$. Then, $f_n \to 0$ pointwise on $[0,1]$. However,

$$f_n'(x) \to g(x) = \begin{cases} 0 & x \in [0,1) \\ 1 & x = 1 \end{cases}.$$

   Thus, $g(x) \neq (0)' = 0$ at $x = 1$.
3. Consider the functions

$$f_n(x) = \begin{cases} 4n^2 x & x \in \left[0, \frac{1}{2n}\right] \\ 4n - 4n^2 x & x \in \left[\frac{1}{2n}, \frac{1}{n}\right] \\ 0 & x \in \left[\frac{1}{n}, 1\right] \end{cases}$$

   as described in the previous lecture. Then, $f_n(x) \to 0$ pointwise on [0,1] as we showed last time. However,

$$\int_0^1 f_n = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2n} \cdot 2n = 1 \not\to 0 = \int_0^1 0.$$

We now prove that the answer to the three questions above is **yes** if convergence is uniform.

Theorem 274
If $f_n : S \to \mathbb{R}$ is continuous for all n, $f : S \to \mathbb{R}$, and $f_n \to f$ uniformly, then f is continuous.

Proof: Let $c \in S$ and let $\epsilon > 0$. Since $f_n \to f$ uniformly, $\exists M \in \mathbb{N}$ such that $\forall n \geq M$, $\forall y \in S$,

$$|f_n(y) - f(y)| < \frac{\epsilon}{3}.$$

Since $f_M : S \to \mathbb{R}$ is continuous, $\exists \delta_0 > 0$ such that $\forall |x - c| < \delta_0$,

$$|f_M(x) - f_M(c)| < \frac{\epsilon}{3}.$$

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Choose $\delta = \delta_0$. If $|x - c| < \delta$, then

$$\begin{aligned} |f(x) - f(c)| &\leq |f(x) + f_M(x)| + |f_M(c) - f(c)| + |f_M(x) - f_M(c)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned}$$

□

Theorem 275

If $f_n : [a, b] \to \mathbb{R}$ is continuous for all n, $f : [a, b] \to \mathbb{R}$ and $f_n \to f$ uniformly, then

$$\int_a^b f_n \to \int_a^b f.$$

Proof: Let $\epsilon > 0$. Since $f_n \to f$ uniform, $\exists M_0 \in \mathbb{N}$ such that $\forall n \geq M_0, \forall x \in [a, b]$,

$$|f_n(x) - f(x)| < \frac{\epsilon}{b - a}.$$

Then, for all $n \geq M = M_0$, we have

$$\left| \int_a^b f_n - \int_a^b f \right| \leq \int_a^b |f_n - f| < \int_a^b \frac{\epsilon}{b - a} = \epsilon.$$

□

Remark 276. *Notationally, this states that*

$$\lim_{n\to\infty} \int_a^b f_n = \int_a^b \lim_{n\to\infty} f_n = \int_a^b f.$$

Theorem 277

If $f_n : [a, b] \to \mathbb{R}$ is continuously differentiable, $f : [a, b] \to \mathbb{R}$, $g : [a, b] \to \mathbb{R}$, and

$$\begin{aligned} f_n &\to f \quad \text{pointwise}, \\ f_n' &\to g \quad \text{uniformly}, \end{aligned}$$

then f is continuously differentiable and $g = f'$.

Proof: By the FTC, $\forall n \forall x \in [a, b]$,

$$f_n(x) - f_n(a) = \int_a^x f_n'.$$

Thus, by the previous two theorems,

$$\begin{aligned} f(x) - f(a) &= \lim_{n\to\infty} (f_n(x) - f_n(a)) \\ &= \lim_{n\to\infty} \int_a^x f_n' \\ &= \int_a^x g. \end{aligned}$$

Therefore, $f(x) = f(a) + \int_a^x g$. Thus, by the FTC, f is differentiable and $f' = (\int_a^x g)' = g$. □

We now return back to our study of power series, answering some questions we asked at the beginning of Lecture 23.

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Theorem 278
Let $\sum_{j=0}^{\infty} a_j(x-x_0)^j$ be a power series of radius of convergence $p \in (0,\infty]$. Then, $\forall r \in (0,p)$, $\sum_{j=0}^{\infty} a_j(x-x_0)^j$ converges uniformly on $[x_0-r, x_0+r]$.

Proof: Let $r \in [0,p)$. Then, $\forall j \in \mathbb{N} \cup \{0\}$, $\forall x \in [x_0-r, x_0+r]$,

$$|a_j(x-x_0)^j| \leq |a_j| r^j =: M_j.$$

Now,

$$\lim j \to \infty M_j^{1/j} = \lim_{j\to\infty} |a_j|^{1/j} r = \begin{cases} \frac{r}{p} & p < \infty \\ 0 & p = \infty \end{cases}$$

since $p^{-1} = \lim_{j\to\infty} |a_j|^{1/j}$. Since $r < p$, we have

$$\lim_{j\to\infty} M_j^{1/j} < 1 \implies \sum_{j=0}^{\infty} M_j \text{ converges.}$$

By the Weierstrass M-test, it follows that $\sum_{j=0}^{\infty} a_j(x-x_0)^j$ converges uniformly on $[x_0-r, x_0+r]$. □

Theorem 279
Let $\sum_{j=0}^{\infty} a_j(x-x_0)^j$ be a power series with radius of convergence $p \in (0,\infty]$. Then,

1. $\forall c \in (x_0-p, x_0+p)$, $\sum_{j=0}^{\infty} a_j(x-x_0)^j$ is differentiable at c and
$$\frac{\mathrm{d}}{\mathrm{d}x} \sum_{j=0}^{\infty} a_j(x-x_0)^j = \sum_{j=0}^{\infty} j a_j (x-x_0)^{j-1}.$$
2. $\forall a, b$ such that $x_0 - p < a < b < x_0 + p$,
$$\int_a^b \sum_{j=0}^{\infty} a_J(x-x_0)^j \,\mathrm{d}x = \sum_{j=0}^{\infty} a_j \left(\frac{(b-x_0)^{j+1}}{j+1} - \frac{(a-x_0)^{j+1}}{j+1} \right).$$

Remark 280. *Since*

$$\lim_{j\to\infty} ((j+1)|a_{j+1}|)^{1/j} = \lim_{j\to\infty} \left(((j+1)|a_{j+1}|^{1/(j+1)} \right)^{(j+1)/j} = \lim_{k\to\infty} |a_k|^{1/k} = p,$$

1. implies $\sum a_j(x-x_0)^j$ is infinitely differentiable and

$$k! a_k = \left(\frac{\mathrm{d}^k}{\mathrm{d}x^k} \sum a_j (x-x_0)^j \right)\Big|_{x=x_0}.$$

Weierstrass Approximation Theorem

Remark 281. *This theorem essentially states: "Every continuous function on $[a,b]$ is almost a polynomial."*

Theorem 282 (Weierstrass Approximation Theorem)
If $f \in C([a,b])$, there exists a sequence of polynomials $\{P_n\}$ such that

$$P_n \to f \text{ uniformly on } [a,b].$$

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The idea of the proof is to choose a suitable sequence of polynomials $\{Q_n\}_n$ such that Q_n behaves like a 'Dirac delta function' as $nto\infty$. Then, the sequence of polynomials $P_n(x) = \int_0^1 Q_n(x-t)f(t)\,\mathrm{d}t$ converges to $f(x)$ as $n \to \infty$. We will prove this momentarily, but first we need to do the ground work.

Notice that we only need to consider $a = 0$ and $b = 1$, with $f(0) = f(1) = 0$. If we prove this case, then for a general $\tilde{f} \in C([0,1])$, $\exists$ a sequence of polynomials

$$P_n(x) \to \tilde{f}(x) - \tilde{f}(0) - x(\tilde{f}(1) - \tilde{f}(0)) \quad \text{uniformly.}$$

Hence,

$$\tilde{P}_n(x) = P_n(x) + \tilde{f}(0) + x(\tilde{f}(1) - \tilde{f}(0)) \to \tilde{f}(x) \quad \text{uniformly.}$$

Theorem 283

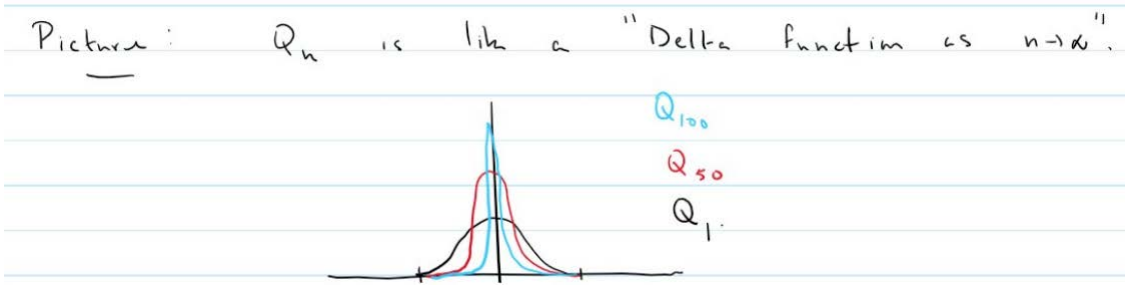
Let $c_n := (\int_{-1}^1 (1-x^2)^n\,\mathrm{d}x)^{-1} > 0$, and let

$$Q_n(x) = c_n(1-x^2)^n.$$

Then,

1. $\forall n$, $\int_{-1}^1 Q_n = 1$.
2. $\forall n$, $Q_n(x) \geq 0$ on $[-1,1]$, and
3. $\forall \delta \in (0,1)$, $Q_n \to 0$ uniformly on $\delta \leq |x| \leq 1$.

Before we prove this, here is a picture of Q_n:



Proof:

2. Immediately clear.
1. $\int_{-1}^1 Q_n = c_n \int_{-1}^1 (1-x^2)^n\,\mathrm{d}x = 1$ by definition of c_n.
3. We first estimate c_n. We have for all $n \in \mathbb{N}$ and $\forall x \in [-1,1]$,

$$(1-x^2)^n \geq 1 - nx^2.$$

We proved this way earlier in the course by induction, but it also follows from the calculus we have proven as

$$g(x) = (1-x^2)^n - (1-nx^2)$$

satisfies $g(0) = 0$, and

$$g'(x) = n \cdot 2x(1 - (1-x^2)^{n-1}) \geq 0$$

in [0,1]. Thus, $g(x) \geq 0$ by the MVT.

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Then,

$$\begin{aligned}\frac{1}{c_n} &= \int_{-1}^{1}(1-x^2)^n\,\mathrm{d}x \\ &= 2\int_{0}^{1}(1-x^2)^n\,\mathrm{d}x \\ &> 2\int_{0}^{1/\sqrt{n}}(1-x^2)^n\,\mathrm{d}x \\ &\geq 2\int_{0}^{1/\sqrt{n}}(1-nx^2)\,\mathrm{d}x \\ &= 2\left(\frac{1}{\sqrt{n}}-\frac{n}{3}\cdot n^{-3/2}\right) \\ &= \frac{4}{3}\sqrt{n} > \sqrt{n}.\end{aligned}$$

Therefore, $c_n < \sqrt{n}$.

Let $\delta > 0$. We note $\lim_{n\to\infty}\sqrt{n}(1-\delta^2)^n = 0$. Then,

$$\begin{aligned}\lim_{n\to\infty}(\sqrt{n}(1-\delta^2)^n)^{1/n} &= \lim_{n\to\infty}(n^{1/n})^{1/2}(1-\delta^2) \\ &= 1-\delta^2 < 1.\end{aligned}$$

Therefore,

$$\lim_{n\to\infty}\sqrt{n}(1-\delta^2)^n = 0.$$

Let $\epsilon > 0$, and choose $M \in \mathbb{N}$ such that for all $n \geq M$,

$$\sqrt{n}(1-\delta^2)^n < \epsilon.$$

Then, $\forall n \geq M$ and $\forall \delta \leq |x| \leq 1$,

$$|c_n(1-x^2)^n| < \sqrt{n}(1-x^2)^n \leq \sqrt{n}(1-\delta^2)^n < \epsilon.$$

□

We now prove the Weierstrass Approximation Theorem.

Proof: Suppose $f \in C([0,1])$, $f(0) = f(1) = 0$. We extend f to an element of $C(\mathbb{R})$ by setting $f(x) = 0$ for all $x \notin [0,1]$. We furthermore define

$$\begin{aligned}P_n(x) &= \int_0^1 f(t)Q_n(t-x)\,\mathrm{d}t \\ &= \int_0^1 f(t)c_n(1-(t-x)^2)^n\,\mathrm{d}t.\end{aligned}$$

Note that $P_n(x)$ is in fact a polynomial.

Furthermore, observe that for $x \in [0,1]$,

$$\begin{aligned}P_n(x) &= \int_0^1 f(t)Q_n(t-x)\,\mathrm{d}t \\ &= \int_{-x}^{1-x} f(x+t)Q_n(t)\,\mathrm{d}t \\ &= \int_{-1}^{1} f(x+t)Q_n(t)\,\mathrm{d}t.\end{aligned}$$

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The second equality is true by a change of variable, and the last equality is true as $f(x+t) = 0$ for $t \notin [-x, 1-x]$.

We now prove $P_n \to f$ uniformly on $[0,1]$. Let $\epsilon > 0$. Since f is uniformly continuous on $[0,1]$, $\exists \delta > 0$ such that $\forall |x-y| \leq \delta$, $|f(x) - f(y)| < \frac{\epsilon}{2}$. Let $C = \sup\{f(x) \mid x \in [0,1]\}$, which exists by the Min/Max theorem i.e. the EVT. Choose $M \in \mathbb{N}$ such that $\forall n \geq M$,

$$\sqrt{n}(1-\delta^2)^n < \frac{\epsilon}{8C}.$$

Thus, $\forall n \geq M, \forall x \in [0,1]$, by the previous theorem,

$$\begin{aligned}
|P_n(x) - f(x)| &= \left| \int_{-1}^{1} (f(x-t) - f(t)) Q_n(t) \, \mathrm{d}t \right| \\
&\leq \int_{-1}^{1} |f(x-t) - f(x)| Q_n(t) \, \mathrm{d}t \\
&\leq \int_{|t| \leq \delta} |f(x-t) - f(x)| Q_n(t) \, \mathrm{d}t + \int_{\delta \leq |t| \leq 1} |f(x-t) - f(x)| Q_n(t) \, \mathrm{d}t \\
&\leq \frac{\epsilon}{2} \int_{|t| \leq \delta} Q_n(t) \, \mathrm{d}t + \sqrt{n}(1-\delta^2)^n \int_{\|\delta \leq |t| \leq 1} 2C \\
&< \frac{\epsilon}{2} + 4C\sqrt{n}(1-\delta^2)^n \\
&< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.
\end{aligned}$$

□

In the last minute of the course, Casey Rodriguez stated: "This was quite an experience; teaching to an empty room. I hope you did get something out of this class. Unfortunately I wasn't able to meet a lot of you, and that's one of the best parts of teaching...."

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TOBIAS HOLCK COLDING

Lecture 1

Two key topics for this class:

- How to write a mathematical proof.
- How to prove theorems.

Here is an example:

Intermediate value theorem:

- Suppose that $f : [a, b] \to \mathbf{R}$ is a continuous functions.
- Assume that $f(a) < 0$ and $f(b) > 0$.

The intermediate value theorem says that there exists a c between a and b where $f(c) = 0$.

- How do we prove this?
- If we draw a picture, then it seems obvious, but how to we actually prove this?

.

That a function is continuous basically means that when you draw the graph of the function the pencil is not allowed to leave the paper.

- How do we make this into a proper proof?
- What properties of the real values are needed for a proof?

This leads to several questions:

- Q1: What is a real number?
- Q2: Why is $\sqrt{2}$ a real number?
- Q3: What is $\sqrt{2}$?

The answer to these questions: $\mathbf{R}$ is a complete ordered field that contains the rational numbers $\mathbf{Q}$.

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Here is some notation:

(1) $\mathbf{N}$ is the natural numbers. This means that $\mathbf{N} = \{1, 2, 3, \cdots\}$.
(2) $\mathbf{Z}$ is the integers. This means that $\mathbf{Z} = \{0, \pm 1, \pm 2, \pm 3, \cdots\}$.
(3) $\mathbf{Q}$ is the rational numbers. So all numbers of the form $\frac{m}{n}$, where $m \in \mathbf{Z}$ and $n \in \mathbf{N}$.

Properties:

Rational numbers:
Rational numbers $\mathbf{Q}$ are numbers of the form $\frac{m}{n}$, where $m \in \mathbf{Z}$ and $n \in \mathbf{N}$.

(1) When are two numbers the same?

$$\frac{m_1}{n_1} = \frac{m_2}{n_2} \iff m_1\, n_2 = m_2\, n_1\,.$$

(2) How do we add two numbers?

$$\frac{m_1}{n_1} + \frac{m_2}{n_2} = \frac{m_1\, n_2 + m_2\, n_1}{n_1\, n_2}\,.$$

(3) How do we multiply two numbers?

$$\frac{m_1}{n_1}\,\frac{m_2}{n_2} = \frac{m_1\, m_2}{n_1\, n_2}\,.$$

(4) When is one number less than another?

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \iff m_1\, n_2 < m_2\, n_1\,.$$

For this to make sense we need (for instance) to show that multiplication is well-defined:

This means that if we have two representations of the same rational number

$$\frac{m_1}{n_1} = \frac{m_2}{n_2}$$

and likewise

$$\frac{p_1}{q_1} = \frac{p_2}{q_2}\,,$$

then

$$\frac{m_1}{n_1}\,\frac{p_1}{q_1} = \frac{m_2}{n_2}\,\frac{p_2}{q_2}\,.$$

Proof. We have that $m_1\, n_2 = m_2\, n_1$ and $p_1\, q_2 = p_2\, q_1$. Therefore,

$$m_1\, p_1\, n_2\, q_2 = m_1\, n_2\, p_1\, q_2 = m_2\, n_1\, p_2\, q_1\,.$$

□

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This illustrate how detailed a proof should be.

A Field:

Definition:
A Field $\mathbf{F}$ is a set with two operations that we are denoting suggestively by "$+$" and "$\cdot$". Those two operations satisfies the following axioms:

Additive properties:

(1) $x, y \in \mathbf{F}$, then $x + y \in \mathbf{F}$.
(2) $x + y = y + x$.
(3) $(x + y) + z = x + (y + z)$.
(4) There exists an element $0 \in \mathbf{F}$ such that $0 + x = x$ for all $x \in \mathbf{F}$.
(5) For all $x \in \mathbf{F}$ there exists an element, suggestively, denoted by $(-x)$ such that $x + (-x) = 0$.

Multiplicative properties:

(1) $x, y \in \mathbf{F}$, then $x\,y \in \mathbf{F}$.
(2) $x\,y = y\,x$.
(3) $(x\,y)\,z = x\,(y\,z)$.
(4) There exists an element, suggestively, denoted by 1 such that $1\,x = x$ for all $x \in \mathbf{F}$.
(5) For all $x \in \mathbf{F} \setminus \{0\}$ there exists an element, suggestively, denoted by $\frac{1}{x}$ such that $x\,\frac{1}{x} = 1$.

The final axion that we need is an axiom that chains addition and multiplication together:

- $$(x + y)\,z = x\,z + y\,z\,.$$

Theorem: For any field 'zero' is unique.

Proof. Suppose there are two. Let us denote them by 0_1 and 0_2. Then

$$0_1 + 0_2 = 0_2$$

since 0_1 is a 'zero' and

$$0_1 + 0_2 = 0_1$$

since 0_2 is a 'zero' so $0_1 = 0_2$. □

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Examples: $\mathbf{Q}$ is a Field, whereas $\mathbf{N}$ and $\mathbf{Z}$ are not Fields.

Ordered set: An ordered set $\mathbf{S}$ is a set with a relation $<$ with the following properties:

(1) For an x, $y \in \mathbf{S}$, one of the following holds: $x < y$ or $y < x$ or $x = y$.
(2) If x, y, $z \in \mathbf{S}$ with $x < y$ and $y < z$, then $x < z$.

Ordered Field:
An ordered Field is an ordered set that is also Field and has the following two additional properties that chains the operations in the Field together with the ordering:

(1) If $x < y$, then $x + z < y + z$.
(2) If $x > 0$ and $y > 0$, then $x\,y > 0$.

Example: $\mathbf{Q}$ is an ordered Field.

Theorem: If $x < y$ and $z > 0$, then $x\,z < y\,z$.

Proof. We need to show that $x\,z < y\,z$ or equivalently $y\,z - x\,z > 0$. The latter can be rewritten as $y\,z - x\,z = (y - x)\,z$. Since $y > x$ we have that $y - x > 0$ and the claim therefore follows since $z > 0$. □

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Lecture 2

The real numbers $\mathbf{R}$ is a complete ordered Field that contains $\mathbf{Q}$.

Question: What is the difference between $\mathbf{R}$ and $\mathbf{Q}$?
One difference is that $\mathbf{R}$ contains $\sqrt{2}$ and $\mathbf{Q}$ does not.

$\sqrt{2}$ is a number x so that $x > 0$ and $x^2 = 2$.

Theorem: There does not exists a rational number x so that $x^2 = 2$.

Proof. We will argue by contradiction. So suppose that there exists a rational number $x = \frac{m}{n}$, where $m \in \mathbf{Z}$ and $n \in \mathbf{N}$, so that $x^2 = \frac{m^2}{n^2} = 2$. We can assume m and n does not have a common factor (other than one). We have that $m^2 = 2\,n^2$ and so 2 is a factor in m^2 and therefore in m itself. This means that $m = 2\,m_1$, where m_1 is also an integer. It follows that $m^2 = 4\,m_1^2 = 2\,n^2$ and therefore $2\,m_1 = n$ and so n is also even. We have now that both m and n are even and so have 2 as a common factor. This is the desired contradiction. This show that there is no rational number x with the property that $x^2 = 2$. □

How do we add $\sqrt{2}$ to the number system?

$\sqrt{2} = 1.4142136\cdots$

So 1, 1.4, 1.41, 1.414, 1.4142, 1.41421, $\to \sqrt{2}$.

$\sqrt{2}$ is the limit of a sequence of numbers.

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Completeness of $\mathbf{R}$. (Least upper bound property.)

Completeness is: If a subset A of $\mathbf{R}$ has an upper bound, then A has a least upper bound.

Suppose that $\mathbf{S}$ is an ordered set and A is a subset of $\mathbf{S}$, then M is an upper bound for A if for all $a \in A$ we have that $a \leq M$.

Example: If $A = \{1, 2, 3\} \subset \mathbf{Z}$, then 4 is an upper bound, whereas 2 is not an upper bound.

Example: If $\mathbf{S} = \mathbf{Q}$, then $\mathbf{N}$ as a subset does not have an upper bound (we will return to this shortly).

Least upper bound: Suppose that $\mathbf{S}$ is an ordered set and A is a subset that has an upper bound. We say that M is a least upper bound for A if M is an upper bound for A and for any other upper bound M_1 we have that $M \leq M_1$.

Complete ordered set: We say that an ordered set is complete if any subset that has an upper bound has a least upper bound.

Theorem: There exists a complete ordered Field that contains $\mathbf{Q}$.

This Field is denoted by $\mathbf{R}$.

We will not prove this, as a proof would take us too far a field, rather we will take it for granted.

Theorem: $\sqrt{2} \in \mathbf{R}$.

Proof. Let $A = (0, \sqrt{2}) \cap \mathbf{Q}$. That is A consists of all the positive rational numbers a so that $a^2 < 2$. Let x be the least upper bound for A. Note that A is nonempty (since $1 \in A$) and that 2 is an upper bound for A. Note also that $x \geq 1 > 0$ since it is an upper bound. We need to show that $x^2 = 2$.

We will first show that $x^2 \leq 2$. Suppose not; so assume that $x^2 > 2$. We will show that this leads to a contradiction. Consider

$$(x - h)^2 = x^2 - 2\,x\,h + h^2 > x^2 - 2\,h\,x\,.$$

As long as $h > 0$ is chosen so that

$$2\,h\,x < x^2 - 2$$

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or, equivalently, that

$$h < \frac{x^2 - 2}{2\,x}$$

then

$$(x - h)^2 > 2$$

and therefore $x - h$ is also an upper bound for A. This contradict that x is the least upper bound. We therefore have that if x is the least upper bound for A, then $x^2 \leq 2$.

To show the reverse inequality (that $x^2 \geq 2$) we argue similarly. Assume that for the least upper bound x we have that $x^2 < 2$. Consider $x + h$, where $0 < h < 1$. We have that

$$(x + h)^2 = x^2 + 2\,x\,h + h^2 < x^2 + 2\,x\,h + h = x^2 + h\,(2\,x + 1)\,.$$

Since we are assuming that $x^2 < 2$ we can choose h positive so that

$$h < \frac{2 - x^2}{2\,x + 1}\,.$$

We therefore have that

$$(x + h)^2 < x^2 + 2 - x^2 < 2\,.$$

This is the desired contradiction and show that $x^2 \geq 2$. Together with the first step we have that $x^2 = 2$. □

Corollary: $\mathbf{Q}$ is not complete.

Proof. If $\mathbf{Q}$ was complete, then $\sqrt{2} \in \mathbf{Q}$ but we have already proven that there is no rational number with the property that $x^2 = 2$. □

Archimedean property: For all $x \in \mathbf{R}$, there exists a natural $n \in \mathbf{N}$ so that $x < n$.

Proof. If this was not the case, then $\mathbf{N}$ would be bounded. To see that $\mathbf{N}$ is not bounded we argue as follows. Assume it is bounded and let α be the least upper bound for $\mathbf{N}$. We would now have that for all $n \in \mathbf{N}$ that $n \leq \alpha$. Since $n + 1$ is also a natural number we would have that $n + 1 \leq \alpha$ as well. So, in fact, $n \leq \alpha - 1$ or in other words, since n was any natural number, $\alpha - 1$ would be an upper bound contradicting that α was the least upper bound. □

As a corollary of the Archimedean property we get the following:

Corollary: If $x < y$, then there exists a rational number $\frac{m}{n}$ such that

$$x < \frac{m}{n} < y\,.$$

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Proof. Set $\beta = \frac{1}{y-x}$. From the Archimedean property we have that there exists a natural number n with $n > \beta$. It follows that

$$0 < \frac{1}{n} < \frac{1}{\beta}\,.$$

Now let $m-1$ be the largest integer so that

$$m - 1 \leq x\, n\,.$$

It follows that $\frac{m}{n}$ has the desired property. □

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[TBB] B.S. Thomson, J.B. Bruckner, and A.M. Bruckner, *Elementary Real Analysis, 2nd edition*
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Lecture 3

Theorem: $\mathbf{R}$ is a complete ordered Field that contains $\mathbf{Q}$.

$\mathbf{S}$ is an ordered set. A non-empty subset A of $\mathbf{S}$ is said to have an upper bound if there exists an $M \in \mathbf{S}$ such that for all $a \in A$ we have that $a \leq M$.

Completeness is the property that every bounded non-empty subset has a least upper bound.

We denote by $\sup A$ the smallest upper bound of A.

Lower bound: A non-empty subset A is said to have a lower bound if there exists $m \in \mathbf{S}$ such that for all $a \in A$ we have that $m \leq a$.

The greatest lower bound is a lower bound that is greater or equal to all other lower bounds.

The greatest lower bound is denoted by $\inf A$.

From now on we will concentrate of the case of $\mathbf{R}$.

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How to write a mathematical proof?

This lecture we will look at how to write a mathematical proof.
We will explain this in two results that we talked about last time.

Let us return to the example of showing that $\sqrt{2} \in \mathbf{R}$.

We already showed this but we did not write it as a "proper proof". That is what we will do next.

Theorem: There exists $\alpha > 0$ such that $\alpha^2 = 2$.

Proof. Define a set A by

$$A = \{x \in \mathbf{R} \,|\, x > 0 \text{ and } x^2 \leq 2\}\,.$$

We will show that A is a non-empty bounded subset and that $\alpha = \sup A$ has the property that $\alpha > 0$ and $\alpha^2 = 2$.

Observe first that $1 \in A$, so A is non-empty. Moreover, 2 is an upper bound for A so A is bounded from above. Let $\alpha = \sup A$, we need to show that $\alpha > 0$ and that $\alpha^2 = 2$. Since $1 \in A$ it follows that $0 < 1 < \alpha$. To show that $\alpha^2 = 2$ we divide the proof into two parts.

Part 1: We will show that $\alpha^2 \leq 2$. Suppose not; we will see that this lead to a contradiction. Indeed, we will show that that if this was the case, then there exists an $0 < \alpha_0 < \alpha$ such that $\alpha_0^2 > 2$ so α_0 is an upper bound that is smaller than α. To show this we set

$$h = \frac{\alpha^2 - 2}{4\,\alpha}$$

and set

$$\alpha_0 = \alpha - h\,.$$

Note that since we are assuming that $\alpha > 2$, then we have that $h > 0$ and therefore $\alpha_0 < \alpha$. Note also that since $1 \leq \alpha \leq 2$ we have that

$$h \leq \frac{1}{2\,\alpha} \leq \frac{1}{2}\,.$$

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In particular, $0 < \alpha_0$. Next

$$\alpha_0^2 = \alpha^2 + h^2 - 2\,h\,\alpha > \alpha^2 - \frac{\alpha^2 - 2}{2} = \frac{\alpha^2}{2} + 1 \geq 2\,.$$

This is the desired contradiction and show that $\alpha^2 \leq 2$.

Part 2: We will next show that $\alpha^2 \geq 2$. Suppose not; we will see that this lead to a contradiction. Indeed, we will show that if this was the case, then there exists an $\alpha_1 > \alpha$ such that $\alpha_1^2 < 2$ contradicting that α was an upper bound for A. So assume that $\alpha^2 < 2$. This time we will set

$$h = \frac{2 - \alpha^2}{4\,\alpha}\,.$$

Note that $1 > h > 0$ (the first inequality follows from that $1 \leq \alpha$). Set $\alpha_1 = \alpha + h$. It follows that

$$\alpha_1^2 = \alpha^2 + h^2 + 2\,h\,\alpha < \alpha^2 + h + \frac{2 - \alpha^2}{2} \leq \alpha^2 + 2\,\frac{2 - \alpha^2}{2} = 2\,.$$

Together parts 1 and 2 show that $\alpha^2 = 2$; completing the proof. □

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Archimedean property:

Formal proof:

Theorem: The set of natural number is not bounded from above.

Proof. If $\mathbf{N}$ is bounded from above, then we can let M be the least upper bound. We now have that for all $n \in \mathbf{N}$

$$n \leq \alpha$$

We claim that also $\alpha - 1$ is an upper bound contradicting that α was the least upper bound. Namely, for a given n since α is an upper bound for all natural numbers we have that

$$n + 1 \leq \alpha$$

but this implies that

$$n \leq \alpha - 1$$

showing that $\alpha - 1$ is an upper bound. That is the desired contradiction. □

Corollary: For any $\epsilon > 0$, there exists an $n \in \mathbf{N}$ such that $\frac{1}{n} < \epsilon$.

Proof. Set $\alpha = \frac{1}{\epsilon}$, By the Archimedean property we know that there exists an $n \in \mathbf{N}$ with $n > \alpha$. It follows that $\frac{1}{n} < \epsilon$. □

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Sequences:

$\sqrt{2}$ can be thought of a limit of a sequence of decimal numbers as follows.

$$1 < 1.4 < 1.41 < 1.414 \cdots .$$

When does a limit exist?

A sequence of real numbers is a function $f : \mathbf{N} \to \mathbf{R}$.

We usually use the notation $a_n = f(n)$.

Example 1: $\sqrt{2}$ is the limit of $a_1 = 1$, $a_2 = 1.4$, $a_3 = 1.41$, $a_4 = 1.414$ etc.

Example 2: $a_n = (-1)^n$. This sequence has NO limit. The a_n's alternates between -1 and 1.

Example 3: The sequence $a_n = \frac{1}{n}$ has zero as its limit.

Limit: Let a_n be a sequence and a a real number. We say that a_n converges to a if for all $\epsilon > 0$, there exists an $N \in \mathbf{N}$ such that if $n \geq N$, then

$$|a_n - a| < \epsilon \, .$$

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Lecture 4

Theorem: $\mathbf{R}$ is a complete ordered Field that contains $\mathbf{Q}$.

$\mathbf{S}$ is an ordered set. A non-empty subset A of $\mathbf{S}$ is said to have an upper bound if there exists an $M \in \mathbf{S}$ such that for all $a \in A$ we have that $a \leq M$.

Completeness is the property that every bounded non-empty subset has a least upper bound.

We denote by $\sup A$ the greatest lower bound of A.

Lower bound: A non-empty subset A is said to have a lower bound if there exists $m \in \mathbf{S}$ such that for all $a \in A$ we have that $m \leq a$.

The greatest lower bound is a lower bound that is greater or equal to all other lower bounds.

Sequences:

$\sqrt{2}$ can be thought of a limit of a sequence of decimal numbers as follows.

$$1 < 1.4 < 1.41 < 1.414 \cdots .$$

A sequence of real numbers is a function $f : \mathbf{N} \to \mathbf{R}$.

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We usually use the notation $a_n = f(n)$.

Example 1: $\sqrt{2}$ is the limit of $a_1 = 1$, $a_2 = 1.4$, $a_3 = 1.41$, $a_4 = 1.414$ etc.

Example 2: $a_n = (-1)^n$. This sequence has NO limit. The a_n's alternates between -1 and 1.

Example 3: The sequence $a_n = \frac{1}{n}$ has zero as its limit.

Limit: Let a_n be a sequence and a a real number. We say that a_n converges to a if for all $\epsilon > 0$, there exists an $N \in \mathbf{N}$ such that if $n \geq N$, then

$$|a_n - a| < \epsilon \, .$$

If this is the case, then we also say that a is the limit of the sequence and we say that the sequence is convergenet.

A sequence that is not covergent is said to be divergent.

Example:

$$0.999999999 \cdots = 1 \, .$$

What does the left hand side mean?

Define a sequence a_n as follows: Set

$$a_1 = 0.9 \, ,$$

$$a_2 = 0.99 \, ,$$

$$a_3 = 0.999 \, ,$$

$$a_4 = 0.9999 \, ,$$

etc.

The left hand side above is then defined as the limit of the sequence a_n.

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Claim:

$$\lim_{n\to\infty} a_n = 1\,.$$

Proof. We need to show that for all $\epsilon > 0$, there exists an $N \in \mathbf{N}$ such that for $n \geq N$ we have that

$$|a_n - 1| < \epsilon\,.$$

By the Archimedean property we can choose N such that $\frac{1}{N} < \epsilon$. We also have that

$$|a_n - 1| = 10^{-n}\,.$$

Therefore, for $n \geq N$ we have that

$$|a_n - 1| = 10^{-n} < \frac{1}{n} \leq \frac{1}{N} < \epsilon\,.$$

This proves the claim. □

Theorem If a_n is a convergent sequence, then the set $\{a_n\}$ is a bounded subset of $\mathbf{R}$.

Proof. Since a_n is convergent to a we can find N such that for $n \geq N$ we have that

$$|a - a_n| < 1\,.$$

Note also that the set $\{a_1, \cdots, a_{N-1}\}$ is bounded so there exists $C \in \mathbf{R}$ such that for $n = 1, \cdots, N-1$ we have that

$$|a_n| \leq C\,.$$

To see that the larger set $\{a_n\}$ is bounded we will use that for $n \geq N$

$$|a_n| \leq |a| + |a_n - a| \leq |a| + 1\,.$$

From this we have that for all n

$$|a_n| \leq \max\{C, |a| + 1\}\,.$$

□

Basic algebraic properties of limits:

Theorem Suppose that a_n and b_n are convergenet sequences with $\lim a_n = a$, $\lim b_n = b$ and $C \in \mathbf{R}$, then

(1) $c_n = C\,a_n$ is convergenet and $\lim_{n\to\infty} c_n = C\,a$.
(2) $c_n = a_n + b_n$ is convergent and $\lim_{n\to\infty} c_n = a + b$.
(3) $c_n = a_n\,b_n$ is convergent with $\lim_{n\to\infty} c_n = a\,b$.
(4) If $b_n \neq 0$, $b \neq 0$ and $c_n = \frac{a_n}{b_n}$, then c_n is convergent and $\lim_{n\to\infty} c_n = \frac{a}{b}$.

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Proof. (of the first property.) If $c = 0$, then the claim is obviously true so we need only show the claim for $C \neq 0$. Given $\epsilon > 0$, there exists an N such that if $n \geq N$, then

$$|a - a_n| < \frac{\epsilon}{|C|}\,.$$

Multiplying both sides by $|C|$ gives that

$$|C\,a - C\,a_n| < \epsilon\,.$$

for $n \geq N$. This show the first property.

(Of the second of these properties.) Observe that

$$|c_n - (a+b)| = |a_n + b_n - (a+b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b|\,.$$

Since $a_n \to a$, given $\epsilon > 0$ we can find a N_a such that if $n \geq N_a$, then

$$|a_n - a| < \frac{\epsilon}{2}\,.$$

Likewise since $b_n \to b$ we can find N_b such that if $n \geq N_b$, then

$$|b_n - b| < \frac{\epsilon}{2}\,.$$

We now set $N = \max\{Na, N_b\}$ and observe that if $n \geq N$, then

$$|c_n - (a+b)| \leq |a_n - a| + |b_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon\,.$$

This proves the second property.

(Outline of how to show the third property.) To prove the third property we will use that

$$|a\,b - a_n\,b_n| \leq |a\,b - a_n\,b| + |a_n\,b - a_n\,b_n| = |b|\,|a - a_n| + |a_n|\,|b - b_n|\,.$$

We then combine it with the theorem above that show that the set

$$\{|a_n|\,|\,n \in \mathbf{N}\}$$

is bounded. This is the main idea of the proof of the third property. There are the details to fill in to make it a proof.

(Outline of how to show the fourth property.) To prove the fourth property we will assume that $a_n = 1$. The general case indeed will follow from this together with the third property. We will use that

$$\frac{1}{b_n} - \frac{1}{b} = \frac{|b - b_n|}{|b|\,|b_n|}\,,$$

together with that

$$|b_n| \leq |b| + |b_n - b|\,.$$

and therefore

$$|b_n| \geq |b| - |b_n - b|\,.$$

We then want to use this to bound the denominator (when n is sufficiently large) from below in

$$\frac{|b - b_n|}{|b|\,|b_n|}\,.$$

Like for the third property there are details to fill in but these are the main ideas. □

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Subsequence:

Example 1: Suppose that $a_n = (-1)^n$.
This is a sequence of 1's and -1's that is alternating between -1 and 1.
The sequence $b_n = 1$ for all n is a subsequence.
Another subsequence is where $c_n = -1$.
Also the sequence $c_n = (-1)^{n+1}$ is a subsequence of a_n.
Another example of a subsequence is

$$1, 1, -1, -1, 1, 1, -1, -1, \cdots .$$

Example 2: Suppose $a_n = n$. So a_n is:

$$1, 2, 3, 4, 5, 6, \cdots .$$

The sequence of increasing odd numbers

$$1, 3, 5, 7, 9, \cdots .$$

is a subsequence.
The sequence of increasing even numbers

$$2, 4, 6, 8, 10, \cdots .$$

is another subsequence.
The sequence

$$1, 1, 2, 2, 3, 3, 4, 4, 5, 5 \cdots ,$$

is NOT s subsequence.

Formel definition: Recall that a sequence a_n is a function $f : \mathbf{N} \to \mathbf{R}$ where we set $a_n = f(n)$. A subsequence b_n of a_n is a composition of functions $f \circ g$ where $g : \mathbf{N} \to \mathbf{N}$ is a strictly increasing function. So $b_n = f(g(n))$. Sometimes a subsequence of the sequence a_n also denoted by a_{n_k}.

Theorem: A sequence a_n is convergent with limit a if and only if all subsequences of a_n are also convergent with limit a.

Proof. We need to show two implications.

First we need to show that is all subsequences of a_n are convergent with limit a, then the sequence a_n is convergent with limit a. However, this is trivially so since a_n itself is a (trivial) subsequence of a_n.

Next we need to show that any subsequence of a convergent sequence is convergent with the same limit. Suppose therefore that $\epsilon > 0$ is given and choose N so large so that for

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$n \geq N$

$$|a_n - a| < \epsilon \, .$$

For $k \geq N$ we have that $n_k \geq k \geq N$ and therefore

$$|a_{n_k} - a| < \epsilon \, .$$

This proves the second implication. □

References

[TBB] B.S. Thomson, J.B. Bruckner, and A.M. Bruckner, *Elementary Real Analysis, 2nd edition*
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Lecture 5

A sequence of real numbers is a function $f : \mathbf{N} \to \mathbf{R}$.

We usually use the notation $a_n = f(n)$.

Limit: Let a_n be a sequence and a a real number. We say that a_n converges to a if for all $\epsilon > 0$, there exists an $N \in \mathbf{N}$ such that if $n \geq N$, then

$$|a_n - a| < \epsilon \, .$$

If this is the case, then we also say that a is the limit of the sequence and we say that the sequence is convergenet. A sequence that is not covergent is said to be divergent.

Theorem If a_n is a convergent sequence, then the set $\{a_n\}$ is a bounded subset of $\mathbf{R}$.

Basic algebraic properties of limits:

Theorem Suppose that a_n and b_n are convergenet sequences with $\lim a_n = a$, $\lim b_n = b$ and $C \in \mathbf{R}$, then

(1) $c_n = C\, a_n$ is convergenet and $\lim_{n\to\infty} c_n = C\, a$.
(2) $c_n = a_n + b_n$ is convergent and $\lim_{n\to\infty} c_n = a + b$.
(3) $c_n = a_n\, b_n$ is convergent with $\lim_{n\to\infty} c_n = a\, b$.
(4) If $b_n \neq 0$, $b \neq 0$ and $c_n = \frac{1}{b_n}$, then c_n is convergent and $\lim_{n\to\infty} c_n = \frac{1}{b}$.

Subsequence: A subsequence b_k of a_n is a composition of functions $f \circ g$ where $g : \mathbf{N} \to \mathbf{N}$ is a strictly increasing function. So $b_k = f(g(k))$. We often write a_{n_k} for b_k.

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Example: A sequence is defined by a_n

$$a_n = \frac{n^2+1}{n^2+n+1}\,.$$

We will show that a_n is convergent with limit 1. We have

$$a_n = \frac{1+\frac{1}{n^2}}{1+\frac{1}{n}+\frac{1}{n^2}}\,.$$

Since $\frac{1}{n} \to 0$ and $\frac{1}{n^2} \to 0$ we have that

$$1+\frac{1}{n^2} \to 1$$

and

$$1+\frac{1}{n}+\frac{1}{n^2} \to 1\,.$$

It now follows from the algebraic properties of limits that

$$a_n \to 1\,.$$

Example: To show that $\sqrt{2}$ is the limit of $a_1 = 1$, $a_2 = 1.4$, $a_3 = 1.41$, $a_4 = 1.414$, ... we need something else. We need the monotone convergence theorem.

Monotone convergence theorem: Increasing version. Let a_n be a monotone increasing sequence. This means that $a_1 \leq a_2 \leq a_3 \leq \cdots$. (Which we can also write this as $a_n \leq a_{n+1}$). If the sequence is bounded from above so that there exists A with

$$a_n \leq A\,,$$

then a_n is convergent with limit $\sup\{a_n\}$.

Monotone convergence theorem: Decreasing version. Similarly, for a bounded monotone decreasing sequence a_n where $a_{n+1} \leq a_n$, we have that a_n converges and

$$\lim_{n\to\infty} a_n = \inf\{a_n\}\,.$$

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Using the monotone convergence theorem we can now show that $\sqrt{2}$ is the limit of $a_1 = 1$, $a_2 = 1.4$, $a_3 = 1.41$, $a_4 = 1.414$, $a_5 = \cdots$.

To show this we argue as follows:

Let

$$a_n = \frac{b_n}{10^{n-1}}$$

where b_n is the largest integer so that

$$b_n^2 \leq 2 \cdot 10^{2\,n-2}\,.$$

We will show that a_n is an increasing and bounded sequence and that the limit a has the property that $a^2 = 2$.

So suppose that b_n is an integer and that

$$b_n^2 \leq 2 \cdot 10^{2\,n-2}\,,$$

then obviously

$$(10\,b_n)^2 \leq 2 \cdot 10^{2\,(n+1)-2}\,.$$

This implies that $b_{n+1} \geq 10\,b_n$; so $a_{n+1} \geq a_n$ and the sequence is increasing.

It is clear that $b_n \leq 2 \cdot 10^{n-1}$, since $(2 \cdot\ 10^{2\,n-2})^2 > 2 \cdot 10^{2\,n-2}$.

Therefore the sequence $a_n = \frac{b_n}{10^{n-1}}$ is bounded and by the monotone convergence theorem $a_n \to a$.

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To show that $a^2 = 2$ we argue as follows:

By the algebraic properties of limits we have that $a^2 = \lim_{n\to\infty} a_n^2$, but for each n we have that $a_n^2\, 10^{2\,n-2} = b_n^2 \leq 2 \cdot 10^{2\,n-2}$ so $a_n^2 \leq 2$.

This show that $a^2 \leq 2$.

Similarly, since $(b_n + 1)^2 > 2 \cdot 10^{2\,n-2}$ we have that $(a_n + 10^{1-n})^2 = \frac{(b_n+1)^2}{10^{2\,n-2}} > 2$.

Therefore, $2 \leq \lim_{n\to\infty}(a_n + 10^{1-n})^2 = \lim_{n\to\infty} a_n^2 = a^2$.

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Proof of the monotone convergence theorem (Increasing version):

The decreasing version is proven similarly.

So suppose that we have a sequence with

$$a_n \leq a_{n+1} \leq A$$

and set

$$a = \sup \{a_n\} .$$

We want to show that

$$a_n \to a .$$

Given $\epsilon > 0$, since $a - \epsilon < a$ we have that $a - \epsilon$ is not an upper bound for the sequence, therefore there exists N such that

$$a_N > a - \epsilon .$$

Since the sequence is increasing we have for $n \geq N$ that

$$a - \epsilon < a_N \leq a_n \leq a$$

Here the last inequality used that a is an upper bound for the sequence.

We now have that for $n \geq N$

$$0 \leq a - a_n < \epsilon .$$

This shows that the sequence is convergent with limit a.

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Cauchy sequence: A sequence a_n is said to be a Cauchy sequence if for all $\epsilon > 0$, there exists an N such that if m, $n \geq N$, then

$$|a_n - a_m| < \epsilon \, .$$

(Tail of the sequence bunch together.)

From Wikipedia: Baron Augustin-Louis Cauchy (1789 – 1857) was a French mathematician, engineer, and physicist. He was one of the first to rigorously state and prove the key theorems of calculus (thereby creating real analysis), pioneered the field complex analysis, and the study of permutation groups in abstract algebra. Cauchy also contributed to a number of topics in mathematical physics, notably continuum mechanics.

Theorem: A sequence is convergent if and only if it is a Cauchy sequence.

Application: Existence of fixed points for a maps. If $T : \mathbf{R} \to \mathbf{R}$ is a map, then $x_0 \in \mathbf{R}$ is a fixed point if

$$T(x_0) = x_0 \, .$$

Definition A contracting map is a map $T : \mathbf{R} \to \mathbf{R}$ such that there exists $c < 1$ so for all x, $y \in \mathbf{R}$ we have that

$$|T(x) - T(y)| \leq c \, |x - y| \, .$$

(Points are squeezed together under the map.)

Contracting mapping theorem: Any contracting map has a fixed point.

Application of contracting mapping theorem: Existence of solutions to ODEs.

More on all of this next time.....

References

[TBB] B.S. Thomson, J.B. Bruckner, and A.M. Bruckner, *Elementary Real Analysis, 2nd edition*
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Lecture 6

Last time:

Basic algebraic properties of limits.

Monotone convergence theorem.

Cauchy sequence.

Cauchy sequence: A sequence a_n is said to be a Cauchy sequence if for all $\epsilon > 0$, there exists an N such that if m, $n \geq N$, then

$$|a_n - a_m| < \epsilon \, .$$

(Tail of the sequence bunch together.)

Theorem (Cauchy convergence theorem): A sequence is convergent if and only if it is a Cauchy sequence.

Application: Existence of fixed points for a maps.

If $T : \mathbf{R} \to \mathbf{R}$ is a map, then $x_0 \in \mathbf{R}$ is a fixed point if

$$T(x_0) = x_0 \, .$$

Definition A contracting map is a map $T : \mathbf{R} \to \mathbf{R}$ such that there exists $c < 1$ so for all x, $y \in \mathbf{R}$ we have that

$$|T(x) - T(y)| \leq c \, |x - y| \, .$$

(Points are squeezed together under the map.)

Contracting mapping theorem: Any contracting map has a fixed point.

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For a contracting map the fix point is unique.

Suppose that x and y are two fixed point we want to show that $x = y$. We have

$$|x - y| = |T(x) - T(y)| \leq c\,|x - y|\,.$$

Since $c < 1$ this implies that $|x - y| = 0$ and so $x = y$.

On Pset 3 you will be asked to show that for a contracting map T and any $a_1 \in \mathbf{R}$ the sequence $a_{n+1} = T(a_n)$ is a Cauchy sequence. By the Cauchy theorem we then have that a_n is convergent.

Let a denote the limit. We claim that $T(a) = a$. Observe that $T(a_n) = a_{n+1} \to a$. If we can show that if $x_n \to x$, then $T(x_n) \to T(x)$, then

$$T(a_n) \to T(a)$$

but we already have that $T(a_n) = a_{n+1} \to a$ so we would have that $T(a) = a$ and thus a is a fixed point.

We need therefore show that if $x_n \to x$, then $T(x_n) \to T(x)$. To do that observe that

$$|T(x_n) - T(x)| \leq c\,|x_n - x|\,.$$

Since $x_n \to x$ we have that $|x_n - x| \to 0$ and so $|T(x_n) - T(x)| \to 0$. It follows that $T(x_n) \to T(x)$. Applying this to the sequence a_n shows that a is a fixed point for T.

Applications of contracting mapping theorem:

Existence of solutions to ODEs. We will return to this later as this needs a version of the contracting mapping theorem where T is defined on a more general space than the real numbers.

Newton's method: Finding a zeroth of a function $f : \mathbf{R} \to \mathbf{R}$. (So find a solution x to $f(x) = 0$.)

Suppose that x_1 is a "good" initial guess, so $f(x_1)$ is sufficiently small. Assume also that $f' \neq 0$. Define a map

$$T(x) = x - \frac{f(x)}{f'(x)}\,.$$

We have

$$T'(x) = 1 - \frac{f'}{f'} + \frac{f\,f''}{(f')^2} = f\,\frac{f''}{(f')^2}\,.$$

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So as long as x stay close to the initial guess and for the initial guess $f(x)$ is small compared with $\frac{f''}{(f')^2}$, then T is a contracting map. By the contracting mapping theorem the sequence $x_{n+1} = T(x_n)$ is a Cauchy sequence that converges to a fixed point of T.

For a fixed point for T we have $T(x) = x$ so $x - \frac{f(x)}{f'(x)} = x$ and therefore $f(x) = 0$.

Back to Cauchy sequences.

Bolzano -Weirstrass theorem: Any bounded sequence has a convergent subsequence.

Once we have the Bolzano-Weirstrass theorem we can prove the Cauchy theorem.

Proof. (of the Cauchy theorem.) So suppose that a_n is a Cauchy sequence. We will first show that a_n is bounded. From the definition of a Cauchy sequence we have that there exists N such that for m, $n \geq N$, then

$$|a_n - a_m| < 1\,.$$

It follows, in particular, that for all $n \geq N$, we have that

$$|a_n - a_N| < 1\,,$$

and so

$$|a_n| = |(a_n - a_N) + a_N| \leq 1 + |a_N|\,.$$

Therefore,

$$|a_n| \leq \max\{|a_N| + 1, |a_1|, \cdots, |a_{N-1}|\}\,.$$

So the sequence is bounded.

From the Bolzano-Weirstrass theorem it follows that a_n has a convergent subsequence a_{n_k} with limit a. We want to show that a_n is convergent with limit a. Given $\epsilon > 0$, there exists an N_1 such that if m, $n \geq N_1$, then

$$|a_n - a_m| < \frac{\epsilon}{2}\,.$$

Moreover, there exist an N_2 such that if $k \geq N_2$, then $|a_{n_k} - a| < \frac{\epsilon}{2}$. Set $N = \max\{N_1, N_2\}$. It follows that if $n \geq N$ and $k \geq N$, then

$$|a_n - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a| < \frac{\epsilon}{2} + \frac{\epsilon}{2}\,.$$

This show that $a_n \to a$ as claimed. □

Another application of the Bolzano-Weirstrass theorem is the Extreme value theorem.

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Before stating this we need another key notion:

A function $f : \mathbf{R} \to \mathbf{R}$ is said to be continuous at a point $x_0 \in \mathbf{R}$, if for all $\epsilon > 0$, there exists a $\delta > 0$ such that if

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon \,.$$

A function is said to be continuous if it is continuous at all points in the domain.

Theorem: If $f : \mathbf{R} \to \mathbf{R}$ is continuous and x_n is a sequence with $x_n \to x_0$, then $f(x_n) \to f(x_0)$.

Proof. Given $\epsilon > 0$, since f is continuous, there exists a $\delta > 0$, such that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon$. Since $x_n \to x_0$, there exists N such that if $n \geq N$, then $|x_n - x_0| < \delta$ and therefore $|f(x_n) - f(x_0)| < \epsilon$. This show that $f(x_n) \to f(x_0)$ as claimed. □

Extreme value theorem: Let f be a continues function on an interval $[a, b]$. The extreme value theorem says that the sup and inf are achieved. That is, there exist $x \in [a, b]$ such that $f(x) = \sup f$. Likewise for $\inf f$.

Proof. We will show that the supremum is achieved. The proof that the infimum is the same with obvious modification. Let $x_n \in [a, b]$ be a sequence where $f(x_n) \to \sup f$. Since the sequence is contained in $[a, b]$ it is bounded and therefore by the Bolzano - Weirstrass theorem has a convergent subsequence $x_{n_k} \to x$. Note that $x \in [a, b]$. By the theorem above $f(x_{n_k}) \to f(x)$ and since we also have that $f(x_{n_k}) \to \sup f$ it follows that $\sup f = f(x)$. This proves that the supremum is achieved. □

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[TBB] B.S. Thomson, J.B. Bruckner, and A.M. Bruckner, *Elementary Real Analysis, 2nd edition*
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Lecture 7

Last time:

Cauchy convergence theorem: Every Cauchy sequence is convergent.

Bolzano-Weirstrass theorem.

Last time we showed that the Bolzano-Weirstrass implies the Cauchy theorem.

Bolzano-Weirstrass theorem: Any bounded sequence has a convergent subsequence.

Proof. (of the Bolzano-Weirstrass theorem.) Suppose that a_n is a bounded sequence. For simplicity assume that $a_n \in [0,1]$.

Defining the subsequence a_{n_k}. Either there are infinite many n such that $a_n \in [0, \frac{1}{2}]$ or there are infinite many such a_n in $[\frac{1}{2}, 1]$ (or both). Assume that there are infinitely many in $[0, \frac{1}{2}]$. Set $a_{n_1} = a_1$. Let a_{n_2} be the next a_n such that $a_n \in [0, \frac{1}{2}]$. We have

$$n_2 > n_1 = 1\,,$$

$$a_{n_1} \in [0,1]\,,$$

$$a_{n_2} \in \left[0, \frac{1}{2}\right]\,.$$

Next either infinitely many a_n lies in $[0, \frac{1}{4}]$ or infinitely many a_n lies in $[\frac{1}{4}, \frac{1}{2}]$. Assume that infinitely many lies in $[\frac{1}{2}, \frac{1}{4}]$. Pick an $n > n_2$ such that $a_n \in [\frac{1}{4}, \frac{1}{2}]$ and set $a_{n_3} = a_n$. We continue this way.

Convergence of a_{n_k}. Note that for $k_1, k_2 \geq k$ we have that

$$|a_{n_{k_1}} - a_{n_{k_2}}| \leq 2^{1-k}\,.$$

Since $2^{-k} \to 0$ as $k \to \infty$ this shows that the subsequence a_{n_k} is a Cauchy sequence. However, more is true. Squeezed between two other sequences $b_k \leq a_{n_k} \leq c_k$. We will define sequences b_k and c_k as follows. The sequence b_k will be the left endpoint of the interval of length 2^{1-k} that all the element a_{n_i} will lie in when $i \geq k$ and c_k will be the right end point of the same interval. We have now that the sequence b_k is increasing and the sequence c_k is decreasing and the a_{n_k} are squeezed between the two. It follows that b_k is convergent (as it is also bounded) and likewise for c_k. Since $c_k - b_k = 2^{1-k}$ it follows that b_k and c_k converges to the

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same number and since the later a_{n_k} are all squeezed between the two they also converges to that same number. □

From Wikipedia: Karl Theodor Wilhelm Weierstrass (1815 – 1897) was a German mathematician often cited as the "father of modern analysis". Despite leaving university without a degree, he studied mathematics and trained as a school teacher, eventually teaching mathematics, physics, botany and gymnastics. Among many other contributions, Weierstrass formalized the definition of the continuity of a function and complex analysis, proved the intermediate value theorem and the Bolzano–Weierstrass theorem, and used the latter to study the properties of continuous functions on closed bounded intervals.

Series: Suppose that a_n is a sequence, we can form a new sequence s_n as follows. We let

$$s_1 = a_1 \,,$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3 \,,$$

and in general set

$$s_n = a_1 + \cdots + a_n = \sum_{i=1}^{n} a_i \,.$$

A series $\sum_{i=1}^{\infty} a_i$ converges if the sequence s_n converges and if it do we also write $\sum_{i=1}^{\infty} a_i$ for the limit.

Geometric series: Suppose now that $a_n = c^n$ so the series is

$$s_n = \sum_{i=0}^{n} c^i \,.$$

This is the geometric series. It is convergent precisely when $|c| < 1$. Moreover, when $|c| < 1$, then the limit (infinite sum) is

$$\sum_{i=0}^{\infty} c^i = \frac{1}{1-c} \,.$$

To see this observe that

$$(1-c)\sum_{i=0}^{n} c^i = 1 - c^{n+1} \,.$$

It follows from this that if $c \neq 1$, then

$$s_n = \sum_{i=0}^{n} c^i = \frac{1-c^{n+1}}{1-c} \,.$$

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Therefore, s_n converges with limit $\frac{1}{1-c}$ if $|c| < 1$ and diverges if $|c| > 1$ or $c = -1$. One easily checks, separately, that it also diverges for $c = 1$.

Harmonic series: For the harmonic series $a_n = \frac{1}{n}$ so the series is $\sum_{i=1}^{\infty} \frac{1}{2}$. This series is divergent. To see this we will show that

$$s_{2^n-1} \geq \frac{n}{2}\,.$$

This is true for $n = 1$ as $s_1 = 1 \geq \frac{1}{2}$.

Assume that it is true for n we will show that it is also true for $n + 1$. Namely,

$$s_{2^{n+1}-1} \geq s_{2^n-1} + \sum_{i=2^n}^{2^{n+1}-1} \frac{1}{i} \geq \frac{n}{2} + 2^n \frac{1}{2^{n+1}-1} \geq \frac{n}{2} + \frac{1}{2} = \frac{n+1}{2}\,.$$

So the formula also holds for $n + 1$ and therefore for all n.

Since $\frac{n}{2} \to \infty$ it follows that the subsequence s_{2^n-1} is divergent and therefore so is the original series.

Absolutely convergent; We say that a series

$$\sum_{n=0}^{\infty} a_n$$

is absolutely convergent if the series

$$\sum_{n=0}^{\infty} |a_n|$$

is convergent. Absolutely convergent implies convergent but not the other way around.

Example: We will see later that the series

$$\sum_{n=1} \frac{(-1)^n}{n}$$

is convergent but if we take the absolute values of the a_n's, then we get the harmonic series which is divergent.

Proof. (of why absolutely convergent implies convergent.) By the Cauchy convergence theorem we only need to show that the sequence

$$\sum_{i=0}^{\infty} a_i$$

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is Cauchy sequence if the series

$$\sum_{n=0}^{\infty} |a_n|$$

is convergent. Set

$$s_n = \sum_{i=0}^{n} a_i \,,$$

$$\mathbf{s}_n = \sum_{i=0}^{n} |a_i| \,.$$

For $m < n$ we have

$$|s_n - s_m| = |a_n + \cdots + a_{m+1}| \leq |a_n| + \cdots + |a_{m+1}| = |\mathbf{s}_n - \mathbf{s}_m| \,.$$

Since the sequence $\mathbf{s}_n$ is a Cauchy sequence it now follows that s_n is. □

Theorem: A series of non-negative numbers

$$\sum_{i=0}^{\infty} a_i \,,$$

where $a_n \geq 0$, is convergent if and only if the sequence s_n is bounded from above.

Proof. The sequence s_n is monotone nondecreasing since

$$s_{n+1} = s_n + a_n \geq s_n \,.$$

The claim now follows from the monotone convergence theorem. □

Example: The series

$$\sum_{i=1}^{\infty} \frac{1}{i^2}$$

is convergent. This is a sequence of non-negative numbers so we only need to show that there exist M such that for all n

$$s_n = \sum_{i=1}^{n} \frac{1}{i^2} \leq M \,.$$

Claim:

$$s_{2^n-1} \leq \sum_{i=0}^{n-1} \left(\frac{1}{2}\right)^i \,.$$

This would be enough because the last is a convergent geometric series so in particular bounded.

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We will show this by induction. For $n = 1$ we have that

$$s_1 = 1 = \left(\frac{1}{2}\right)^0 .$$

So it is correct for $n = 1$. Assume next that it is true for n; we will show that it also holds for $n + 1$.

$$s_{2^{n+1}-1} = s_{2^n-1} + \sum_{i=2^n}^{2^{n+1}-1} \frac{1}{i^2} \leq \sum_{i=0}^{n-1} \left(\frac{1}{2}\right)^i + 2^n \frac{1}{(2^n)^2}$$
$$\leq \sum_{i=0}^{n-1} \left(\frac{1}{2}\right)^i + \frac{1}{2^n} = \sum_{i=0}^{n} \left(\frac{1}{2}\right)^i .$$

This show the induction step and completes the proof.

To help determine whether or not a series converges there are a number of tests:

- Comparison test.
- Ratio test.
- Root test.

Comparison test; version 1: Suppose that a_n and b_n are two sequences with

$$0 \leq a_n \leq b_n .$$

If

$$\sum_{n=1}^{\infty} b_n$$

is convergent, then so is

$$\sum_{n=1}^{\infty} a_n .$$

Example: The series

$$\sum_{n=1}^{\infty} \frac{2^{-n}}{n}$$

is convergent. Namely, if we set

$$a_n = \frac{2^{-n}}{n}$$

and

$$b_n = 2^{-n} ,$$

then $0 \leq a_n \leq b_n$ and since the series $\sum_{n=1}^{\infty} b_n$ is convergent, then by the comparison test so is the series $\sum_{n=1}^{\infty} a_n$.

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Comparison test; version 2: Suppose that a_n and b_n are two sequences with $b_n \neq 0$ and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L \neq 0\,,$$

The series

$$\sum_{n=1}^{\infty} a_n$$

is convergent if and only if

$$\sum_{n=1}^{\infty} b_n$$

is.

Example: The series

$$\sum_{n=2}^{\infty} \frac{1}{n^2-1}$$

is convergent since

$$\frac{n^2}{n^2-1} \to 1\,,$$

and the series

$$\sum_{n=2}^{\infty} \frac{1}{n^2}$$

is convergent.

Ratio test: Let $a_n \geq 0$ and assume that

$$\frac{a_{n+1}}{a_n} \to a$$

.

If

- $a < 1$, then the series $\sum a_n$ is convergent.
- $a > 1$, then the series $\sum a_n$ is divergent.
- $a = 1$, it is inconclusive.

Example 1:

$$a_n = \frac{1}{n}\,.$$

In this case

$$\frac{a_{n+1}}{a_n} \to 1$$

so the test is inconclusive, but the series is divergent.

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Example 2:

$$a_n = \frac{1}{n^2}\,.$$

In this case

$$\frac{a_{n+1}}{a_n} \to 1$$

so the test is inconclusive, but the series is convergent.

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Lecture 8

Series: Suppose that a_n is a sequence, we can form a new sequence s_n as follows. We let

$$s_1 = a_1\,,$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3\,,$$

and in general set

$$s_n = a_1 + \cdots + a_n = \sum_{i=1}^{n} a_i\,.$$

A series $\sum_{i=1}^{\infty} a_i$ converges if the sequence s_n converges and if it do we also write $\sum_{i=1}^{\infty} a_i$ for the limit.

Geometric series:

$$\sum_{i=0}^{\infty} c^i n\,.$$

Convergent precisely when $|c| < 1$.

Harmonic series:

$$\sum_{i=1}^{\infty} \frac{1}{n}\,.$$

This series is divergent.

Absolutely convergent; We say that a series

$$\sum_{n=0}^{\infty} a_n$$

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is absolutely convergent if the series

$$\sum_{n=0}^{\infty} |a_n|$$

is convergent. Absolutely convergent implies convergent but not the other way around.

Theorem: A series of non-negative numbers $a_n \geq 0$

$$\sum_{i=0}^{\infty} a_i \,,$$

is convergent if and only if the sequence s_n is bounded from above.

To help determine whether or not a series converges there are a number of tests:

- Comparison test.
- Ratio test.
- Root test.
- Other tests that we will discuss later.

Comparison test; version 1: Suppose that a_n and b_n are two sequences with

$$0 \leq a_n \leq b_n \,.$$

If

$$\sum_{n=1}^{\infty} b_n$$

is convergent, then so is

$$\sum_{n=1}^{\infty} a_n \,.$$

Example: The series

$$\sum_{n=1}^{\infty} \frac{2^{-n}}{n}$$

is convergent. Namely, if we set

$$a_n = \frac{2^{-n}}{n}$$

and

$$b_n = 2^{-n} \,,$$

then $0 \leq a_n \leq b_n$ and since the series $\sum_{n=1}^{\infty} b_n$ is convergent, then by the comparison test so is the series $\sum_{n=1}^{\infty} a_n$.

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Comparison test; version 2: Suppose that a_n and b_n are two sequences with $b_n \neq 0$ and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L \neq 0\,,$$

The series

$$\sum_{n=1}^{\infty} a_n$$

is convergent if and only if

$$\sum_{n=1}^{\infty} b_n$$

is.

Example: The series

$$\sum_{n=2}^{\infty} \frac{1}{n^2-1}$$

is convergent since

$$\frac{n^2}{n^2-1} \to 1\,,$$

and the series

$$\sum_{n=2}^{\infty} \frac{1}{n^2}$$

is convergent.

Ratio test: Let $a_n \geq 0$ and assume that

$$\frac{a_{n+1}}{a_n} \to a$$

.

If

- $a < 1$, then the series $\sum a_n$ is convergent.
- $a > 1$, then the series $\sum a_n$ is divergent.
- $a = 1$, it is inconclusive.

Example 1:

$$a_n = \frac{1}{n}\,.$$

In this case

$$\frac{a_{n+1}}{a_n} \to 1$$

so the test is inconclusive, but the series is divergent.

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Example 2:

$$a_n = \frac{1}{n^2}\,.$$

In this case

$$\frac{a_{n+1}}{a_n} \to 1$$

so the test is inconclusive, but the series is convergent.

Root test: Let $a_n \geq 0$ be a sequence of non-negative numbers. Suppose $\lim_{n\to\infty}(a_n)^{\frac{1}{n}} = r$. If

- $r < 1$, then the series $\sum_{n=0}^{\infty} a_n$ is convergent.
- $r > 1$, then the series $\sum_{n=0}^{\infty} a_n$ is divergent.
- $r = 1$, then it is inconclusive.

Proof. (of root test.) Suppose that $r < 1$. It follows that for $r < r_0 < 1$, there exists N such that if $n \geq N$, then

$$(a_n)^{\frac{1}{n}} \leq r_0\,.$$

Therefore,

$$0 \leq a_n \leq r_0^n\,.$$

However, the series $\sum_{n=0}^{\infty} r_0^n$ is a geometric series that is convergent since $r_0 < 1$. We now have by the first version of the comparison test that also the series $\sum_{n=0}^{\infty} a_n$ is convergent.

Suppose that $r > 1$. In that case we have that for $1 < r_0 < r$, there exists N such that if $n \geq N$, then

$$(a_n)^{\frac{1}{n}} \geq r_0\,.$$

Hence, for $n \geq N$

$$a_n \geq r_0^n\,,$$

where, the series $\sum_{n=0}^{\infty} r_0^n$ is a divergent geometric series. Therefore, by the comparison test the original series is divergent. □

Power series:

- $$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}\,.$$
- $$\sum_{n=0}^{\infty} \frac{x^n}{n!} = \exp\, x\,.$$

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- $$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2\,n}}{(2\,n)!} = \cos\, x\,.$$
- $$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2\,n+1}}{(2\,n+1)!} = \sin\, x\,.$$

Formel definition: Let c_n be a sequence, then $\sum_{n=0}^{\infty} c_n\, x^n$ is a power series.

When does a power series converge?

Why does it give familiar functions?

We will answer the second question next time for the exponential function.

The answer to the first question comes from the root test or the ratio test.

Example: Consider the power series:

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}\,.$$

By the ratio test with $a_n = \frac{x^n}{n!}$ we have

$$\frac{a_{n+1}}{a_n} = \frac{n!\, x^{n+1}}{(n+1)!\, x^n} = \frac{x}{n+1} \to 0\,.$$

It follows that the power series is convergent for all x.

Example:

$$\sum_{n=0}^{\infty} x^n\,.$$

This series is convergent for $|x| < 1$ and divergent otherwise.

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To talk about convergence of a general power series we need the notion of lim sup of a sequence. This is defined as follows.

Let a_n be a sequence. If it is not bounded from above, then we set $\limsup a_n$ to be ∞. Otherwise we will define a new sequence b_n from a_n as follows.

$$b_n = \sup\{a_n, a_{n+1}, a_{n+2}, \cdots\}\,.$$

Note that since we are assuming that the a_n's are bounded from above the b_n's are real numbers and the sequence b_n is decreasing. – It is decreasing since

$$b_n = \sup\{a_n, a_{n+1}, a_{n+2}, \cdots\} \geq \sup\{a_{n+1}, a_{n+2}, \cdots\} = b_{n+1}\,.$$

(For b_{n+1} supremum is taken over a smaller set.)

Since the sequence b_n is decreasing it is converging with limit b that possibly could be $-\infty$ if the sequence b_n is not bounded from below.

Definition (of lim sup):

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} b_n = b\,.$$

Back to power series. Suppose that

$$\sum_{n=0}^{\infty} a_n\, x^n$$

is a power series. Set

$$R = \frac{1}{\limsup_{n\to\infty} |a_n|^{\frac{1}{n}}}\,.$$

R is said to be the radius of convergence.

Convention: If $\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} = 0$, then the radius of convergence is said to be ∞. If $\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} = \infty$, then we set $R = 0$.

From the root test one can now show the following:

The power series in convergent if $|x| < R$ and divergent if $|x| > R$.

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The case of where $|x| = R$ has to be examined on a case by case basis.

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TOBIAS HOLCK COLDING

Lecture 9

Power series: Suppose that a_n is a sequence, for each x we can form a series

$$\sum_{n=0}^{\infty} a_n \, x^n \, .$$

Exponential map as a power series: Define $E(x)$ as the power series

$$E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \, .$$

Step 0: The power series converges for all x. Namely, since

$$\frac{|a_{n+1} \, x^{n+1}|}{|a_n \, x^n|} = \frac{|n! \, x^{n+1}|}{|(n+1)! \, x^n|} = \frac{x}{n+1} \to 0 \, ,$$

the claim follows from the ratio test.

Step 1: Define $\mathrm{e} = E(1)$ and $\mathrm{e}^2 = \mathrm{e}\,\mathrm{e}$, $\mathrm{e}^3 = \mathrm{e}\,\mathrm{e}\,\mathrm{e}$ etc. This way e^k is defined for all $k \in \mathbf{N}$.

We also define $\mathrm{e}^{-k} = \frac{1}{\mathrm{e}^k}$. (The idea is that we would like to have $\mathrm{e}^{x+y} = \mathrm{e}^x \, \mathrm{e}^y$.)

We set $\mathrm{e}^0 = 1$.

If $q = \frac{m}{n} \in \mathbf{Q}$, where $m \in \mathbf{Z}$ and $n \in \mathbf{N}$, then we let e^q be the positive number α so that $\alpha^n = \mathrm{e}^m$. (Again the idea is that we would like to have that $\mathrm{e}^{x+y} = \mathrm{e}^x \, \mathrm{e}^y$.)

This way e^q is defined for all rational numbers.

What about the irrational numbers like $\sqrt{2}$?

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Step 2: Next time we will show that

$$E(x+y) = E(x)\,E(y)\,.$$

Here we claim that $E(x) > 0$ for all x. If $x \geq 0$, then this is clear since

$$E(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \geq 1\,.$$

If $x < 0$, then using the formula we will show next time we have that

$$1 = E(0) = E(x)\,E(-x)\,.$$

Therefore,

$$E(x) = \frac{1}{E(-x)} > 0\,.$$

Using that $E(x+y) = E(x)\,E(y)$ we now claim that $E(q) = \mathrm{e}^q$ for all rational numbers q.

For integers m this is how we defined e^m. For $m = -k$, where $k \in \mathbf{N}$ we defined

$$\mathrm{e}^{-k} = \frac{1}{\mathrm{e}^k} = \frac{1}{E(k)} = E(-k)\,.$$

For a general rational number $q = \frac{m}{n}$, where $m \in \mathbf{N}$ and $n \in \mathbf{Z}$, we have

$$\left(E\left(\frac{m}{n}\right)\right)^n = E\left(\frac{m}{n}\right)\cdots E\left(\frac{m}{n}\right) = E(m)\,,$$

and

$$E\left(\frac{m}{n}\right) > 0\,.$$

This gives us that $E(q) = \mathrm{e}^q$ for all rational numbers q.

Step 3: We now have $E(x)$ is defined for all x whereas e^x is defined for all rational numbers.

What other properties would we want of the exponential function?

We would want it to be continuous!

Reminder: A function $f : A \to \mathbf{R}$ on some set $A \subset \mathbf{R}$ is said to be continuous if for all $x_0 \in A$ we have:

For all $\epsilon > 0$, there exists a $\delta = \delta(x_0) > 0$ such that if $|x - x_0| < \delta$ $(x \in A)$, then $|f(x) - f(x_0)| < \epsilon$.

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On Pset 5 you will be asked to show that $E(x)$ is continuous at all points.

Step 4: We will next show that $E(x)$ is the unique continuous function where $E(q) = \mathrm{e}^q$ for all rational numbers q.

Theorem: Let f and g be two continuous function on $\mathbf{R}$ that agrees on all rational numbers, then $f = g$.

We will show this theorem next time. For now here are some more about what it means for a function to be continuous.

Example 1: Suppose $f(x) = c$, where c is a constant. We will show that f is continuous. Given $x_0 \in \mathbf{R}$ and $\epsilon > 0$, set $\delta = 1$. We then have that if $|x - x_0| < \delta = 1$, then $|f(x) - f(x_0)| = 0 < \epsilon$. This show that f is continuous.

Example 2: Suppose $f(x) = x$, we will show that f is continuous. Given $x_0 \in \mathbf{R}$ and $\epsilon > 0$, set $\delta = \epsilon$. We then have that if $|x - x_0| < \delta = \epsilon$, then $|f(x) - f(x_0)| = |x - x_0| < \epsilon$. This show that f is continuous.

Algebraic properties of continuous functions:

- If f and g are continuous functions, then so is $f + g$.
- If f is continuous and c is a constant, then $c\,f$ is continuous.
- If f and g are continuous, then $f\,g$ is also continuous.
- If f is continuous and $f \neq 0$, then $\frac{1}{f}$ is continuous.
- If $f(x)$ and $g(x)$ are continuous, then $f(g(x))$ is continuous.

Proof. (the proof is very similar to the one we gave for the algebraic properties of limits.) □

Theorem: All polynomials are continuous.

Example 3: If $f(x) = x^2 + 1$, then f is continuos. We have already proven that $g(x) = x$ is continuous so by the algebraic properties we have that $x^2 = g\,g$ is continuous. We have also already shown that the constant functions are continuous so $h(x) = 1$ is continuous and therefore by the algebraic properties we have that $f = g^2 + h$ is continuous.

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TOBIAS HOLCK COLDING

Lecture 10

Power series: Suppose that a_n is a sequence. For each x we can form a series

$$\sum_{n=0}^{\infty} a_n \, x^n \, .$$

Exponential function as a power series: Define $E(x)$ by the power series

$$E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \, .$$

Step 0: The power series converges for all x.

Step 1: Define e^q for all rational numbers q.

Step 2: Need to show that

$$E(x+y) = E(x) \, E(y) \, .$$

Step 3: $E(x)$ is defined for all x, whereas e^x is defined for all rational numbers, and $E(q) = \mathrm{e}^q$ for all rational numbers.

Step 4: E is continuous on all of $\mathbf{R}$. (Pset 5.)

Step 5: If f and g are continuous functions on $\mathbf{R}$ that agrees on $\mathbf{Q}$, then $f = g$ everywhere.

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Suppose that we have two convergent series

$$\sum_{n=0}^{\infty} a_n \text{ and } \sum_{n=0}^{\infty} b_n$$

of non-negative numbers a_n, $b_n \geq 0$.

Form the "product series"

$$\sum_{n=0}^{\infty} c_n \,,$$

where

$$c_n = \sum_{i=0}^{n} a_i \, b_{n-i} \,.$$

Note that each $c_n \geq 0$ so by the monotone convergence theorem the series

$$\sum_{n=0}^{\infty} c_n$$

is convergent if it is bounded.

Theorem 1: If $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ are as above, then the series

$$\sum_{n=0}^{\infty} c_n$$

is convergent with limit

$$\sum_{n=0}^{\infty} a_n \sum_{n=0}^{\infty} b_n \,.$$

Proof. Denote

$$s^a_n = \sum_{i=0}^{n} a_i \text{ and } s^b_n = \sum_{i=0}^{n} b_i \text{ and } s^c_n = \sum_{i=0}^{n} c_i \,.$$

The idea here is that

$$(*) \qquad \left(\sum_{\ell=0}^{n} a_\ell \right) \left(\sum_{\ell=0}^{n} b_\ell \right) = \sum_{k=0}^{n} \sum_{i+j=k} a_i \, b_j + \sum_{i+j>n \text{ and } i,j \leq n} a_i \, b_j \leq \sum_{k=0}^{2\,n} \sum_{i+j=k} a_i \, b_j \,.$$

In other words

$$(**) \qquad s^c_n \leq s^a_n \, s^b_n \leq s^c_{2\,n} \,.$$

This is because $(*)$ is

$$s^a_n \, s^b_n = s^c_n + \sum_{i+j>n \text{ and } i,j \leq n} a_i \, b_j \leq s^c_{2\,n} \,,$$

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and

$$0 \leq \sum_{i+j>n \text{ and } i,j\leq n} a_i\, b_j\,.$$

Note that the first inequality in $(**)$ implies that the sequence s_n^c is bounded and therefore since a_n, b_n, $c_n \geq 0$ we have that

$$s_n^a \uparrow s^a\,, \qquad s_n^b \uparrow s^b\,, \qquad s_n^c \uparrow s^c$$

by the monotone convergence theorem for sequences. Since the product $s_n^a\, s_n^b$ is squeezed between s_n^c and $s_{2\,n}^c$ by $(**)$ we have that

$$s^c \leq s^a\, s^b \leq s^c\,.$$

From this the claim follows. □

Applying Theorem 1 to the power series $E(x)$ we can now prove the following:

Theorem 2:

$$E(x+y) = E(x)\, E(y)\,.$$

Proof. We will show this assuming that x, $y \geq 0$. Once we have shown the theorem for x, $y \geq 0$ the general case is not too difficult but we will not prove that here. The idea is that $E(x+y)$ will play the role of

$$\sum_{n=0}^{\infty} c_n$$

above. So set

$$c_n = \frac{(x+y)^n}{n!}\,,$$

By the "binomial" formula

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i\, y^{n-i}\,.$$

So

$$c_n = \frac{1}{n!} \sum_{i=0}^{n} \binom{n}{i} x^i\, y^{n-i}\,.$$

Since

$$\binom{n}{i} = \frac{n!}{i!\,(n-i)!}\,,$$

we have that

$$c_n = \sum_{i=0}^{n} \frac{x^i}{i!}\, \frac{y^{n-i}}{(n-i)!}\,.$$

This shows that

$$c_n = \sum_{i=0}^{n} a_i\, b_{n-i}\,,$$

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where

$$a_i = \frac{x^i}{i!}$$

and

$$b_i = \frac{y^i}{i!}\,.$$

The claim now follows from Theorem 1. □

Coming back to the functions E and e. We have that they agree on all rational numbers and that E is defined for all real numbers.

We would want the exponential function to be continuous!

Reminder: A function $f : A \to \mathbf{R}$ on some set $A \subset \mathbf{R}$ is said to be continuous if for all $x_0 \in A$ we have:

For all $\epsilon > 0$, there exists a $\delta = \delta(x_0) > 0$ such that if $|x - x_0| < \delta$ $(x \in A)$, then $|f(x) - f(x_0)| < \epsilon$.

On Pset 5 you will be asked to show that $E(x)$ is continuous at all points.

Step 5: We will show that $E(x)$ is the unique continuous function where $E(q) = \mathrm{e}^q$ for all rational numbers q.

Theorem 3: (On Pset 5.) Let f and g be two continuous function on $\mathbf{R}$ that agrees on all rational numbers, then $f = g$.

We will next see that there are functions on $\mathbf{R}$ that are not continuous at any point!

Before defining such a function recall that we already proved that $\sqrt{2}$ is a irrational number and thus for all $\delta > 0$, there exists an N such that if $n \geq N$, then

$$0 < \frac{\sqrt{2}}{n} < \delta\,.$$

So arbitrarily close to zero there are irrational numbers. Likewise by the Archimedean property we have that arbitrarily close to any irrational number there is a rational number.

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On $\mathbf{R}$ define a function f as follows

$$f(x) = \begin{cases} 1 & x \in \mathbf{Q} \\ 0 & \text{otherwise} \end{cases}$$

We claim that f is nowhere continuous. Suppose first that x_0 is rational and let $0 < \epsilon < 1$. We have that $f(x_0) = 1$ and for any $\delta > 0$, there exists a irrational number x with $|x - x_0| < \delta$ but we also have that

$$\epsilon < 1 = |f(x) - f(x_0)| .$$

This show that f is discontinuous at x_0.

Likewise suppose x_0 is an irrational number. We have that $f(x_0) = 0$. Given $0 < \epsilon < 1$ for any $\delta > 0$, there exists a rational number x with $|x - x_0| < \delta$. On the other hand

$$\epsilon < 1 = |f(x) - f(x_0)| .$$

This show that f is discontinuous at x_0.

This gives an example of a function that is discontinuous at all points. On the other hand recall from last time how to generate continuous functions from known continuous functions:

Algebraic properties of continuous functions:

- If f and g are continuous functions, then so is $f + g$.
- If f is continuous and c is a constant, then $c\,f$ is continuous.
- If f and g are continuous, then $f\,g$ is also continuous.
- If f is continuous and $f \neq 0$, then $\frac{1}{f}$ is continuous.
- If $f(x)$ and $g(x)$ are continuous, then $f(g(x))$ is continuous.

Proof. (The proof is very similar to the one we gave for the algebraic properties of limits of sequences.) □

Theorem: All polynomials are continuous.

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Lecture 11

Reminder: A function $f : A \to \mathbf{R}$ on some set $A \subset \mathbf{R}$ is said to be continuous if for all $x_0 \in A$ we have:

For all $\epsilon > 0$, there exists a $\delta = \delta(x_0) > 0$ such that if $|x - x_0| < \delta$ $(x \in A)$, then $|f(x) - f(x_0)| < \epsilon$.

Two theorems about continuous functions:

Extreme Value Theorem: Suppose that $f : [a, b] \to \mathbf{R}$ is a continuous function, then there exist $x_M \in [a, b]$ such that $f(x_M) \geq f(x)$ for all $x \in [a, b]$. Similarly, there exists $x_m \in [a, b]$ such that $f(x_m) \leq f(x)$ for all $x \in [a, b]$.

Intermediate Value Theorem: Suppose that $f : [a, b] \to \mathbf{R}$ is a continuous function, then for all y between $f(a)$ and $f(b)$, there exists $x \in [a, b]$ such that $f(x) = y$.

We will show these theorems using a lemma that connects sequences and continuous functions. This is the following:

Lemma: Suppose that $f : [a, b] \to \mathbf{R}$ is a continuous function and $x_n \to x_\infty$ a sequence, then $f(x_n) \to f(x_\infty)$. We can also write this as

$$\lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right) .$$

Proof. To show that $f(x_n) \to f(x_\infty)$ let $\epsilon > 0$ be given. Since f is continuous at x_∞, there exists $\delta > 0$ such that if $|x - x_\infty| < \delta$, then $|f(x) - f(x_\infty)| < \epsilon$. Since $x_n \to x_\infty$ there exists N such that if $n \geq N$, then $|x_n - x_\infty| < \delta$ and therefore $|f(x_n) - f(x_\infty)| < \epsilon$. This shows the lemma. □

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Using this lemma we can now prove the extreme value theorem:

Proof. (of EVT.) Let $E = f([a,b])$ and set $M = \sup E$. We will show that $M < \infty$ and that $M = f(x)$ for some $x \in [a,b]$. We show first that M is finite. Otherwise for each n there exists an $x_n \in [a,b]$ such that $f(x_n) > n$. Since the sequence $\{x_n\}$ is bounded by the Bolzano-Weirstrass theorem it has a convergent subsequence. Let us denote that by x_{n_k}. We have $x_{n_k} \to x_\infty \in [a,b]$. By the lemma above $f(x_{n_k}) \to f(x_\infty)$ but we assumed that the sequence $f(x_{n_k})$ is unbounded which is the desired contradiction.

For each integer n we can now choose $x_n \in [a,b]$ such that $f(x_n) > M - \frac{1}{n}$. Again since this sequence is bounded by the Bolzano-Weirstrass theorem it has a convergent subsequence $x_{n_k} \to x_\infty \in [a,b]$. By the lemma above $f(x_{n_k}) \to f(x_\infty) \geq M$. Since $M = \sup f([a,b])$ we have that $f(x_\infty) = M$. This show the EVT. □

Proof. (of IVT.) We will assume that $f(a) < 0 < f(b)$ and show that there exists $x \in [a,b]$ such that $f(x) = 0$. The general case is similar. Let $A = \{y \mid \text{ for all } x \leq y \text{ we have that } f(x) \leq 0\}$. Note that $a \in A$ so the set is non-empty. Set $M = \sup A$ and let x_n be a sequence with $x_n < M$ and $x_n \to M$. It follows that $f(x_n) < 0$ and so by the lemma above we must have that $f(M) \leq 0$. We are done if $f(M) = 0$ so assume that $f(M) < 0$. We have that $M < b$ and by continuity there exist a whole interval around M where $f < 0$. This contradict that M was the supremum of the set A. Showing the IVT. □

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Abstract metric space.

Definition: Metric space A metric space is a set X with a function $d : X \times X \to \mathbf{R}$ with the following three properties:

(1) $d(x, y) \geq 0$ for all x, $y \in X$ and $d(x, y) = 0$ if and only if $x = y$. (Distances ≥ 0.)
(2) $d(x, y) = d(y, x)$. (Symmetric.)
(3) $d(x, z) \leq d(x, y) + d(y, z)$. (Triangle inequality.)

Examples:

(1) $X = \mathbf{R}$ and
$$d(x, y) = |x - y| .$$
(2) $X = \mathbf{R}^2$ and for $x = (x_1, x_2)$ and $y = (y_1, y_2)$
$$d(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} .$$
(3) $X = \mathbf{R}^3$ and for $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$
$$d(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2 + |x_3 - y_3|^2} .$$

Example: Continuous function on an interval $[a, b]$. Let $X = C([a, b])$ where $C([a, b])$ is the set of continuous functions on $[a, b]$. The distance between two continuous functions f and g is then
$$d(f, g) = \max_{x \in [a,b]} |f(x - g(x)| .$$
Since $f - g$ is also a continuous function the EVT theorem guarantees that the max is achieved for some $x \in [a, b]$.

Metric spaces plays the role of generalised real numbers. A lot of the discussion that we have had in the class holds also for metric spaces and this is useful in many circumstances. For instance, we will see in a later class that we can use it to solve ODEs.

Sequences in a metric space: A sequence in a metric space (X, d) is a map $f : \mathbf{N} \to X$. We typically denote the image $f(n)$ by x_n. Similarly we define a **subsequence** as the composition of a strictly increasing map $g : \mathbf{N} \to \mathbf{N}$ with f and $x_{n_k} = f(g(k))$.

It is not all results that we know from $\mathbf{R}$ that generalises to general metric spaces. For instance, in general there are no algebraic properties, no squeeze theorem, no monotone

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convergence theorem. On the other hand the statement of both the Cauchy convergence theorem and the Bolzano-Weirstrass theorems makes sense in a general metric space.

Example (Box distance): The space is $X = \mathbf{R}^2$ and if $\underline{x} = (x_1, x_2)$ and $\underline{y} = (y_1, y_2)$, then

$$d(\underline{x}, \underline{y}) = |x_1 - y_1| + |x_2 - y_2| \, .$$

Example (Strange metric on integers): The space is $X = \mathbf{N}$ and if m, n are integers, then

$$d(m, n) = \left| \frac{1}{n} - \frac{1}{m} \right| \, .$$

Here is a wild example of a metric space:

Example (French railway metric): The space is $X = \mathbf{R}^2$ and if $\underline{x} = (x_1, x_2)$ and $\underline{y} = (y_1, y_2)$, then

$$d(\underline{x}, \underline{y}) = \begin{cases} |\underline{x} - \underline{y}| & \text{if } \underline{x} = c\,\underline{y} \text{ or } \underline{y} = c\,\underline{x} \text{ for some } c \in \mathbf{R} \\ |\underline{x}| + |\underline{y}| & \text{otherwise} \end{cases} .$$

Here

$$|\underline{x} - \underline{y}| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} \, ,$$

and likewise for $|\underline{x}|$ and $|\underline{y}|$.

Definition: Convergent sequence in a metric space If (X, d) is a metric space and x_n is a sequence in X, then we say that x_n converges to x and write $x_n \to x$ or $x = \lim_{n \to \infty} x_n$ if for all $\epsilon > 0$, there exists an N such that if $n \geq N$, then

$$d(x, x_n) < \epsilon \, .$$

This is equivalent to that the sequence $d(x_n, x_\infty) \to 0$.

Definition: Cauchy sequence in a metric space If (X, d) is a metric space and x_n is a sequence in X, then we say that x_n is a Cauchy sequence if for all $\epsilon > 0$, there exists an N, such that if m, $n \geq N$, then

$$d(x_m, x_n) < \epsilon \, .$$

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Theorem: In any metric space (X, d) a convergent sequence is also a Cauchy sequence.

Proof. So suppose that $x_n \in X$ is a sequence and $x_n \to x$. Given $\epsilon > 0$, convergence means that there exists N such that if $n \geq N$, then $d(x, x_n) < \frac{\epsilon}{2}$. If both m, $n \geq N$, then we have by the triangle inequality that

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon\,.$$

This show the theorem. □

The converse is not always the case: If $X = (0, 1) \subset \mathbf{R}$ with $d(x, y) = |x - y|$, then the sequence $x_n = \frac{1}{n}$ is a Cauchy sequence but since 0 is not in X, it is not convergent. We sometimes express this by saying that in this case X is not Cauchy complete.

Definition: Continuous function on a metric space (X, d) Suppose that $F : X \to \mathbf{R}$ is a function. We say that f is continuous at $x_0 \in X$, if for all $\epsilon > 0$, there exists a $\delta > 0$, such that if $x \in X$ with $d(x, x_0) < \delta$, then

$$|F(x) - F(x_0)| < \epsilon\,.$$

Example: Let again $X = C([0, 1])$ be the set of continuous functions on $[0, 1]$. Equip X with the distance described above. So the distance between to continuous functions f and g is then

$$d(f, g) = \max_{x \in [a,b]} |f(x - g(x)|\,.$$

Define F on X to be the function $F(f) = f(0)$ where $f \in C([0, 1])$. F is easily seen to be a continuous function on the metric space X.

We can now extend one of the earlier lemmas to general metric spaces.

Lemma: Let (X, d) be a general metric space. Suppose that $f : X \to \mathbf{R}$ is a continuous function and x_n is a sequence in X with $x_n \to x_\infty$, then $f(x_n) \to f(x_\infty)$.

Proof. To show that $f(x_n) \to f(x_\infty)$ let $\epsilon > 0$ be given. Since f is continuous at x_∞, there exists $\delta > 0$ such that if $d(x, x_\infty) < \delta$, then $|f(x) - f(x_\infty)| < \epsilon$. Since $x_n \to x_\infty$ there exists N such that if $n \geq N$, then $d(x_n, x_\infty) < \delta$ and therefore $|f(x_n) - f(x_\infty)| < \epsilon$. This shows the lemma. □

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Lecture 12

Review (discussed in lectures so far):

(1) **R** is the complete ordered field that contains **Q**.
(2) Sequences and limits.
(3) Series.
(4) Continuous functions.
(5) Metric spaces.

Definition: **Metric space.** A metric space is a set X with a function $d : X \times X \to \mathbf{R}$ with the following three properties:

(1) $d(x, y) \geq 0$ for all x, $y \in X$ and $d(x, y) = 0$ if and only if $x = y$. (Distances ≥ 0.)
(2) $d(x, y) = d(y, x)$. (Symmetric.)
(3) $d(x, z) \leq d(x, y) + d(y, z)$. (Triangle inequality.)

Examples (Euclidean distance):

(1) $X = \mathbf{R}$ and
$$d(x, y) = |x - y| .$$
(2) $X = \mathbf{R}^2$ and for $x = (x_1, x_2)$ and $y = (y_1, y_2)$
$$d(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} .$$
(3) $X = \mathbf{R}^3$ and for $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$
$$d(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2 + |x_3 - y_3|^2} .$$

Example: Continuous function on an interval $[a, b]$. Let $X = C([a, b])$ where $C([a, b])$ is the set of continuous functions on $[a, b]$. The distance between two continuous functions f and g is then
$$d(f, g) = \max_{x \in [a,b]} |f(x - g(x)| .$$
Since $f - g$ is also a continuous function the EVT theorem guarantees that the max is achieved for some $x \in [a, b]$.

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Example (Box distance): The space is $X = \mathbf{R}^2$ and if $\underline{x} = (x_1, x_2)$ and $\underline{y} = (y_1, y_2)$, then

$$d(\underline{x}, \underline{y}) = |x_1 - y_1| + |x_2 - y_2| \, .$$

Sequences in a metric space: A sequence in a metric space (X, d) is a map $f : \mathbf{N} \to X$. We typically denote the image $f(n)$ by x_n. Similarly we define a **subsequence** as the composition of a strictly increasing map $g : \mathbf{N} \to \mathbf{N}$ with f and $x_{n_k} = f(g(k))$.

It is not all results that we know from $\mathbf{R}$ that generalises to general metric spaces. For instance, in general there are no algebraic properties, no squeeze theorem, no monotone convergence theorem. On the other hand the statement of both the Cauchy convergence theorem and the Bolzano-Weirstrass theorems makes sense in a general metric space.

Definition: Convergent sequence in a metric space If (X, d) is a metric space and x_n is a sequence in X, then we say that x_n converges to x and write $x_n \to x$ or $x = \lim_{n\to\infty} x_n$ if for all $\epsilon > 0$, there exists an N such that if $n \geq N$, then

$$d(x, x_n) < \epsilon \, .$$

This is equivalent to that the sequence $d(x_n, x) \to 0$.

Definition: Cauchy sequence in a metric space If (X, d) is a metric space and x_n is a sequence in X, then we say that x_n is a Cauchy sequence if for all $\epsilon > 0$, there exists an N, such that if m, $n \geq N$, then

$$d(x_m, x_n) < \epsilon \, .$$

Theorem: In any metric space (X, d) a convergent sequence is also a Cauchy sequence.

Proof. So suppose that $x_n \in X$ is a sequence and $x_n \to x$. Given $\epsilon > 0$, convergence means that there exists N such that if $n \geq N$, then $d(x, x_n) < \frac{\epsilon}{2}$. If both m, $n \geq N$, then we have by the triangle inequality that

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \, .$$

This show the theorem. □

The converse is not always the case: If $X = (0, 1) \subset \mathbf{R}$ with $d(x, y) = |x - y|$, then the sequence $x_n = \frac{1}{n}$ is a Cauchy sequence but since 0 is not in X, it is not convergent. We sometimes express this by saying that in this case X is not Cauchy complete.

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A metric space is said to be Cauchy complete if every Cauchy sequence is convergent.

Definition: (metric) ball. If (X, d) is a metric space, $x \in X$ and $r > 0$, then

$$B_r(x) = \{y \in X \,|\, d(x, y) < r\}$$

is said to be the ball with center x and radius r.

Definition: Bounded subset. If (X, d) is a metric space and $A \subset X$, then we say that A is bounded if A is contained in some metric ball $B_r(x)$.

Theorem: In a metric space (X, d) any Cauchy sequence is bounded.

Proof. Suppose that x_n is a Cauchy sequence. By definition of a Cauchy sequence, there exists some N such that if m, $n \geq N$, then

$$d(x_n, x_m) < 1\,.$$

Set

$$r = 1 + \max\{d(x_N, x_i) \,|\, i < N\}\,.$$

We claim that

$$\{x_n\} \subset B_r(x_N)\,.$$

Since $r \geq 1$ and $d(x_N, x_n) < 1$ for $n \geq N$ we only need to see that $x_n \in B_r(x_N)$ for $n < N$. This follows from that $d(x_N, x_n) < r$ when $n < N$ by definition of r. □

Bolzano - Weirstrass theorem: Any bounded sequence of real numbers have a convergent subsequence. This theorem does not hold for a general metric space but it holds if the metric space is compact. To discuss this we need the notion of what an open subset of a metric space is.

Definition (Open subset): Let (X, d) be a metric space. We say that O is an open subset of X if for all $x \in O$, there exists an $r > 0$ such that $B_r(x) \subset O$.

Note that $\emptyset$ (the empty set) and X are both open.

On subsets of a set X we have the following operations.

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- Union of two or more subsets.
  If U_1 and U_2 are subsets, then $U_1 \cup U_2$ is the union. So
  $$U_1 \cup U_2 = \{x \in X \,|\, x \in U_1 \text{ or } x \in U_2 \text{ or both}\}.$$
  Similarly, for union of more than two subsets.
- Intersection of two or more subsets.
  If U_1 and U_2 are subsets, then $U_1 \cap U_2$ is the intersection. So
  $$U_1 \cap U_2 = \{x \in X \,|\, x \in U_1 \text{ and } x \in U_2\}.$$
  Similarly, for intersection of more than two subsets.
- Complement of a subset U.
  $X \setminus U$ is all the elements of X that are not in U.

Example: $X = \mathbf{R}$, $A = (0,3)$, $B = (-1,2)$ and $C = (0,2)$.
$A \cup B = (0,3)$.
$A \cap B = (0,2)$.
$X \setminus A = (-\infty, 0] \cup [3, \infty)$.
$C \subset B$.

Union and intersections of families of subsets

- Union of families.
  If U_α is a family of subsets, then $\cup_\alpha U_\alpha$ is the union of all the subsets. So
  $$\cup_\alpha U_\alpha = \{x \in X \,|\, x \in U_\alpha \text{ for some } \alpha\}.$$
- Intersection of families.
  If U_α is a family of subsets, then $\cap_\alpha U_\alpha$ is the intersection of all the subsets. So
  $$\cap_\alpha U_\alpha = \{x \in X \,|\, x \in U_\alpha \text{ for all } \alpha\}.$$

Example: $X = \mathbf{R}$, $U_n = (-\frac{1}{n}, \frac{1}{n})$, where $n \in \mathbf{N}$, then
$$\cup_n U_n = (-1,1) \text{ and } \cap_n U_n = \{0\}.$$

Lemma: For a set X and subsets A, B we have $A = B$ if and only if $A \subset B$ and $B \subset A$.

Lemma: For a set and subset A, B and A_α we have
(1) $X \setminus (X \setminus A) = A$.
(2) $X \setminus \cup_\alpha A_\alpha = \cap_\alpha (X \setminus A_\alpha)$.
(3) $X \setminus \cap_\alpha A_\alpha = \cup_\alpha (X \setminus A_\alpha)$.

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Proof. To prove the first of these claim that $X \setminus (X \setminus A) = A$ we need to show two directions. Suppose $x \in A$, then $x \notin X \setminus A$ and therefore $x \in X \setminus (X \setminus A)$. Conversely, if $x \in X \setminus (X \setminus A)$, then $x \notin X \setminus A$ and therefore $x \in A$.

To prove the second claim observe that if $x \in X \setminus \cup_\alpha A_\alpha$, then $x \notin \cup_\alpha A_\alpha$ so x is not in any of the A_α's. Therefore x must be in all the $X \setminus A_\alpha$ and hence in the intersection of those so $x \in \cap_\alpha (X \setminus A_\alpha)$. This show that $X \setminus \cap_\alpha A_\alpha \subset \cap_\alpha (X \setminus A_\alpha)$. To show the other direction suppose that $x \in \cap_\alpha (X \setminus A_\alpha)$. This means that for all α we have that $x \notin A_\alpha$. Therefore, $x \notin \cup_\alpha A_\alpha$ and hence $x \in X \setminus (\cup_\alpha A_\alpha)$. This show the other direction.

Finally, to prove the third claim observe that if $x \in X \setminus \cap_\alpha A_\alpha$, then $x \notin \cap_\alpha A_\alpha$ and so there exists some α so that $x \in X \setminus A_\alpha$. In other words, $x \in \cup_\alpha (X \setminus A_\alpha)$. This show one direction. To see the other direction observe that if $x \in \cup_\alpha (X \setminus A_\alpha)$, then there exists some α so that $x \in X \setminus A_\alpha$. It follows that $x \notin A_\alpha$ and hence $x \notin \cap_\alpha A_\alpha$ but instead $x \in X \setminus \cap_\alpha A_\alpha$. This show the other direction and completes the proof of the lemma. □

Lemma: In a metric space any ball $B_r(x)$ is an open subset.

Proof. Suppose that $y \in B_r(x)$, and let $s = r - d(x, y)$. Note that since $y \in B_r(x)$ we have that $d(x, y) < r$ and so $s > 0$. We will show that $B_s(y) \subset B_r(x)$. To see that assume that $z \in B_s(y)$ we then have that $d(y, z) < s$ and so by the triangle inequality

$$d(z, x) \leq d(z, y) + d(y, x) < s + d(y, x) = (r - d(x, y)) + d(y, x) = r\,.$$

This shows the claim. □

Lemma: In a metric space if O_α are open subsets, then

$$\cup_\alpha O_\alpha$$

is open.

Proof. See Pset. □

Lemma: In a metric space if $O_1, \cdots, O_n$ are finitely many open subsets, then

$$O_1 \cap \cdots \cap O_n$$

is open.

Proof. Suppose that $x \in O_1 \cap \cdots \cap O_n$, then x lies in each O_i. For each i, there exists an $r_i > 0$, such that $B_{r_i}(x) \subset O_i$. Let $r = \min r_i$, then for each i we have that $B_r(x) \subset B_{r_i}(x) \subset O_i$ so $B_r(x)$ is a subset of each O_i and hence $B_r(x) \subset O_1 \cap \cdots \cap O_n$. This shows the claim. □

Warning: Intersection of infinitely many open subsets may not be open!!!!

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Example: $X = \mathbf{R}$ and for each natural number let O_n be the open set $O_n = (-\frac{1}{n}, \frac{1}{n})$, then

$$\cap_n O_n = \{0\}.$$

So the intersection of these infinitely many open subsets is not open.

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Lecture 13

Definition (Closed subsets): Let (X, d) be a metric space. We say that C is a closed subset of X if the complement $X \setminus C$ is open.

Note that $\emptyset$ (the empty set) and X are both closed.

Examples:

- $(0, 1)$ is not a closed subset of $\mathbf{R}$.
- $\{0\}$ is a closed subset of $\mathbf{R}$.
- $[0, 1]$ is a closed subset of $\mathbf{R}$.
- $[0, 1] \times [0, 1]$ is a closed subset of $\mathbf{R}^2$.

Lemma: Let (X, d) be a metric space and $r > 0$, then

$$A_r = \{y \,|\, d(x, y) > r\}$$

is open. Equivalently, $\bar{B}_r(x) = \{y \,|\, d(x, y) \leq r\}$ is closed.

Proof. Suppose that $y \in A_r$, then $d(y, x) > r$ and if we set $s = d(y, x) - r$, then $s > 0$. Moreover, if $z \in B_s(y)$, then by the triangle inequality

$$d(x, y) \leq d(y, z) + d(z, x)\,.$$

So

$$r < r + s - d(y, z) \leq d(x, y) - d(y, z) \leq d(z, x)\,.$$

This show that $B_s(z) \subset A_r$ and so A_r is open. □

There is an equivalent way of defining closed subsets and that comes from the next theorem.

Theorem: A subset C of a metric space (X, d) is closed if and only if for all convergent sequences x_n with all x_n in C also the limit is in C.

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Proof. Suppose first that A is closed and let x_n be a convergent sequence is A with limit x we need to show that $x \in A$. Since A is closed the complement is open and if $x \in X \setminus A$, then there exists some $r > 0$ so $B_r(x) \subset X \setminus A$ and therefore for all $y \in A$ we would have that $d(y, x) \geq r$. This contradict that $x_n \to x$ and $x_n \in A$.

We also need to show the converse. So suppose that A is a subset with the property that for all sequences in A that are convergent in X the limit is in A. We will show that A is closed or equivalent that the complement is open. If the complement is not open, then there exists an $x \in X \setminus A$ such that no ball around x is entirely contained in the complement. Therefore for each n there exists an $x_n \in A$. This sequence converges to x which was assumed not to be in A contradicting that A contained all limits of sequences in A and therefore the complement must be open and A itself closed. □

For union and intersection of closed subsets we have the following:

Theorem:

- Union: If C_α is a family of closed subsets, then $\cap_\alpha C_\alpha$ is also closed.
- Intersection: If $C_1, \cdots, C_n$ are closed subsets, then $C_1 \cup \cdots \cup C_n$ is also closed.

Proof. There are several ways of proving this. The easiest is probably straight from the definition using the operations on sets. For the first claim we need to show that the complement of $\cap_\alpha C_\alpha$ is open. Using the operations of sets we have that

$$X \setminus \cap_\alpha C_\alpha = \cup_\alpha (X \setminus C_\alpha) .$$

Since each $X \setminus C_\alpha$ are open this is the union of open sets and therefore open. This shows the first claim.

To see the second claim we argue similarly. We want to show that $C_1 \cup \cdots \cup C_n$ is closed or, equivalently, $X \setminus (C_1 \cup \cdots \cup C_n)$ is open. However,

$$X \setminus (C_1 \cup \cdots \cup C_n) = (X \setminus C_1) \cap \cdots \cap (X \setminus C_n) ,$$

where the last is the intersection of finitely many open sets and therefore open. This show the second claim. □

Warning: Union of infinitely many closed sets may not be closed!!!

Definition (Cover, open cover and finite sub-cover): If A is a subset of X, then a cover of A is a collection collection of subsets U_α of X so that

$$A \subset \cup_\alpha U_\alpha .$$

We say that a $U_{\alpha_1}, \cdots, U_{\alpha_n}$ is a finite sub-cover if also $\{U_{\alpha_i}\}_i$ is a cover.

If (X, d) is a metric space and all the U_α are open, then we say that $\{U_\alpha\}_\alpha$ is an open cover.

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Example: If $X = \mathbf{R} \times \mathbf{R}$, then $A_n = (-n, n) \times (-n, n)$ is an open cover of X.

Example: Note that in the example where $X = \mathbf{R} \times \mathbf{R}$ and $A_n = (-n, n) \times (-n, n)$ is an open cover, then there is no finite sub-cover. On the other hand if $A \subset \mathbf{R} \times \mathbf{R}$ is bounded, then for n sufficiently large $A \subset A_n$ so for A, there is a finite sub-cover of this cover.

Definition (Compact subset): If (X, d) is a metric space and A is a subset, then we say that A is compact if each open cover has a finite sub-cover.

Example: If (X, d) is $\mathbf{R}$ with the usual metric and $A = (0, 1)$, then $A_n = (\frac{1}{n}, 1)$ is an open cover of A but there is no finite sub-cover of $\{A_n\}_n$ that covers A.

Theorem: $[a, b] \subset \mathbf{R}$ is compact.

Proof. We will show this next time. □

Theorem: If (X, d) is a metric space and A a compact subset, then A is closed and bounded.

Proof. Suppose first that A is not closed. We will show that this leads to a contradiction. If it is not closed, then there exists a convergent sequence $x_n \in A$ with limit x not in A. Set

$$O_n = \left\{ y \,|\, d(x, y) > \frac{1}{n} \right\}$$

. By the earlier lemma these are open sets. Since $\cup_n A_n = X \setminus \{x\}$ and x is assumed not to be in A we indeed have that A_n is an open cover of A. Since $A_n \subset A_{n+1}$ any finite cover of A_n's would be contained in A_N for some large N but this would imply that for all $y \in A$ we would have that $d(x, y) > \frac{1}{N}$ contradicting that $x_n \in A$ and $x_n \to x$. This show that the limit x is in A.

Since A is compact,

$$X = \cup_y B_r(y)$$

and each $B_1(x)$ is open, then finitely many of these covers A. Say $A \subset B_1(y_1) \cup \cdots \cup B_1(y_n)$. Set $r = 1 + \max_i\{d(y_1, y_i)\}$. It follows by the triangle inequality that $A \subset B_r(y_1)$. Hence, A is bounded. □

Warning: The converse is not the case!!! There are closed a bounded subsets of metric spaces that are not compact.

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If $(X, d) = (0, 1)$ with the usual metric, then X is closed and bounded but it is not compact.

Here is a more illuminating example:

Example: Let $X = C([0, 1])$ be the set of continuous functions on the unit interval $[0, 1]$. We equip X with the metric where

$$d(f, g) = \max_x |f(x) - g(x)| .$$

Let $f_n(x)$ be the sequence of continuous functions on $[0, 1]$ given by that

$$f_n(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq \frac{1}{n+1} \\ 1 - n\,(n+1)\,\left(x - \frac{1}{n+1}\right) & \text{if } \frac{1}{n+1} \leq x \leq \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

We have the f_n is a bounded sequence. After all they all lies in the metric ball $B_2(0)$ where 0 is the zero function. That is, the function on $[0, 1]$ that is identically equal to zero. However, the sequence f_n does not have a convergent subsequence (and does not even have a subsequence that is a Cauchy sequence). Indeed, for any $m \neq m$ we have that

$$d(f_m, f_n) = 1 .$$

Note also that the (closed) ball $A = \bar{B}_1(0)$ is closed and bounded but not compact. It is not compact because for the balls $\cup_f B_{\frac{1}{2}}(f)$ finitely many does not cover A. If finitely many did cover A, then for one such ball say $B_{\frac{1}{2}}(f)$ infinitely many f_n's would lie in it but any two elements in such a ball would have distance < 1 showing that there could at most be one f_n in such a ball.

Theorem: If (X, d) is a metric space and A a compact subset, then any closed subset C contained in A is also compact.

Proof. Let O_α be a open cover of C. Since C is closed $X \setminus C$ is open and so $\{O_\alpha\}$ together with $X \setminus C$ is an open cover of A and hence finitely many of those say $O_1, \cdots, O_n, X \setminus C$ covers A. Since $X \setminus C$ contains no elements in C it follows that $C \subset O_1 \cup \cdots \cup O_n$ and thus C is compact. □

Bolzano-Weirstrass theorem for metric spaces.

Theorem: If (X, d) is a metric space and A a compact subset, then any sequence in A has a convergent subsequence.

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Proof. We will show this next time. □

References

[TBB] B.S. Thomson, J.B. Bruckner, and A.M. Bruckner, *Elementary Real Analysis, 2nd edition*
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Lecture 14

Definition (Compact subset): If (X, d) is a metric space and A is a subset, then we say that A is compact if each open cover has a finite sub-cover.

Theorem 0: If (X, d) is a metric space and A a compact subset, then A is closed and bounded.

Warning: The converse is not the case!!! There are closed a bounded subsets of metric spaces that are not compact.

Example: If $(X, d) = (0, 1)$ with the usual metric, then X is closed and bounded but it is not compact.

Here is a more illuminating example:

Example: Let $X = C([0, 1])$ be the set of continuous functions on the unit interval $[0, 1]$. We equip X with the metric where

$$d(f, g) = \max_x |f(x) - g(x)| .$$

Let $f_n(x)$ be the sequence of continuous functions on $[0, 1]$ given by that

$$f_n(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq \frac{1}{n+1} \\ 1 - n\,(n+1)\,\left(x - \frac{1}{n+1}\right) & \text{if } \frac{1}{n+1} \leq x \leq \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

We have the f_n is a bounded sequence. After all they all lies in the metric ball $B_2(0)$ where 0 is the zero function. That is, the function on $[0, 1]$ that is identically equal to zero. However, the sequence f_n does not have a convergent subsequence (and does not even have

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a subsequence that is a Cauchy sequence). Indeed, for any $m \neq m$ we have that

$$d(f_m, f_n) = 1\,.$$

Note also that the (closed) ball $A = \bar{B}_1(0)$ is closed and bounded but not compact. It is not compact because for the balls $\cup_f B_{\frac{1}{2}}(f)$ finitely many does not cover A. If finitely many did cover A, then for one such ball say $B_{\frac{1}{2}}(f)$ infinitely many f_n's would lie in it but any two elements in such a ball would have distance < 1 showing that there could at most be one f_n in such a ball.

Using what we have shown in earlier lectures one can show the following:

Theorem 1: In $\mathbf{R}^n$, a subset is compact if and only if it is closed and bounded.

In a general metric space this is not the case as the above examples shows.

We won't show this theorem here but instead we will show a version of the Bolzano-Weirstrass theorem for metric spaces. This is the next theorem.

Theorem 2: If (X, d) is a metric space and A a compact subset, then any sequence in A has a convergent subsequence.

Before proving Theorem 2 we will need some results:

Lemma: Let (X, d) be a compact metric space if C_α is a family of closed (decreasing) nested subsets. That is, closed subsets so that $C_{n+1} \subset C_n$. If all C_n are non-empty, then

$$\cap_n C_n \neq \emptyset\,.$$

Proof. Set $O_\alpha = X \setminus C_\alpha$, then each O_α is open. If $\cap_n C_n \neq \emptyset$, then

$$\cup_n O_\alpha = X\,.$$

Therefore, finitely many of the O_n's cover X by compactness. Denote these by O_i for $i = 1, \cdots, k$. Since

$$O_1 \cup \cdots \cup O_k = X$$

it follows that

$$C_1 \cap \cdots \cap C_k = \emptyset\,.$$

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However, by the nested property one of these k closed subsets is the smallest, say C_k and therefore $C_1 \cap \cdots \cap C_k = C_k$. Contradicting that the intersection is empty. $\square$

Before stating the next results recall that in a metric space (X, d) the set $\bar{B}_r(x) = \{y \in X \,|\, d(x, y) \leq r\}$ is closed and is referred to as the closed ball. The above lemma gives the following useful corollary:

Corollary: Let (X, d) be a compact metric space and suppose that $B_{r_n}(x_n)$ is a family of balls with centres x_n and radii $r_n > 0$, where $r_n \to 0$ and $\bar{B}_{r_{n+1}}(x_{n+1}) \subset \bar{B}_{r_n}(x_n)$. Then

$$\cap_n \bar{B}_{r_n}(x_n) = \{x\}\,.$$

That is, the intersection is non-empty and consists of a single point.

Proof. Set

$$A = \cap_n \bar{B}_{r_n}(x_n)\,.$$

Observe first that for each n we have that $x_n \in \bar{B}_{r_n}(x_n)$ so from the lemma above we have that A is non-empty. We claim that A consists of just one element. Suppose that $x,\, y \in A$, for any integer n we have that

$$x,\, y \in \bar{B}_{r_n}(x_n)\,,$$

and so by the triangle inequality

$$d(x, y) \leq d(x, x_n) + d(x_n, y) \leq r_n + r_n = 2\, r_n\,.$$

Since this holds for all n we see that $d(x, y) = 0$ and so there is at most one such point. $\square$

Proof. (of Theorem 2.) Suppose that x_n is a sequence in a compact subset A of a metric space. Fix $r > 0$ and write

$$A \subset_{x \in A} B_r(x)\,.$$

Since A is compact finitely many of these cover A. This means that in one of these balls, say $B_r(y_1)$, there are infinitely many x_n's. From here on and out we will focus on this ball. Since $A \cap \bar{B}_r(y)$ is a closed subset of a compact set we can now cover $\bar{B}_r(y)$ by balls of radius $\frac{r}{4}$. By compactness finitely many of these sub-balls cover the ball $\bar{B}_r(y)$. In one of those sub-balls there are also infinitely many x_n's. Fix such a sub-ball and call it $\bar{B}_{\frac{r}{4}}(y_2)$. We have that

$$\bar{B}_{\frac{r}{2}}(y_2) \subset \bar{B}_{2\,r}(y_1)$$

and that infinitely many x_n's belongs to $B_{\frac{r}{4}}(y_2)$. If the original $r = 1$ gives after repeating this process i times balls $B_{4^{1-i}}(y_i)$ so that

$$\cdots \subset \bar{B}_{2\,4^{1-i}}(y_i) \subset \cdots \bar{B}_{2\,4^{-1}}(y_2) \subset \bar{B}_2(y_1)\,.$$

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where each of these balls contains infinitely many elements from the original sequence. Since the radii of this sequence converges to zero this sequence satisfies the assumptions of the corollary we have from the corollary that

$$\cap B_{2\,4^{1-i}}(y_i) = \{x\}\,.$$

Moreover, we can pick a subsequence x_{n_k} of the original sequence such that

$$x_{n_k} \subset \bar{B}_{2\,4^{1-k}}(y_k)\,.$$

It follows that this subsequence converges to x. □

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TOBIAS HOLCK COLDING

Lecture 15

Definition: If $f : \mathbf{R} \to \mathbf{R}$ is a function, then we say that f is differentiable at x_0 if the limit

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. (Note that in this fraction x is assumed to be $\neq x_0$.) When the limit exists, then we say that the function f is differentiable at x_0 and that its derivative at x_0 is the limit. In this case we denote the derivative at x_0 by $f'(x_0)$.

Examples:

(1) Constant functions. Suppose that $f(x) = c$ for some constant $c \in \mathbf{R}$, then

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{c - c}{x - x_0} = 0\,.$$

It follows that the limit exists and is zero and so f is differentiable at all points and the derivative is zero.

(2) Linear functions. Suppose that $f(x) = x$, then

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x - x_0}{x - x_0} = 1\,.$$

It follows that the limit exists and is one and so f is differentiable at all points and the derivative is one.

These are just two examples where we computed the derivative directly from the definition. How do we compute the derivative of a general function?

For that there are some tools:

- Sum rule.
- Product rule.
- Quotient rule.
- Chain rule.

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Once we know how to compute the derivative of a function, then we would like to understand the function using information about its derivative. For that we have the following tools:

- Mean value theorem.
- L'Hopital's rule.
- Taylor expansion.

Before getting to how to use the derivative we need to be able to compute it. For that it will be useful to note the following:

Lemma: If f is differentiable at x_0, then f is continuous at x_0.

Proof. Since f is differentiable at x_0 we have that

$$\frac{f(x) - f(x_0)}{x - x_0} \to f'(x_0)\,.$$

Therefore, there exist $\delta_1 > 0$ such that if $|x - x_0| < \delta_1$, then

$$\frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \ < 1$$

or, equivalently,

$$|f(x) - f(x_0) - f'(x_0)\,(x - x_0)| < |x - x_0|\,.$$

Therefore, for $|x - x_0| < \delta_1$ we have

$$|f(x) - f(x_0)| < (|f'(x_0)| + 1)\,|x - x_0|\,.$$

Given $\epsilon > 0$, set

$$\delta = \min\left\{\delta_1, \frac{\epsilon}{|f'(x_0)| + 1}\right\}\,.$$

It follows that if $|x - x_0| < \delta$, then

$$|f(x) - f(x_0)| < \epsilon\,.$$

This show that f is continuous at x_0. □

Example: On the real line suppose that f is the function given by that $f(0) = 0$ and for all other x

$$f(x) = x\,\sin\frac{1}{x}\,.$$

This is an example of a function that is continuous at zero but not differentiable at zero. It is not differentiable at zero because it fluctuate too much near zero. To see that it is continuous at zero we will use that $|\sin t| \leq 1$ for all t. Indeed using that it is easy to see

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that f is continuous at zero. Next we will see that it is not differentiable at zero. To see that we look at the difference quotient

$$\frac{f(x)-f(0)}{x-0}=\frac{x\,\sin\frac{1}{x}-0}{x-0}=\sin\frac{1}{x}\,.$$

As $x\to 0$ this function fluctuate between -1 and 1 so it does not have a limit and therefore the original function f is not differentiable at zero.

Example: If we dampen the fluctuation of the function given in the previous example further, then we get a differentiable function at zero even if it still fluctuate but just not as much. This is done in the following example. Suppose that f is the function that is given by that $f(0)=0$ and for all other x

$$f(x)=x^2\,\sin\frac{1}{x}\,.$$

Again we form the difference quotient

$$\frac{f(x)-f(0)}{x-0}=\frac{x^2\,\sin\frac{1}{x}-0}{x-0}=x\,\sin\frac{1}{x}\,.$$

In this case we see that as $x\to 0$, then $x\,\sin\frac{1}{x}\to 0$ and so the function is differentiable at zero and the derivative there is zero.

The following is very useful to compute the derivative of many functions:

Theorem: If f, g are functions on $\mathbf{R}$ that both are differentiable at x_0, then

- (Sum rule.)
$$(f+g)'(x_0)=f'(x_0)+g'(x_0)\,.$$
- (Leibniz's rule.)
$$(f\,g)(x_0)=f'(x_0)\,g(x_0)+f(x_0)\,g'(x_0)\,.$$
- (Quotient rule.) If also $g(x_0)\neq 0$, then
$$\left(\frac{f}{g}\right)'(x_0)=\frac{f'(x_0)\,g(x_0)-f(x_0)\,g'(x_0)}{g^2(x_0)}\,.$$

Proof. To prove the sum rule consider the difference quotient

$$\frac{(f+g)(x)-(f+g)(x_0)}{x-x_0}=\frac{f(x)-f(x_0)}{x-x_0}+\frac{g(x)-g(x_0)}{x-x_0}\to f'(x_0)+g'(x_0)$$

This show the sum rule.

To prove the Leibniz rule we form the difference quotient

$$\frac{(f\,g)(x)-(f\,g)(x_0)}{x-x_0}\,.$$

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We rewrite this using a trick we have used before in other settings. Namely, we can write this as

$$\frac{(f\,g)(x)-(f\,g)(x_0)}{x-x_0}=\frac{f(x)\,g(x)-f(x)\,g(x_0)+f(x)\,g(x_0)-f(x_0)\,g(x_0)}{x-x_0}$$

$$=f(x)\,\frac{g(x)-g(x_0)}{x-x_0}+\frac{f(x)-f(x_0)}{x-x_0}\,g(x_0)\to f(x_0)\,g'(x_0)+f'(x_0)\,g(x_0)\,.$$

(Here we used that by the continuity lemma above $f(x)\to f(x_0)$.) This proves Leibniz's rule.

Finally, to prove the quotient rule we observe first that since g is differentiable at x_0 it is continuous at x_0 and therefore (since $g(x_0)\neq 0$) when x is close to x_0 we have that $g(x)\neq 0$. Moreover, we have that

$$\frac{\frac{f(x)}{g(x)}-\frac{f(x_0)}{g(x_0)}}{x-x_0}=\frac{f(x)\,g(x_0)-f(x_0)\,g(x)}{(x-x_0)\,g(x)\,g(x_0)}$$

$$=\frac{f(x)\,g(x_0)-f(x_0)\,g(x_0)}{(x-x_0)\,g(x)\,g(x_0)}+\frac{f(x_0)\,g(x_0)-f(x_0)\,g(x)}{(x-x_0)\,g(x)\,g(x_0)}$$

$$=\frac{1}{g(x)}\,\frac{f(x)-f(x_0)}{x-x_0}+\frac{f(x_0)}{g(x)\,g(x_0)}\,\frac{g(x)-g(x)}{x-x_0}$$

$$\to\frac{f'(x_0)}{g(x_0)}+\frac{f(x_0)\,g'(x_0)}{g^2(x_0)}\,.$$

From this the claim easily follows. □

Leibniz's rule is named after Gottfried Wilhelm Leibniz (1646 - 1716). Leibniz [from Wikipedia] was a German polymath active as a mathematician, philosopher, scientist and diplomat who is credited, alongside Sir Isaac Newton, with the creation of calculus in addition to many other branches of mathematics, such as binary arithmetic and statistics. Leibniz has been called the "last universal genius" due to his vast expertise across fields, which became a rarity after his lifetime with the coming of the Industrial Revolution and the spread of specialised labor.

Theorem: (Chain rule.) If $f:[a,b]\to[c,d]$ and $g:[c,d]\to\mathbf{R}$ are functions, where f is differentiable at x_0 and g differentiable at $y_0=f(x_0)$, then the composition $g\circ f$ is differentiable at x_0 and the derivative at x_0 is

$$(g\circ f)'(x_0)=g'(y_0)\,f'(x_0)\,.$$

Proof. Set $y=f(x)$ and $y_0=f(x_0)$. Assume first that $f'(x_0)\neq 0$. In this case for $x\neq x_0$ but close to x_0 we have that $y\neq y_0$ and we can write the difference quotient as follows. We have that

$$\frac{g(f(x))-g(f(x_0))}{x-x_0}=\frac{g(y)-g(y_0)}{y-y_0}\,\frac{f(x)-f(x_0)}{x-x_0}\,.$$

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Since f is differentiable at x_0 as $x \to x_0$ we have that

$$\frac{f(x) - f(x_0)}{x - x_0} \to f'(x_0)\,.$$

Moreover, when $x \to x_0$ we have that $f(x) = y \to f(x_0) = y_0$ by the continuity lemma above. It follows that when $x \to x_0$ we have that

$$\frac{g(y) - g(y_0)}{y - y_0} \to g'(y_0)\,.$$

Combining this gives that

$$\frac{g(f(x)) - g(f(x_0))}{x - x_0} = \frac{g(y) - g(y_0)}{y - y_0}\,\frac{f(x) - f(x_0)}{x - x_0} \to g'(y_0)\,f'(x_0)\,.$$

This proves the chain rule when $f'(x_0) \neq 0$. When $f'(x_0) = 0$ we argue as above but have to be more careful as in this case we can have that $y = y_0$ even when $x \neq x_0$. For x where $y = y_0$ the difference quotient is zero and where $y \neq y_0$ we can argue as above and rewrite the difference quotient as the product of two factors. In either case we get that the limit is zero proving the remaining case of the chain rule. □

Lemma: Let $f : [a, b] \to \mathbf{R}$ be a differentiable function and suppose that $a < x_0 < b$ and that f has a local maximum or minimum at x_0, then

$$f'(x_0) = 0\,.$$

Proof. Suppose that x_0 is a local maximum. The proof when x_0 is a local minimum. It follows from the assumption that for all x near x_0

$$f(x) - f(x_0) \leq 0\,.$$

Therefore, when $x > x_0$ we have that

$$\frac{f(x) - f(x_0)}{x - x_0} \leq 0\,,$$

whereas when $x < x_0$ we have that for the difference quotient

$$\frac{f(x) - f(x_0)}{x - x_0} \geq 0\,.$$

Since the limit is the same whether x converges to x_0 from the the left (negative side) or from the right (positive side) it follows that $f'(x_0) = 0$ as claimed.

□

Theorem: (Rolle's theorem.) Let $f : [a, b] \to \mathbf{R}$ be a differentiable function with $f(a) = f(b)$, then there exists a x_0 between a and b such that

$$f'(x_0) = 0\,.$$

Proof. There are three cases to consider:

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(1) f is constant equal to $f(a)$.
(2) For some x between a and b we have that $f(x) > f(a)$.
(3) For some x between a and b we have that $f(x) < f(a)$.

In the first case the function is constant and the derivate is zero everywhere. The second and third cases are similar so we will just argue in the second case. In the second case by the extreme value theorem there exists some x_0 such that $f(x_0) = \max f > f(a)$. It now follows from the previous lemma that $f'(x_0) = 0$. □

Theorem: (Mean value theorem.) Let $f : [a, b] \to \mathbf{R}$ be a differentiable function, then there exists a x_0 between a and b such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}\,.$$

.

Proof. Consider the function g given by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}\,(x - a)\,.$$

Observe that for g we have $g(a) = g(b)$ and so Rolle's theorem applies and we have that there exists some x_0 where $g'(x_0) = 0$. Since

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}\,,$$

the claim follows. □

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TOBIAS HOLCK COLDING

Lecture 16

Last time we defined what it means for a function to be differentiable. This is the following:

Definition: If $f : \mathbf{R} \to \mathbf{R}$ is a function, then we say that f is differentiable at x_0 if the limit

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. (Note that in this fraction x is assumed to be $\neq x_0$.) When the limit exists, then we say that the function f is differentiable at x_0 and that its derivative at x_0 is the limit. In this case we denote the derivative at x_0 by $f'(x_0)$.

One of the first things we showed about differentiable function was that they are continuous:

Lemma: If f is differentiable at x_0, then f is continuous at x_0.

We also established some very useful rules for computing the derivative of functions that are constructed from other functions whose derivative we know:

Theorem: If f, g are functions on $\mathbf{R}$ that both are differentiable at x_0, then

- (Sum rule.)
$$(f + g)'(x_0) = f'(x_0) + g'(x_0)\,.$$
- (Leibniz's rule.)
$$(f\,g)(x_0) = f'(x_0)\,g(x_0) + f(x_0)\,g'(x_0)\,.$$
- (Quotient rule.) If also $g(x_0) \neq 0$, then
$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)\,g(x_0) - f(x_0)\,g'(x_0)}{g^2(x_0)}\,.$$

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Finally, for the composition of functions we have the chain rule:

Theorem: (Chain rule.) If $f : [a, b] \to [c, d]$ and $g : [c, d] \to \mathbf{R}$ are functions, where f is differentiable at x_0 and g differentiable at $y_0 = f(x_0)$, then the composition $g \circ f$ is differentiable at x_0 and the derivative at x_0 is

$$(g \circ f)'(x_0) = g'(y_0)\, f'(x_0)\,.$$

Now that we know how to compute the derivative of many functions we will be interested in using the derivative to describe the growth or decay of a function. The first step towards this is the next lemma.

Before stating it recall that a function $f : \mathbf{R} \to \mathbf{R}$ has a local maximum at x_0 if there exists a $\delta > 0$ such that

$$f(x_0) = \max_{[x_0-\delta, x_0+\delta]} f\,,$$

and similarly for a local minimum.

Lemma: Let $f : [a, b] \to \mathbf{R}$ be a differentiable function and suppose that $a < x_0 < b$ and that f has a local maximum or minimum at x_0, then

$$f'(x_0) = 0\,.$$

Proof. Suppose that x_0 is a local maximum. The proof when x_0 is a local minimum is similar. It follows from the assumption that for all x near x_0

$$f(x) - f(x_0) \leq 0\,.$$

Therefore, when $x > x_0$ we have that

$$\frac{f(x) - f(x_0)}{x - x_0} \leq 0\,,$$

whereas when $x < x_0$ we have that for the difference quotient

$$\frac{f(x) - f(x_0)}{x - x_0} \geq 0\,.$$

Since the limit is the same whether x converges to x_0 from the the left (negative side) or from the right (positive side) it follows that $f'(x_0) = 0$ as claimed. □

We can now use this lemma to establish the following very useful result:

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Theorem: (Rolle's theorem.) Let $f : [a, b] \to \mathbf{R}$ be a differentiable function with $f(a) = f(b)$, then there exists a x_0 between a and b such that

$$f'(x_0) = 0\,.$$

Proof. There are three cases to consider:

(1) f is constant equal to $f(a)$.
(2) For some x between a and b we have that $f(x) > f(a)$.
(3) For some x between a and b we have that $f(x) < f(a)$.

In the first case the function is constant and the derivate is zero everywhere. The second and third cases are similar so we will just argue in the second case. In the second case by the extreme value theorem there exists some x_0 such that $f(x_0) = \max f > f(a)$. It now follows from the previous lemma that $f'(x_0) = 0$. □

Rolle's theorem can then be used to show both the mean value theorem and the Cauchy mean value theorem:

Theorem: (Mean value theorem.) Let $f : [a, b] \to \mathbf{R}$ be a differentiable function, then there exists a x_0 between a and b such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}\,.$$

.

Proof. Consider the function g given by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}\,(x - a)\,.$$

Observe that for g we have $g(a) = g(b)$ and so Rolle's theorem applies and we have that there exists some x_0 where $g'(x_0) = 0$. Since

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}\,,$$

the claim follows. □

Theorem: (Cauchy mean value theorem.) Let f, $g : [a, b] \to \mathbf{R}$ be differentiable functions, then there exists a x_0 between a and b such that

$$f'(x_0)\,[g(b) - g(a)] = g'(x_0)\,[f(b) - f(a)]\,.$$

In particular, if $g(b) - g(a) \neq 0$, then

$$\frac{f'(x_0)}{g'(x_0)} = \frac{f(b) - f(a)}{g(b) - g(a)}\,.$$

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Proof. Consider the function

$$h(x) = f(x)\,[g(b) - g(a)] - g(x)\,[f(b) - f(a)]\,.$$

Note that

$$h(a) = f(a)\,[g(b) - g(a)] - g(a)\,[f(b) - f(a)] = f(a)\,g(b) - g(a)\,f(b)\,.$$

$$h(b) = f(b)\,[g(b) - g(a)] - g(b)\,[f(b) - f(a)] = f(a)\,g(b) - g(a)\,f(b)\,.$$

Therefore, by Rolle's theorem, there exists x_0 between a and b such that $h'(x_0) = 0$. Since

$$h'(x) = f'(x)\,[g(b) - g(a)] - g'(x)\,[f(b) - f(a)]$$

this shows the claim. □

We observe that the Cauchy mean value theorem implies the earlier mean value theorem. Namely, if we let the second function g be $g(x) = x$, then $g'(x) = 1$ and $g(b) - g(a) = b - a$. Therefore, the Cauchy mean value theorem becomes

$$f'(x_0)\,(b - a) = f'(x_0)\,(g(b) - g(a)) = g'(x_0)\,(f(b) - f(a)) = f(b) - f(a)\,,$$

which is the earlier mean value theorem.

The next two rules are useful to establishing the limit of a faction of function when the denominator either tend to zero or infinity.

Theorem: (L'Hopital's rule, version 1.) Let $f,\ g : (a, b) \to \mathbf{R}$ be differentiable functions with $g(x) \neq 0$ and $g'(x) \neq 0$ for all x, assume that

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0\,.$$

If

$$\lim_{x\to a} \frac{f'(x)}{g'(x)}$$

exists, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}\,.$$

Proof. We will see that this is an easy consequence of the Cauchy mean value theorem. By assumption given $\epsilon > 0$, there exists $\delta > 0$ such that if $a < x < \delta$, then

$$\frac{f'(x)}{g'(x)} - L\ < \epsilon\,.$$

By the Cauchy mean value theorem we have for any y with $a < y < x$ that there exist z with $y < z < x$ so that

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(z)}{g'(z)}\,.$$

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We therefore have that

$$\frac{f(x)-f(y)}{g(x)-g(y)} - L \ < \epsilon \, .$$

By letting $y \to 0$ we see that

$$\frac{f(x)}{g(x)} - L \ \leq \epsilon \, .$$

Since this holds for all ϵ we get that

$$\lim_{x \to a} \frac{f(x)}{g(x)} = L$$

as claimed. □

Theorem: (L'Hopital's rule, version 2.) Let f, $g : (a,b) \to \mathbf{R}$ be differentiable functions with $g(x) \neq 0$ and $g'(x) \neq 0$ for all x, assume that

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = \infty \, .$$

If

$$\lim_{x \to a} \frac{f'(x)}{g'(x)}$$

exists, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)} \, .$$

Proof. Given $\epsilon > 0$, since $\frac{f'(x)}{g'(x)} \to L$ as as $x \to a$ we have that there exists a $\delta > 0$ such that if $a < x < a + 2\,\delta$, then

$$\frac{f'(x)}{g'(x)} - L \ < \epsilon \, .$$

Set $x_1 = a + \delta$. For a given $x \in (a, x_1)$, there exists $x_0 \in (x, x_1)$ such that

$$\frac{f'(x_0)}{g'(x_0)} = \frac{f(x_1)-f(x)}{g(x_1)-g(x)} \, .$$

It follows that

$$\frac{f(x_1)-f(x)}{g(x_1)-g(x)} - L \ < \epsilon \, .$$

By dividing the nominator and denominator of the fraction in this expression by $g(x)$ we get

$$\frac{\frac{f(x)}{g(x)} - \frac{f(x_1)}{g(x)}}{1 - \frac{g(x_1)}{g(x)}} - L \ < \epsilon \, .$$

This implies that

$$\frac{f(x)}{g(x)} - \frac{f(x_1)}{g(x_1)} - L \left(1 - \frac{g(x_1)}{g(x)}\right) \ < \epsilon \left(1 - \frac{g(x_1)}{g(x)}\right) .$$

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Since this holds for all $x \in (a, a+\delta)$ and $g(x) \to \infty$ as $x \to a$ we have that for $x > a$ but sufficiently close to a that

$$\frac{f(x)}{g(x)} - L \ \leq \epsilon \, .$$

Since this holds for all ϵ we see that

$$\lim_{x \to a} \frac{f(x)}{g(x)} = L$$

This proves the claim. □

Finally, we have the following key fact that show that any differentiable function can be approximated by a polynomial and give a way of estimate the difference between the function and the approximating polynomial.

Theorem: (Taylor expansion.) Let $f : [a, b] \to \mathbf{R}$ be a function and k a positive integer. Assume that f, f', $f^{(2)}, \cdots, f^{(k-1)}$ exists on $[a, b]$ and are continuous and that $f^{(k)}$ is defined on (a, b), then there exists c between a and b such that

$$f(b) = f(a) + f'(a)\,(b-a) + \frac{f^{(2)}(a)}{2}\,(b-a)^2 + \cdots + \frac{f^{(k-1)}(a)}{(k-1)!}\,(b-a)^{k-1}$$

$$+ \frac{f^{(k)}(c)}{(k)!}\,(b-a)^k \, .$$

Proof. Define the Taylor polynomial by

$$P(x) = f(a) + f'(a)\,(x-a) + \frac{f^{(2)}(a)}{2}\,(x-a)^2 + \cdots + \frac{f^{(k-1)}(a)}{(k-1)!}\,(x-a)^{k-1}$$

and define a number M by that

$$f(b) = P(b) + \frac{M}{k!}\,(b-a)^k \, .$$

We want to show that there exists some c between a and b such that

$$M = f^{(k)}(c) \, .$$

To do that we set

$$R(x) = f(x) - P(x) - \frac{M}{k!}\,(x-a)^k \, .$$

We have that $R(a) = R(b) = 0$ and so by Rolle's theorem, there exists some c_1 between a and b with $R'(c_1) = 0$. Next observe that $R'(a) = R'(c_1) = 0$ and so again by Rolle's theorem, there exists c_2 between a and c_1 with $R^{(2)}(c_2) = 0$. Since $R^{(i)}(a) = 0$ for $i = 0, \cdots, k-1$ we can continue this process k times and find some $c = c_k$ such that $R^{(k)}(c) = 0$. However, $0 = R^{(k)}(c) = f^k(c) - M$ Therefore, $M = f^{(k)}(c)$ as claimed. □

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Lecture 17

Recall that last time we showed the Taylor expansion theorem:

Theorem: (Taylor expansion.) Let $f : [a, b] \to \mathbf{R}$ be a function and k a positive integer. Assume that f, f', $f^{(2)}, \cdots, f^{(k-1)}$ exists on $[a, b]$ and are continuous and that $f^{(k)}$ is defined on (a, b), then there exists c between a and b such that

$$f(b) = f(a) + f'(a)\,(b-a) + \frac{f^{(2)}(a)}{2}\,(b-a)^2 + \cdots + \frac{f^{(k-1)}(a)}{(k-1)!}\,(b-a)^{k-1}$$

$$+\frac{f^{(k)}(c)}{(k)!}\,(b-a)^k\,.$$

For and infinitely differentiable function f on $\mathbf{R}$ we define the $(k-1)$ Taylor polynomial at a by

$$P_{k-1}(x) = f(a) + f'(a)\,(x-a) + \frac{f^{(2)}(a)}{2}\,(x-a)^2 + \cdots + \frac{f^{(k-1)}(a)}{(k-1)!}\,(x-a)^{k-1}\,.$$

Question: One naturally wonders how well does this polynomial approximate f when x is near a?

Answer: This depend on the value of the remainder

$$R_k(x) = \frac{f^{(k)}(c)}{k!}\,(x-a)^k\,.$$

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Example 1: Suppose that $f(x) = \mathrm{e}^x$ so $f^{(k)}(x) = f(x)$ for all k. This means that the Taylor expansion near $a = 0$ becomes

$$P_{k-1}(x) = \sum_{i=0}^{k-1} \frac{x^i}{i!}\,.$$

By the Taylor expansion theorem we have that

$$f(x) = P_{k-1}(x) + \frac{f^{(k)}(c)}{k!}\, x^k\,.$$

Since $f^{(k)}(x) = f(x)$ for all k, it follows from the Taylor expansion theorem that we have

$$|f(x) - P_{k-1}(x)| \leq \frac{\mathrm{e}^{|x|}}{k!}\,.$$

We conclude that for k large the polynomial P_{k-1} gives a pretty good approximation to f. For instance, if $|x| \leq 1$, then we have that

$$|f(x) - P_{k-1}(x)| \leq \frac{\mathrm{e}}{k!}\,.$$

Example 2: On $\mathbf{R}$ define a function f by

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ \mathrm{e}^{-\frac{1}{x^2}} & \text{otherwise} \end{cases}$$

It is easy to see that f is infinitely differentiable and that $f^{(k)}(0) = 0$ for all k. It follows that for all k the Taylor polynomial at 0 is $P_{k-1} \equiv 0$. Thus in this case $f(x) = R_k(x)$.

Riemann integrals

Partition: Let $[a, b]$ be an interval. A partition $\mathcal{P}$ of the interval $[a, b]$ is a number of sub-divisions x_i such that

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b\,.$$

The partition is then the sub-intervals $[x_{i-1}, x_i]$. We will set $\Delta\, x_i = x_i - x_{i-1}$.

Upper and lower sums: Suppose now that $f : [a, b] \to \mathbf{R}$ is a bounded function and that $\mathcal{P} = \{x_i\}$ is a partition of the interval $[a, b]$. We define upper and lower sums as follows. Set

$$M_i = \sup_{[x_{i-1}, x_i]} f\,,$$

$$m_i = \inf_{[x_{i-1}, x_i]} f\,,$$

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and upper $U(f, \mathcal{P})$ and lower sums $L(f, \mathcal{P})$ by

$$U(f, \mathcal{P}) = \sum_{i=1}^{n} M_i \, \Delta \, x_i \, ,$$

$$L(f, \mathcal{P}) = \sum_{i=1}^{n} m_i \, \Delta \, x_i \, .$$

Example 3: Suppose that the function is $f(x) = x^2 + 1$ on the interval $[-2, 2]$ and that the partition is $\mathcal{P}$ is $\{-2, -1, 0, 1, 2\}$. We have

$$m_1 = 2 \text{ and } M_1 = 5 \, ,$$
$$m_2 = 1 \text{ and } M_2 = 2 \, ,$$
$$m_3 = 1 \text{ and } M_3 = 2 \, ,$$
$$m_4 = 2 \text{ and } M_4 = 5 \, .$$

For the lower and upper sums we have

$$L(f, \mathcal{P}) = 2 + 1 + 1 + 2 = 6 \, ,$$
$$U(f, \mathcal{P}) = 5 + 2 + 2 + 5 = 14 \, .$$

The following lemma is immediate (from that $M_i \geq m_i$):

Lemma 1: We always have that

$$U(f, \mathcal{P}) \geq L(f, \mathcal{P}) \, .$$

Sub-partition: Let $[a, b]$ be an interval and $\mathcal{P}_1$ and $\mathcal{P}_2$ two partitions of the interval $[a, b]$. We say that $\mathcal{P}_2$ is a sub-partition (or refinement) of $\mathcal{P}_1$ if all the dividing points in $\mathcal{P}_1$ are also in $\mathcal{P}_1$ (and then presumable some additional dividing points).

Example 4: Suppose that the interval is $[-2, 2]$ and the given partition $\mathcal{P}_1$ is

$$\{-2, -1, 0, 1, 2\} \, .$$

Then the partition

$$\mathcal{P}_2 = \left\{ -2, -1\,\frac{1}{2}, -1, 0, \frac{1}{2}, 1, 2 \right\}$$

is a refinement (or sub-division) of $\mathcal{P}_1$. Indeed, $\mathcal{P}_2$ has the same dividing points as $\mathcal{P}_1$ in addition to some more.

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We now have the following:

Lemma 2: Suppose now that $f : [a, b] \to \mathbf{R}$ is a bounded function and that $\mathcal{P}_1$ is a partition of the interval $[a, b]$ and $\mathcal{P}_2$ is a refinement of $\mathcal{P}_1$, then

$$L(f, \mathcal{P}_1) \leq L(f, \mathcal{P}_2) \leq U(f, \mathcal{P}_2) \leq U(f, \mathcal{P}_1)\,.$$

Proof. The middle inequality is the previous lemma. The inequality to the right follows from that if $\mathcal{P}_2$ is a subdivision of $\mathcal{P}_1$. Namely, suppose that $a = x_0 < x_1 < \cdots < x_n = b$ are the dividing points for $\mathcal{P}_1$ and that between say x_{i-1} and x_i there is an extra dividing point in $\mathcal{P}_2$ say y so $x_{i-1} < y < x_i$, then we have

$$\sup_{[x_{i-1}, y]} f \leq M_i$$

and

$$\sup_{[y, x_i]} f \leq M_i$$

so

$$[\sup_{[x_{i-1}, y]} f]\,(y - x_{i-1}) + [\sup_{[y, x_i]} f]\,(x_i - y) \leq M_i\,\Delta\, x_i\,.$$

From this it follows easily that

$$U(f, \mathcal{P}_2) \leq U(f, \mathcal{P}_1)\,.$$

Similarly, for the inequality to the left. □

Upper and lower integrals: Suppose now that $f : [a, b] \to \mathbf{R}$ is a bounded function. Define the upper Riemann integral of f by

$$\overline{\int_a^b} f\, dx = \inf_{\mathcal{P}} U(f, \mathcal{P})\,.$$

Here the infimum is taken over all partitions of $[a, b]$. Likewise, we define the lower Riemann integral by

$$\underline{\int_a^b} f\, dx = \sup_{\mathcal{P}} L(f, \mathcal{P})\,.$$

Riemann integral: Suppose that $f : [a, b] \to \mathbf{R}$ is a bounded function, then we say that f is Riemann integrable if

$$\overline{\int_a^b} f\, dx = \underline{\int_a^b} f\, dx\,.$$

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If the function is Riemann integrable, then the Riemann integral is

$$\int_a^b f\,dx = \overline{\int_a^b} f\,dx = \underline{\int_a^b} f\,dx\,.$$

The Riemann integrable functions is denoted by $\mathcal{R}([a,b])$.

From Wikipedia: Georg Friedrich Bernhard Riemann (1826 – 1866) was a German mathematician who made profound contributions to analysis, number theory, and differential geometry. Riemann held his first lectures in 1854, which founded the field of Riemannian geometry and thereby set the stage for Albert Einstein's general theory of relativity. In the field of real analysis, he is mostly known for the first rigorous formulation of the integral, the Riemann integral, and his work on Fourier series. His contributions to complex analysis include most notably the introduction of Riemann surfaces, breaking new ground in a natural, geometric treatment of complex analysis. His 1859 paper on the prime-counting function, containing the original statement of the Riemann hypothesis, is regarded as a foundational paper of analytic number theory. He is considered by many to be one of the greatest mathematicians of all time.

Example 5: Let $f : [0,1] \to \mathbf{R}$ be given by

$$f(x) = \begin{cases} 0 & \text{if } x \in [0,1] \cap \mathbf{Q} \\ 1 & \text{otherwise} \end{cases}$$

For this function and all partitions $\mathcal{P}$ we have that

$$L(f,\mathcal{P}) = 0 \text{ and } U(f,\mathcal{P}) = 1\,.$$

Thus, f is not Riemann integrable.

We will be interested in the questions: "What kind of functions are Riemann integrable?"

.... and "How do we compute the integral?"

The answer to the second question will be the fundamental theorem of calculus. This will be the topic of a later lecture.

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Lemma 3: Suppose now that $f : [a, b] \to \mathbf{R}$ is a bounded function, then $f \in \mathcal{R}([a, b])$ if and only if for all $\epsilon > 0$, there exists a partition $\mathcal{P}$ such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \epsilon \, .$$

Proof. Suppose that $f \in \mathcal{R}([a, b])$, then

$$\sup_{\mathcal{P}} L(f, \mathcal{P}) = \underline{\int_a^b} f \, dx = \overline{\int_a^b} f \, dx = \inf_{\mathcal{P}} U(f, \mathcal{P}) \, .$$

This means that given $\epsilon > 0$, there exists partitions $\mathcal{P}_1$ and $\mathcal{P}_2$ such that

$$\int_a^b f \, dx - \frac{\epsilon}{2} < L(f, \mathcal{P}_1)$$

and

$$U(f, \mathcal{P}_2) \leq \int_a^b f \, dx + \frac{\epsilon}{2} \, .$$

Let $\mathcal{P}$ be the partition that has all the dividing points of both $\mathcal{P}_1$ and $\mathcal{P}_2$. So $\mathcal{P}$ is a refinement of both $\mathcal{P}_1$ and $\mathcal{P}_2$. It follows that

$$\int_a^b f \, dx - \frac{\epsilon}{2} < L(f, \mathcal{P}_1) \leq L(f, \mathcal{P}) \leq U(f, \mathcal{P}) \leq U(f, \mathcal{P}_2) \leq \int_a^b f \, dx + \frac{\epsilon}{2} \, .$$

This proves the claim.

To see the converse, suppose that for some $\epsilon > 0$, there exists a partition $\mathcal{P}$ such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \epsilon \, .$$

Since

$$L(f, \mathcal{P}) \leq \underline{\int_a^b} f \, dx$$

and

$$\overline{\int_a^b} f \, dx \leq U(f, \mathcal{P})$$

we have that

$$\overline{\int_a^b} f \, dx - \underline{\int_a^b} f \, dx \leq U(f, \mathcal{P}) - L(f, \mathcal{P}) < \epsilon \, .$$

Since this holds for all $\epsilon > 0$ we get the claim. □

We now get to a key theorem that gives a simple criterium for a function to be Riemann integrable:

Theorem: Any continuous function on $[a, b]$ is in $\mathcal{R}([a, b])$.

Proof. We will show this next time once we have shown that a continuous function on a closed and bounded interval is, in fact, uniformly continuous. □

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The proof of this theorem needs the following key concept.

Definition: Uniformly continuous. Suppose that $f : I \to \mathbf{R}$ is a function, where I is an interval. We say that f is uniformly continuous if for all $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon \text{ if } |x - y| < \delta \,.$$

Note that being uniformly continuous is stronger than being continuous. It means that for a given $\epsilon > 0$, the same δ can be used for all x.

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Lecture 18

Definition: Uniformly continuous. Suppose that $f : I \to \mathbf{R}$ is a function, where I is an interval. We say that f is uniformly continuous if for all $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon \text{ if } |x - y| < \delta \, .$$

Note that being uniformly continuous is stronger than being continuous. It means that for a given $\epsilon > 0$, the same δ can be used for all x.

Example 1: Suppose that

$$f(x) = x^2$$

on $\mathbf{R}$, then f is NOT uniformly continuous. To see this, let $\epsilon > 0$ be given if f was uniformly continuous, then there would exists $\delta > 0$ such that

$$f(x + \delta) - f(x) < \epsilon \, ,$$

for all x. This would mean that

$$2 \, \delta \, x < (x + \delta)^2 - x^2 < \epsilon$$

for all x, which is clearly not the case.

Example 2: Suppose that

$$f(x) = \frac{1}{x}$$

on $(0, 1]$, then f is NOT uniformly continuous. To see this, consider $x_n = \frac{1}{n}$ and $y_n = \frac{1}{2\,n}$, then

$$|f(x_n) - f(y_n)| = n$$

and

$$|x_n - y_n| < \frac{1}{n} \, .$$

From this it easily follows that f is not uniformly continuous.

Theorem 1: Any continuous function on $[a, b]$ is uniformly continuous.

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Proof. Suppose not; then there exists $\epsilon > 0$ such that for all $n > 0$, there are x_n and y_n with

$$|x_n - y_n| < \frac{1}{n}$$

and so that

$$|f(x_n) - f(y_n)| \geq \epsilon \, .$$

Since the interval $[a, b]$ is compact we can choose a subsequence of x_n say x_{n_k} so that

$$x_{n_k} \to x \, .$$

Since

$$|x - y_{n_k}| \leq |x - x_{n_k}| + |x_{n_k} - y_{n_k}|$$

we have that $y_{n_k} \to x$ as well. Since f is continuous we have that $f(x_{n_k}) \to f(x)$ and $f(y_{n_k}) \to f(x)$. However, this contradict that

$$|f(x_{n_k} - f(y_{n_k})| \geq \epsilon \, ,$$

. □

We now get to a key theorem that gives a simple criterium for a function to be Riemann integrable:

Theorem 2: Any continuous function on $[a, b]$ is in $\mathcal{R}([a, b])$.

Proof. Given $\epsilon > 0$, since f is uniformly continuous by Theorem 1 it follows that there exists $\delta > 0$ such that if $|x - y| < \delta$, then

$$|f(x) - f(y)| < \frac{\epsilon}{b - a} \, .$$

Let $\mathcal{P}$ be a partition so that for all i we have $\Delta x_i < \delta$, then on each interval of the partition of the form $[x_{i-1}, x_i]$ we have that

$$M_i - m_i < \frac{\epsilon}{b - a} \, .$$

It follows that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) = \sum_i M_i \, \Delta x_i - \sum_i m_i \, \Delta x_i$$

$$= \sum_i [M_i - m_i] \, \Delta x_i < \frac{\epsilon}{b - a} \sum_i \Delta x_i = \epsilon \, .$$

Since this holds for all $\epsilon > 0$ we have that f is integrable. □

Basic properties of integrals.

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Theorem 3: We have the following basic formulas for integrals:

(1) If $f \in \mathcal{R}([a,b])$ and $c \in \mathbf{R}$, then $c\,f \in \mathcal{R}([a,b])$ and

$$\int_a^b (c\,f)\,dx = c\int_a^b f\,dx\,.$$

(2) If $f,\ g \in \mathcal{R}([a,b])$, then $f+g \in \mathcal{R}([a,b])$ and

$$\int_a^b (f+g)\,dx = \int_a^b f\,dx + \int_a^b g\,dx\,.$$

(3) If $f,\ g \in \mathcal{R}([a,b])$ and $f \le g$, then

$$\int_a^b f\,dx \le \int_a^b g\,dx\,.$$

(4) If $f \in \mathcal{R}([a,b])$ and $c \in (a,b)$, then $f \in \mathcal{R}([a,c])$ and $f \in \mathcal{R}([c,b])$ and

$$\int_a^c f\,dx + \int_c^b f\,dx = \int_a^b f\,dx\,.$$

Proof. The first claim follow from that if $\mathcal{P}$ is a partition, then

$$L(c\,f,\mathcal{P}) = c\,L(f,\mathcal{P})$$

and

$$U(c\,f,\mathcal{P}) = c\,U(f,\mathcal{P})\,.$$

To prove the second claim. Given $\epsilon > 0$, let $\mathcal{P}_1$ and $\mathcal{P}_2$ be partitions so that

$$U(f,\mathcal{P}_1) - L(f,\mathcal{P}_1) < \frac{\epsilon}{2}$$

and

$$U(g,\mathcal{P}_2) - L(g,\mathcal{P}_2) < \frac{\epsilon}{2}\,.$$

Let $\mathcal{P}$ be the partition that has the combined dividing points of $\mathcal{P}_1$ and $\mathcal{P}_2$. It follows that

$$U(f,\mathcal{P}) - L(f,\mathcal{P}) < \frac{\epsilon}{2}$$

and

$$U(g,\mathcal{P}) - L(g,\mathcal{P}) < \frac{\epsilon}{2}\,.$$

Therefore,

$$U(f+g,\mathcal{P}) - L(f+g,\mathcal{P}) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon\,.$$

From this the second claim follows.

To see the third claim let $\mathcal{P}$ be any partition of $[a,b]$. It follows that

$$U(g,\mathcal{P}) \le U(f,\mathcal{P})\,.$$

Since

$$\int_a^b g\,dx = \inf_{\mathcal{P}} U(g,\mathcal{P})$$

and likewise for f the claim now follows.

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Finally, to see the fourth claim. Let $\mathcal{P}$ be any partition of $[a,b]$ and let $\mathcal{P}_0$ be the refinement of $\mathcal{P}$ that in addition to the dividing points of $\mathcal{P}$ also have c as a dividing point. It follows that

$$U(f,\mathcal{P}_0\cap[a,c])+U(f,\mathcal{P}_0\cap[c,b])=U(f,\mathcal{P}_0)\,.$$

Likewise,

$$L(f,\mathcal{P}_0\cap[a,c])+L(f,\mathcal{P}_0\cap[c,b])=L(f,\mathcal{P}_0)\,.$$

Therefore,

$$\begin{aligned}U(f,\mathcal{P}_0\cap[a,c])-L(f,\mathcal{P}_0\cap[a,c])+U(f,\mathcal{P}_0\cap[c,b])-L(f,\mathcal{P}_0\cap[c,b])\\ =U(f,\mathcal{P}_0)-L(f,\mathcal{P}_0)\,.\end{aligned}$$

From this the fourth claim easily follows. □

Corollary: Suppose that f, $|f|\in\mathcal{R}([a,b])$, then

$$\int_x^b f\,dx\ \le\int_a^b|f|\,dx\,.$$

Proof. This follows from the lemma since $f\le|f|$ and $-f\le|f|$. Namely, from the first of these inequalities together with the lemma we get that

$$\int_a^b f\,dx\le\int_a^b|f|\,dx\,,$$

whereas from the second we get that

$$-\int_a^b f\,dx=\int_a^b(-f)\,dx\le\int_a^b|f|\,dx\,.$$

Together these gives the claim. □

Fundamental theorem of calculus, version 1: Let f be a continuous function on $[a,b]$ and define F on $[a,b]$ by

$$F(x)=\int_a^x f(s)\,ds\,.$$

The function F is differentiable with derivative f.

Proof. Fix $x_0\in[a,b]$ and assume first that $x>x_0$. We then have that

$$F(x)=\int_a^x f(s)\,ds=\int_a^{x_0}f(s)\,ds+\int_{x_0}^x f(s)\,ds=F(x_0)+\int_{x_0}^x f(s)\,ds\,.$$

It follows that

$$F(x)-F(x_0)=\int_{x_0}^x f(s)\,ds\,.$$

Therefore,

$$(x-x_0)\min_{[x_0,x]}f\le F(x)-F(x_0)\le(x-x_0)\max_{[x_0,x]}f$$

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and hence

$$\min_{[x_0,x]} f \leq \frac{F(x) - F(x_0)}{x - x_0} \leq \max_{[x_0,x]} f\,.$$

Since f is continuous at x_0 as $x \to x_0$ both the left and right hand side of this string of inequalities converges to $f(x_0)$. This proves the claim when $x > x_0$. When $x < x_0$ we can write $F(x)$ as

$$F(x) + \int_x^{x_0} f(s)\,ds = F(x_0)\,.$$

Therefore,

$$F(x) - F(x_0) = -\int_x^{x_0} f(s)\,ds\,.$$

Arguing as above gives the claim also in this case. □

Fundamental theorem of calculus, version 2: Suppose that $F : [a, b] \to \mathbf{R}$ is differentiable and that $F' = f \in \mathcal{R}([a, b])$, then

$$F(b) - F(a) = \int_a^b f(s)\,ds\,.$$

Proof. Since f is integrable, then for all $\epsilon > 0$, there exists a partition $\mathcal{P}$ of $[a, b]$ such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \epsilon\,.$$

For a given partition $\mathcal{P}$ with dividing points x_i we have

$$L(f, \mathcal{P}) = \sum_i m_i\,(x_i - x_{i-1})\,,$$

$$U(f, \mathcal{P}) = \sum_i M_i\,(x_i - x_{i-1})\,,$$

Moreover, by the mean value inequality

$$F(x_i) - F(x_{i-1}) = f(y_i)\,(x_i - x_{i-1})\,.$$

We now have that

$$m_i\,(x_i - x_{i-1}) \leq F(x_i) - F(x_{i-1}) \leq M_i\,(x_i - x_{i-1})\,.$$

It follows that

$$L(f, \mathcal{P}) \leq \sum_{i=1}^{n} [F(x_i) - F(x_{i-1})] \leq U(f, \mathcal{P})\,.$$

Finally, the claim follows from the above by observing that

$$F(b) - F(a) = \sum_{i=1}^{n} [F(x_i) - F(x_{i-1})]\,.$$

□

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Example 3: To compute

$$\int_0^1 x^2\,dx\,,$$

we use the second version of the fundamental theorem of calculus. Namely, observe that the derivative of the function

$$F(x) = \frac{x^3}{3}$$

is x^2 and therefore, by the second version of the fundamental theorem of calculus we have that

$$\int_0^1 x^2\,dx = F(1) - F(0) = \frac{1}{3} - 0 = \frac{1}{3}\,.$$

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TOBIAS HOLCK COLDING

Lecture 19

Question: What kind of functions are integrable?

Theorem: Any continuous function on $[a, b]$ is in $\mathcal{R}([a, b])$.

Basic properties of integrals.

Theorem: We have the following basic formulas for integrals:

(1) If $f \in \mathcal{R}([a, b])$ and $c \in \mathbf{R}$, then $c\, f \in \mathcal{R}([a, b])$ and

$$\int_a^b (c\, f)\, dx = c \int_a^b f\, dx\,.$$

(2) If $f,\, g \in \mathcal{R}([a, b])$, then $f + g \in \mathcal{R}([a, b])$ and

$$\int_a^b (f + g)\, dx = \int_a^b f\, dx + \int_a^b g\, dx\,.$$

(3) If $f,\, g \in \mathcal{R}([a, b])$ and $f \leq g$, then

$$\int_a^b f\, dx \leq \int_a^b g\, dx\,.$$

(4) If $f \in \mathcal{R}([a, b])$ and $c \in (a, b)$, then $f \in \mathcal{R}([a, c])$ and $f \in \mathcal{R}([c, b])$ and

$$\int_a^c f\, dx + \int_c^b f\, dx = \int_a^b f\, dx\,.$$

Corollary: Suppose that $f,\, |f| \in \mathcal{R}([a, b])$, then

$$\int_x^b f\, dx \ \leq \int_a^b |f|\, dx\,.$$

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Fundamental theorem of calculus, version 1: Let f be a continuous function on $[a,b]$ and define F on $[a,b]$ by

$$F(x) = \int_a^x f(s)\,ds\,.$$

The function F is differentiable with derivative f.

Fundamental theorem of calculus, version 2: Suppose that $F : [a,b] \to \mathbf{R}$ is differentiable and that $F' = f \in \mathcal{R}([a,b])$, then

$$F(b) - F(a) = \int_a^b f(s)\,ds\,.$$

Application of integrals: arclength.

Suppose that f and $g : [a,b] \to \mathbf{R}$ are differentiable functions and their derivatives are continuous, then we define the arclength of the curve

$$s \to (f(s), g(s))$$

by

$$L = \int_a^b \sqrt{(f'(s))^2 + (g'(s))^2}\,ds\,.$$

Example 1: Suppose that $f(s) = s$ and $g(s) = s^2$, then $f' = 1$ and $s' = 2\,s$. Therefore, the arclength of the curve (s, s^2), where $s \in [0,1]$ is

$$L = \int_0^1 \sqrt{1 + (2\,s)^2}\,ds = \int_0^1 \sqrt{1 + 4\,s^2}\,ds\,.$$

Improper integrals.

Unbounded interval.

Suppose that $f \in \mathcal{R}([a,b])$ for all $b > a$. If

$$\lim_{b\to\infty} \int_a^b f(x)\,dx$$

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exists, then we say that the improper integral

$$\int_a^\infty f(x)\,dx$$

exists and that

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$$

Example 2: On $[1,\infty)$, set

$$f(x) = \frac{1}{x^2}\,,$$

then

$$\int_1^c \frac{1}{x^2}\,dx = \left[-\frac{1}{x}\right]_c^1 = -\frac{1}{c} + 1\,.$$

Since $-\frac{1}{c} + 1 \to 1$ as $c \to \infty$, the improper integral

$$\int_1^\infty \frac{1}{x^2}\,dx$$

exist and is equation to 1.

Example 3: On $[1,\infty)$, set

$$f(x) = \frac{1}{x}\,,$$

then

$$\int_1^c \frac{1}{x}\,dx = [\log x]_1^c = \log c\,.$$

The improper integral

$$\int_1^\infty \frac{1}{x}\,dx$$

does not exist.

Unbounded function.

Suppose that $f \in \mathcal{R}([c,b])$ for all $c > a$. If

$$\lim_{c\to a}\int_c^b f(x)\,dx$$

exists, then we say that the improper integral

$$\int_a^b f(x)\,dx$$

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exists and that

$$\int_a^b f(x)\,dx = \lim_{c\to a}\int_c^b f(x)\,dx$$

Example 4: On $(0,1]$, set

$$f(x) = \frac{1}{\sqrt{x}}\,,$$

then

$$\int_c^1 \frac{1}{\sqrt{x}}\,dx = \left[2\sqrt{x}\right]_c^1 = 2 - 2\sqrt{c}\,.$$

Since $2-2\sqrt{c}\to 2$ as $c\to 0$, the improper integral exists and is equal to

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = 2\,.$$

Example 5: On $(0,1]$, set

$$f(x) = \frac{1}{x}\,,$$

then

$$\int_c^1 \frac{1}{x}\,dx = \left[\log x\right]_c^1 = -\log c\,.$$

Note that $-\log c\to\infty$ as $c\to 0$ so the improper integral does not exist.

Question: How do we define angle?

Answer: We define it through arclength.

On the unit circle

$$\{(x,y)\,|\,x^2+y^2=1\}$$

we define angle and the arclength. That is, suppose that (x,y) lies on the unit circle. The angle θ between $(1,0)$ and (x,y) is the arclength of the part of the unit circle from $(1,0)$ to (x,y). This part of the circle is parametrized by $(f(s),g(s)) = (s,\sqrt{1-s^2})$ and where $x\le s\le 1$. Since $f'(s)=1$ and $g'(s) = -\frac{s}{\sqrt{1-s^2}}$ we get that

$$\theta = \int_x^1 \sqrt{1+\frac{s^2}{1-s^2}}\,ds = \int_x^1 \frac{1}{\sqrt{1-s^2}}\,ds\,.$$

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The function $\arcsin x$ is defined by

$$\arcsin x = \int_0^x \frac{1}{\sqrt{1-s^2}}\, ds\,.$$

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TOBIAS HOLCK COLDING

Lecture 20

Application of integrals: arclength.

Suppose that f and $g : [a, b] \to \mathbf{R}$ are differentiable functions and their derivatives are continuous, then we define the arclength of the curve

$$s \to (f(s), g(s))$$

by

$$L = \int_a^b \sqrt{(f'(s))^2 + (g'(s))^2}\, ds\,.$$

Example 1: Suppose that $f(s) = s$ and $g(s) = s^2$, then $f' = 1$ and $s' = 2\,s$. Therefore, the arclength of the curve (s, s^2), where $s \in [0, 1]$ is

$$L = \int_0^1 \sqrt{1 + (2\,s)^2}\, ds = \int_0^1 \sqrt{1 + 4\,s^2}\, ds\,.$$

Question: How do we define angle?

Answer: We define it through arclength.

On the unit circle

$$\{(x, y) \,|\, x^2 + y^2 = 1\}$$

we define angle and the arclength. That is, suppose that (x, y) lies on the unit circle. The angle θ between $(1, 0)$ and (x, y) is the arclength of the part of the unit circle from $(1, 0)$ to (x, y). This part of the circle is parametrized by $(f(s), g(s)) = (s, \sqrt{1 - s^2})$ and where $x \leq s \leq 1$. Since $f'(s) = 1$ and $g'(s) = -\frac{s}{\sqrt{1-s^2}}$ we get that

$$\theta = \int_x^1 \sqrt{1 + \frac{s^2}{1 - s^2}}\, ds = \int_x^1 \frac{1}{\sqrt{1 - s^2}}\, ds\,.$$

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The function $\arccos x$ is defined by

$$\arccos x = \int_x^1 \frac{1}{\sqrt{1-s^2}}\, ds\,.$$

By the fundamental theorem of calculus we see that

$$\arccos x = -\frac{1}{\sqrt{1-x^2}}\,.$$

Pointwise convergence: Suppose that f_n is a sequence of functions on an interval I, then we say that f_n convergences pointwise to a function f if for all x we have

$$f_n(x) \to f(x)\,.$$

Example 1: Suppose that $f_n(x) = x^n$ on $[0,1]$, then f_n converges pointwise to f where

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1\,. \end{cases}$$

Suppose first that $0 \leq x < 1$, then $f_n(x) = x^n \to 0$. If $x = 1$, then $f_n(x) = 1$ for all n and so $f_n(x) \to 1$. This show the claim.

Example 2: If $E_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$, then $E_n(x) \to \exp x$ pointwise. We have already proven that the radius of convergence for the power series

$$\sum_{k=0}^{\infty} \frac{x^k}{k!}$$

is infinity. From this the claim follows.

Uniform convergence: Suppose that f_n is a sequence of functions on an interval I, then we say that f_n convergences uniformly to a function f if for all $\epsilon > 0$, there exists an N such that if $n \geq N$, then for all x

$$|f(x) - f_n(x)| < \epsilon\,.$$

Lemma 1: Suppose that I is an interval and f_n is a sequence of functions on I that converges uniformly to a function f, then f_n also converges pointwise to f.

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Proof. This is immediate from the definition of uniform convergence. □

Example 1A: Suppose again that $f_n(x) = x^n$ on $[0,1]$, then f_n converges pointwise but NOT uniformly to f where

$$f(x) = \begin{cases} 0 & \text{if } 0 \le x < 1 \\ 1 & \text{if } x = 1\,. \end{cases}$$

To see this observe that for each n,since f_n is continuous by the intermediate value theorem there exists x_n with $0 < x_n < 1$ such that $f_n(x) = \frac{1}{2}$. It now follows that

$$\frac{1}{2} = |f(x_n) - f_n(x_n)| \le \sup_{x\in[0,1]} |f(x) - f_n|\,.$$

Thus we see that the convergence is not uniform. We already saw in Example 1 that the convergence is pointwise.

Example 2A: If $E_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}$, then $E_n(x) \to \exp x$ uniformly on any interval of the form $[-L, L]$. This will be a consequence of of Weirstrass M-test that we will discuss next.

Lemma 2 [Weirstrass M-test]: Suppose that I is an interval and f_n is a sequence of functions on I. Suppose also that M_n is a sequence of non-negative numbers with

$$|f_n(x)| \le M_n \text{ for all } x \in I\,.$$

If the series

$$\sum_{n=1}^{\infty} M_n$$

converges, then the sequence of functions

$$S_n(x) = \sum_{k=0}^{n} f_k(x)$$

converges uniformly.

Proof. For each fixed x we have that that the sequence

$$\sum_{k=0}^{\infty} f_k(x)$$

converges. Moreover, we have that for all x and $m < n$ we have

$$|S_n(x) - S_m(x)| \le |f_n(x)| + |f_{n-1}(x)| + \cdots + |f_{m+1}(x)| \le M_n + \cdots + M_{m+1}\,.$$

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For m fixed and since $S_n(x) \to S(x)$ it follows that

$$|S(x) - S_m(x)| \leq \sum_{k=m+1}^{\infty} M_k \,.$$

Since $\sum_{k=0} M_k$ is convergent it implies that given $\epsilon > 0$, there exists N such that if $m \geq N$, then $\sum_{k=m+1}^{\infty} M_k < \epsilon$. Therefore, for $m \geq N$ and all x

$$|S(x) - S_m(x)| < \epsilon \,.$$

This proves the claim. □

Example 2A: On the interval $I = [-L, L]$ suppose

$$f_n = \frac{x^n}{n!} \,.$$

Then

$$|f_n| \leq \frac{L^n}{n!} \,.$$

Since

$$\sum_n \frac{L^n}{n!}$$

is convergent Weirstrass M-test gives that the series

$$\sum_{n=0}^{\infty} f_n$$

converges uniformly on I.

Theorem: If

$$\sum_{k=0}^{\infty} a_k \, x^k$$

is a power series and R is its radius of convergence. Then it converges uniformly on any (finite) interval of the form $[-L, L]$ where $L < R$.

Proof. Recall that if $M = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}$, then the radius of convergence is $R = \frac{1}{M}$. It follows that if $|x| \leq L < R$, then

$$\limsup |a_n \, x^n|^{\frac{1}{n}} = |x| \, \limsup |a_n|^{\frac{1}{n}} \leq L \, M < 1 \,.$$

Choose $1 > \alpha > L \, M$. For n sufficiently large $|a_n \, x^n| \leq M_n = \alpha^n$. Since the geometric series $\sum_n \alpha^n$ is convergent, Weirstrass M-test gives the claim. □

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Example 3: The geometric power series

$$\sum_{k=0}^{\infty} x^k$$

converges uniformly to $\frac{1}{1-x}$ on all intervals of the form $[-L, L]$ where $L < 1$. Since the radius of convergence of the power series is one the claim therefore follows from the theorem above.

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TOBIAS HOLCK COLDING

Lecture 21

Theorem 1: Suppose that I is an interval and f_n is a sequence of continuous functions on I. If f_n converges uniformly to f, then f is also continuous.

Proof. Let x_0 in I be arbitrary but fixed. We will show that f is continuous at x_0. Given $\epsilon > 0$, since $f_n \to f$ uniformly, there exists a N such that if $n \geq N$, then for all x in I

$$|f(x) - f_n(x)| < \frac{\epsilon}{3}\,.$$

Since f_N is continuous at x_0, there exists $\delta > 0$ such that if $|x - x_0| < \delta$, then

$$|f_N(x) - f_N(x_0)| < \frac{\epsilon}{3}\,.$$

Combining this gives that for $|x - x_0| < \delta$

$$|f(x) - f(x_0)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon\,.$$

This gives the claim. □

Example 1: Set

$$E_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}\,,$$

$$E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}\,.$$

In the previous lecture we showed that Weirstrass M-test implies that $E_n \to E$ uniformly on $[-L, L]$. Since each E_n is continuous we have from Theorem 1 that E is continuous.

Here is another useful way of thinking of uniform convergence. Recall that on the space of continuous functions $C(I)$ on an interval $I = [a, b]$ there is a natural metric given by that

$$d(f, g) = \max_{x \in I} \{|f(x) - g(x)| \,|\, x \in I\}\,.$$

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We have the following:

Proposition: Let I be an interval $[a,b]$ and f_n, $f \in C(I)$, then $f_n \to f$ in the metric space if and only if f_n converges to f uniformly.

Proof. To see this note that

$$|f(x) - f_n(x)| \leq \epsilon \text{ for all } x \in I$$

if and only if

$$d(f,f_n) \leq \epsilon\,.$$

To say that $f_n \to f$ uniformly is therefore equivalent to that $d(f, f_n) \to 0$ giving the claim. □

From this we get:

Corollary: $C([a,b])$ is Cauchy complete.

Proof. Suppose that f_n is a Cauchy sequence in $C([a,b])$ we need to find a $f \in C([a,b])$ such that $f_n \to f$ uniformly. For each x fixed, the sequence $f_n(x)$ is a Cauchy sequence in $\mathbf{R}$. This follows since

$$|f_n(x) - f_m(x)| \leq d(f_n, f_m)\,.$$

Therefore, since $\mathbf{R}$ is Cauchy complete, for each x there exists a $f(x)$ such that $f_n(x) \to f(x)$. This defines the function f and show that $f_n \to f$ converges pointwise. We need to show that the convergence is uniform. To see that observe that given $\epsilon > 0$ since f_n is a Cauchy sequence, there exists N such that if n and $m \geq N$, then

$$|f_n(x) - f_m(x)| < \frac{\epsilon}{2} \text{ for all } x \in I\,.$$

Therefore, for $f(x) = \lim_{m\to\infty} f_m(x)$ we have

$$|f_n(x) - f(x)| \leq \frac{\epsilon}{2} < \epsilon \text{ for all } x \in I\,.$$

This show that the convergence is uniform. □

Theorem 2: If $f_n \in \mathcal{R}([a,b])$ and $f_n \to f$ uniformly, then $f \in \mathcal{R}([a,b])$ and

$$\int_a^b f_n\,dx \to \int_a^b f\,dx\,.$$

Proof. We need to first show that $f \in \mathcal{R}([a,b])$ and so we need to show that given $\epsilon > 0$, there exists a partition $\mathcal{P}$ of the interval $[a,b]$ such that

$$U(f,\mathcal{P}) - L(f,\mathcal{P}) < \epsilon\,.$$

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Since $f_n \to f$ uniformly we have that there exists a N such that if $n \geq N$, then

$$|f(x) - f_n(x)| < \frac{\epsilon}{3\,(b-a)}\,.$$

We have therefore that for any partition $\mathcal{P}$ that

$$|m_i^{f_n} \leq m_i^f| \leq \frac{\epsilon}{3\,(b-a)}\,,$$

$$|M_i^{f_n} \leq M_i^f| \leq \frac{\epsilon}{3\,(b-a)}\,,$$

It follows that for any partition when $n \geq N$, then

$$|U(f,\mathcal{P}) - U(f_n,\mathcal{P})| < \frac{\epsilon}{3}\,,$$

$$|L(f,\mathcal{P}) - L(f_n,\mathcal{P})| < \frac{\epsilon}{3}\,.$$

We can now use that since $f_N \in \mathcal{R}\,([a,b])$ we have that there exists a partition $\mathcal{P}$ such that

$$U(f_N,\mathcal{P}) - L(f_N,\mathcal{P}) < \frac{\epsilon}{3}\,.$$

Combining it all gives that

$$\begin{aligned} U(f,\mathcal{P}) - L(f,\mathcal{P}) &< U(f,\mathcal{P}) - U(f_N,\mathcal{P}) + U(f_N,\mathcal{P}) - L(f_N,\mathcal{P}) + L(f_N,\mathcal{P}) - L(f,\mathcal{P}) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3}\,. \end{aligned}$$

This show that $f \in \mathcal{R}\,([a,b])$. We also need to see that

$$\int_a^b f\,dx = \lim_{n\to\infty} \int_a^b f_n\,dx\,.$$

This, however, follows from that

$$L(f,\mathcal{P}) \leq \int_a^b f\,dx \leq U(f,\mathcal{P})\,,$$

$$L(f_n,\mathcal{P}) \leq \int_a^b f_n\,dx \leq U(f_n,\mathcal{P})\,.$$

and that for $n \geq N$

$$|U(f,\mathcal{P}) - U(f_n,\mathcal{P})| < \frac{\epsilon}{3}\,,$$

$$|L(f,\mathcal{P}) - L(f_n,\mathcal{P})| < \frac{\epsilon}{3}\,.$$

Namely, we now have that also

$$L(f,\mathcal{P}) - \frac{\epsilon}{3} \leq \int_a^b f_n\,dx \leq U(f,\mathcal{P}) + \frac{\epsilon}{3}$$

and therefore

$$\int_a^b f\,dx - \int_a^b f\,dx \ < \epsilon\,.$$

□

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Example 1A: Set

$$E_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}\,,$$
$$E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}\,.$$

Then we have from Example 1 that $E_n \to E$ uniformly on $[-L, L]$. We now have from Theorem 2 that

$$\sum_{k=0}^{n} \int_0^1 \frac{x^k}{k!}\, dx \to \int_0^1 E(x)\, dx\,.$$

Theorem 3: Suppose that f_n are differentiable functions on $[a, b]$ and $x_0 \in [a, b]$. If

- $f_n(x_0) \to c$,
- $f_n' \to g$ uniformly,
- f_n' are continuous on $[a, b]$,

then there exists a differentiable function f with

- $f_n \to f$ uniformly,
- $f_n' \to f'$ uniformly.

Proof. Define a function F on $[a, b]$ by

$$f(x) = c + \int_{x_0}^{x} g\, dx\,,$$

and note since f_n' are continuous and that $f_n' \to g$ uniformly, it follows from Theorem 1 that g is also continuous. Therefore, by the fundamental theorem of calculus f is differentiable and $f' = g$. Moreover, by the fundamental theorem of calculus we have that

$$f_n(x) = f_n(x_0) + \int_{x_0}^{x} f_n'\, dx\,.$$

We are done provided we can show that $f_n \to f$. To see that note that

$$\begin{aligned} |f(x) - f_n(x)| = \; & c + \int_{x_0}^{x} g\, dx - f_n(x_0) - \int_{x_0}^{x} f_n'\, dx \\ \leq |c - f_n(x_0)| + \; & \int_{x_0}^{x} (g - f_n')\, dx \;\leq |c - f_n(x_0)| + \int_{x_0}^{x} |g - f_n'|\, dx \\ & \leq |c - f_n(x_0)| + (b - a)\, d(g, f_n')\,. \end{aligned}$$

The claim now follows since $f_n(x_0) \to c$ and $d(g, f_n') \to 0$. □

Example 1B: Set

$$E_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}\,,$$

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$$E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}\,.$$

Then

$$E_n' = \sum_{k=1}^{n} k\,\frac{x^{k-1}}{k!} = \sum_{k=0}^{n-1} \frac{x^{k-1}}{(k-1)!} = E_{n-1}\,.$$

From Example 1 that $E_{n-1} \to E$ uniformly on $[-L, L]$ and each E_n are continuous. Moreover, for all n we have that

$$E_n(0) = 1 = E(0)\,.$$

It follows therefore from Theorem 3 that

$$E_n' = E_{n-1} \to E'$$

uniformly and since $E_n' = E_{n-1}$, then we have that $E' = E$.

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Lecture 22

Suppose that a_n is a sequence and

$$\sum_{n=0}^{\infty} a_n \, x^n \, ,$$

is a power series, the radius of convergence R is

$$R = \frac{1}{M} \text{ where } M = \limsup |a_n|^{\frac{1}{n}} \, .$$

Lemma: The radius of convergence is the same for the power series

$$\sum_{n=0}^{\infty} a_n \, x^n$$

as the power series

$$\sum_{n=1}^{\infty} n \, a_n \, x^{n-1} \, .$$

.

Proof. Since

$$n^{\frac{1}{n-1}} = \mathrm{e}^{\frac{\log n}{n-1}} \to 1 \, ,$$

and

$$\limsup_{n \to \infty} |a_n|^{\frac{1}{n}} = \limsup_{n \to \infty} |a_n|^{\frac{1}{n-1}}$$

we have that

$$\limsup_{n \to \infty} |n \, a_n|^{\frac{1}{n-1}} = \limsup_{n \to \infty} |a_n|^{\frac{1}{n}} \, .$$

From this the claim follows. □

Iterating this gives:

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Corollary: The power series

$$\sum_{n=0}^{\infty} a_n \, x^n$$

has the same radius of convergence as the power series

$$\sum_{n=k}^{\infty} \frac{n!}{(n-k)!} \, a_n \, x^{n-k} \, .$$

We now get the following:

Theorem: Suppose that

$$f(x) = \sum_{n=0}^{\infty} a_n \, x^n \, ,$$

is a power series with radius of convergence R, then

$$f^{(k)}(x) = \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} \, a_n \, x^{n-k}$$

and

$$\int f(x) \, dx = \sum_{n=1}^{\infty} \frac{a_{n-1}}{n} \, x^n \, .$$

Proof. Let us first argue for $= 1$. We will see that this is a consequence of Theorem 3 from Lecture 21. Set

$$f_n(x) = \sum_{k=0}^{n} a_k \, x^k$$

and

$$f(x) = \sum_{k=0}^{\infty} a_k \, x^k \, .$$

Moreover, let R be the radius of convergence for the power series f. We have the following three properties

(1)

$$f_n(0) = a_0 = f(0) \, .$$

(2) On each interval $[-L, L]$, where $L < R$, we have uniform convergence

$$f_n' \to \sum_{k=1}^{\infty} k \, a_k \, x^{k-1} \, .$$

(3) Each f_n' is continuous.

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We see that Theorem 3 applies and show that

$$f' = \sum_{k=1}^{\infty} k\, a_k\, x^{k-1}\,.$$

Iterating this gives the claim for all k. Finally, the claim about the integral

$$\int f(x)\, dx$$

follows easily from Theorem 2 from Lecture 21. □

Ordinary differential equations: A differential equation is an equation that involves an unknown function and its derivative.

Example: Here are some examples of differential equations

$$f'(x) = x\,.$$
$$f'(x) - f^2(x) = 0\,.$$
$$f(x)\, f'(x)\, f''(x) = 1\,.$$

For the first of these and each constant c, the function

$$f_c(x) = \frac{1}{2}\, x^2 + c$$

is a solution. For the second

$$f(x) = \frac{1}{1-x}$$

is a solution. For the third $y = 0$ is a solution and so is $y = x$.

We will be interested in an ordinary differential equation (ODE) of the form

$$y' = f(y) + g(x)\,.$$

Here $y = y(x)$ is the unknown function and f, g are given functions. Note that while g only depend on x the function f also depend on the unknown function y.

We are interested in whether there exist solutions and when they exist if they are unique.

More precisely, suppose that we have the following:

- f be a continuously differentiable function on $\mathbf{R}$.
- g be a continuous function on $\mathbf{R}$.
- a is a real number.

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We are intersted in existence and uniqueness of the ODE:

$$\begin{cases} y'(x) &= f(y(x)) + g(x) \\ y(0) &= a\,. \end{cases}$$

We will show next time the following:

Picard-Lindelöf theorem: There exists $\delta > 0$ such that there is a unique solution to this ODE on $(-\delta, \delta)$.

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TOBIAS HOLCK COLDING

Lecture 23

Ordinary differential equations: A differential equation is an equation that involves an unknown function and its derivative.

Suppose that we have the following:

- f be a continuously differentiable function on $\mathbf{R}$.
- g be a continuous function on $\mathbf{R}$.
- a is a real number.

We will be intersted in existence and uniqueness of the ODE:

$$(\dagger) \quad \begin{cases} y'(x) &= f(y(x)) + g(x)\,, \\ y(0) &= a\,. \end{cases}$$

We say that this is a first order equation since it only involves the function and its derivative and not higher derivatives.

The following theorem gives a satisfying answer to the question of existence and uniqueness for this ODE:

Picard-Lindelöf theorem: There exists $\delta > 0$ such that there is a unique solution to $(\dagger)$ on $(-\delta, \delta)$.

Before we prove this theorem let us recall a result that we have proven earlier. Suppose that $[a, b]$ is an interval and let $C([a, b])$ be the space of continuous functions on $[a, b]$. We equip this space with the metric d given by that if h_1, $h_2 \in C([a, b])$, then

$$d(h_1, h_2) = \max_{x \in [a,b]} |h_1(x) - h_2(x)|\,.$$

We proved earlier the following theorem:

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Theorem 1: The metric space $(C([a,b]), d)$ is Cauchy complete.

We will also need to recall what it means for a map from a metric space to itself is contracting. A map T is a contracting map on a metric space (X, d) if for some $c < 1$ and all x, $y \in X$

$$d(T(x), T(y)) \leq c\, d(x, y)\,.$$

We shall also use that we have proven the following fact:

Theorem 2: If (X, d) is a Cauchy complete metric space and $T : X \to X$ is a contracting map, then T has a unique fix point.

Indeed this theorem was proven by showing that for any $x \in X$, the sequence x, $T(x)$, $T^2(x)$, $T^3(x), \cdots$ is a Cauchy sequence and the limit is the unique fix point of T. The proof of this used that

$$d(T^{n+1}(x), T^n(x)) \leq c^n\, d(T(x), x)\,,$$

and therefore by the triangle inequality

$$d(T^{n+k}(x), T^n(x)) \leq \sum_{i=1}^{k} d(T^{i+n}(x), T^{i-1+n}(x)) \leq c^{i-1+n}\, d(T(x), x)\,.$$

Which is easily seen to imply that the sequence $T^n(x)$ is a Cauchy sequence.

We will also use the following lemma:

Lemma 1: Suppose that u_1 and u_2 are continuous functions on an interval I. Assume also that

- $u_1(x_0) = u_2(x_0)$.
- If $u_1(x) = u_2(x)$, then $u_1 = u_2$ in a neighborhood of x.

then $u_1 = u_2$.

Proof. Let

$$J_+ = \{z \in I \,|\, z \geq x_0 \text{ and } u_1(x) = u_2(x) \text{ for all } x \in [x_0, z]\}\,.$$

Then $x_0 \in J_+$ so $J_+ \neq \emptyset$. Let $z_0 = \sup J_+$, if $z_0 \in I$, then $u_1(z_0) = u_2(z_0)$ by continuity. Since also u_1 and u_2 agrees in a neighborhood of z_0 it follows that z_0 must be the right end point of I. Similarly one can show that $u_1 = u_2$ everywhere to the left of z_0. This proves the lemma. □

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Finally, in the proof of the Picard-Lindelöf theorem we will also need the next lemma. In this lemma f is a function on $\mathbf{R}$ as above, so f differentiable and the derivative of f is continuous and g will be a continuous function on $\mathbf{R}$. For $\delta > 0$, on the space of continuous functions on $[-\delta, \delta]$ we define a map T on functions y as follows

$$T(y)(x) = a + \int_0^x [f(z(s)) + g(s)]\, ds\,.$$

Note that when y is continuous, then so is $T(y)$.

Lemma 2: Let a be a constant and set $R = |a| + 2$. There exists a $\delta > 0$ such that:

- The map T maps the ball (in the metric space $(C([-\delta, \delta]), d)$) of radius R and with center the constant function zero into itself. We write $B_R(0)$ for this ball and so have that $T : B_R(0) \to B_R(0)$.
- The map T is contracting on $B_R(0)$.

Proof. Let

$$L_1 = \max_{|z| \le R} |f(z)|\,,$$
$$L_2 = \max_{|x| \le 1} |g(x)|\,.$$

We will first show that if that if we choose $\delta_0 > 0$ small enough, then T maps $B_R(0)$ into itself. That is, we will show that if $|y| \le R$ on $[-\delta_0, \delta_0]$, then

$$|T(y)| \le R\,.$$

To see this set

$$\delta_0 = \min\left\{1, \frac{1}{L_1 + 1}, \frac{1}{L_2 + 1}\right\}\,.$$

Now suppose that $|y| \le R$ and $|x| \le \delta_0$, then

$$\begin{aligned} |T(y)(x)| &\le |a| + \int_0^x |f(y(s))|\, ds + \int_0^x |g(s)|\, ds \\ &\le |a| + \delta_0\, L_1 + \delta_0\, L_2 \le |a| + 2 = R\,. \end{aligned}$$

This show that T maps $B_R(0)$ into itself.

Next set

$$M = \max_{|z| \le R} |f'(z)|\,,$$

and

$$\delta = \min\left\{\delta_0, \frac{1}{2\,M + 1}\right\}\,.$$

Suppose that y_1 and y_2 are two continuous functions on $[-\delta, \delta]$ in $B_R(0)$, then

$$|T(y_1)(x) - T(y_2)(x)| = \int_0^x [f(y_1(s)) - f(y_2(s))]\, ds \quad .$$

By the mean value theorem applied to f for each s we have a z_s between $y_1(s)$ and $y_2(s)$ such that

$$f(y_1(s)) - f(y_2(s)) = f'(z_s)\,(y_1(s) - y_2(s))\,.$$

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Since $|y_i| \leq R$ we have that we have that for each s

$$|f(y_1(s)) - f(y_2(s))| \leq M\,|y_1(s) - y_2(s)| \leq M\,\max\,|y_1 - y_2| = M\,d(y_1, y_2)\,,$$

and therefore

$$|T(y_1)(x) - T(y_2)(x)| = \int_0^x [f(y_1(s)) - f(y_2(s))]\,ds \;\leq M\;\delta\,d(y_1, y_2) < \frac{1}{2}\,d(y_1, y_2)\,.$$

□

We are now ready to show the Picard-Lindelöf theorem:

Proof. (of the Picard-Lindelöf theorem.) Let T be defined as above and R and $\delta > 0$ be given by Lemma 2. A fixed point for T is a function y such that $T(y) = y$. By the fundamental theorem of calculus if y is a fix point of T, then we have that

$$y'(x) = (T(y))'(x) = f(y(x)) + g(x)\,.$$

Moreover, $y(0) = a$. In other words any fix point of T is a solution to the ODE.

We need to show that the solution is unique. Suppose that y^* is any other solution, then by the fundamental theorem of calculus

$$y^*(x) = a + \int_0^x (y^*)'(s)\,ds = T(y^*(s))\,.$$

Note that this holds even if the interval I that y^* is defined on (containing 0) is different from $[-\delta, \delta]$. We have from this that any solution is a fix point of T. Since T is contracting on $B_R(0)$ it follows that for any fix point with $|y| \leq R$, then y is unique. In general, suppose that y_1 and y_2 are two solutions defined on intervals I_1 and I_2 both containing 0. We have from the above that they agree in a neighborhood of 0. The argument in Lemma 2 that proved uniqueness in a small neigborhood of 0 works equally well in a neighborhood of any other point. It now follow from Lemma 1 that y_1 and y_2 agrees everywhere. □

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TOBIAS HOLCK COLDING

Review for the Final

Definition (Closed subsets): Let (X, d) be a metric space. We say that C is a closed subset of X if the complement $X \setminus C$ is open.

Note that $\emptyset$ (the empty set) and X are both closed.

Lemma: Let (X, d) be a metric space and $r > 0$, then

$$A_r = \{y \,|\, d(x, y) > r\}$$

is open. Equivalently, $\bar{B}_r(x) = \{y \,|\, d(x, y) \leq r\}$ is closed.

Theorem: A subset C of a metric space (X, d) is closed if and only if for all convergent sequences x_n with all x_n in C also the limit is in C.

Theorem:

- Union: If C_α is a family of closed subsets, then $\cap_\alpha C_\alpha$ is also closed.
- Intersection: If $C_1, \cdots, C_n$ are closed subsets, then $C_1 \cup \cdots \cup C_n$ is also closed.

Warning: Union of infinitely many closed sets may not be closed!!!

Definition (Cover, open cover and finite sub-cover): If A is a subset of X, then a cover of A is a collection collection of subsets U_α of X so that

$$A \subset \cup_\alpha U_\alpha \, .$$

We say that a $U_{\alpha_1}, \cdots, U_{\alpha_n}$ is a finite sub-cover if also $\{U_{\alpha_i}\}_i$ is a cover.

If (X, d) is a metric space and all the U_α are open, then we say that $\{U_\alpha\}_\alpha$ is an open cover.

Definition (Compact subset): If (X, d) is a metric space and A is a subset, then we say that A is compact if each open cover has a finite sub-cover.

Theorem: (Heine-Borel.) $[a, b]^n \subset \mathbf{R}^n$ is compact.

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Theorem: If (X, d) is a metric space and A a compact subset, then A is closed and bounded.

Warning: The converse is not the case!!! There are closed a bounded subsets of metric spaces that are not compact.

Theorem: If (X, d) is a metric space and A a compact subset, then any closed subset C contained in A is also compact.

Theorem: If (X, d) is a metric space and A a compact subset, then any sequence in A has a convergent subsequence.

Definition: If $f : \mathbf{R} \to \mathbf{R}$ is a function, then we say that f is differentiable at x_0 if the limit

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. (Note that in this fraction x is assumed to be $\neq x_0$.) When the limit exists, then we say that the function f is differentiable at x_0 and that its derivative at x_0 is the limit. In this case we denote the derivative at x_0 by $f'(x_0)$.

Lemma: If f is differentiable at x_0, then f is continuous at x_0.

Theorem: If f, g are functions on $\mathbf{R}$ that both are differentiable at x_0, then

- (Sum rule.)
$$(f + g)'(x_0) = f'(x_0) + g'(x_0)\,.$$
- (Leibniz's rule.)
$$(f\,g)(x_0) = f'(x_0)\,g(x_0) + f(x_0)\,g'(x_0)\,.$$
- (Quotient rule.) If also $g(x_0) \neq 0$, then
$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)\,g(x_0) - f(x_0)\,g'(x_0)}{g^2(x_0)}\,.$$

Theorem: (Chain rule.) If $f : [a, b] \to [c, d]$ and $g : [c, d] \to \mathbf{R}$ are functions, where f is differentiable at x_0 and g differentiable at $y_0 = f(x_0)$, then the composition $g \circ f$ is differentiable at x_0 and the derivative at x_0 is

$$(g \circ f)'(x_0) = g'(y_0)\,f'(x_0)\,.$$

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Lemma: Let $f : [a, b] \to \mathbf{R}$ be a differentiable function and suppose that $a < x_0 < b$ and that f has a local maximum or minimum at x_0, then

$$f'(x_0) = 0\,.$$

Theorem: (Mean value theorem.) Let $f : [a, b] \to \mathbf{R}$ be a differentiable function, then there exists a x_0 between a and b such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}\,.$$

.

Theorem: (Cauchy mean value theorem.) Let f, $g : [a, b] \to \mathbf{R}$ be differentiable functions, then there exists a x_0 between a and b such that

$$f'(x_0)\,[g(b) - g(a)] = g'(x_0)\,[f(b) - f(a)]\,.$$

In particular, if $g(b) - g(a) \neq 0$, then

$$\frac{f'(x_0)}{g'(x_0)} = \frac{f(b) - f(a)}{g(b) - g(a)}\,.$$

Theorem: (L'Hopital's rule, version 1.) Let f, $g : (a, b) \to \mathbf{R}$ be differentiable functions with $g(x) \neq 0$ and $g'(x) \neq 0$ for all x, assume that

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0\,.$$

If

$$\lim_{x \to a} \frac{f'(x)}{g'(x)}$$

exists, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}\,.$$

Theorem: (L'Hopital's rule, version 2.) Let f, $g : (a, b) \to \mathbf{R}$ be differentiable functions with $g(x) \neq 0$ and $g'(x) \neq 0$ for all x, assume that

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = \infty\,.$$

If

$$\lim_{x \to a} \frac{f'(x)}{g'(x)}$$

exists, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}\,.$$

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Theorem: (Taylor expansion.) Let $f : [a, b] \to \mathbf{R}$ be a function and k a positive integer. Assume that f, f', $f^{(2)}, \cdots, f^{(k-1)}$ exists on $[a, b]$ and are continuous and that $f^{(k)}$ is defined on (a, b), then there exists c between a and b such that

$$f(b) = f(a) + f'(a)\,(b-a) + \frac{f^{(2)}(a)}{2}\,(b-a)^2 + \cdots + \frac{f^{(k-1)}(a)}{(k-1)!}\,(b-a)^{k-1} + \frac{f^{(k)}(c)}{(k)!}\,(b-a)^k\,.$$

For and infinitely differentiable function f on $\mathbf{R}$ we define the $(k-1)$ Taylor polynomial at a by

$$P_{k-1}(x) = f(a) + f'(a)\,(x-a) + \frac{f^{(2)}(a)}{2}\,(x-a)^2 + \cdots + \frac{f^{(k-1)}(a)}{(k-1)!}\,(x-a)^{k-1}\,.$$

Riemann integrals

Partition: Let $[a, b]$ be an interval. A partition $\mathcal{P}$ of the interval $[a, b]$ is a number of sub-divisions x_i such that

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b\,.$$

The partition is then the sub-intervals $[x_{i-1}, x_i]$. We will set $\Delta\, x_i = x_i - x_{i-1}$.

Upper and lower sums: Suppose now that $f : [a, b] \to \mathbf{R}$ is a bounded function and that $\mathcal{P} = \{x_i\}$ is a partition of the interval $[a, b]$. We define upper and lower sums as follows. Set

$$M_i = \sup_{[x_{i-1}, x_i]} f\,,$$

$$m_i = \inf_{[x_{i-1}, x_i]} f\,,$$

and upper $U(f, \mathcal{P})$ and lower sums $L(f, \mathcal{P})$ by

$$U(f, \mathcal{P}) = \sum_{i=1}^{n} M_i\ \Delta\, x_i\,,$$

$$L(f, \mathcal{P}) = \sum_{i=1}^{n} m_i\ \Delta\, x_i\,.$$

Upper and lower integrals: Suppose now that $f : [a, b] \to \mathbf{R}$ is a bounded function. Define the upper Riemann integral of f by

$$\overline{\int_a^b} f\, dx = \inf_{\mathcal{P}} U(f, \mathcal{P})\,.$$

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Here the infimum is taken over all partitions of $[a, b]$. Likewise, we define the lower Riemann integral by

$$\underline{\int_a^b} f\,dx = \sup_{\mathcal{P}} L(f, \mathcal{P})\,.$$

Riemann integral: Suppose that $f : [a, b] \to \mathbf{R}$ is a bounded function, then we say that f is Riemann integrable if

$$\overline{\int_a^b} f\,dx = \underline{\int_a^b} f\,dx\,.$$

If the function is Riemann integrable, then the Riemann integral is

$$\int_a^b f\,dx = \overline{\int_a^b} f\,dx = \underline{\int_a^b} f\,dx\,.$$

The Riemann integrable functions is denoted by $\mathcal{R}([a, b])$.

Theorem: Any continuous function on $[a, b]$ is in $\mathcal{R}([a, b])$.

Definition: Uniformly continuous. Suppose that $f : I \to \mathbf{R}$ is a function, where I is an interval. We say that f is uniformly continuous if for all $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon \text{ if } |x - y| < \delta\,.$$

Note that being uniformly continuous is stronger than being continuous. It means that for a given $\epsilon > 0$, the same δ can be used for all x.

Basic properties of integrals.

Theorem: We have the following basic formulas for integrals:

(1) If $f \in \mathcal{R}([a, b])$ and $c \in \mathbf{R}$, then $c\,f \in \mathcal{R}([a, b])$ and

$$\int_a^b (c\,f)\,dx = c \int_a^b f\,dx\,.$$

(2) If f, $g \in \mathcal{R}([a, b])$, then $f + g \in \mathcal{R}([a, b])$ and

$$\int_a^b (f + g)\,dx = \int_a^b f\,dx + \int_a^b g\,dx\,.$$

(3) If f, $g \in \mathcal{R}([a, b])$ and $f \le g$, then

$$\int_a^b f\,dx \le \int_a^b g\,dx\,.$$

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(4) If $f \in \mathcal{R}([a,b])$ and $c \in (a,b)$, then $f \in \mathcal{R}([a,c])$ and $f \in \mathcal{R}([c,b])$ and

$$\int_a^c f\,dx + \int_c^b f\,dx = \int_a^b f\,dx\,.$$

Corollary: Suppose that f, $|f| \in \mathcal{R}([a,b])$, then

$$\int_x^b f\,dx \;\leq \int_a^b |f|\,dx\,.$$

Fundamental theorem of calculus, version 1: Let f be a continuous function on $[a,b]$ and define F on $[a,b]$ by

$$F(x) = \int_a^x f(s)\,ds\,.$$

The function F is differentiable with derivative f.

Fundamental theorem of calculus, version 2: Suppose that $F : [a,b] \to \mathbf{R}$ is differentiable and that $F' = f \in \mathcal{R}([a,b])$, then

$$F(b) - F(a) = \int_a^b f(s)\,ds\,.$$

Suppose that f and $g : [a,b] \to \mathbf{R}$ are differentiable functions and their derivatives are continuous, then we define the arclength of the curve

$$s \to (f(s), g(s))$$

by

$$L = \int_a^b \sqrt{(f'(s))^2 + (g'(s))^2}\,ds\,.$$

Improper integrals.

Unbounded interval.

Suppose that $f \in \mathcal{R}([a,b])$ for all $b > a$. If

$$\lim_{b\to\infty} \int_a^b f(x)\,dx$$

exists, then we say that the improper integral

$$\int_a^\infty f(x)\,dx$$

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exists and that

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$$

Unbounded function.

Suppose that $f \in \mathcal{R}([c,b])$ for all $c > a$. If

$$\lim_{c\to a}\int_c^b f(x)\,dx$$

exists, then we say that the improper integral

$$\int_a^b f(x)\,dx$$

exists and that

$$\int_a^b f(x)\,dx = \lim_{c\to a}\int_c^b f(x)\,dx$$

Pointwise convergence: Suppose that f_n is a sequence of functions on an interval I, then we say that f_n convergences pointwise to a function f if for all x we have

$$f_n(x) \to f(x)\,.$$

Uniform convergence: Suppose that f_n is a sequence of functions on an interval I, then we say that f_n convergences uniformly to a function f if for all $\epsilon > 0$, there exists an N such that if $n \geq N$, then for all x

$$|f(x) - f_n(x)| < \epsilon\,.$$

Lemma 1: Suppose that I is an interval and f_n is a sequence of functions on I that converges uniformly to a function f, then f_n also converges pointwise to f.

Lemma [Weirstrass M-test]: Suppose that I is an interval and f_n is a sequence of functions on I. Suppose also that M_n is a sequence of non-negative numbers with

$$|f_n(x)| \leq M_n \text{ for all } x \in I\,.$$

If the series

$$\sum_{n=1}^\infty M_n$$

converges, then the sequence of functions

$$S_n(x) = \sum_{k=0}^n f_k(x)$$

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converges uniformly.

Theorem: If

$$\sum_{k=0}^{\infty} a_k \, x^k$$

is a power series and R is its radius of convergence. Then it converges uniformly on any (finite) interval of the form $[-L, L]$ where $L < R$.

Theorem: Suppose that I is an interval and f_n is a sequence of continuous functions on I. If f_n converges uniformly to f, then f is also continuous.

Proposition: Let I be an interval $[a, b]$ and f_n, $f \in C(I)$, then $f_n \to f$ in the metric space $(C(I), d)$ if and only if f_n converges to f uniformly.

Corollary: $C([a, b])$ is Cauchy complete.

Theorem: If $f_n \in \mathcal{R}([a, b])$ and $f_n \to f$ uniformly, then $f \in \mathcal{R}([a, b])$ and

$$\int_a^b f_n \, dx \to \int_a^b f \, dx \, .$$

Theorem: Suppose that f_n are differentiable functions on $[a, b]$ and $x_0 \in [a, b]$. If

- $f_n(x_0) \to c$,
- $f_n' \to g$ uniformly,
- f_n' are continuous on $[a, b]$,

then there exists a differentiable function f with

- $f_n \to f$ uniformly,
- $f_n' \to f'$ uniformly.

Suppose that a_n is a sequence and

$$\sum_{n=0}^{\infty} a_n \, x^n \, ,$$

is a power series, the radius of convergence R is

$$R = \frac{1}{M} \text{ where } M = \limsup |a_n|^{\frac{1}{n}} \, .$$

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Corollary: The power series

$$\sum_{n=0}^{\infty} a_n \, x^n$$

has the same radius of convergence as the power series

$$\sum_{n=k}^{\infty} \frac{n!}{(n-k)!} \, a_n \, x^{n-k} \, .$$

Theorem: Suppose that

$$f(x) = \sum_{n=0}^{\infty} a_n \, x^n \, ,$$

is a power series with radius of convergence R, then

$$f^{(k)}(x) = \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} \, a_n \, x^{n-k}$$

and

$$\int f(x) \, dx = \sum_{n=1}^{\infty} \frac{a_{n-1}}{n} \, x^n \, .$$

Ordinary differential equations: Sppose that we have the following:

- f be a continuously differentiable function on $\mathbf{R}$.
- g be a continuous function on $\mathbf{R}$.
- a is a real number.

We are intersted in existence and uniqueness of the ODE:

$$\begin{cases} y'(x) & = f(y(x)) + g(x) \\ y(0) & = a \, . \end{cases}$$

Picard-Lindelöf theorem: There exists $\delta > 0$ such that there is a unique solution to this ODE on $(-\delta, \delta)$.

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Lecture 1

1 Review of Topology

1.1 Metric Spaces

Definition 1.1. Let X be a set. Define the Cartesian product $X \times X = \{(x, y) : x, y \in X\}$.

Definition 1.2. Let $d : X \times X \to \mathbb{R}$ be a mapping. The mapping d is a *metric on* X if the following four conditions hold for all $x, y, z \in X$:

(i) $d(x, y) = d(y, x)$,

(ii) $d(x, y) \geq 0$,

(iii) $d(x, y) = 0 \iff x = y$, and

(iv) $d(x, z) \leq d(x, y) + d(y, z)$.

Given a metric d on X, the pair (X, d) is called a *metric space*.

Suppose d is a metric on X and that $Y \subseteq X$. Then there is an automatic metric d_Y on Y defined by restricting d to the subspace $Y \times Y$,

$$d_Y = d|Y \times Y. \tag{1.1}$$

Together with Y, the metric d_Y defines the automatic metric space (Y, d_Y).

1.2 Open and Closed Sets

In this section we review some basic definitions and propositions in topology. We review open sets, closed sets, norms, continuity, and closure. Throughout this section, we let (X, d) be a metric space unless otherwise specified.

One of the basic notions of topology is that of the *open* set. To define an open set, we first define the ϵ-neighborhood.

Definition 1.3. Given a point $x_o \in X$, and a real number $\epsilon > 0$, we define

$$U(x_o, \epsilon) = \{x \in X : d(x, x_o) < \epsilon\}. \tag{1.2}$$

We call $U(x_o, \epsilon)$ the *ϵ-neighborhood of x_o in X*.

Given a subset $Y \subseteq X$, the ϵ-neighborhood of x_o in Y is just $U(x_o, \epsilon) \cap Y$.

Definition 1.4. A subset U of X is *open* if for every $x_o \in U$ there exists a real number $\epsilon > 0$ such that $U(x_o, \epsilon) \subseteq U$.

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We make some propositions about the union and intersections of open sets. We omit the proofs, which are fairly straightforward.

The following Proposition states that arbitrary unions of open sets are open.

Proposition 1.5. *Let $\{U_\alpha, \alpha \in I\}$ be a collection of open sets in X, where I is just a labeling set that can be finite or infinite. Then, the set*

$$\bigcup_{\alpha \in I} U_\alpha \text{ is open.}$$

The following Corollary is an application of the above Proposition.

Corollary 1. *If $Y \subset X$ and A is open in Y (w.r.t. d_Y), then there exists on open set U in X such that $U \cap Y = A$.*

Proof. The set A is open in Y. So, for any $p \in A$ there exists an $\epsilon_p > 0$ such that $U(p, \epsilon_p) \cap Y \subseteq A$. We construct a set U containing A by taking the union of the sets $U(p, \epsilon_p)$ over all p in A,

$$U = \bigcup_{p \in A} U(p, \epsilon_p). \tag{1.3}$$

For every $p \in A$, we have $U(p, \epsilon_p) \cap Y \subseteq A$, which shows that $U \cap Y \subseteq A$. Furthermore, the union is over all $p \in A$, so $A \subseteq U$, which implies that $A \subseteq U \cap Y$. This shows that $U \cap Y = A$. To conclude the proof, we see that U is open by the openness of the $U(p, \epsilon_p)$ and the above theorem. □

The following Proposition states that finite intersections of open sets are open.

Proposition 1.6. *Let $\{U_i, i = 1, \ldots, N\}$ be a finite collection of open sets in X. Then the set*

$$\bigcap_{i=1}^{i=N} U_i \text{ is open.}$$

Definition 1.7. Define the *complement of A in X* to be $A^c = X - A = \{x \in X : x \notin A\}$.

We use the complement to define *closed* sets.

Definition 1.8. The set A is *closed in X* if A^c is open in X.

1.3 Metrics on $\mathbb{R}^n$

For most of this course, we will only consider the case $X = \mathbb{R}^n$ or X equals certain subsets of $\mathbb{R}^n$ called manifolds, which we will define later.

There are two interesting metrics on $\mathbb{R}^n$. They are the *Euclidean metric* and the *sup metric*, and are defined in terms of the *Euclidean norm* and the *sup norm*, respectively.

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Definition 1.9. Let $x \in \mathbb{R}^n$, written out in component form as $x = (x_1, x_2, \ldots, x_n)$. The *Euclidean norm* of x is

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2},$$

and the the *sup norm* of x is

$$|x| = \max_i |x_i|.$$

From these norms we obtain the Euclidean distance function

$$\|x - y\| \tag{1.4}$$

and the sup distance function

$$|x - y|, \tag{1.5}$$

respectively.

These two distance functions are related in several ways. In particular,

$$|x - y| \leq \|x - y\| \leq \sqrt{n}|x - y|.$$

These distance functions are also related by the following Proposition, which will sometimes come in handy.

Proposition 1.10. *A subset U of $\mathbb{R}^n$ is open w.r.t. the $\|\ \|$ distance function if and only if it is open w.r.t. the $|\ |$ distance function.*

So, these two distance functions give the same topologies of $\mathbb{R}^n$.

1.4 Continuity

Consider two metric spaces (X, d_X) and (Y, d_Y), a function $f : X \to Y$, and a point $x_o \in X$.

Definition 1.11. The function f is *continuous at x_o* if for every $\epsilon > 0$ there exists a $\delta > 0$ such that

$$d_X(x, x_o) < \delta \implies d_Y(f(x), f(x_o)) < \epsilon. \tag{1.6}$$

By definition, a function is continuous if it is continuous at all points in its domain.

Definition 1.12. The function f is *continuous* if f is continuous at every point $x_o \in X$.

There is an alternative formulation of continuity that we present here as a theorem.

Theorem 1.13. *The function f is continuous if and only if for every open subset U of Y, the pre-image $f^{-1}(U)$ is open in X.*

Continuous functions can often be combined to construct other continuous functions. For example, if $f, g : X \to \mathbb{R}$ are continuous functions, then $f + g$ and fg are continuous functions.

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1.5 Limit Points and Closure

As usual, let (X, d) be a metric space.

Definition 1.14. Suppose that $A \subseteq X$. The point $x_o \in X$ is a *limit point* of A if for every ϵ-neighborhood $U(x_o, \epsilon)$ of x_o, the set $U(x_o, \epsilon)$ is an infinite set.

Definition 1.15. The *closure* of A, denoted by $\bar{A}$, is the union of A and the set of limit points of A,

$$\bar{A} = A \cup \{x_o \in X : x_o \text{ is a limit point of } A\}. \tag{1.7}$$

Now we define the interior, exterior, and the boundary of a set in terms of open sets. In the following, we denote the complement of A by $A^c = X - A$.

Definition 1.16. The set

$$\text{Int } A \equiv (\bar{A^c})^c \tag{1.8}$$

is called the *interior* of A.

It follows that

$$x \in \text{Int } A \iff \exists \epsilon > 0 \text{ such that } U(x, \epsilon) \subset A. \tag{1.9}$$

Note that the interior of A is open.
We define the exterior of a set in terms of the interior of the set.

Definition 1.17. The *exterior* of A is defined to be Ext $A \equiv$ Int A^c.

The boundary of a set is the collection of all points not in the interior or exterior.

Definition 1.18. The *boundary* of A is defined to be Bd $A \equiv X - ((\text{Ext } A) \cup (\text{Int } A))$.

Always, we have $X = \text{Int } A \cup \text{Ext } A \cup \text{Bd } A$.

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Lecture 2

1.6 Compactness

As usual, throughout this section we let (X, d) be a metric space. We also remind you from last lecture we defined the open set

$$U(x_o, \lambda) = \{x \in X : d(x, x_o) < \lambda\}. \tag{1.10}$$

Remark. If $U(x_o, \lambda) \subseteq U(x_1, \lambda_1)$, then $\lambda_1 > d(x_o, x_1)$.

Remark. If $A_i \subseteq U(x_o, \lambda_i)$ for $i = 1, 2$, then $A_1 \cup A_2 \subseteq U(x_o, \lambda_1 + \lambda_2)$.

Before we define compactness, we first define the notions of boundedness and covering.

Definition 1.19. A subset A of X is *bounded* if $A \subseteq U(x_o, \lambda)$ for some λ.

Definition 1.20. Let $A \subseteq X$. A collection of subsets $\{U_\alpha \subseteq X, \alpha \in I\}$ is a *cover* of A if

$$A \subset \bigcup_{\alpha \in I} U_\alpha.$$

Now we turn to the notion of compactness. First, we only consider compact sets as subsets of $\mathbb{R}^n$.

For any subset $A \subseteq \mathbb{R}^n$,

$$A \text{ is compact } \iff A \text{ is closed and bounded.}$$

The above statement holds true for $\mathbb{R}^n$ but not for general metric spaces. To motivate the definition of compactness for the general case, we give the Heine-Borel Theorem.

Heine-Borel (H-B) Theorem. *Let $A \subseteq \mathbb{R}^n$ be compact and let $\{U_\alpha, \alpha \in I\}$ be a cover of A by open sets. Then a finite number of U_α's already cover A.*

The property that a finite number of the U_α's cover A is called the Heine-Borel (H-B) property. *So, the H-B Theorem can be restated as follows: If A is compact in $\mathbb{R}^n$, then A has the H-B property.*

Sketch of Proof. First, we check the H-B Theorem for some simple compact subsets of $\mathbb{R}^n$. Consider rectangles $Q = I_1 \times \cdots \times I_n \subset \mathbb{R}^n$, where $I_k = [a_k, b_k]$ for each k. Starting with one dimension, it can by shown by induction that these rectangles have the H-B property.

Too prove the H-B theorem for general compact subsets, consider any closed and bounded (and therefore compact) subset A of $\mathbb{R}^n$. Since A is bounded, there exists a rectangle Q such that $A \subseteq Q$. Suppose that the collection of subsets $\{U_\alpha, \alpha \in I\}$ is

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an open cover of A. Then, define $U_o = \mathbb{R}^n - A$ and include U_o in the open cover. The rectangle Q has the H-B property and is covered by this new cover, so there exists a finite subcover covering Q. Furthermore, the rectangle Q contains A, so the finite subcover also covers A, proving the H-B Theorem for general compact subsets. □

The following theorem further motivates the general definition for compactness.

Theorem 1.21. *If $A \subseteq \mathbb{R}^n$ has the H-B property, then A is compact.*

Sketch of Proof. We need to show that the H-B property implies A is bounded (which we leave as an exercise) and closed (which we prove here).

To show that A is closed, it is sufficient to show that A^c is open. Take any $x_o \in A^c$, and define

$$C_N = \{x \in \mathbb{R}^n : d(x, x_o) \leq 1/N\}, \tag{1.11}$$

and

$$U_N = C_N^c. \tag{1.12}$$

Then,

$$\bigcap C_N = \{x_o\} \tag{1.13}$$

and

$$\bigcup U_N = \mathbb{R}^n - \{x_o\}. \tag{1.14}$$

The U_N's cover A, so the H-B Theorem implies that there is a finite subcover $\{U_{N_1}, \ldots, U_{N_k}\}$ of A. We can take $N_1 < N_2 < \cdots < N_k$, so that $A \subseteq U_{N_k}$. By taking the complement, it follows that $C_{N_k} \subseteq A^c$. But $U(x_o, 1/N_k) \subseteq C_{N_k}$, so x_o is contained in an open subset of A^c. The above holds for any $x_o \in A^c$, so A^c is open. □

Let us consider the above theorem for arbitrary metric space (X, d) and $A \subseteq X$.

Theorem 1.22. *If $A \subseteq X$ has the H-B property, then A is closed and bounded.*

Sketch of Proof. The proof is basically the same as for the previous theorem. □

Unfortunately, the converse is not always true. Finally, we come to our general definition of compactness.

Definition 1.23. A subset $A \subseteq X$ is *compact* if it has the H-B property.

Compact sets have many useful properties, some of which we list here in the theorems that follow.

Theorem 1.24. *Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a continuous map. If A is a compact subset of X, then $f(A)$ is a compact subset of Y.*

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Proof. Let $\{U_\alpha, \alpha \in I\}$ be an open covering of $f(A)$. Each pre-image $f^{-1}(U_\alpha)$ is open in X, so $\{f^{-1}(U_\alpha) : \alpha \in I\}$ is an open covering of A. The H-B Theorem says that there is a finite subcover $\{f^{-1}(U_{\alpha_i}) : 1 \leq i \leq N\}$. It follows that the collection $\{U_{\alpha_i} : 1 \leq i \leq N\}$ covers $f(A)$, so $f(A)$ is compact. □

A special case of the above theorem proves the following theorem.

Theorem 1.25. *Let A be a compact subset of X and $f : X \to \mathbb{R}$ be a continuous map. Then f has a maximum point on A.*

Proof. By the above theorem, $f(A)$ is compact, which implies that $f(a)$ is closed and and bounded. Let $a =$ l.u.b. of $f(a)$. The point a is in $f(A)$ because $f(A)$ is closed, so there exists an $x_o \in A$ such that $f(x_o) = a$. □

Another useful property of compact sets involves the notion of uniform continuity.

Definition 1.26. Let $f : X \to \mathbb{R}$ be a continuous function, and let A be a subset of X. The map f is *uniformly continuous on A* if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$d(x, y) < \delta \implies |f(x) - f(y)| < \epsilon,$$

for all $x, y \in A$.

Theorem 1.27. *If $f : X \to Y$ is continuous and A is a compact subset of X, then f is uniformly continuous on A.*

Proof. Let $p \in A$. There exists a $\delta_p > 0$ such that $|f(x) - f(p)| < \epsilon/2$ for all $x \in U(p, \delta_p)$. Now, consider the collection of sets $\{U(p, \delta_p/2) : p \in A\}$, which is an open cover of A. The H-B Theorem says that there is a finite subcover $\{U(p_i, \delta_{p_i}/2) : 1 \leq i \leq N\}$. Choose $\delta \leq \min \delta_{p_i}/2$. The following claim finishes the proof.

Claim. *If $d(x, y) < \delta$, then $|f(x) - f(y)| < \epsilon$.*

Proof. Given x, choose p_i such that $x \in U(p_i, \delta_{p_i}/2)$. So, $d(p_i, x) < \delta_{p_i}/2$ and $d(x, y) < \delta < \delta_{p_i}/2$. By the triangle inequality we conclude that $d(p_i, y) < \delta_{p_i}$. This shows that $x, y \in U(p_i, \delta_{p_i})$, which implies that $|f(x) - f(p_i)| < \epsilon/2$ and $|f(y) - f(p_i)| < \epsilon/2$. Finally, by the triangle inequality, $|f(x) - f(y)| < \epsilon$, which proves our claim.

□

1.7 Connectedness

As usual, let (X, d) be a metric space.

Definition 1.28. The metric space (X, d) is *connected* if it is impossible to write X as a disjoint union $X = U_1 \cup U_2$ of non-empty open sets U_1 and U_2.

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Note that disjoint simply means that $U_1 \cap U_2 = \phi$, where ϕ is the empty set.

A few simple examples of connected spaces are $\mathbb{R}$, $\mathbb{R}^n$, and $I = [a, b]$. The following theorem shows that a connected space gets mapped to a connected subspace by a continuous function.

Theorem 1.29. *Given metric spaces (X, d_X) and (Y, d_Y), and a continuous map $f : X \to Y$, it follows that*

$$X \text{ is connected} \implies f(X) \text{ is connected.}$$

Proof. Suppose $f(X)$ can be written as a union of open sets $f(X) = U_1 \cup U_2$ such that $U_1 \cap U_2 = \phi$. Then $X = f^{-1}(U_1) \cup f^{-1}(U_2)$ is a disjoint union of open sets. This contradicts that X is connected. □

The intermediate-value theorem follows as a special case of the above theorem.

Intermediate-value Theorem. *Let (X, d) be connected and $f : X \to \mathbb{R}$ be a continuous map. If $a, b \in f(X)$ and $a < r < b$, then $r \in f(X)$.*

Proof. Suppose $r \notin f(X)$. Let $A = (-\infty, r)$ and $B = (r, \infty)$. Then $X = f^{-1}(A) \cup f^{-1}(B)$ is a disjoint union of open sets, a contradiction. □

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Lecture 3

2 Differentiation

2.1 Differentiation in n dimensions

We are setting out to generalize to n dimensions the notion of differentiation in one-dimensional calculus. We begin with a review of elementary one-dimensional calculus.

Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$ be a map, and let $a \in I$.

Definition 2.1. The *derivative of f at a* is

$$f'(a) = \lim_{t \to 0} \frac{f(a+t) - f(a)}{t}, \tag{2.1}$$

provided that the limit exists. If the limit exists, then *f is differentiable at a.*

There are half a dozen or so possible reasonable generalizations of the notion of derivative to higher dimensions. One candidate generalization which you have probably already encountered is the directional derivative.

Definition 2.2. Given an open set U in $\mathbb{R}^n$, a map $f : U \to \mathbb{R}^m$, a point $a \in U$, and a point $u \in \mathbb{R}^n$, the *directional derivative of f in the direction of u at a* is

$$D_u f(a) = \lim_{t \to 0} \frac{f(a+tu) - f(a)}{t}, \tag{2.2}$$

provided that the limit exists.

In particular, we can calculate the directional derivatives in the direction of the standard basis vectors $e_1, \ldots, e_n$ of $\mathbb{R}^n$, where

$$\begin{aligned} e_1 &= (1, 0, \ldots, 0), && (2.3)\\ e_2 &= (0, 1, 0, \ldots, 0), && (2.4)\\ &\vdots && (2.5)\\ e_n &= (0, \ldots, 0, 1). && (2.6) \end{aligned}$$

Notation. The directional derivative in the direction of a standard basis vector e_i of $\mathbb{R}^n$ is denoted by

$$D_i f(a) = D_{e_i} f(a) = \frac{\partial}{\partial x_i} f(a). \tag{2.7}$$

We now try to answer the following question: What is an adequate definition of differentiability at a point a for a function $f : U \to \mathbb{R}^m$?

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- Guess 1: Require that $\frac{\partial f}{\partial x_i}(a)$ exists.

  However, this requirement is inadequate. Consider the function defined by

$$f(x_1, x_2) = \begin{cases} 0, & \text{if } (x_1, x_2) \text{ lies on the } x_1\text{-axis or the } x_2\text{-axis,} \\ 1, & \text{otherwise.} \end{cases} \tag{2.8}$$

  Then, both

$$\frac{\partial f}{\partial x_1}(0) = 0 \text{ and } \frac{\partial f}{\partial x_2}(0) = 0, \tag{2.9}$$

  but the function f is not differentiable at $(0, 0)$ along any other direction.

- Guess 2: Require that all directional derivatives exist at a.

  Unfortunately, this requirement is still inadequate. For example (from Munkres chapter 5), consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2+y^4}, & (x, y) \neq (0, 0) \\ 0, & x = y = 0. \end{cases} \tag{2.10}$$

  Claim. *The directional derivative $D_u f(0)$ exists for all u.*

  Proof. Let $u = (h, k)$. Then

$$\begin{aligned} \lim_{t\to 0} \frac{f(tu) - f(0)}{t} &= \lim_{t\to 0} \frac{f(tu)}{t} \\ &= \lim_{t\to 0} \left(\frac{t^3 hk^2}{t^2h^2 + t^4k^4} \right) \frac{1}{t} \\ &= \begin{cases} 0, & h = 0 \\ k^2/h, & h \neq 0. \end{cases} \end{aligned} \tag{2.11}$$

  So the limit exists for every u. □

  However, the function is a non-zero constant on a parabola passing through the origin: $f(t^2, t) = \frac{t^4}{2t^4} = \frac{1}{2}$, except at the origin where $f(0, 0) = 0$. The function f is discontinuous at the origin despite the existence of all directional derivatives.

- Guess 3. This guess will turn out to be correct.

  Remember than in one-dimensional calculus we defined

$$f'(a) = \lim_{t\to 0} \frac{f(a + t) - f(a)}{t}, \tag{2.12}$$

  for a function $f : I \to \mathbb{R}$ and a point $a \in I$. Now consider the function $\lambda : \mathbb{R} \to \mathbb{R}$ defined by

$$\lambda(t) = f'(a)t. \tag{2.13}$$

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Then,

$$\begin{aligned}\lim_{t\to 0}\frac{f(a+t)-f(a)-\lambda(t)}{t} &= \lim_{t\to 0}\frac{f(a+t)-f(a)}{t} - f'(a) \\ &= 0.\end{aligned} \tag{2.14}$$

So, $\lambda(t) \approx f(a+t) - f(a)$ when t is small.

Now we generalize to n dimensions.

Definition 2.3. Given an open subset U of $\mathbb{R}^n$, a map $f : U \to \mathbb{R}^m$, and a point $a \in U$, the function f is *differentiable at* a if there exists a linear mapping $B : \mathbb{R}^n \to \mathbb{R}^m$ such that for every $h \in \mathbb{R}^n - \{0\}$,

$$\frac{f(a+h)-f(a)-Bh}{|h|} \to 0 \text{ as } h \to 0. \tag{2.15}$$

That is, $f(a+h) - f(a) \approx Bh$ when h is small.

Theorem 2.4. *If f is differentiable at a, then for every u the directional derivative of f in the direction of u at a exists.*

Proof. The function f is differentiable at a, so

$$\frac{f(a+tu)-f(a)-B(tu)}{|tu|} \to 0 \text{ as } t \to 0. \tag{2.16}$$

Furthermore,

$$\begin{aligned}\frac{f(a+tu)-f(a)-B(tu)}{|tu|} &= \frac{t}{|tu|}\frac{f(a+tu)-f(a)-B(tu)}{t} \\ &= \frac{t}{|t|}\frac{1}{|u|}\left(\frac{f(a+tu)-f(a)}{t} - Bu\right) \\ &\to 0,\end{aligned} \tag{2.17}$$

as $t \to 0$, so

$$\frac{f(a+tu)-f(a)}{t} \to Bu \text{ as } t \to 0. \tag{2.18}$$

□

Furthermore, the linear map B is unique, so the following definition is well-defined.

Definition 2.5. The *derivative of f at a* is $Df(a) = B$, where B is the linear map defined above.

Note that $Df(a) : \mathbb{R}^n \to \mathbb{R}^m$ is a linear map.

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Theorem 2.6. *If f is differentiable at a, then f is continuous at a.*

Sketch of Proof. Note that for $h \neq 0$ in $\mathbb{R}^n$,

$$\frac{f(a+h) - f(a) - Bh}{|h|} \to 0 \text{ as } h \to 0 \tag{2.19}$$

implies that

$$f(a+h) - f(a) - Bh \to 0 \text{ as } h \to 0. \tag{2.20}$$

From this you can conclude that f is continuous at a. □

Remark. Let $L : \mathbb{R}^n \to \mathbb{R}^m$ be a linear map and $a \in \mathbb{R}^n$. The point a can be written as a sum $a = \sum_{j=1}^n a_j e_j = (a_1, \ldots, a_n)$. The point La can be written as the sum $La = \sum a_j L e_j$, and L can be written out in components as $L = (L_1, \ldots, L_m)$, where each $L_j : \mathbb{R}^n \to \mathbb{R}$ is a linear map. Then $Le_j = (L_1, e_j, \ldots, L_m e_j)$, and $L_i e_j = \ell_{i,j}$. The numbers $\ell_{i,j}$ form an $n \times n$ matrix denoted by $[\ell_{i,j}]$.

Remark. Let $U \subseteq \mathbb{R}^n$, and let $f_1 : \mathbb{R}^n \to \mathbb{R}^{m_1}$ and $f_2 : \mathbb{R}^n \to \mathbb{R}^{m_2}$ be differentiable maps. Let $m = m_1 + m_2$, so that $\mathbb{R}^{m_1} \times \mathbb{R}^{m_2} = \mathbb{R}^m$. Now, construct a function $f : \mathbb{R}^n \to \mathbb{R}^m$ defined in component form by $f = (f_1, f_2)$. The derivative of f at a is

$$Df(a) = (Df_1(a), Df_2(a)). \tag{2.21}$$

Remark. Let $f : U \to \mathbb{R}^m$ be a map. The action of f on input x written out in component form is $f(x) = (f_1(x), \ldots, f_m(x))$. So, the map can be represented in component form as $f = (f_1, \ldots, f_m)$, where each f_i as a map of the form $f_i : U \to \mathbb{R}$. The derivative of f acting on the standard basis vector e_j is

$$\begin{aligned} Df(a)e_j &= (Df_1(a)e_j, \ldots, Df_m(a)e_j) \\ &= (\frac{\partial f_1}{\partial x_j}(a), \ldots, \frac{\partial f_m}{\partial x_j}(a)). \end{aligned} \tag{2.22}$$

So, the derivative $(Df)(a)$ can be represented by an $m \times n$ matrix

$$(Df)(a) \cong J_f(a) = \left[\frac{\partial f_i}{\partial x_j}(a)\right] \tag{2.23}$$

called the Jacobian matrix of f at a, which you probably recognize.

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Lecture 4

2.2 Conditions for Differentiability

In this lecture we will discuss conditions that guarantee differentiability. First, we begin with a review of important results from last lecture.

Let U be an open subset of $\mathbb{R}^n$, let $f : U \to \mathbb{R}^n$ be a map, and let $a \in U$.

We defined f to be differentiable at a if there exists a linear map $B : \mathbb{R}^n \to \mathbb{R}^m$ such that for $h \in \mathbb{R}^n - \{0\}$,

$$\frac{f(a+h) - f(a) - Bh}{|h|} \to 0 \text{ as } h \to 0. \tag{2.24}$$

If such a B exists, then it is unique and $B = Df(a)$. The matrix representing B is the Jacobian matrix $J_f(a) = \left[\frac{\partial f_i}{\partial x_j}(a)\right]$, where $f = (f_1, \ldots, f_m)$.

Note that the mere existence of all of the partial derivatives in the Jacobian matrix does not guarantee differentiability.

Now we discuss conditions that guarantee differentiability.

Theorem 2.7. *Suppose that all of the partial derivatives $\frac{\partial f_i}{\partial x_j}$ in the Jacobian matrix exist at all points $x \in U$, and that all of the partial derivatives are continuous at $x = a$. Then f is differentiable at a.*

Sketch of Proof. This theorem is very elegantly proved in Munkres, so we will simply give the general ideas behind the proof here.

First, we look at the case $n = 2, m = 1$. The main ingredient in the proof is the Mean Value Theorem from 1-D calculus, which we state here without proof.

Mean Value Theorem. *Given an interval $[a, b] \subseteq \mathbb{R}$ and a map $\phi : [a, b] \to \mathbb{R}$, if ϕ is continuous on $[a, b]$ and differentiable on (a, b), then there exists a point $c \in (a, b)$ such that $\phi(b) - \phi(a) = \phi'(c)(b - a)$.*

Now we continue with the proof. Let f be a map $f : U \to \mathbb{R}$, where $U \subseteq \mathbb{R}^2$. So, f is a function of two variables $f = f(x_1, x_2)$. Consider a point $a = (a_1, a_2) \in U$ and any point $h \in \mathbb{R}^2 - \{0\}$ "close" to zero, where by close we mean $a + h \in U$. We want to compute $f(a + h) - f(a)$.

$$\begin{aligned} f(a+h) - f(a) &= f(a_1 + h_1, a_2 + h_2) - f(a_1, a_2) \\ &= f(a_1 + h_1, a_2 + h_2) - f(a_1, a_2 + h_2) \\ &\quad + f(a_1, a_2 + h_2) - f(a_1, a_2). \end{aligned} \tag{2.25}$$

Thinking of the first two terms as functions of the first argument only, and thinking of the last two terms as functions of the second term only, and applying the Mean

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Value Theorem to each pair of terms, we obtain

$$\begin{aligned} f(a+h) - f(a) = & \frac{\partial f}{\partial x_1}(c_1, a_2 + h_2)h_1 \\ & + \frac{\partial f}{\partial x_2}(a_1, d_2)h_2, \end{aligned} \tag{2.26}$$

where $a_1 < c_1 < a_1 + h_1$ and $a_2 < d_2 < a_2 + h_2$. This can be rewritten as

$$f(a+h) - f(a) = \frac{\partial f}{\partial x_1}(c)h_1 + \frac{\partial f}{\partial x_2}(d)h_2, \tag{2.27}$$

where $c = (c_1, a_2 + h_2)$ and $d = (a_1, d_2)$.

We want to show that $(f(a+h) - f(a) - Df(a)h)/|h| \to 0$ as $h \to 0$, where $Df(a) = \left[\frac{\partial f}{\partial x_1}(a), \frac{\partial f}{\partial x_2}(a)\right]$. Using our previously derived expression for $f(a+h)-f(a)$, we find that

$$\begin{aligned} f(a+h) - f(a) - Df(a)h &= f(a+h) - f(a) - \frac{\partial f}{\partial x_1}(a)h_1 - \frac{\partial f}{\partial x_2}(a)h_2 \\ &= \left(\frac{\partial f}{\partial x_1}(c) - \frac{\partial f}{\partial x_1}(a)\right)h_1 + \left(\frac{\partial f}{\partial x_2}(d) - \frac{\partial f}{\partial x_2}(a)\right)h_2. \end{aligned} \tag{2.28}$$

We can use the sup norm to show that

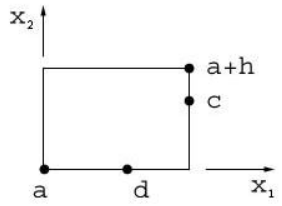
$$|f(a+h)-f(a)-Df(a)h| \leq \left|\frac{\partial f}{\partial x_1}(c) - \frac{\partial f}{\partial x_1}(a)\right| |h_1| + \left|(\frac{\partial f}{\partial x_2}(d) - \frac{\partial f}{\partial x_2}(a)\right| |h_2|, \tag{2.29}$$

from which it follows that

$$\frac{|f(a+h) - f(a) - Df(a)h|}{|h|} \leq \left|\frac{\partial f}{\partial x_1}(c) - \frac{\partial f}{\partial x_1}(a)\right| + \left|(\frac{\partial f}{\partial x_2}(d) - \frac{\partial f}{\partial x_2}(a)\right|, \tag{2.30}$$

where we used the fact that $|h| = \max(|h_1|, |h_2|)$.

Notice that as $h \to 0$, both $c \to a$ and $d \to a$, as can be easily seen using the following diagram. This means that the r.h.s. of Equation (2.30) goes to zero as h



goes to zero, because the partial derivatives are continuous. It follows that the l.h.s. goes to zero, which completes our proof.

The proof in n dimensions is similar to the above proof, but the details are harder to follow. □

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We now introduce a useful class of functions.

Definition 2.8. Given $U \subseteq \mathbb{R}^n$ and $f : U \to \mathbb{R}$, we define

$$f \in \mathcal{C}^1(U) \iff \frac{\partial f}{\partial x_i}, i = 1, \ldots, n \text{ exist and are continuous at all points } x \in U. \tag{2.31}$$

Similarly, we define

$$f \in \mathcal{C}^2(U) \iff \frac{\partial f}{\partial x_i} \in \mathcal{C}^1(U), i = a, \ldots, n. \tag{2.32}$$

$$f \in \mathcal{C}^k(U) \iff \frac{\partial f}{\partial x_i} \in \mathcal{C}^{k-1}(U), i = a, \ldots, n. \tag{2.33}$$

$$f \in \mathcal{C}^\infty(U) \iff f \in \mathcal{C}^k(U) \forall k. \tag{2.34}$$

If f is multiply differentiable, then you can perform higher order mixed partial derivatives.

One of the fundamental theorems of calculus is that the order of the partial derivatives can be taken in any order. For example,

$$\frac{\partial}{\partial x_i}\left(\frac{\partial}{\partial x_j}\right) = \frac{\partial}{\partial x_j}\left(\frac{\partial}{\partial x_i}\right) \equiv \frac{\partial^2 f}{\partial x_i \partial x_j} \tag{2.35}$$

Let's do the proof for this case. Let $U \subseteq \mathbb{R}^2$ and $f = f(x_1, x_2)$. We prove the following claim:

Claim.

$$\frac{\partial}{\partial x_i}\left(\frac{\partial}{\partial x_j}\right) = \frac{\partial}{\partial x_j}\left(\frac{\partial}{\partial x_i}\right). \tag{2.36}$$

Proof. Take $a \in U$ written in components as $a = (a_1, a_2)$, and take $h = (h_1, h_2) \in \mathbb{R}^2 - \{0\}$ such that $a + h \in U$. That is, take $h \approx 0$.

Define

$$\Delta(h) = f(a_1 + h_1, a_2 + h_2) - f(a_1, a_2 + h_2) - f(a_1 + h_1, a_2) + f(a_1, a_2), \tag{2.37}$$

and define

$$\phi(s) = f(a_1 + h_1, s) - f(a_1, s), \tag{2.38}$$

where $a_2 \leq s \leq a_2 + h_2$. We find that

$$\begin{aligned} \Delta(h) &= \phi(a_2 + h_2) - \phi(a_2) \\ &= \phi'(c_2)h_2, a_2 < c_2 < a_2 + h_2, \end{aligned} \tag{2.39}$$

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by the Mean Value Theorem. Writing out ϕ' using partial derivatives of f, and using the Mean Value Theorem again, we find

$$\begin{aligned}\Delta(h) &= \left(\frac{\partial f}{\partial x_2}(a_1+h_1,c_2)-\frac{\partial f}{\partial x_1}(a_1,c_2)\right)h_2\\ &= \left(\frac{\partial}{\partial x_1}\left(\frac{\partial f}{\partial x_2}(c_1,c_2)h_1\right)\right)h_2, a_1<c_1<a_1+h_1\\ &= \left(\frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_2}f\right)\right)(c)h_1h_2\\ &= \left(\frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_1}f\right)\right)(d)h_1h_2,\end{aligned} \tag{2.40}$$

where we obtained the last line by symmetry. This shows that

$$\frac{\partial}{\partial x_1}\left(\frac{\partial f}{\partial x_2}\right)(c)=\frac{\partial}{\partial x_2}\left(\frac{\partial f}{\partial x_1}\right)(d). \tag{2.41}$$

As $h\to 0, c\to a$ and $d\to a$, so

$$\frac{\partial}{\partial x_1}\left(\frac{\partial f}{\partial x_2}\right)(a)=\frac{\partial}{\partial x_2}\left(\frac{\partial f}{\partial x_1}\right)(a), \tag{2.42}$$

for any $a\in U$. □

The above argument can be iterated for $f\in\mathcal{C}^3(U)$, $f\in\mathcal{C}^3(4)$, etc.

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Lecture 5

2.3 Chain Rule

Let U and v be open sets in $\mathbb{R}^n$. Consider maps $f : U \to V$ and $g : V \to \mathbb{R}^k$. Choose $a \in U$, and let $b = f(a)$. The composition $g \circ f : U \to \mathbb{R}^k$ is defined by $(g \circ f)(x) = g(f(x))$.

Theorem 2.9. *If f is differentiable at a and g is differentiable at b, then $g \circ f$ is differentiable at a, and the derivative is*

$$(Dg \circ f)(a) = (Dg)(b) \circ Df(a). \tag{2.43}$$

Proof. This proof follows the proof in Munkres by breaking the proof into steps.

- Step 1: Let $h \in \mathbb{R}^n - \{0\}$ and $h \doteq 0$, by which we mean that h is very close to zero. Consider $\Delta(h) = f(a+h) - f(a)$, which is continuous, and define

$$F(h) = \frac{f(a+h) - f(a) - Df(a)h}{|a|}. \tag{2.44}$$

Then f is differentiable at a if and only if $F(h) \to 0$ as $h \to 0$.

$$F(h) = \frac{\Delta(h) - Df(a)h}{|h|}, \tag{2.45}$$

so

$$\Delta(h) = Df(a)h + |h|F(h). \tag{2.46}$$

Lemma 2.10.

$$\frac{\Delta(h)}{|h|} \text{ is bounded.} \tag{2.47}$$

Proof. Define

$$|Df(a)| = \sup_i \left| \frac{\partial f}{\partial x_i}(a) \right|, \tag{2.48}$$

and note that

$$\frac{\partial f}{\partial x_i}(a) = Df(a)e_i, \tag{2.49}$$

where the e_i are the standard basis vectors of $\mathbb{R}^n$. If $h = (h_1, \ldots, h_n)$, then $h = \sum h_i e_i$. So, we can write

$$Df(a)h = \sum h_i Df(a)e_i = \sum h_i \frac{\partial f}{\partial x_i}(a). \tag{2.50}$$

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It follows that

$$\begin{aligned} |Df(a)h| &\leq \sum_{i=1}^{m} h_i \left| \frac{\partial f}{\partial x_i}(a) \right| \\ &\leq m|h||Df(a)|. \end{aligned} \tag{2.51}$$

By Equation 2.46,

$$|\Delta(h)| \leq m|h||Df(a)| + |h|F(h), \tag{2.52}$$

so

$$\frac{|\Delta(h)|}{|h|} \leq m|Df(a)| + F(h). \tag{2.53}$$

□

- Step 2: Remember that $b = f(a)$, $g : V \to \mathbb{R}^k$, and $b \in V$. Let $k \doteq 0$. This means that $k \in \mathbb{R}^n - \{0\}$ and that k is close to zero. Define

$$G(k) = \frac{g(b+k) - g(b) - (Dg)(b)k}{|k|}, \tag{2.54}$$

so that

$$g(b+k) - g(b) = Dg(b)k + |k|G(k). \tag{2.55}$$

We proceed to show that $g \circ f$ is differentiable at a.

$$\begin{aligned} g \circ f(a+h) - g \circ f(a) &= g(f(a+h)) - g(f(a)) \\ &= g(b + \Delta(h)) - g(b), \end{aligned} \tag{2.56}$$

where $f(a) = b$ and $f(a+h) = f(a) + \Delta(h) = b + \Delta(h)$. Using Equation 2.55 we see that the above expression equals

$$Dg(b)\Delta(h) + |\Delta(h)|G(\Delta(h)). \tag{2.57}$$

Substituting in from Equation 2.46, we obtain

$$\begin{aligned} g \circ f(a+h) - g \circ f(a) &= \ldots \\ &= Dg(b)(Df(a)h + |h|F(h)) + \ldots \\ &= Dg(b) \circ Df(a)h + |h|Dg(b)F(h) + |\Delta(h)|G(\Delta(h)) \end{aligned} \tag{2.58}$$

This shows that

$$\frac{g \circ f(a+h) - g \circ f(a) - Dg(b) \circ Df(a)h}{|h|} = Dg(b)F(h) + \frac{\Delta(h)}{|h|}G(\Delta(h)). \tag{2.59}$$

We see in the above equation that $g \circ f$ is differentiable at a if and only if the l.h.s. goes to zero as $h \to 0$. It suffices to show that the r.h.s. goes to zero as $h \to 0$, which it does: $F(h) \to 0$ as $h \to 0$ because f is differentiable at a; $G(\Delta(h)) \to 0$ because g is differentiable at b; and $\Delta(h)/|h|$ is bounded.

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□

We consider the same maps g and f as above, and we write out f in component form as $f = (f_1, \ldots, f_n)$ where each $f_i : U \to \mathbb{R}$. We say that f is a $\mathcal{C}^r$ map if each $f_i \in \mathcal{C}^r(U)$. We associate $Df(x)$ with the matrix

$$Df(x) \sim \left[\frac{\partial f_i}{\partial x_j}(x)\right]. \tag{2.60}$$

By definition, f is $\mathcal{C}^r$ (that is to say $f \in \mathcal{C}^r(U)$) if and only if Df is $\mathcal{C}^{r-1}$.

Theorem 2.11. *If $f : U \to V \subseteq \mathbb{R}^n$ is a $\mathcal{C}^r$ map and $g : V \to \mathbb{R}^p$ is a $\mathcal{C}^r$ map, then $g \circ f : U \to \mathbb{R}^p$ is a $\mathcal{C}^r$ map.*

Proof. We only prove the case $r = 1$ and leave the general case, which is inductive, to the student.

- Case $r = 1$:

$$Dg \circ f(x) = Dg(f(x)) \circ Df(x) \sim \left[\frac{\partial g_i}{\partial x_j} f(x)\right]. \tag{2.61}$$

  The map g is $\mathcal{C}^1$, which implies that $\partial g_i / \partial x_j$ is continuous. Also,

$$Df(x) \sim \left[\frac{\partial f_i}{\partial x_j}\right] \tag{2.62}$$

  is continuous. It follows that $Dg \circ f(x)$ is continuous. Hence, $g \circ f$ is $\mathcal{C}^1$.

□

2.4 The Mean-value Theorem in n Dimensions

Theorem 2.12. *Let U be an open subset of $\mathbb{R}^n$ and $f : U \to \mathbb{R}$ a $\mathcal{C}^1$ map. For $a \in U$, $h \in \mathbb{R}^n$, and $h \doteq 0$,*

$$f(a + h) - f(a) = Df(c)h, \tag{2.63}$$

where c is a point on the line segment $a + th, 0 \leq t \leq 1$, joining a to $a + h$.

Proof. Define a map $\phi : [0, 1] \to \mathbb{R}$ by $\phi(t) = f(a + th)$. The Mean Value Theorem implies that $\phi(1) - \phi(0) = \phi'(c) = (Df)(c)h$, where $0 < c < 1$. In the last step we used the chain rule. □

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2.5 Inverse Function Theorem

Let U and V be open sets in $\mathbb{R}^n$, and let $f : U \to V$ be a $\mathcal{C}^1$ map. Suppose there exists a map $g : V \to U$ that is the inverse map of f (which is also $\mathcal{C}^1$). That is, $g(f(x)) = x$, or equivalently $g \circ f$ equals the identity map.

Using the chain rule, if $a \in U$ and $b = f(a)$, then

$$(Dg)(b) = \text{ the inverse of } Df(a). \tag{2.64}$$

That is, $Dg(b) \circ Df(a)$ equals the identity map. So,

$$Dg(b) = (Df(a))^{-1} \tag{2.65}$$

However, this is not a trivial matter, since we do not know if the inverse exists. That is what the inverse function theorem is for: if $Df(a)$ is invertible, then g exists for some neighborhood of a in U and some neighborhood of $f(a)$ in V. We state this more precisely in the following lecture.

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Lecture 6

We begin with a review of some earlier definitions.
Let $\delta > 0$ and $a \in \mathbb{R}^n$.

$$\text{Euclidean ball: } B_\delta(a) = \{x \in \mathbb{R}^n : \| x - a \| < \delta\} \tag{2.66}$$

$$\begin{aligned}\text{Supremum ball: } R_\delta(a) &= \{x \in \mathbb{R}^n : |x - a| < \delta\} \\ &= I_1 \times \cdots \times I_n,\ I_j = (a_j - \delta, a_j + \delta).\end{aligned} \tag{2.67}$$

Note that the supremum ball is actually a rectangle. Clearly, $B_\delta(a) \subseteq R_\delta(a)$. We use the notation $B_\delta = B_\delta(0)$ and $R_\delta = R_\delta(0)$.

Continuing with our review, given U open in $\mathbb{R}^n$, a map $f : U \to \mathbb{R}^k$, and a point $a \in U$, we defined the derivate $Df(a) : \mathbb{R}^n \to \mathbb{R}^k$ which we associated with the matrix

$$Df(a) \sim \left[\frac{\partial f_i}{\partial x_j}(a)\right], \tag{2.68}$$

and we define

$$|Df(a)| = \sup_{i,j} \left|\frac{\partial f_i}{\partial x_j}(a)\right|. \tag{2.69}$$

Lastly, we define $U \subseteq \mathbb{R}^n$ to be *convex* if

$$a, b \in U \implies (1 - t)a + tb \in U \text{ for all } 0 \leq t \leq 1. \tag{2.70}$$

Before we state and prove the Inverse Function Theorem, we give the following definition.

Definition 2.13. Let U and V be open sets in $\mathbb{R}^n$ and $f : U \to V$ a $\mathcal{C}^r$ map. The map f is is a $\mathcal{C}^r$ *diffeomorphism* if it is bijective and $f^{-1} : V \to U$ is also $\mathcal{C}^r$.

Inverse Function Theorem. *Let U be an open set in $\mathbb{R}^n$, $f : U \to \mathbb{R}^n$ a $\mathcal{C}^r$ map, and $a \in U$. If $Df(a) : \mathbb{R}^n \to \mathbb{R}^n$ is bijective, then there exists a neighborhood U_1 of a in U and a neighborhood V of $f(a)$ in $\mathbb{R}^n$ such that $F|U_1$ is a $\mathcal{C}^r$ diffeomorphism of U_1 at V.*

Proof. To prove this we need some elementary multi-variable calculus results, which we provide with the following lemmas.

Lemma 2.14. *Let U be open in $\mathbb{R}^n$ and $F : U \to \mathbb{R}^k$ be a $\mathcal{C}^1$ mapping. Also assume that U is convex. Suppose that $|Df(a)| \leq c$ for all $A \in U$. Then, for all $x, y \in U$,*

$$|f(x) - f(y)| \leq nc|x - y|. \tag{2.71}$$

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Proof. Consider any $x, y \in U$. The Mean Value Theorem says that for every i there exists a point c on the line joining x to y such that

$$f_i(x) - f_i(y) = \sum_j \frac{\partial f_i}{\partial x_j}(d)(x_j - y_j). \tag{2.72}$$

It follows that

$$\begin{aligned} |f_i(x) - f_i(y)| &\leq \sum_j \left| \frac{\partial f_i}{\partial x_j}(d) \right| |x_j - y_j| \\ &\leq \sum_j c|x_j - y_j| \\ &\leq nc|x - y| \end{aligned} \tag{2.73}$$

This is true for each i, so $|f(x) - f(y)| \leq nc|x - y|$ □

Lemma 2.15. *Let U be open in $\mathbb{R}^n$ and $f : U \to \mathbb{R}$ a $\mathcal{C}^1$ map. Suppose f takes a minimum value at some point $b \in U$. Then*

$$\frac{\partial f}{\partial x_i}(b) = 0, \ i = 1, \ldots, n. \tag{2.74}$$

Proof. We reduce to the one-variable result. Let $b = (b_1, \ldots, b_n)$ and let $\phi(t) = f(b_1, \ldots, b_{i-1}, t, b_{i+1}, \ldots, b_n)$, which is $\mathcal{C}^1$ near b_1 and has a minimum at b_i. We know from one-variable calculus that this implies that $\frac{\partial \phi}{\partial t}(b_i) = 0$. □

In our proof of the Inverse Function Theorem, we want to show that f is locally a diffeomorphism at a. We will make the following simplifying assumptions:

$$a = 0, \ \ f(a) = 0, \ \ Df(0) = I \text{ (identity)}. \tag{2.75}$$

Then, we define a map $g : U \to \mathbb{R}^n$ by $g(x) = x - f(x)$, so that we obtain the further simplification

$$Dg(0) = Df(0) - I = 0. \tag{2.76}$$

Lemma 2.16. *Given $\epsilon > 0$, there exists $\delta > 0$ such that for any $x, y \in R_\delta$,*

$$|g(x - g(y)| < \epsilon|x - y|. \tag{2.77}$$

Proof. The result that $Dg(0) = 0$ implies that there exists $\delta > 0$ such that for any $x \in R_\delta$, $|Dg(x)| \leq \epsilon/n$. Applying the first lemma, the proof is complete. □

Now, remember that $g(x) = x - f(x)$. Take any $x, y \in R_\delta$. Then

$$\begin{aligned} x - y &= x - f(x) + f(x) - f(y) + f(y) - y \\ &= g(x) - g(y) + f(x) - f(y). \end{aligned} \tag{2.78}$$

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Using the Triangle Inequality we obtain

$$|x-y| \leq |g(x)-g(y)| + |f(x)-f(y)| \tag{2.79}$$

Using the previous lemma, we find that

$$(1-\epsilon)|x-y| \leq |f(x)-f(y)|. \tag{2.80}$$

We choose δ such that $\epsilon > 1/2$, so that

$$|x-y| \leq 2|f(x)-f(y)|. \tag{2.81}$$

This proves that $f : R_\delta \to \mathbb{R}^n$ is one-to-one.

We also want to prove that f is onto. We have $Df(0) = I$, so $\det(\frac{\partial f_i}{\partial x_j}(0)) = 1$. We can choose δ such that for any $x \in R_\delta$,

$$\det\left(\frac{\partial f_i}{\partial x_j}(x)\right) > \frac{1}{2}. \tag{2.82}$$

Lemma 2.17. *If $y \in B_{\delta/4}$, than there exists a point $c \in R_\delta$ such that $f(c) = y$.*

Proof. Let $h : \bar{R}_\delta \to \mathbb{R}$ be a map defined by $h(x) = \| f(x) - y \|^2$. The domain $\bar{R}_\delta$ is compact, so h has a minimum at some point $c \in \bar{R}_\delta$.

Claim. *The point c is an interior point. That is, $c \in R_\delta$.*

Proof. For any $x \in \bar{R}_\delta$, $|x| = \delta$ implies that $|f(x) - f(0)| = |f(x)| \geq \delta/2$

$$\begin{aligned} &\implies \| f(x) \| \geq \frac{\delta}{2} \\ &\implies \| f(x) - y \| \geq \frac{\delta}{4}, \text{ when } x \in \text{Bd } R_\delta. \\ &\implies h(x) \geq \left(\frac{\delta}{4}\right)^2. \end{aligned} \tag{2.83}$$

At the origin, $h(0) = \| f(0) - y \|^2 = \| y \|^2 < (\delta/4)^2$, since $y \in B_{\delta/4}$. So, $h(0) \leq h$ on Bd R_δ, which means that the minimum point c of h is in R_δ. This ends the proof of the claim. □

Now that we know that the minimum point c occurs in the interior, we can apply the second lemma to h to obtain

$$\frac{\partial h}{\partial x_j}(c) = 0, \quad j = 1, \ldots, n. \tag{2.84}$$

From the definition of h,

$$h(x) = \sum_{i=1}^{n} (f_i(c) - y_i)\frac{\partial f_i}{\partial x_j}(c) = 0, \ i = 1, \ldots, n, \tag{2.85}$$

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so

$$\frac{\partial h}{\partial x_j}(c) = 2\sum_{i-1}^{n}(f_i(c) - y_i)\frac{\partial f_i}{\partial x_j}(c) = 0,\ i = 1, \ldots, n. \tag{2.86}$$

Note that

$$\det\left[\frac{\partial f_i}{\partial x_j}(c)\right] \neq 0, \tag{2.87}$$

so, by Cramer's Rule,

$$f_i(c) - y_i = 0,\ i = 1, \ldots, n. \tag{2.88}$$

□

Let $U_1 = R_\delta \sim f^{-1}(B_{\delta/4})$, where we have chosen $V = B_{\delta/4}$. We have shown that f is a bijective map.

Claim. *The map $f^{-1} : V \to U_1$ is continuous.*

Proof. Let $a, b \in V$, and define $x = f^{-1}(a)$ and $y = f^{-1}(b)$. Then $a = f(x)$ and $b = f(y)$.

$$|a - b| = |f(x) - f(y)| \geq \frac{\partial |x - y|}{\partial 2}, \tag{2.89}$$

so

$$|a - b| \geq \frac{1}{2}|f^{-1}(a) - f^{-1}(b)|. \tag{2.90}$$

This shows that f^{-1} is continuous on $V = B_{\delta/4}$. □

As a last item for today's lecture, we show the following:

Claim. *The map f^{-1} is differentiable at 0, and $Df^{-1}(0) = I$.*

Proof. Let $k \in \mathbb{R}^n - \{0\}$ and choose $k \doteq 0$. We are trying to show that

$$\frac{f^{-1}(0 + k) - f^{-1}(0) - Df^{-1}(0)k}{|k|} \to 0 \text{ as } k \to 0. \tag{2.91}$$

We simplify

$$\frac{f^{-1}(0 + k) - f^{-1}(0) - Df^{-1}(0)k}{|k|} = \frac{f^{-1}(k) - k}{|k|}. \tag{2.92}$$

Define $h = f^{-1}(k)$ so that $k = f(h)$ and $|k| \leq 2|h|$. To show that

$$\frac{f^{-1}(k) - k}{|k|} \to 0 \text{ as } k \to 0, \tag{2.93}$$

it suffices to show that

$$\frac{f^{-1}(k) - k}{|h|} \to 0 \text{ as } h \to 0. \tag{2.94}$$

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That is, it suffices to show that

$$\frac{h - f(h)}{|h|} \to 0 \text{ as } h \to 0. \tag{2.95}$$

But this is equal to

$$-\frac{f(h) - f(0) - Df(0)h}{|h|}, \tag{2.96}$$

which goes to zero as $h \to 0$ because f is differentiable at zero. □

The proof of the Inverse Function Theorem continues in the next lecture.

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Lecture 7

We continue our proof of the Inverse Function Theorem.

As before, we let U be an open set in $\mathbb{R}^n$, and we assume that $0 \in U$. We let $f : U \to \mathbb{R}^n$ be a $\mathcal{C}^1$ map, and we assume $f(0) = 0$ and that $Df(0) = I$. We summarize what we have proved so far in the following theorem.

Theorem 2.18. *There exists a neighborhood U_0 of 0 in U and a neighborhood V of 0 in $\mathbb{R}^n$ such that*

1. *f maps U_0 bijectively onto V*
2. *$f^{-1} : V \to U_0$ is continuous,*
3. *f^{-1} is differentiable at 0.*

Now, we let U be an open set in $\mathbb{R}^n$, and we let $f : U \to \mathbb{R}^n$ be a $\mathcal{C}^2$ map, as before, but we return to our original assumptions that $a \in U$, $b = f(a)$, and $Df(a) : \mathbb{R}^n \to \mathbb{R}^n$ is bijective. We prove the following theorem.

Theorem 2.19. *There exists a neighborhood U_0 of a in U and a neighborhood V of b in $\mathbb{R}^n$ such that*

1. *f maps U_0 bijectively onto V*
2. *$f^{-1} : V \to U_0$ is continuous,*
3. *f^{-1} is differentiable at b.*

Proof. The map $f : U \to \mathbb{R}^n$ maps a to b. Define $U' = U - a = \{x - a : x \in U\}$. Also define $f_1 : U' \to \mathbb{R}^n$ by $f_1(x) = f(x + a) - b$, so that $f_1(0) = 0$ and $Df_1(0) = Df(a)$ (using the Chain Rule).

Let $A = Df(a) = Df_1(0)$. We know that A is invertible.

Now, define $f_2 : U' \to \mathbb{R}^n$ by $f_2 = A^{-1} f_1$, so that $f_2(0) = 0$ and $Df_2(0) = I$. The results from last lecture show that the theorem at hand is true for f_2. Because $f_1 = A \circ f_2$, the theorem is also true for f_1. Finally, because $f(x) = f_1(x - a) + b$, the theorem is true for f. □

So, we have a bijective map $f : U_0 \to V$. Let us take $c \in U_0$ and look at the derivative

$$Df(c) \sim \left[\frac{\partial f_i}{\partial x_j}(c)\right] = J_f(c). \tag{2.97}$$

Note that

$$Df(c) \text{ is bijective} \iff \det\left[\frac{\partial f_i}{\partial x_j}(c)\right] \neq 0. \tag{2.98}$$

Because f is $\mathcal{C}^1$, the functions $\frac{\partial f_i}{\partial x_j}$ are continuous on U_0. If $\det J_f(a) \neq 0$, then $\det J_f(c) \neq 0$ for c close to a. We can shrink U_0 and V such that $\det J_f(c) \neq 0$ for

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all $c \in U_0$, so for every $c \in U_0$, the map f^{-1} is differentiable at $f(c)$. That is, f^{-1} is differentiable at all points of V.

We have thus improved the previous theorem. We can replace the third point with

$$3.\ f^{-1} \text{ is differentiable at all points of } V. \tag{2.99}$$

Let $f^{-1} = g$, so that $g \circ f =$ identity map. The Chain Rule is used to show the following. Suppose $p \in U_0$ and $q = f(p)$. Then $Dg(q) = Df(p)^{-1}$, so $J_g(q) = J_f(p)^{-1}$. That is, for all $x \in V$,

$$\left[\frac{\partial g_i}{\partial x_j}(x)\right] = \left[\frac{\partial f_i}{\partial x_j}(g(x))\right]^{-1}. \tag{2.100}$$

The function f is $\mathcal{C}^1$, so $\frac{\partial f_i}{\partial x_j}$ is continuous on U_0. It also follows that g is continuous, so $\frac{\partial f_i}{\partial x_j}(g(x))$ is continuous on V.

Using Cramer's Rule, we conclude that the entries of matrix on the r.h.s. of Equation 2.100 are continuous functions on V. This shows that $\frac{\partial f_i}{\partial x_j}$ is continuous on V, which implies that g is a $\mathcal{C}^1$ map.

We leave as an exercise to show that $f \in \mathcal{C}^r$ implies that $g \in \mathcal{C}^r$ for all r. The proof is by induction.

This concludes the proof of the Inverse Function Theorem, signifying the end of this section of the course.

□

3 Integration

3.1 Riemann Integral of One Variable

We now begin to study the next main topic of this course: integrals. We begin our discussion of integrals with an 18.100 level review of integrals.

We begin by defining the Riemann integral (sometimes written in shorthand as the R. integral).

Let $[a, b] \subseteq \mathbb{R}$ be a closed interval in $\mathbb{R}$, and let P be a finite subset of $[a, b]$. Then P is a *partition* if $a, b \in P$ and if all of the elements $t_i, \ldots, t_N$ in P can be arranged such that $t_1 = a < t_2 < \cdots < t_n = b$. We define $I_i = [t_i, t_{i+1}]$, which are called the subintervals of $[a, b]$ belonging to P.

Let $f : [a, b] \to \mathbb{R}$ be a bounded function, and let I_i be a subinterval belonging to P. Then we define

$$\begin{aligned} m_i &= \inf f : I_i \to \mathbb{R} \\ M_i &= \sup f : I_i \to \mathbb{R}, \end{aligned} \tag{3.1}$$

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from which we define the *lower* and *upper* Riemann sums

$$\begin{aligned} L(f,P) &= \sum_i m_i \times (\text{length of } I_i) \\ U(f,P) &= \sum_i M_i \times (\text{length of } I_i), \end{aligned} \tag{3.2}$$

respectively.

Clearly,

$$L(f,P) \leq U(f,P). \tag{3.3}$$

Now, let P and P' be partitions.

Definition 3.1. The partition P is a *refinement* of P if $P' \supset P$.

If P' is a refinement of P, then

$$\begin{aligned} &L(f,P') \geq L(f,P), \text{ and} \\ &U(f,P') \leq U(f,P). \end{aligned} \tag{3.4}$$

If P and P' are *any* partitions, then you can take $P'' = P \cup P'$, which is a refinement of both P and P'. So,

$$L(f,P) \leq L(f,P'') \leq U(f,P'') \leq U(f,P') \tag{3.5}$$

for any partitions P, P'. That is, the lower Riemann sum is always less than or equal to the upper Riemann sum, regardless of the partitions used.

Now we can define the Lower and Upper Riemann integrals

$$\begin{aligned} \underline{\int}_{[a,b]} f &= \text{l.u.b. } \{L(f,P) | P \text{ a partition of } [a,b]\} \\ \overline{\int}_{[a,b]} f &= \text{g.l.b. } \{U(f,P) | P \text{ a partition of } [a,b]\} \end{aligned} \tag{3.6}$$

We can see from the above that

$$\underline{\int} f \leq \overline{\int} f. \tag{3.7}$$

Claim. *If f is continuous, then*

$$\underline{\int} f = \overline{\int} f. \tag{3.8}$$

Definition 3.2. For any bounded function $f : [a,b] \to \mathbb{R}$, the function f is *(Riemann) integrable* if

$$\underline{\int}_{[a,b]} f = \overline{\int}_{[a,b]} f. \tag{3.9}$$

In the next lecture we will begin to generalize these notions to multiple variables.

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Lecture 8

3.2 Riemann Integral of Several Variables

Last time we defined the Riemann integral for one variable, and today we generalize to many variables.

Definition 3.3. A *rectangle* is a subset Q of $\mathbb{R}^n$ of the form

$$Q = [a_1, b_1] \times \cdots \times [a_n, b_n], \tag{3.10}$$

where $a_i, b_i \in \mathbb{R}$.

Note that $x = (x_1, \ldots, x_n) \in Q \iff a_i \leq x_i \leq b_i$ for all i. The volume of the rectangle is

$$v(Q) = (b_1 - a_1) \cdots (b_n - a_n), \tag{3.11}$$

and the width of the rectangle is

$$\text{width}(Q) = \sup_i (b_i - a_i). \tag{3.12}$$

Recall (stated informally) that given $[a, b] \in \mathbb{R}$, a finite subset P of $[a, b]$ is a partition of $[a, b]$ if $a, b \in P$ and you can write $P = \{t_i : i = 1, \ldots, N\}$, where $t_1 = a < t_2 < \ldots < t_N = b$. An interval I belongs to P if and only if I is one of the intervals $[t_i, t_{i+1}]$.

Definition 3.4. A *partition P of Q* is an n-tuple $(P_1, \ldots, P_n)$, where each P_i is a partition of $[a_i, b_i]$.

Definition 3.5. A rectangle $R = I_1 \times \cdots \times I_n$ *belongs to P* if for each i, the interval I_i belongs to P_i.

Let $f : Q \to \mathbb{R}$ be a bounded function, let P be a partition of Q, and let R be a rectangle belonging to P.

We define

$$\begin{aligned} m_R f &= \inf_R f = \text{g.l.b. } \{f(x) : x \in \mathbb{R}\} \\ M_R f &= \sup_R f = \text{l.u.b. } \{f(x) : x \in \mathbb{R}\}, \end{aligned} \tag{3.13}$$

from which we define the lower and upper Riemann sums,

$$\begin{aligned} L(f, P) &= \sum_R m_R(f) v(R) \\ U(f, P) &= \sum_R M_R(f) v(R). \end{aligned} \tag{3.14}$$

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It is evident that

$$L(f, P) \leq U(f, P). \tag{3.15}$$

Now, we will take a sequence of partitions that get finer and finer, and we will define the integral to be the limit.

Let $P = (P_1, \ldots, P_n)$ and $P' = (P'_1, \ldots, P'_n)$ be partitions of Q. We say that P' refines P if $P'_i \supset P_i$ for each i.

Claim. *If P' refines P, then*

$$L(f, P') \geq L(f.P). \tag{3.16}$$

Proof. We let $P_j = P'_j$ for $j \neq i$, and we let $P'_i = P_i \cup \{a\}$, where $a \in [a_i, b_i]$. We can create any refinement by multiple applications of this basic refinement. If R is a rectangle belonging to P, then either

1. R belongs to P', or
2. $R = R' \cup R''$, where R', R'' belong to P'.

In the first case, the contribution of R to $L(f, P')$ equals the contribution of R to $L(f, P)$, so the claim holds.

In the second case,

$$m_R v(R) = m_R(v(R') + v(R'')) \tag{3.17}$$

and

$$m_r = \inf_R f \leq \inf_{R'} f, \inf_{R''} f. \tag{3.18}$$

So,

$$m_R \leq m_{R'}, m_{R''} \tag{3.19}$$

Taken altogether, this shows that

$$m_R v(R) \leq m_{R'} v(R') + m_{R''} v(R'') \tag{3.20}$$

Thus, R' and R'' belong to P'. □

Claim. *If P' refines P, then*

$$U(f, P') \leq U(f, P) \tag{3.21}$$

The proof is very similar to the previous proof. Combining the above two claims, we obtain the corollary

Corollary 2. *If P and P' are partitions, then*

$$U(f, P') \geq L(f, P) \tag{3.22}$$

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Proof. Define $P'' = (P_1 \cup P_1', \ldots, P_n \cup P_n')$. So, P'' refines P and P'. We have shown that

$$\begin{aligned} U(f, P'') &\leq U(f, P) \\ L(f, P') &\leq L(f, P'') \\ L(f, P'') &\leq U(f, P''). \end{aligned} \tag{3.23}$$

Together, these show that

$$U(f, P) \geq L(f, P'). \tag{3.24}$$

□

With this result in mind, we define the lower and upper Riemann integrals:

$$\begin{aligned} \underline{\int_Q} f &= \sup_P L(f, P) \\ \overline{\int_Q} f &= \inf_P U(f, P). \end{aligned} \tag{3.25}$$

Clearly, we have

$$\underline{\int_Q} f \leq \overline{\int_Q} f, \tag{3.26}$$

Finally, we define Riemann integrable.

Definition 3.6. A function f is *Riemann integrable over* Q if the lower and upper Riemann integrals coincide (are equal).

3.3 Conditions for Integrability

Our next problem is to determine under what conditions a function is (Riemann) integrable.

Let's look at a trivial case:

Claim. *Let $F : Q \to \mathbb{R}$ be the constant function $f(x) = c$. Then f is R. integrable over Q, and*

$$\int_Q c = cv(Q). \tag{3.27}$$

Proof. Let P be a partition, and let R be a rectangle belonging to P. We see that $m_R(f) = M_R(f) = c$, so

$$\begin{aligned} U(f, P) &= \sum_R M_R(f) v(R) = c \sum_R v(R) \\ &= cv(Q). \end{aligned} \tag{3.28}$$

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Similarly,

$$L(f, P) = cv(Q). \tag{3.29}$$

□

Corollary 3. *Let Q be a rectangle, and let $\{Q_i : i = 1, \ldots, N\}$ be a collection of rectangles covering Q. Then*

$$v(Q) \leq \sum v(Q_i). \tag{3.30}$$

Theorem 3.7. *If $f : Q \to \mathbb{R}$ is continuous, then f is R. integrable over Q.*

Proof. We begin with a definition

Definition 3.8. Given a partition P of Q, we define

$$\text{mesh width}(P) = \sup_R \text{width}(R). \tag{3.31}$$

Remember that

$$Q \text{ compact } \implies f : Q \to \mathbb{R} \text{ is uniformly continuous.} \tag{3.32}$$

That is, given $\epsilon > 0$, there exists $\delta > 0$ such that if $x, y \in Q$ and $|x - y| < \delta$, then $|f(x) - f(y)| < \epsilon$.

Choose a partition P of Q with mesh width less than δ. Then, for every rectangle R belonging to P and for every $x, y \in R$, we have $|x - y| < \delta$. By uniform continuity we have, $M_R(f) - m_R(f) \leq \epsilon$, which is used to show that

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_R (M_R(f) - m_R(f))v(R) \\ &\leq \epsilon \sum v(R) \\ &\leq \epsilon v(Q). \end{aligned} \tag{3.33}$$

We can take $\epsilon \to 0$, so

$$\sup_P L(f, P) = \inf_P U(f, P), \tag{3.34}$$

which shows that f is integrable. □

We have shown that continuity is sufficient for integrability. However, continuity is clearly not necessary. What is the general condition for integrability? To state the answer, we need the notion of *measure zero*.

Definition 3.9. Suppose $A \subseteq \mathbb{R}^n$. The set A is of *measure zero* if for every $\epsilon > 0$, there exists a countable covering of A by rectangles $Q_1, Q_2, Q_3, \ldots$ such that $\sum_i v(Q_i) < \epsilon$.

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Theorem 3.10. *Let $f : Q \to \mathbb{R}$ be a bounded function, and let $A \subseteq Q$ be the set of points where f is* not *continuous. Then f is R. integrable if and only if A is of measure zero.*

Before we prove this, we make some observations about sets of measure zero:

1. Let $A, B \subseteq \mathbb{R}^n$ and suppose $B \subset A$. If A is of measure zero, then B is also of measure zero.

2. Let $A_i \subseteq \mathbb{R}^n$ for $i = 1, 2, 3, \ldots$, and suppose the A_i's are of measure zero. Then $\cup A_i$ is also of measure zero.

3. Rectangles are *not* of measure zero.

We prove the second observation:

For any $\epsilon > 0$, choose coverings $Q_{i,1}, Q_{i,2}, \ldots$ of A_i such that each covering has total volume less than $\epsilon/2^i$. Then $\{Q_{i,j}\}$ is a countable covering of $\cup A_i$ of total volume

$$\sum_{i=1}^{\infty} \frac{\epsilon}{2^i} = \epsilon. \tag{3.35}$$

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Lecture 9

We quickly review the definition of measure zero.

A set $A \subseteq \mathbb{R}^n$ is of *measure zero* if for every $\epsilon > 0$, there exists a covering of A by rectangles $Q_1, Q_2, Q_3, \ldots$ such that the total volume $\sum v(Q_i) < \epsilon$.

Remark. In this definition we can replace "rectangles" by "open rectangles." To see this, given any $\epsilon > 0$ let $Q_1, Q_2, \ldots$ be a cover of A with volume less than $\epsilon/2$. Next, choose Q_i' to be rectangles such that Int $Q_i' \supset Q_i$ and $v(Q_i') < 2v(Q_i)$. Then Int Q_1', Int $Q_2', \ldots$ cover A and have total volume less than ϵ.

We also review the three properties of measure zero that we mentioned last time, and we prove the third.

1. Let $A, B \subseteq \mathbb{R}^n$ and suppose $B \subset A$. If A is of measure zero, then B is also of measure zero.

2. Let $A_i \subseteq \mathbb{R}^n$ for $i = 1, 2, 3, \ldots$, and suppose the A_i's are of measure zero. Then $\cup A_i$ is also of measure zero.

3. Rectangles are *not* of measure zero.

We prove the third property:

Claim. *If Q is a rectangle, then Q is* not *of measure zero.*

Proof. Choose $\epsilon < v(Q)$. Suppose $Q_1, Q_2, \ldots$ are rectangles such that the total volume is less than ϵ and such that Int Q_1, Int $Q_2, \ldots$ cover Q.

The set Q is compact, so the H-B Theorem implies that the collection of sets Int $Q_1, \ldots,$ Int Q_N cover Q for N sufficiently large. So,

$$Q \subseteq \bigcup_{i=1}^{N} Q_i, \tag{3.36}$$

which implies that

$$v(Q) \leq \sum_{i=1}^{N} v(Q_i) < \epsilon < v(Q), \tag{3.37}$$

which is a contradiction. □

We then have the following simple result.

Claim. *If* Int *A is non-empty, then A is* not *of measure zero.*

Proof. Consider any $p \in$ Int A. There exists a $\delta > 0$ such that $U(p, \delta) = \{x : |x - p| < \delta\}$ is contained in A. Then let $Q = \{x : |x - p| \leq \delta\}$. It follows that if A is of measure zero, then Q is of measure zero, by the first property. We know that Q is not of measure zero by the third property. □

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We restate the necessary and sufficient condition for R. integrability from last time, and we now prove the theorem.

Theorem 3.11. *Let Q be a rectangle and $f : Q \to \mathbb{R}$ be a bounded function. Let D be the set of points in Q where f is not continuous. Then f is R. integrable if and only if D is of measure zero.*

Proof. First we show that

$$D \text{ is of measure zero} \implies f \text{ is R. integrable} \tag{3.38}$$

Lemma 3.12. *Let $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$, and let $Q^\alpha, \alpha = 1, \ldots, N$, be a covering of Q by rectangles. Then there exists a partition P of Q such that every rectangle R belonging to P is contained in Q^α for some α.*

Proof. Write out $Q^\alpha = I_1^\alpha \times \cdots \times I_n^\alpha$, and let

$$P_j = \left(\bigcup_\alpha \text{Endpoints of } I_j^\alpha \right) \cap [a_j, b_j] \cup \{a_j, b_j\}. \tag{3.39}$$

One can show that P_j is a partition of $[a_j, b_j]$, and $P = (P_1, \ldots, P_n)$ is a partition of Q with the above properties. □

Let $f : Q \to \mathbb{R}$ be a bounded function, and let D be the set of points at which f is discontinuous. Assume that D is of measure zero. We want to show that f is R. integrable.

Let $\epsilon > 0$, and let $Q_i', i = 1, 2, 3, \ldots$ be a collection of rectangles of total volume less than ϵ such that Int $Q_1', Q_2', \ldots$ cover D.

If $p \in Q - D$, we know that f is continuous at p. So, there exists a rectangle Q_p with $p \in$ Int Q_p and $|f(x) - f(p)| < \epsilon/2$ for all $x \in Q_p$ (for example, $Q_p = \{x || x - p| \leq \delta\}$ for some δ). Given any $x, y \in Q_p$, we find that $|f(x) - f(y)| < \epsilon$.

The rectangles Int $Q_p, p \in Q - D$ along with the rectangles Int $Q_i', i = 1, 2, \ldots$ cover Q. The set Q is compact, so the H-B Theorem implies that there exists a finite open subcover:

$$Q_i \equiv \text{Int } Q_{p_i}, i = 1, \ldots, \ell; \quad \text{Int } Q_j', j = 1, \ldots, \ell. \tag{3.40}$$

Using the lemma, there exists a partition P of Q such that every rectangle belonging to P is contained in a Q_i or a Q_j'.

We now show that f is R. integrable.

$$\begin{aligned} U(f, P) - L(f, P) = &\sum_R (M_R(f) - m_R(f)) v(R) \\ &+ \sum_{R'} (M_{R'}(f) - m_{R'}(f)) v(R'), \end{aligned} \tag{3.41}$$

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where each R in the first sum belongs to a Q_i, and each R' in the second sum belongs to a Q'_j.

We look at the first sum. If $x, y \in R \subseteq Q_i$, then $|f(x) - f(y)| \leq \epsilon$. So, $M_R(f) - m_R(f) \leq \epsilon$. It follows that

$$\begin{aligned} \sum_R (M_R(f) - m_R(f))v(R) &\leq \epsilon \sum_R v(R) \\ &\leq \epsilon v(Q). \end{aligned} \tag{3.42}$$

We now look at the second sum. The function $f : Q \to \mathbb{R}$ is bounded, so there exists a number c such that $-c \leq f(x) \leq c$ for all $x \in Q$. Then, $M_{R'}(f) - m_{R'}(f) \leq 2c$ so

$$\begin{aligned} \sum_{R'} (M_{R'}(f) - f_{R'}(f))v(R') &\leq 2c \sum_{R'} v(R') \\ &= 2c \sum_{i=1}^{\ell} \sum_{R' \subseteq Q'_i} v(R') \\ &\leq 2c \sum_i v(Q'_i) \\ &\leq 2c\epsilon. \end{aligned} \tag{3.43}$$

Substituting back into Equation 3.41, we get

$$U(f, P) - L(f, P) \leq \epsilon(v(Q) + 2c). \tag{3.44}$$

So,

$$\overline{\int_Q} f - \underline{\int_Q} f \leq \epsilon(v(Q) + 2c), \tag{3.45}$$

because

$$U(f, P) \geq \overline{\int_Q} f \text{ and } L(f, P) \leq \underline{\int_Q} f. \tag{3.46}$$

Letting ϵ go to zero, we conclude that

$$\overline{\int_Q} f = \underline{\int_Q} f, \tag{3.47}$$

which shows that f is Riemann integrable.

This concludes the proof in one direction. We do not prove the other direction. □

Corollary 4. *Suppose $f : Q \to \mathbb{R}$ is R. integrable and that $f \geq 0$ everywhere. If $\int_Q f = 0$, then $f = 0$ except on a set of measure zero.*

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Proof. Let D be the set of points where f is discontinuous. The function f is R. integrable, so D is of measure zero.

If $p \in Q - D$, then $f(p) = 0$. To see this, suppose that $f(p) = \delta > 0$. The function f is continuous at p, so there exists a rectangle R_0 centered at p such that $f \geq n\delta/2$ on R_0. Choose a partition P such that R_0 is a rectangle belonging to P. On any rectangle R belonging to P, $f \geq 0$, so $m_R(f) \geq 0$. This shows that

$$\begin{aligned} L(f, P) &= m_{R_0}(f)v(R_0) + \sum_{R \neq R_0} m_R(f)v(R) \\ &\geq \frac{\delta}{2}v(R_0) + 0. \end{aligned} \tag{3.48}$$

But we assumed that $\int_Q f = 0$, so we have reached a contradiction. So $f = 0$ at all points $p \in Q - D$. □

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Lecture 10

We begin today's lecture with a simple claim.

Claim. *Let $Q \subseteq \mathbb{R}^n$ be a rectangle and $f, g : Q \to \mathbb{R}$ be bounded functions such that $f \leq g$. Then*

$$\underline{\int_Q} f \leq \underline{\int_Q} g. \tag{3.49}$$

Proof. Let P be a partition of Q, and let R be a rectangle belonging to P. Clearly, $m_R(f) \leq m_R(g)$, so

$$L(f, P) = \sum_R m_R(f) v(R) \tag{3.50}$$

$$L(g, P) = \sum_R m_R(g) v(R) \tag{3.51}$$

$$\implies L(f, P) \leq L(g, P) \leq \underline{\int_Q} g, \tag{3.52}$$

for all partitions P. The lower integral

$$\underline{\int_Q} f \tag{3.53}$$

is the l.u.b. of $L(f, P)$, so

$$\underline{\int_Q} f \leq \underline{\int_Q} g. \tag{3.54}$$

□

Similarly,

$$\overline{\int_Q} f \leq \overline{\int_Q} g. \tag{3.55}$$

It follows that if $f \leq g$, then

$$\int_Q f \leq \int_Q g. \tag{3.56}$$

This is the *monotonicity* property of the R. integral.

3.4 Fubini Theorem

In one-dimensional calculus, when we have a continuous function $f : [a, b] \to \mathbb{R}$, then we can calculate the R. integral

$$\int_a^b f(x)dx = F(b) - F(a), \tag{3.57}$$

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where F is the anti-derivative of f.

When we integrate a continuous function $f : Q \to \mathbb{R}$ over a two-dimensional region, say $Q = [a_1, b_1] \times [a_2, b_2]$, we can calculate the R. integral

$$\int_Q f = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x, y) dx dy = \int_{a_1}^{b_1} \left(\int_{a_2}^{b_2} f(x, y) dx dy \right) \tag{3.58}$$

That is, we can break up Q into components and integrate separately over those components. We make this more precise in the following Fubini Theorem.

First, we define some notation that will be used.

Let $n = k + \ell$ so that $\mathbb{R}^n = \mathbb{R}^l \times \mathbb{R}^\ell$. Let $c = (c_1, \ldots, c_n) \in \mathbb{R}^n$. We can write $c = (a, b)$, where $a = (c_1, \ldots, c_) \in \mathbb{R}^k$ and $b = (c_{k+1}, \ldots, c_{k+\ell}) \in \mathbb{R}^\ell$. Similarly, let $Q = I_1 \times \cdots I_n$ be a rectangle in $\mathbb{R}^n$. Then we can write $Q = A \times B$, where $A = I_1 \times \cdots \times I_k \in \mathbb{R}^k$ and $B = I_{k+1} \times \cdots \times I_{k+\ell} \in \mathbb{R}^\ell$. Along the same lines, we can write a partition $P = (P_1, \ldots, P_n)$ as $P = (P_A, P_B)$, where $P_A = (P_1, \ldots, P_k)$ and $P_B = (P_{k+1}, \ldots, P_{k+\ell})$.

Fubini Theorem. *Let $f : Q \to \mathbb{R}$ be a bounded function and $Q = A \times B$ a rectangle as defined above. We write $f = f(x, y)$, where $x \in A$, and $y \in B$. Fixing $x \in A$, we can define a function $f_x : B \to \mathbb{R}$ by $f_x(y) = f(x, y)$. Since this function is bounded, we can define new functions $g, h : A \to \mathbb{R}$ by*

$$g(x) = \underline{\int_B} f_x, \tag{3.59}$$

$$h(x) = \overline{\int_B} f_x. \tag{3.60}$$

Note that $g \leq h$. The Fubini Theorem concludes the following: If f is integrable over Q, then g and h are integrable over A and

$$\int_A g = \int_A h = \int_Q f. \tag{3.61}$$

Proof. Let $P = (P_A, P_B)$ be a partition of Q, and let $R = R_A \times R_B$ be a rectangle belonging to P (so R_A belongs to P_A and R_B belongs to P_B). Fix $x_0 \in A$.

First, we claim that

$$m_{R_A \times R_B}(f) \leq m_{R_b}(f_{x_0}), \tag{3.62}$$

the proof of which is straightforward.

Next,

$$\begin{aligned} \sum_{R_B} m_{R_A \times R_B}(f) v(R_B) &\leq \sum_{R_B} m_{R_B}(f_{x_0}) v(R_B) \\ &= L(f_{x_0}, P_B) \\ &\leq \underline{\int_B} f_{x_0} = g(x_0). \end{aligned} \tag{3.63}$$

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So,

$$\sum_{R_B} m_{R_A \times R_B}(f) v(R_B) \ \leq g(x_0) \tag{3.64}$$

for all $x_0 \in R_A$. The above equation must hold for the infimum of the r.h.s, so

$$\sum_{R_B} m_{R_A \times R_B}(f) v(R_B) \ \leq m_{R_A}(g). \tag{3.65}$$

Observe that $v(R_A \times R_B) = v(R_A)v(R_B)$, so

$$\begin{aligned} L(f,P) &= \sum_{R_A \times R_B} m_{R_A \times R_B}(f) v(R_A \times R_B) \\ &\leq \sum_{R_A} m_{R_A}(g) v(R_A) \\ &\leq \underline{\int}_A g. \end{aligned} \tag{3.66}$$

We have just shown that for any partition $P = (P_A, P_B)$,

$$L(f,P) \leq L(g,P_A) \leq \underline{\int}_A g, \tag{3.67}$$

so

$$\underline{\int}_Q f \leq \underline{\int}_A g. \tag{3.68}$$

By a similar argument, we can show that

$$\overline{\int}_A h \leq \overline{\int}_Q f. \tag{3.69}$$

Summarizing, we have shown that

$$\underline{\int}_Q f \leq \underline{\int}_A g \leq \overline{\int}_A h \leq \overline{\int}_Q f, \tag{3.70}$$

where we used monotonicity for the middle inequality. Since f is R. integrable,

$$\underline{\int}_Q f = \overline{\int}_Q f, \tag{3.71}$$

so all of the inequalities are in fact equalities. □

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Remark. Suppose that for every $x \in A$, that $f_x : B \to \mathbb{R}$ is R. integrable. That's the same as saying $g(x) = h(x)$. Then

$$\begin{aligned} \int_A \left(\int_B f_x \right) &= \int_A dx \left(\int_B f(x,y) dy \right) \\ &= \int_{A \times B} f(x,y) dx dy, \end{aligned} \tag{3.72}$$

using standard notation from calculus.

Remark. In particular, if f is continuous, then f_x is continuous. Hence, the above remark holds for all continuous functions.

3.5 Properties of Riemann Integrals

We now prove some standard calculus results.

Theorem 3.13. *Let $Q \subseteq \mathbb{R}^n$ be a rectangle, and let $f, g : Q \to \mathbb{R}$ be R. integrable functions. Then, for all $a, b \in \mathbb{R}$, the function $af + bg$ is R. integrable and*

$$\int_Q af + bg = a \int_Q f + b \int_Q g. \tag{3.73}$$

Proof. Let's first assume that $a, b \leq 0$. Let P be a partition of Q and R a rectangle belonging to P. Then

$$a m_R(f) + b m_R(g) \leq m_R(af + bg), \tag{3.74}$$

so

$$\begin{aligned} aL(f,P) + bL(g,P) \ &\leq L(af + bg, P) \\ &\leq \underline{\int_Q} af + bg. \end{aligned} \tag{3.75}$$

Claim. *For any pair of partitions P' and P'',*

$$aL(f,P') + bL(g,P'') \leq \underline{\int_Q} af + bg. \tag{3.76}$$

To see that the claim is true, take P to be a refinement of P' and P'', and apply Equation 3.75. Thus,

$$a \underline{\int_Q} f + b \underline{\int_Q} g \leq \underline{\int_Q} af + bg. \tag{3.77}$$

Similarly, we can show that

$$\overline{\int_Q} af + bg \leq a \overline{\int_Q} f + b \overline{\int_Q} g. \tag{3.78}$$

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Since f and g are R. integrable, we know that

$$\overline{\int}_Q f = \underline{\int}_Q f, \ \overline{\int}_Q g = \underline{\int}_Q g. \tag{3.79}$$

These equalities show that the previous inequalities were in fact equalities, so

$$\int_Q af + bg = a\int_Q f + b\int_Q g. \tag{3.80}$$

However, remember that we assumed that $a, b \geq 0$. To deal with the case of arbitrary a, b, it suffices to check what happens when we change the sign of a or b.

Claim.

$$\int_Q -f = -\int_Q f. \tag{3.81}$$

Proof Hint. Let P be any partition of Q. Then $L(f, P) = -U(-f, P)$. □

You should check this claim, and then use it to complete the proof. □

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Lecture 11

We review some basic properties of the Riemann integral.

Let $Q \subseteq \mathbb{R}^n$ be a rectangle, and let $f, g : q \to \mathbb{R}$ be bounded functions. Assume that f, g are R. integrable. We have the following properties of R. integrals:

- Linearity: $a, b \in \mathbb{R} \implies af + bg$ is R. integrable and

$$\int_Q af + bg = a \int_Q f + b \int_Q g. \tag{3.82}$$

- Monotonicity: If $f \leq g$, then

$$\int_Q f \leq \int_Q g. \tag{3.83}$$

- Maximality Property: Let $h : Q \to \mathbb{R}$ be a function defined by $h(x) = \max(f(x), g(x))$.

  Theorem 3.14. *The function h is R. integrable and*

$$\int_Q h \geq \max\left(\int_Q f, \int_Q g\right). \tag{3.84}$$

  Proof. We need the following lemma.

  Lemma 3.15. *If f and g are continuous at some point $x_0 \in Q$, then h is continuous at x_0.*

  Proof. We consider the case $f(x_0) = g(x_0) = h(x_0) = r$. The functions f and g are continuous at x_0 if and only if for every $\epsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - f(x_0)| < \epsilon$ and $|g(x) - g(x_0)| < \epsilon$ whenever $|x - x_0| < \delta$.

  Substitute in $f(x_0) = g(x_0) = r$. The value of $h(x)$ is either $f(x)$ or $g(x)$, so $|h(x) - r| < \epsilon$ for $|x - x_0| < \delta$. That is $|h(x) - h(x_0)| < \epsilon$ for $|x - x_0| < \delta$, so h is continuous at x_0.

  The proofs of the other cases are left to the student.

  □

  We defined $h = \max(f, g)$. The lemma tells is that h is integrable.

  Define E, F, and G as follows:

$$E = \text{ Set of points in } Q \text{ where } f \text{ is discontinuous,} \tag{3.85}$$
$$F = \text{ Set of points in } Q \text{ where } g \text{ is discontinuous,} \tag{3.86}$$
$$G = \text{ Set of points in } Q \text{ where } h \text{ is discontinuous.} \tag{3.87}$$

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The functions f, g are integrable over Q if and only if E, F are of measure zero. The lemma shows that $G \subseteq E \cup F$, so h is integrable over Q. To finish the proof, we notice that

$$h = \max(f, g) \geq f, g. \tag{3.88}$$

Then, by monotonicity,

$$\int_Q h \geq \max\left(\int_Q f, \int_Q g\right). \tag{3.89}$$

□

Remark. Let $k = \min(f, g)$. Then $k = -\max(-f, -g)$. So, the maximality property also implies that k is integrable and

$$\int_Q k \leq \min\left(\int_Q f, \int_Q g\right). \tag{3.90}$$

A useful trick for when dealing with functions is to change the sign. The preceding remark and the following are examples where such a trick is useful.

Let $f : Q \to \mathbb{R}$ be a R. integrable function. Define

$$f_+ = \max(f, 0), \quad f_- = \max(-f, 0). \tag{3.91}$$

Both of these functions are R. integrable and non-negative: $f_+, f_- \geq 0$. Also note that $f = f_+ - f_-$. This decomposition is a trick we will use over and over again.

Also note that $|f| = f_+ + f_-$, so $|f|$ is R. integrable. By monotonicity,

$$\begin{aligned} \int_Q |f| &= \int_Q f_+ + \int_Q f_- \\ &\geq \int_Q f_+ - \int_Q f_- \\ &= \int_Q f. \end{aligned} \tag{3.92}$$

By replacing f by $-f$, we obtain

$$\begin{aligned} \int_Q |f| &\geq \int_Q -f \\ &= -\int_Q f. \end{aligned} \tag{3.93}$$

Combining these results, we arrive at the following claim

Claim.

$$\int_Q |f| \geq \left|\int_Q f\right| \tag{3.94}$$

Proof. The proof is above. □

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3.6 Integration Over More General Regions

So far we've been defining integrals over rectangles. Let us now generalize to other sets.

Let S be a bounded set in $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be a bounded function. Let $f_S : \mathbb{R}^n \to \mathbb{R}$ be the map defined by

$$f_S(x) = \begin{cases} f(x) & \text{if } x \in S, \\ 0 & \text{if } x \notin S. \end{cases} \tag{3.95}$$

Let Q be a rectangle such that $\operatorname{Int} Q \supset \bar{S}$.

Definition 3.16. The map f is *Riemann integrable over* S if f_S is Riemann integrable over Q. Additionally,

$$\int_S f = \int_Q f_S. \tag{3.96}$$

One has to check that this definition does not depend on the choice of Q, but we do not check this here.

Claim. *Let S be a bounded set in $\mathbb{R}^n$, and let $f, g : S \to \mathbb{R}$ be bounded functions. Assume that f, g are R. integrable over S. Then the following properties hold:*

- *Linearity: If $a, b \in \mathbb{R}$, then $af + bg$ is R. integrable over S, and*

$$\int_S af + bg = a \int_S f + b \int_S g. \tag{3.97}$$

- *Monotonicity: If $f \leq g$, then*

$$\int_S f \leq \int_S g. \tag{3.98}$$

- *Maximality: If $h = \max(f, g)$ (over the domain S), then h is R. integrable over S, and*

$$\int_S h \geq \max\left(\int_S f, \int_S g \right). \tag{3.99}$$

Proof. The proofs are easy, and we outline them here.

- Linearity: Note that $af_S + bg_S = (af + bg)_S$, so

$$\begin{aligned} \int_S af + bg &= \int_Q (af + bg)_S \\ &= a \int_Q f_S + b \int_Q g_S \\ &= a \int_S f + b \int_S g. \end{aligned} \tag{3.100}$$

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- Monotonicity: Use $f \leq g \implies f_S \leq g_S$.
- Maximality: Use $h = \max(f, g) \implies h_S = \max(f_S, g_S)$.

□

Let's look at some nice set theoretic properties of the Riemann integral.

Claim. *Let S, T be bounded subsets of $\mathbb{R}^n$ with $T \subseteq S$. Let $f : S \to \mathbb{R}$ be bounded and non-negative. Suppose that f is R. integrable over both S and T. Then*

$$\int_T f \leq \int_S f. \tag{3.101}$$

Proof. Clearly, $f_T \leq f_S$. Let Q be a rectangle with $\bar{S} \supseteq \text{Int } Q$. Then

$$\int_Q f_T \leq \int_Q f_S. \tag{3.102}$$

□

Claim. *Let S_1, S_2 be bounded subsets of $\mathbb{R}^n$, and let $f : S_1 \cup S_2 \to \mathbb{R}$ be a bounded function. Suppose that f is R. integrable over both S_1 and S_2. Then f is R. integrable over $S_1 \cap S_2$ and over $S_1 \cup S_2$, and*

$$\int_{S_1 \cup S_2} f = \int_{S_1} f + \int_{S_2} f - \int_{S_1 \cap S_2} f. \tag{3.103}$$

Proof. Use the following trick. Notice that

$$f_{S_1 \cup S_2} = \max(f_{S_1}, f_{S_2}), \tag{3.104}$$
$$f_{S_1 \cap S_2} = \min(f_{S_1}, f_{S_2}). \tag{3.105}$$

Now, choose Q such that

$$\text{Int } Q \supset \overline{S_1 \cup S_2}, \tag{3.106}$$

so $f_{S_1 \cup S_2}$ and $f_{S_1 \cap S_2}$ are integrable over Q.

Note the identity

$$f_{S_1 \cup S_2} = f_{S_1} + f_{S_2} - f_{S_1 \cap S_2}. \tag{3.107}$$

So,

$$\int_Q f_{S_1 \cup S_2} = \int_Q f_{S_1} + \int_Q f_{S_2} - \int_Q f_{S_1 \cap S_2}, \tag{3.108}$$

from which it follows that

$$\int_{S_1 \cup S_2} f = \int_{S_1} f + \int_{S_2} f - \int_{S_1 \cap S_2} f. \tag{3.109}$$

□

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Lecture 12

So far, we have been studying only the Riemann integral. However, there is also the Lebesgue integral. These are the two basic integral theories. The Riemann integral is very intuitive and is usually adequate for problems that usually come up. The Lebesgue integral is not as intuitive, but it can handle more general problems. We do not encounter these problems in geometry or physics, but we would in probability and statistics. You can learn more about Lebesgue integrals by taking Fourier Analysis (18.103) or Measure and Integration (18.125). We do not study the Lebesgue integral.

Let S be a bounded subset of $\mathbb{R}^n$.

Theorem 3.17. *If the boundary of S is of measure zero, then the constant function 1 is R. integrable over S. The converse is also true.*

Proof. Let Q be a rectangle such that Int $Q \supset \bar{S}$. Define

$$1_S(x) = \begin{cases} 1 & \text{if } x \in S, \\ 0 & \text{if } x \notin S. \end{cases} \tag{3.110}$$

The constant function 1 is integrable over S if and only if the function 1_S is integrable over Q. The function 1_S is integrable over Q if the set of points D in Q where 1_S is discontinuous is of measure zero. If so, then

$$\int_Q 1_S = \int_S 1. \tag{3.111}$$

Let $x \in Q$.

1. If $x \in$ Int S, then $1_S = 1$ in a neighborhood of x, so 1_S is continuous at x.
2. If $x \in$ Ext S, then $1_S = 0$ in a neighborhood of x, so 1_S is continuous at x.
3. If $x \in$ Bd X, then in every neighborhood U of x there exists points in Ext S where $1_S = 0$ and points in Int S where $1_S = 1$. So, 1_S is discontinuous at x.

Thus, D is the boundary of S, $D =$ Bd S. Therefore, the function 1_S is integrable if and only if Bd S is of measure zero. □

3.7 Improper Integrals

Definition 3.18. The set S is *rectifiable* if the boundary of S is of measure zero. If S is rectifiable, then

$$v(S) = \int_S 1. \tag{3.112}$$

Let us look at the properties of $v(S)$:

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1. Monotonicity: If S_1 and S_2 are rectifiable and $S_1 \subseteq S_2$, then $v(S_1) \leq v(S_2)$.

2. Linearity: If S_1, S_2 are rectifiable, then $S_1 \cup S_2$ and $S_1 \cap S_2$ are rectifiable, and

$$v(S_1 \cup S_2) = v(S_1) + v(S_2) - v(S_1 \cap S_2). \tag{3.113}$$

3. If S is rectifiable, then $v(S) = 0$ if and only if S is of measure zero.

4. Let $A = \text{Int } S$. If S is rectifiable, then A is rectifiable, and $v(S) = v(A)$.

The first two properties above are special cases of the theorems that we proved last lecture:

1.
$$\int_{S_1} 1 \leq \int_{S_2} \text{ if } S_1 \subseteq S_2. \tag{3.114}$$

2.
$$\int_{S_1 \cup S_2} 1 = \int_{S_1} 1 + \int_{S_2} 1 - \int_{S_1 \cap S_2} 1. \tag{3.115}$$

To see the the third and fourth properties are true, we use some previous results. Let Q be a rectangle, and let $f : Q \to \mathbb{R}$ be R. integrable. We proved the following two theorems:

Theorem A. *If $f \geq 0$ and $\int_Q f = 0$, then $f = 0$ except on a set of measure zero.*

Theorem B. *If $f = 0$ except on a set of measure zero, then $\int_S f = 0$.*

Property 3. above is a consequence of Theorem A with $f = 1_S$.

Property 4. above is a consequence of Theorem B with $f = 1_S - 1_A$.

We are still lacking some simple criteria for a bounded set to be integrable. Let us now work on that.

Let S be a bounded set, and let $f : S \to \mathbb{R}$ be a bounded function. We want simple criteria on S and f such that f to be integrable over S.

Theorem 3.19. *If S is rectifiable and $f : S \to \mathbb{R}$ is bounded and continuous, then f is R. integrable over S.*

Proof. Let Q be a rectangle such that Int $Q \supset \bar{S}$. Define $f_S : Q \to \mathbb{R}$ by

$$f_S(x) = \begin{cases} f(x) & \text{if } x \in S, \\ 0 & \text{if } x \notin S. \end{cases} \tag{3.116}$$

By definition, f is integrable over S if and only if f_S is integrable over Q. If so then $\int_S f = \int_Q f_S$.

Let D be the set of points in Q where f_S is discontinuous. Then f_S is integrable over Q if and only if D is of measure zero. What is D?

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1. If $x \in \text{Int } S$, then $f_S = f$ in a neighborhood of x, so f_S is continuous at x.

2. If $x \in \text{Ext } S$, then $f_S = 0$ in a neighborhood of x, so f_S is continuous at x.

So, we know that $D \subseteq \text{Bd } S$. Because S is rectifiable, the boundary of S has measure zero, so D has measure zero. Thus, f_S is R. integrable, and therefore so is f. □

Theorem 3.20. *Let A be an open subset of $\mathbb{R}^n$. There exists a sequence of compact rectifiable sets $C_N, N = 1, 2, 3, \ldots$ such that*

$$C_N \subseteq \text{Int } C_{N+1} \tag{3.117}$$

and

$$\bigcup C_N = A. \tag{3.118}$$

Definition 3.21. The set $\{C_N\}$ is called an *exhaustion* of A.

Proof. Take the complement of A, namely $B = \mathbb{R}^n - A$. Define $d(x, B) = \inf_{y \in B}\{|x - y|\}$. The function $d(x, B)$ is a continuous function of x (the theorem for this is in section 4 of Munkres). Let

$$D_N = \{x \in A : d(x, B) \geq 1/N \text{ and } |x| \leq N\}. \tag{3.119}$$

The set D_N is compact. It is easy to check that $D_N \subseteq \text{Int } D_{N+1}$.

Claim.

$$\bigcup D_N = A. \tag{3.120}$$

Proof. Let $x \in A$. The set A is open, so there exists $\epsilon > 0$ such that the set $\{y \in \mathbb{R}^n : |y - x| \leq \epsilon\}$ is contained in A. So, $d(x, B) \geq \epsilon$.

Now, choose N such that $1/N < \epsilon$ and such that $|x| < N$. Then, by definition, $x \in D_N$. Therefore $\cup D_N = A$. □

So, the D_N's satisfy the right properties, except they are not necessarily rectifiable. We can make them rectifiable as follows.

For every $p \in D_N$, let Q_p be a rectangle with $p \in \text{Int } Q_p$ and $Q_p \subseteq \text{Int } D_{N+1}$. Then the collection of sets $\{\text{Int } Q_p : p \in D_N\}$ is an open cover of D_N. By the H-B Theorem, there exists a finite subcover $\text{Int } Q_{p_1}, \ldots, \text{Int } Q_{p_r}$. Now, let

$$C_N = Q_{p_1} \cup \cdots \cup Q_{p_r}. \tag{3.121}$$

Then $C_N \subseteq \text{Int } D_N \subseteq \text{Int } C_{N+1}$. □

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Lecture 13

Let A be an open set in $\mathbb{R}^n$, and let $f : A \to \mathbb{R}$ be a continuous function. For the moment, we assume that $f \geq 0$. Let $D \subseteq A$ be a compact and rectifiable set. Then $f|D$ is bounded, so $\int_D f$ is well-defined. Consider the set of all such integrals:

$$\# = \{\int_D f : D \subseteq A, D \text{ compact and rectifiable}\}. \tag{3.122}$$

Definition 3.22. The *improper integral of f over A* exists if $*$ is bounded, and we define the improper integral of f over A to be its l.u.b.

$$\int_A^{\#} f \equiv \text{l.u.b. } \int_D f = \text{ improper integral of } f \text{ over } A. \tag{3.123}$$

Claim. *If A is rectifiable and $f : A \to \mathbb{R}$ is bounded, then*

$$\int_A^{\#} f = \int_A f. \tag{3.124}$$

Proof. Let $D \subseteq A$ be a compact and rectifiable set. So,

$$\int_D f \leq \int_A f \tag{3.125}$$

$$\implies \sup_D \int_D f \leq \int_A f \tag{3.126}$$

$$\implies \int_A^{\#} f \leq \int_A f. \tag{3.127}$$

The proof of the inequality in the other direction is a bit more complicated.

Choose a rectangle Q such that $\bar{A} \subseteq \text{Int } Q$. Define $f_A : Q \to \mathbb{R}$ by

$$f_A(x) = \begin{cases} f(x) & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases} \tag{3.128}$$

By definition,

$$\int_A f = \int_Q f_A. \tag{3.129}$$

Now, let P be a partition of Q, and let $R_1, \ldots, R_k$ be rectangles belonging to a partition of A. If R is a rectangle belonging to P not contained in A, then $R - A \neq \phi$. In such a case, $m_R(f_A) = 0$. So

$$L(f_A, P) = \sum_{i=1}^{k} m_{R_i}(f_A) v(R_i). \tag{3.130}$$

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On the rectangle R_i,

$$f_A = f \geq m_{R_i}(f_A). \tag{3.131}$$

So,

$$\begin{aligned} \sum_{i=1}^{k} m_{R_i}(f_A) v(R_i) &\leq \sum \int_{R_i} f \\ &= \int_D f \\ &\leq \int_A^{\#}, \end{aligned} \tag{3.132}$$

where $D = \bigcup R_i$, which is compact and rectifiable.

The above was true for all partitions, so

$$\int_Q f_A \leq \int_Z^{\#} f. \tag{3.133}$$

We proved the inequality in the other direction, so

$$\int_A f = \int_A^{\#} f. \tag{3.134}$$

□

3.8 Exhaustions

Definition 3.23. A sequence of compact sets $C_i, i = 1, 2, 3 \ldots$ is an *exhaustion of* A if $C_i \subseteq \text{Int } C_{i_1}$ for every i, and $\bigcup C_i = A$.

It is easy to see that

$$\bigcup \text{Int } C_i = A. \tag{3.135}$$

Let $C_i,\ i = 1, 2, 3, \ldots$ be an exhaustion of A by compact rectifiable sets. Let $f : A \to \mathbb{R}$ be continuous and assume that $f \geq 0$. Note that

$$\int_{C_i} f \leq \int_{C_{i=1}} f, \tag{3.136}$$

since $C_{i=1} \supset C_i$. So

$$\int_{C_i} f,\ i = 1, 2, 3 \ldots \tag{3.137}$$

is an increasing (actually, non-decreasing) sequence. Hence, either $\int_{C_i} f \to \infty$ as $i \to \infty$, or it has a finite limit (by which we mean $\lim_{i\to\infty} \int_{C_i} f$ exists).

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Theorem 3.24. *The following two properties are equivalent:*

1. $\int_A^{\#} f$ *exists,*
2. $\lim_{i\to\infty} \int_{C_i} f$ *exists.*

Moreover, if either (and hence both) property holds, then

$$\int_A^{\#} f = \lim_{i\to\infty} \int_{C_i} f. \tag{3.138}$$

Proof. The set C_i is a compact and rectifiable set contained in A. So, if

$$\int_A^{\#} f \text{ exists, then} \tag{3.139}$$

$$\int_{C_i} f \leq \int_A^{\#} f. \tag{3.140}$$

That shows that the sets

$$\int_{C_i} f,\ i = 1, 2, 3 \dots \tag{3.141}$$

are bounded, and

$$\lim_{i\to\infty} \int_{C_i} f \leq \int_A^{\#} f. \tag{3.142}$$

Now, let us prove the inequality in the other direction.

The collection of sets $\{\text{Int } C_i : i = 1, 2, 3 \dots\}$ is an open cover of A. Let $D \subseteq A$ be a compact rectifiable set contained in A. By the H-B Theorem,

$$D \subseteq \bigcup_{i=1}^{N} \text{Int } C_i, \tag{3.143}$$

for some N. So, $D \subseteq \text{Int } C_N \subseteq C_N$. For all such D,

$$\int_D f \leq \int_{C_i} f \leq \lim_{i\to\infty} \int_{C_i} f. \tag{3.144}$$

Taking the infimum over all D, we get

$$\int_A^{\#} f \leq \lim_{i\to\infty} \int_{C_i} f. \tag{3.145}$$

We have proved the inequality in both directions, so

$$\int_A^{\#} f = \lim_{i\to\infty} \int_{C_i} f. \tag{3.146}$$

□

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A typical illustration of this theorem is the following example.

Consider the integral

$$\int_0^1 \frac{dx}{\sqrt{x}}, \tag{3.147}$$

which we wrote in the normal integral notation from elementary calculus. In *our* notation, we would write this as

$$\int_{(0,1)} \frac{1}{\sqrt{x}}. \tag{3.148}$$

Let $C_N = [\frac{1}{N}, 1 - \frac{1}{N}]$. Then

$$\begin{aligned} \int_{(0,1)}^{\#} \frac{1}{\sqrt{x}} &= \lim_{N\to\infty} \int_{C_N} \frac{A}{\sqrt{x}} \\ &= 2\sqrt{x}|_{1/N}^{1-1/N} \to 2 \text{ as } N \to \infty. \end{aligned} \tag{3.149}$$

So,

$$\int_{(0,1)}^{\#} \frac{1}{\sqrt{x}} = 2. \tag{3.150}$$

Let us now remove the assumption that $f \geq 0$. Let $f : A \to \mathbb{R}$ be any continuous function on A. As before, we define

$$f_+(x) = \max\{f(x), 0\}, \tag{3.151}$$

$$f_-(x) = \max\{-f(x), 0\}. \tag{3.152}$$

We can see that f_+ and f_- are continuous.

Definition 3.25. The improper R. integral of f over A exists if and only if the improper R. integral of f_+ and f_- over A exist. Moreover, we define

$$\int_A^{\#} f = \int_A^{\#} f_+ - \int_A^{\#} f_-. \tag{3.153}$$

We compute the integral using an exhaustion of A.

$$\begin{aligned} \int_A^{\#} f &= \lim_{N\to\infty} \left(\int_{C_N} f_+ - \int_{C_N} f_- \right) \\ &= \lim_{N\to\infty} \int_{C_N} f. \end{aligned} \tag{3.154}$$

Note that $|f| = f_+ + f_-$, so

$$\lim_{N\to\infty} \left(\int_{C_N} f_+ + \int_{C_N} f_- \right) = \lim_{N\to\infty} \int_{C_N} |f|. \tag{3.155}$$

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Therefore, the improper integral of f exists if and only if the improper integral of $|f|$ exists.

Define a function $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ e^{-1/x} & \text{if } x > 0. \end{cases} \tag{3.156}$$

This is a $\mathcal{C}^\infty(\mathbb{R})$ function. Clearly, $f'(x) = f''(x) = \ldots = 0$ when $x = 0$, so in the Taylor series expansion of f at zero,

$$\sum a_n x^n = 0, \tag{3.157}$$

all of the coefficients a_n are zero. However, f has a non-zero value in every neighborhood of zero.

Take $a \in \mathbb{R}$ and $\epsilon > 0$. Define a new function $g_{a,a+\epsilon} : \mathbb{R} \to \mathbb{R}$ by

$$g_{a,a+\epsilon}(x) = \frac{f(x-a)}{f(x-a) + f(a+\epsilon-x)}. \tag{3.158}$$

The function $g_{a,a+\epsilon}$ is a $\mathcal{C}^\infty(\mathbb{R})$ function. Notice that

$$g_{a,a+\epsilon} = \begin{cases} 0 & \text{if } x \leq a, \\ 1 & \text{if } x \geq a+\epsilon. \end{cases} \tag{3.159}$$

Take b such that $a < a + \epsilon < b - \epsilon < b$. Define a new function $h_{a,b} \in \mathcal{C}^\infty(\mathbb{R})$ by

$$h_{a,b}(x) = g_{a,a+\epsilon}(x)(1 - g_{a-\epsilon,b}(x)). \tag{3.160}$$

Notice that

$$h_{a,b} = \begin{cases} 0 & \text{if } x \leq a, \\ 1 & \text{if } a+\epsilon \leq x \leq b-\epsilon, \\ 0 & \text{if } b \leq x. \end{cases} \tag{3.161}$$

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Lecture 14

As before, let $f : \mathbb{R} \to \mathbb{R}$ be the map defined by

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ e^{-1/x} & \text{if } x > 0. \end{cases} \tag{3.162}$$

This is a $Cinf(\mathbb{R})$ function. Take the interval $[a, b] \in \mathbb{R}$ and define the function $f_{a,b} : \mathbb{R} \to \mathbb{R}$ by $f_{a,b}(x) = f(x - a)f(b - x)$. Note that $f_{a,b} > 0$ on (a, b), and $f_{a,b} = 0$ on $\mathbb{R} - (a, b)$.

We generalize the definition of f to higher dimensions. Let $Q \subseteq \mathbb{R}^n$ be a rectangle, where $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$. Define a new map $f_Q : \mathbb{R}^n \to \mathbb{R}$ bye

$$f_Q(x_1, \ldots, x_n) = f_{a_1,b_1}(x_1) \ldots f_{a_n,b_n}(x_n). \tag{3.163}$$

Note that $f_Q > 0$ on Int Q, and that $f_Q = 0$ on $\mathbb{R}^n -$ Int Q.

3.9 Support and Compact Support

Now for some terminology. Let U be an open set in $\mathbb{R}^n$, and let $f : U \to \mathbb{R}$ be a continuous function.

Definition 3.26. The *support* of f is

$$\text{supp } f = \overline{\{x \in U : f(x) \neq 0\}}. \tag{3.164}$$

For example, supp $f_Q = Q$.

Definition 3.27. Let $f : U \to \mathbb{R}$ be a continuous function. The function f is *compactly supported* if supp f is compact.

Notation.

$$\mathcal{C}_0^k(U) = \text{The set of compactly supported } \mathcal{C}^k \text{ functions on } U. \tag{3.165}$$

Suppose that $f \in \mathcal{C}_0^k(U)$. Define a new set $U_1 = (\mathbb{R}^n - \text{supp } f)$. Then $U \cup U_1 = \mathbb{R}^n$, because supp $f \subseteq U$.

Define a new map $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ by

$$\tilde{f} = \begin{cases} f & \text{on } U, \\ 0 & \text{on } U_1. \end{cases} \tag{3.166}$$

The function $\tilde{f}$ is $\mathcal{C}^k$ on U and $\mathcal{C}^k$ on U_1, so $\tilde{f}$ is in $\mathcal{C}_0^k(\mathbb{R}^n)$.

So, whenever we have a function $f \in \mathcal{C}^k$ is compactly supported on U, we can drop the tilde and think of f as in $\mathcal{C}_0^k(\mathbb{R}^n)$.

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3.10 Partitions of Unity

Let $\{U_\alpha : \alpha \in I\}$ be a collection of of open subsets of $\mathbb{R}^n$ such that $U = \cup_\alpha U_\alpha$.

Theorem 3.28. *There exists a sequence of rectangles $Q_i,\ i = 1, 2, 3, \dots$ such that*

1. *Int $Q_i,\ i = 1, 2, 3 \dots$ is a cover of U,*
2. *Each $Q_i \subset I_\alpha$ for some α,*
3. *For every point $p \in U$, there exists a neighborhood U_p of p such that $U_p \cap Q_i = \phi$ for all $i > N_p$.*

Proof. Take an exhaustion $A_1, A_2, A_3, \dots$ of U. By definition, the exhaustion satisfies

$$\begin{cases} A_i \subseteq \text{Int } A_{i+1} \\ A_i \text{ is compact} \\ \cup A_i = U. \end{cases}$$

We previously showed that you can always find an exhaustion.

Let $B_i = A_i - \text{Int } A_{i-1}$. For each $x \in B_i$, let Q_x be a rectangle with $x \in \text{Int } Q_x$ such that $Q_x \subseteq U_\alpha$, for some alpha, and $Q_x \subset \text{Int } A_{i+1} - A_{i-2}$. Then, the collection of sets $\{\text{Int } Q_x : x \in B_i\}$ covers B_i. Each set B_i is compact, so, by the H-B Theorem, there exists a finite subcover $\text{Int } Q_{x_r} \equiv \text{Int } Q_{i,r},\ r = 1, \dots, N_i$.

The rectangles $Q_{i,r}, 1 \leq r \leq N_i, i = 1, 2, 3 \dots$ satisfy the hypotheses of the theorem, after relabeling the rectangles in linear sequence Q_1, Q_2, Q_3, etc. (you should check this). □

The following theorem is called the Partition of Unity Theorem.

Theorem 3.29. *There exist functions $f_i \subseteq \mathcal{C}_0^\infty(U)$ such that*

1. *$f_1 \geq 0$,*
2. *supp $f_i \subseteq U_\alpha$, for some α,*
3. *For every $p \in U$, there exists a neighborhood U_p of p such that $U_p \cup \text{supp } f_i = \phi$ for all $i > N_p$,*
4. *$\sum f_i = 1$.*

Proof. Let $Q_i, i = 1, 2, 3, \dots$ be a collection of rectangles with the properties of the previous theorem. Then the functions $f_{Q_i}, i = 1, 2, 3, \dots$ have all the properties presented in the theorem, except for property 4. We now prove the fourth property. We now that $f_{Q_i} > 0$ on Int Q_i, and $\{\text{Int } Q_i : i = 1, 2, 3, \dots\}$ is a cover of U. So, for every $p \in U, f_{Q_i}(p) > 0$ for some i. So

$$\sum f_{Q_i} > 0. \tag{3.167}$$

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We can divide by a nonzero number, so we can define

$$f_i = \frac{f_{Q_i}}{\sum_{i=1}^{\infty} f_{Q_i}}. \tag{3.168}$$

This new function satisfies property 4. Note that the infinite sum converges because the sum has only a finite number of nonzero terms. □

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Lecture 15

We restate the partition of unity theorem from last time. Let $\{U_\alpha : \alpha \in I\}$ be a collection of open subsets of $\mathbb{R}^n$ such that

$$U = \bigcup_{\alpha \in I} U_\alpha. \tag{3.169}$$

Theorem 3.30. *There exist functions $f_i \subseteq \mathcal{C}_0^\infty(U)$ such that*

1. $f_1 \geq 0$,
2. $\text{supp } f_i \subseteq U_\alpha$, *for some* α,
3. *For every $p \in U$, there exists a neighborhood U_p of p such that $U_p \cap \text{supp } f_i = \phi$ for all $i > N_p$,*
4. $\sum f_i = 1$.

Remark. Property (4) makes sense because of property (3), because at each point it is a finite sum.

Remark. A set of functions satisfying properties (1), (3), and (4) is called a *partition of unity.*

Remark. Property (2) can be restated as "the partition of unity is subordinate to the cover $\{U_\alpha : \alpha \in I\}$."

Let us look at some typical applications of partitions of unity.

The first application is to improper integrals. Let $\phi : U \to \mathbb{R}$ be a continuous map, and suppose

$$\int_U \phi \tag{3.170}$$

is well-defined. Take a partition of unity $\sum f_i = 1$. The function $f_i\phi$ is continuous and compactly supported, so it bounded. Let $\text{supp } f_i \subseteq Q_i$ for some rectangle Q_i. Then,

$$\int_{Q_i} f_i\phi \tag{3.171}$$

is a well-defined R. integral. It follows that

$$\int_U f_i\phi = \int_{Q_i} f_i\phi. \tag{3.172}$$

It follows that

$$\int_U \phi = \sum_{i=1}^{\infty} \int_{Q_i} f_i\phi. \tag{3.173}$$

This is proved in Munkres.

The second application of partitions of unity involves *cut-off functions.*

Let $f_i \in \mathcal{C}_0^\infty(U),\ i = 1, 2, 3, \ldots$ be a partition of unity, and let $A \subseteq U$ be compact.

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Lemma 3.31. *There exists a neighborhood U' of A in U and a number $N > 0$ such that $A \cup \text{supp } f_i = \phi$ for all $i > N$.*

Proof. For any $p \in A$, there exists a neighborhood U_p of p and a number N_p such that $U' \cup \text{supp } f_i = \phi$ for all $i > N_p$. The collection of all these U_p is a cover of A. By the H-B Theorem, there exists a finite subcover $U_{p_i},\ i = 1, 2, 3, \ldots$ of A. Take $U_p = \cup U_{p_i}$ and take $N = \max\{N_{p_i}\}$. □

We use this lemma to prove the following theorem.

Theorem 3.32. *Let $A \subseteq \mathbb{R}^n$ be compact, and let U be an open set containing A. There exists a function $f \in \mathcal{C}_0^\infty(U)$ such that $f \equiv 1$ (identically equal to 1) on a neighborhood $U' \subset U$ of A.*

Proof. Choose U' and N as in the lemma, and let

$$f = \sum_{i=1}^{N} f_i. \tag{3.174}$$

Then supp $f_i \cap U' = \phi$ for all $i > N$. So, on U',

$$f = \sum_{i=1}^{\infty} f_i = 1. \tag{3.175}$$

□

Such an f can be used to create cut-off functions. We look at an application.

Let $\phi : U \to \mathbb{R}$ be a continuous function. Define $\psi = f\phi$. The new function ψ is called a cut-off function, and it is compactly supported with supp $\phi \subseteq U$. We can extend the domain of ψ by defining $\psi = 0$ outside of U. The extended function $\psi : \mathbb{R}^n \to \mathbb{R}$ is still continuous, and it equals ϕ on a neighborhood of A.

We look at another application, this time to *exhaustion functions*.

Definition 3.33. Given an open set U, and a collection of compact subsets $A_i\, i = 1, 2, 3, \ldots$ of U, the sets A_i form an *exhaustion of U* if $A_i \subseteq \text{Int } A_{i+1}$ and $\cup A_i = U$ (this is just a quick reminder of the definition of exhaustion).

Definition 3.34. A function $\phi \in \mathcal{C}^\infty(U)$ is an *exhaustion function* if

1. $\phi > 0$,
2. the sets $A_i = \phi^{-1}([0, 1])$ are compact.

Note that this implies that the $A_i's$ are an exhaustion.

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We use the fact that we can always find a partition of unity to show that we can always find exhaustion functions.

Take a partition of unity $f_i \in \mathcal{C}^\infty(U)$, and define

$$\phi = \sum_{i=1}^{\infty} i f_i. \tag{3.176}$$

This sum converges because only finitely many terms are nonzero.

Consider any point

$$p \notin \bigcup_{j \le i} \text{supp } f_j. \tag{3.177}$$

Then,

$$\begin{aligned} 1 &= \sum_{k=1}^{\infty} f_k(p) \\ &= \sum_{k>i} f_k(p), \end{aligned} \tag{3.178}$$

so

$$\begin{aligned} \sum_{\ell=1}^{\infty} \ell f_\ell(p) &= \sum_{\ell>i} \ell f_\ell \\ &\ge i \sum_{\ell>i} f_\ell \\ &= i. \end{aligned} \tag{3.179}$$

That is, if $p \notin \cup_{j \le i} \text{supp } f_j$, then $f(p) > i$. So,

$$\phi^{-1}([0,i]) \subseteq \bigcup_{j \le i} \text{supp } f_j, \tag{3.180}$$

which you should check yourself. The compactness of the r.h.s. implies the compactness of the l.h.s.

Now we look at problem number 4 in section 16 of Munkres. Let A be an arbitrary subset of $\mathbb{R}^n$, and let $g : A \to \mathbb{R}^k$ be a map.

Definition 3.35. The function g *is* $\mathcal{C}^k$ *on* A if for every $p \in A$, there exists a neighborhood U_p of p in $\mathbb{R}^n$ and a $\mathcal{C}^k$ map $g^p : U_p \to \mathbb{R}^k$ such that $g^p|U_p \cap A = g|U_p \cap A$.

Theorem 3.36. *If $g : A \to \mathbb{R}^k$ is $\mathcal{C}^k$, then there exists a neighborhood U of A in $\mathbb{R}^n$ and a $\mathcal{C}^k$ map $\tilde{g} : U \to \mathbb{R}^k$ such that $\tilde{g} = g$ on A.*

Proof. This is a very nice application of partition of unity. Read Munkres for the proof. □

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Lecture 16

4 Multi-linear Algebra

4.1 Review of Linear Algebra and Topology

In today's lecture we review chapters 1 and 2 of Munkres. Our ultimate goal (not today) is to develop vector calculus in n dimensions (for example, the generalizations of grad, div, and curl).

Let V be a vector space, and let $v_i \in V, i = 1, \ldots, k$.

1. The $v_i's$ are *linearly independent* if the map from $\mathbb{R}^k$ to V mapping $(c_1, \ldots, c_k)$ to $c_1v_1 + \ldots + c_kv_k$ is injective.

2. The $v_i's$ *span* V if this map is surjective (onto).

3. If the $v_i's$ form a basis, then $\dim V = k$.

4. A subset W of V is a *subspace* if it is also a vector space.

5. Let V and W be vector spaces. A map $A : V \to W$ is *linear* if $A(c_1v_1 + c_2v_2) = c_1A(v_1) + c_2A(v_2)$.

6. The *kernel* of a linear map $A : V \to W$ is
$$\ker A = \{v \in V : Av = 0\}. \tag{4.1}$$

7. The *image of* A is
$$\text{Im } A = \{Av : v \in V\}. \tag{4.2}$$

8. The following is a basic identity:
$$\dim \ker A + \dim \text{Im } A = \dim V. \tag{4.3}$$

9. We can associate linear mappings with matrices. Let $v_1, \ldots, v_n$ be a basis for V, and let $w_1, \ldots, w_m$ be a basis for W. Let
$$Av_j = \sum_{i=1}^{m} a_{ij}w_j. \tag{4.4}$$
Then we associate the linear map A with the matrix $[a_{ij}]$. We write this $A \sim [a_{ij}]$.

10. If $v_1, \ldots, v_n$ is a basis for V and $u_j = \sum a_{ij}w_j$ are n arbitrary vectors in W, then there exists a unique linear mapping $A : V \to W$ such that $Av_j = u_j$.

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11. Know all the material in Munkres section $\oint 2$ on matrices and determinants.

12. The quotient space construction. Let V be a vector space and W a subspace. Take any $v \in V$. We define $v + W \equiv \{v + w : w \in W\}$. Sets of this form are called W-*cosets*. One can check that given $v_1 + W$ and $v_2 + W$,

    (a) If $v_1 - v_2 \in W$, then $v_1 + W = v_2 + W$.

    (b) If $v_1 - v_2 \notin W$, then $(v_1 + W) \cap (v_2 + W) = \phi$.

    So every vector $v \in V$ belongs to a unique W-coset.

    The *quotient space* V/W is the set of all W=cosets.

    For example, let $V = \mathbb{R}^2$, and let $W = \{(a, 0) : a \in \mathbb{R}\}$. The W-cosets are then vertical lines.

    The set V/W is a vector space. It satisfies vector addition: $(v_1+W)+(v_2+W) = (v_1 + v_2) + W$. It also satisfies scaler multiplication: $\lambda(v + W) = \lambda v + W$. You should check that the standard axioms for vector spaces are satisfied.

    There is a natural projection from V to V/W:

$$\pi : V \to V/W,\ v \to v + W. \tag{4.5}$$

    The map π is a linear map, it is surjective, and $\ker \pi = W$. Also, Im $\pi = V/W$, so

$$\begin{aligned} \dim V/W &= \dim \text{Im } \pi \\ &= \dim V - \dim \ker \pi \\ &= \dim V - \dim W. \end{aligned} \tag{4.6}$$

4.2 Dual Space

13. The *dual space* construction: Let V be an n-dimensional vector space. Define V^* to be the set of all linear functions $\ell : V \to \mathbb{R}$. Note that if $\ell_1, \ell_2 \in V^*$ and $\lambda_1, \lambda_2 \in \mathbb{R}$, then $\lambda_1\ell_1 + \lambda_2\ell_2 \in V^*$, so V^* is a vector space.

    What does V^* look like? Let $e_1, \ldots, e_n$ be a basis of V. By item (9), there exists a unique linear map $e_i^* \in V^*$ such that

$$\begin{cases} e_i^*(e_i) = 1, \\ e_i^*(e_j) = 0, \text{ if } j \neq i. \end{cases}$$

    Claim. *The set of vectors $e_1^*, \ldots, e_n^*$ is a basis of V^*.*

    Proof. Suppose $\ell = \sum c_i e_i^* = 0$. Then $0 = \ell(e_j) = \sum c_i e_i^*(e_j) = c_j$, so $c_1 = \ldots = c_n = 0$. This proves that the vectors e_i^* are linearly independent. Now, if $\ell \in V^*$ and $\ell(e_i) = c_j$ one can check that $\ell = \sum c_i e_i^*$. This proves that the vectors e_i^* span V^*. □

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The vectors $e_1^*, \ldots, e_n^*$ are said to be a *basis of V^* dual to* $e_1, \ldots, e_n$.

Note that $\dim V^* = \dim V$.

Suppose that we have a pair of vectors spaces V, W and a linear map $A : V \to W$. We get another map

$$A^* : W^* \to V^*, \tag{4.7}$$

defined by $A^*\ell = \ell \circ A$, where $\ell \in W^*$ is a linear map $\ell : W \to \mathbb{R}$. So $A^*\ell$ is a linear map $A^*\ell : V \to \mathbb{R}$. You can check that $A^* : W^* \to V^*$ is linear.

We look at the matrix description of A^*. Define the following bases:

$$e_1, \ldots, e_n \text{ a basis of } V \tag{4.8}$$
$$f_1, \ldots, f_n \text{ a basis of } W \tag{4.9}$$
$$e_1^*, \ldots, e_n^* \text{ a basis of } V^* \tag{4.10}$$
$$f_1^*, \ldots, f_n^* \text{ a basis of } W^*. \tag{4.11}$$

Then

$$\begin{aligned} A^* f_j^*(e_i) &= f_j^*(Ae_i) \\ &= f_j^*(\sum_k a_{ki} f_k) \\ &= a_{ji} \end{aligned} \tag{4.12}$$

So,

$$A^* f_j = \sum_k a_{jk} e_k^*, \tag{4.13}$$

which shows that $A^* \sim [a_{ji}] = [a_{ij}]^t$, the transpose of A.

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Lecture 17

Today we begin studying the material that is also found in the Multi-linear Algebra Notes. We begin with the theory of *tensors*.

4.3 Tensors

Let V be a n-dimensional vector space. We use the following notation.

Notation.

$$V^k = \underbrace{V \times \cdots \times V}_{k \text{ times}}. \tag{4.14}$$

For example,

$$V^2 = V \times V, \tag{4.15}$$
$$V^3 = V \times V \times V. \tag{4.16}$$

Let $T : V^k \to \mathbb{R}$ be a map.

Definition 4.1. The map T is *linear in its ith factor* if for every sequence $v_j \in V, 1 \leq j \leq n, j \neq i$, the function mapping $v \in V$ to $T(v_1, \ldots, v_{i-1}, v, v_{i+1}, \ldots, v_k)$ is linear in v.

Definition 4.2. The map T is *k-linear* (or is a *k-tensor*) if it is linear in all k factors.

Let T_1, T_2 be k-tensors, and let $\lambda_1, \lambda_2 \in \mathbb{R}$. Then $\lambda_1 T_1 + \lambda_2 T_2$ is a k-tensor (it is linear in all of its factors).

So, the set of all k-tensors is a vector space, denoted by $\mathcal{L}^k(V)$, which we sometimes simply denote by $\mathcal{L}^k$.

Consider the special case $k = 1$. The the set $\mathcal{L}^1(V)$ is the set of all linear maps $\ell : V \to \mathbb{R}$. In other words,

$$\mathcal{L}^1(V) = V^*. \tag{4.17}$$

We use the convention that

$$\mathcal{L}^0(V) = \mathbb{R}. \tag{4.18}$$

Definition 4.3. Let $T_i \in \mathcal{L}^{k_i}, i = 1, 2$, and define $k = k_1 + k_2$. We define the *tensor product* of T_1 and T_2 to be the tensor $T_1 \otimes T_2 : V^k \to \mathbb{R}$ defined by

$$T_1 \otimes T_2(v_1, \ldots, v_k) = T_1(v_1, \ldots, v_{k_1}) T_2(v_{k_1+1}, \ldots, v_k). \tag{4.19}$$

We can conclude that $T_1 \otimes T_2 \in \mathcal{L}^k$.

We can define more complicated tensor products. For example, let $T_i \in \mathcal{L}^{k_i}, i = 1, 2, 3$, and define $k = k_1 + k_2 + k_3$. Then we have the tensor product

$$\begin{aligned} &T_1 \otimes T_2 \otimes T_3(v_1, \ldots, v_k) \\ &\qquad = T_1(v_i, \ldots, v_{k_1}) T_2(v_{k_1+1}, \ldots, v_{k_1+k_2}) T_3(v_{k_1+k_2+1}, \ldots, v_k). \end{aligned} \tag{4.20}$$

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Then $T_1 \otimes T_2 \otimes T_3 \in \mathcal{L}^k$. Note that we could have simply defined

$$\begin{aligned} T_1 \otimes T_2 \otimes T_3 &= (T_1 \otimes T_2) \otimes T_3 \\ &= T_1 \otimes (T_2 \otimes T_3), \end{aligned} \tag{4.21}$$

where the second equality is the associative law for tensors. There are other laws, which we list here.

- Associative Law: $(T_1 \otimes T_2) \otimes T_3 = T_1 \otimes (T_2 \otimes T_3)$.
- Right and Left Distributive Laws: Suppose $T_i \in \mathcal{L}^{k_i}, i = 1, 2, 3$, and assume that $k_1 = k_2$. Then
  - Left: $(T_1 + T_2) \otimes T_3 = T_1 \otimes T_3 + T_2 \otimes T_3$.
  - Right: $T_3 \otimes (T_1 + T_2) = T_3 \otimes T_1 + T_3 \otimes T_2$.
- Let λ be a scalar. Then

$$\lambda(T_1 \otimes T_2) = (\lambda T_1) \otimes T_2 = T_1 \otimes (\lambda T_2). \tag{4.22}$$

Now we look at an important class of k-tensors. Remember that $\mathcal{L}^1(V) = V^*$, and take any 1-tensors $\ell_i \in V^*, i = 1, \ldots, k$.

Definition 4.4. The tensor $T = \ell_1 \otimes \cdots \otimes \ell_k$ is a *decomposable k-tensor*.

By definition, $T(v_1, \ldots, v_k) = \ell_1(v_1) \ldots \ell_k(v_k)$. That is, $\ell_1 \otimes \cdots \otimes \ell_k(v_1, \ldots, v_k) = \ell_1(v_1) \ldots \ell_k(v_k)$.

Now let us go back to considering $\mathcal{L}^k = \mathcal{L}^k(V)$.

Theorem 4.5.

$$\dim \mathcal{L}^k = n^k. \tag{4.23}$$

Note that for $k = 1$, this shows that $\mathcal{L}^1(V) = V^*$ has dimension n.

Proof. Fix a basis $e_1, \ldots, e_n$ of V. This defines a dual basis $e_i^*, \ldots, e_n^*$ of V^*, $e_i^* : V \to \mathbb{R}$ defined by

$$e_i^*(e_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \tag{4.24}$$

Definition 4.6. A *multi-index I* of length k is a set of integers $(i_1, \ldots, i_k), 1 \leq i_r \leq n$. We define

$$e_I^* = e_{i_1}^* \otimes \cdots \otimes e_{i_k}^* \in \mathcal{L}^k. \tag{4.25}$$

Let $J = (j_1, \ldots, j_k)$ be a multi-index of length k. Then

$$e_I^*(e_{j_1}, \ldots, e_{j_k}) = e_{i_1}^*(e_{j_1}) \ldots e_{i_k}^*(e_{j_k}) = \begin{cases} 1 & \text{if } I = J, \\ 0 & \text{if } I \neq J. \end{cases} \tag{4.26}$$

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Claim. *The k-tensors e_I^* are a basis of $\mathcal{L}^k$.*

Proof. To prove the claim, we use the following lemma.

Lemma 4.7. *Let T be a k-tensor. Suppose that $T(e_{i_1}, \ldots, e_{i_k}) = 0$ for all multi-indices I. Then $T = 0$.*

Proof. Define a $(k-1)$-tensor $T_i : V^{k-1} \to \mathbb{R}$ by setting

$$T_i(v_1, \ldots, v_{k-1}) = T(v_1, \ldots, v_{k-1}, e_j), \tag{4.27}$$

and let $v_k = \sum a_i e_i$. By linearity, $T(v_1, \ldots, v_k) = \sum a_i T_i(v_1, \ldots, v_{k-1})$. So, if the lemma is true for the T_i's, then it is true for T by an induction argument (we leave this to the student to prove). □

With this lemma we can prove the claim.

First we show that the e_I^*'s are linearly independent. Suppose that

$$0 = T = \sum c_I e_I^*. \tag{4.28}$$

For any multi-index J of length k,

$$\begin{aligned} 0 &= T(e_{j_1}, \ldots, e_{j_k}) \\ &= \sum c_I e_I^*(e_{j_1}, \ldots, e_{j_k}) \\ &= c_J \\ &= 0. \end{aligned} \tag{4.29}$$

So the e_I^*'s are linearly independent.

Now we show that the e_I^*'s span $\mathcal{L}^k$. Let $T \in \mathcal{L}^k$. For every I let $T_I = T(e_{i_1}, \ldots, e_{i_l})$, and let $T' = \sum T_I e_I^*$. One can check that $(T - T')(e_{j_1}, \ldots, e_{j_k}) = 0$ for all multi-indices J. Then the lemma tells us that $T = T'$, so the e_I^*'s span $\mathcal{L}^k$, which proves our claim. □

Since the e_I^*'s are a basis of $\mathcal{L}^k$, we see that

$$\dim \mathcal{L}^k = n^k, \tag{4.30}$$

which proves our theorem. □

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4.4 Pullback Operators

Let V, W be vector spacers, and let $A : V \to W$ be a linear map. Let $T \in \mathcal{L}^k(W)$, and define a new map $A^*T \in \mathcal{L}^k(V)$ (called the "pullback" tensor) by

$$A^*T(v_1, \ldots, v_k) = T(Av_1, \ldots, Av_k). \tag{4.31}$$

You should prove the following claims as an exercise:

Claim. *The map $A^* : \mathcal{L}^k(W) \to \mathcal{L}^k(V)$ is a linear map.*

Claim. *Let $T_i \in \mathcal{L}^{k_i}(W), i = 1, 2$. Then*

$$A^*(T_1 \otimes T_2) = A^*T_1 \otimes A^*T_2. \tag{4.32}$$

Now, let $A : V \to W$ and $B : W \to U$ be maps, where U is a vector space. Given $T \in \mathcal{L}^k(U)$, we can "pullback" to W by B^*T, and then we can "pullback" to V by $A^*(B^*T) = (B \circ A)^*T$.

4.5 Alternating Tensors

In this course we will be restricting ourselves to *alternating tensors*.

Definition 4.8. A permutation of order k is a bijective map

$$\sigma : \{1, \ldots, k\} \to \{1, \ldots, k\}. \tag{4.33}$$

The map is a bijection, so σ^{-1} exists.

Given two permutations σ_1, σ_2, we can construct the composite permutation

$$\sigma_1 \circ \sigma_2(i) = \sigma_1(\sigma_2(i)). \tag{4.34}$$

We define

$$S_k \equiv \text{ The set of all permutations of } \{1, \ldots, k\}. \tag{4.35}$$

There are some special permutations. Fix $1 \leq i < j \leq k$. Let τ be the permutation such that

$$\tau(i) = j \tag{4.36}$$
$$\tau(j) = i \tag{4.37}$$
$$\tau(\ell) = \ell, \ell \neq i, j. \tag{4.38}$$

The permutation τ is called a *transposition*.

Definition 4.9. The permutation τ is an elementary transposition if $j = i + 1$.

We state without proof two very useful theorems.

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Theorem 4.10. *Every permutation can be written as a product $\sigma = \tau_1 \circ \tau_2 \circ \cdots \circ \tau_m$, where each τ_i is an elementary transposition.*

Theorem 4.11. *Every permutation σ can be written either as a product of an even number of elementary transpositions or as a product of an odd number of elementary transpositions, but not both.*

Because of the second theorem, we can define an important invariant of a permutation: the *sign of the permutation.*

Definition 4.12. If $\sigma = \tau_1 \circ \cdots \circ \tau_m$, where the τ_i's are elementary transpositions, then the *sign of* σ is

$$\text{sign of } \sigma = (-1)^\sigma = (-1)^m. \tag{4.39}$$

Note that if $\sigma = \sigma_1 \circ \sigma_2$, then $(-1)^\sigma = (-1)^{\sigma_1}(-1)^{\sigma_2}$. We can see this by letting $\sigma_1 = \tau_1 \circ \cdots \circ \tau_{m_1}$, and $\sigma_2 = \tau'_1 \circ \cdots \circ \tau'_{m_2}$, and noting that $\sigma_1 \circ \sigma_2 = \tau_1 \circ \cdots \circ \tau_{m_1} \circ \tau'_1 \circ \cdots \circ \tau'_{m_2}$.

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Lecture 18

We begin with a quick review of permutations (from last lecture).

A permutation of order k is a bijective map $\sigma : \{1, \ldots, k\} \to \{1, \ldots, k\}$. We denote by S_k the set of permutations of order k.

The set S_k has some nice properties. If $\sigma \in S_k$, then $\sigma^{-1} \in S_k$. The inverse permutation σ^{-1} is defined by $\sigma^{-1}(j) = i$ if $\sigma(i) = j$. Another nice property is that if $\sigma, \tau \in S_k$, then $\sigma\tau \in S_k$, where $\sigma\tau(i) = \sigma(\tau(i))$. That is, if $\tau(i) = j$ and $\sigma(j) = k$, then $\sigma\tau(i) = k$.

Take $1 \leq i < j \leq k$, and define

$$\tau_{i,j}(i) = j \tag{4.40}$$
$$\tau_{i,j}(j) = i \tag{4.41}$$
$$\tau_{i,j}(\ell) = \ell, \ell \neq i, j. \tag{4.42}$$

The permutation $\tau_{i,j}$ is a transposition. It is an elementary transposition of $j = i+1$. Last time we stated the following theorem.

Theorem 4.13. *Every permutation σ can be written as a product*

$$\sigma = \tau_1 \tau_2 \cdots \tau_r, \tag{4.43}$$

where the τ_i's are elementary transpositions.

In the above, we removed the symbol $\circ$ denoting composition of permutations, but the composition is still implied.

Last time we also defined the sign of a permutation

Definition 4.14. The sign of a permutation σ is $(-1)^\sigma = (-1)^r$, where r is as in the above theorem.

Theorem 4.15. *The above definition of sign is well-defined, and*

$$(-1)^{\sigma\tau} = (-1)^\sigma (-1)^\tau. \tag{4.44}$$

All of the above is discussed in the Multi-linear Algebra Notes.

Part of today's homework is to show the following two statements:

1. $|S_k| = k!$. The proof is by induction.
2. $(-1)^{\tau_{i,j}} = -1$. Hint: use induction and $\tau_{i,j} = (\tau_{j-1,j})(\tau_{i,j-1})(\tau_{j-1,j})$, with $i < j$.

We now move back to the study of tensors. Let V be an n-dimensional vector space. We define

$$V^k = \underbrace{V \times \cdots \times V}_{k \text{ factors}}. \tag{4.45}$$

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We define $\mathcal{L}^k(v)$ to be the space of all k-linear functions $T : V^k \to \mathbb{R}$. If $T_i \in \mathcal{L}^{k_i}, i = 1, 2$, and $k = k_1 + k_2$, then $T_1 \otimes T_2 \in \mathcal{L}^k$. Decomposable k-tensors are of the form $T = \ell_1 \otimes \cdots \otimes \ell_k$, where each $\ell_i \in \mathcal{L}^1 = V^*$. Note that $\ell_1 \otimes \cdots \otimes \ell_k(v_1, \ldots, v_k) = \ell_1(v_1) \ldots \ell_k(v_k)$.

We define a permutation operation on tensors. Take $\sigma \in S_k$ and $T \in \mathcal{L}^k(V)$.

Definition 4.16. We define the map $T^\sigma : V^k \to \mathbb{R}$ by

$$T^\sigma(v_1, \ldots, v_k) = T(v_{\sigma^{-1}(1)}, \ldots, v_{\sigma^{-1}(k)}). \tag{4.46}$$

Clearly, $T^\sigma \in \mathcal{L}^k(V)$. We have the following useful formula:

Claim.

$$(T^\sigma)^\tau = T^{\tau\sigma}. \tag{4.47}$$

Proof.

$$\begin{aligned} T^{\tau\sigma}(v_1, \ldots, v_k) &= T(v_{\sigma^{-1}(\tau^{-1}(1))}, \ldots, v_{\sigma^{-1}(\tau^{-1}(k))}) \\ &= T^\sigma(v_{\tau^{-1}(1)}, \ldots, v_{\tau^{-1}(k)}) \\ &= (T^\sigma)^\tau(v_1, \ldots, v_k). \end{aligned} \tag{4.48}$$

□

Let us look at what the permutation operation does to a decomposable tensor $T = \ell_1 \otimes \cdots \otimes \ell_k$.

$$T^\sigma(v_1, \ldots, v_k) = \ell_1(v_{\sigma^{-1}(1)}) \ldots \ell_k(v_{\sigma^{-1}(k)}). \tag{4.49}$$

The ith factor has the subscript $\sigma^{-1}(i) = j$, where $\sigma(j) = i$, so the the ith factor is $\ell_{\sigma(j)}(v_j)$. So

$$\begin{aligned} T^\sigma(v_1, \ldots, v_k) &= \ell_{\sigma(1)}(v_1) \ldots \ell_{\sigma(k)}(v_k) \\ &= (\ell_{\sigma(1)} \otimes \cdots \otimes \ell_{\sigma(k)})(v_1, \ldots, v_k). \end{aligned} \tag{4.50}$$

To summarize,

$$\begin{cases} T = \ell_1 \otimes \cdots \otimes \ell_k \\ T^\sigma = \ell_{\sigma(1)} \otimes \cdots \otimes \ell_{\sigma(k)}. \end{cases} \tag{4.51}$$

Proposition 4.17. *The mapping $T \in \mathcal{L}^k \to T^\sigma \in \mathcal{L}^k$ is linear.*

We leave the proof of this as an exercise.

Definition 4.18. A tensor $T \in \mathcal{L}^k(V)$ is *alternating* if $T^\sigma = (-1)^\sigma T$ for all $\sigma \in S_k$.

Definition 4.19. We define

$$\mathcal{A}^k(V) = \text{ the set of all alternating } k\text{-tensors.} \tag{4.52}$$

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By our previous claim, $\mathcal{A}^k$ is a vector space.

The *alternating operator* Alt can be used to create alternating tensors.

Definition 4.20. Given a k-tensor $T \in \mathcal{L}^k(V)$, we define the *alternating operator* $\text{Alt} : \mathcal{L}^k(V) \to \mathcal{A}^k(V)$ by

$$\text{Alt}(T) = \sum_{\tau \in S_k} (-10)^\tau T^\tau. \tag{4.53}$$

Claim. *The alternating operator has the following properties:*

1. $\text{Alt}(T) \in \mathcal{A}^k(V)$,
2. *If* $T \in \mathcal{A}^k(V)$, *then* $\text{Alt}(T) = k!T$,
3. $\text{Alt}(T^\sigma) = (-1)^\sigma \text{Alt}(T)$,
4. *The map* $\text{Alt} : \mathcal{L}^k(V) \to \mathcal{A}^k(V)$ *is linear.*

Proof. 1.

$$\text{Alt}(T) = \sum_\tau (-1)^\tau T^\tau, \tag{4.54}$$

so

$$\begin{aligned} \text{Alt}(T)^\sigma &= \sum_\tau (-1)^\tau (T^\tau)^\sigma \\ &= \sum_\tau (-1)^\tau T^{\sigma\tau} \\ &= (-1)^\sigma \sum_{\sigma\tau} (-1)^{\sigma\tau} T^{\sigma\tau} \\ &= (-1)^\sigma \text{Alt}(T). \end{aligned} \tag{4.55}$$

2.

$$\text{Alt}(T) = \sum_\tau (-1)^\tau T^\tau, \tag{4.56}$$

but $T^\tau = (-1)^\tau T$, since $T \in \mathcal{A}^k(V)$. So

$$\begin{aligned} \text{Alt}(T) &= \sum_\tau (-1)^\tau (-1)^\tau T \\ &= k!T. \end{aligned} \tag{4.57}$$

3.

$$\begin{aligned} \text{Alt}(T^\sigma) &= \sum_\tau (-1)^\tau (T^\sigma)^\tau \\ &= \sum_\tau (-1)^\tau T^{\tau\sigma} \\ &= (-1)^\sigma \sum_{\tau\sigma} (-1)^{\tau\sigma} T^{\tau\sigma} \\ &= (-1)^\sigma \text{Alt}(T). \end{aligned} \tag{4.58}$$

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4. We leave the proof as an exercise.

□

Now we ask ourselves: what is the dimension of $\mathcal{A}^k(V)$? To answer this, it is best to write a basis.

Earlier we found a basis for $\mathcal{L}^k$. We defined $e_1, \ldots, e_n$ to be a basis of V and $e_1^*, \ldots, e_n^*$ to be a basis of V^*. We then considered multi-indices $I = (i_1, \ldots, i_k), 1 \leq i_r \leq n$ and defined $\{e_I^* = e_{i_1}^* \otimes \cdots \otimes e_{i_k}^*,\ I \text{ a multi-index}\}$ to be a basis of $\mathcal{L}^k$. For any multi-index $J = (j_1, \ldots, j_k)$, we had

$$e_I^*(e_{j_1}, \ldots, e_{j_k}) = \begin{cases} 1 & \text{if } I = J, \\ 0 & \text{if } I \neq J. \end{cases} \tag{4.59}$$

Definition 4.21. A multi-index $I = (i_1, \ldots, i_k)$ is *repeating* if $i_r = i_s$ for some $r < s$.

Definition 4.22. The multi-index I is *strictly increasing* if $1 \leq i_1 < \ldots < i_k \leq n$.

Notation. Given $\sigma \in S_k$ and $I = (i_1, \ldots, i_k)$, we denote $I^\sigma = (i_{\sigma(1)}, \ldots, i_{\sigma(k)})$.

Remark. If J is a non-repeating multi-index, then there exists a permutation σ such that $J = I^\sigma$, where I is strictly increasing.

$$e_J^* = e_{I^\sigma}^* = e_{\sigma(i_1)}^* \otimes \cdots \otimes e_{\sigma(i_k)}^* = (e_I^*)^\sigma. \tag{4.60}$$

Define $\psi_I = \operatorname{Alt}(e_I^*)$.

Theorem 4.23. *1. $\psi_{I^\sigma} = (-1)^\sigma \psi_I$,*

2. If I is repeating, then $\psi_I = 0$,

3. If I, J are strictly increasing, then

$$\psi_I(e_{j_1}, \ldots, e_{j_k}) = \begin{cases} 1 & \text{if } I = J, \\ 0 & \text{if } I \neq J. \end{cases} \tag{4.61}$$

Proof. 1.

$$\begin{aligned} \psi_{I^\sigma} &= \operatorname{Alt} e_{I^\sigma}^* \\ &= \operatorname{Alt}((e_I^*)^\sigma) \\ &= (-1)^\sigma \operatorname{Alt} e_I^* \\ &= (-1)^\sigma \psi_I. \end{aligned} \tag{4.62}$$

2. Suppose that I is repeating. Then $I = I^\tau$ for some transposition τ. So $\psi_I = (-1)^\tau \psi_I$. But (as you proved in the homework) $(-1)^\tau = -1$, so $\psi_I = 0$.

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3.

$$\begin{aligned}\psi_I &= \operatorname{Alt}(e_I^*) \\ &= \sum_\tau (-1)^\tau e_{I^\tau}^*,\end{aligned} \tag{4.63}$$

so

$$\psi_I(e_{j_1}, \ldots, e_{j_k}) = \sum_\tau (-1)^\tau \underbrace{e_{I^\tau}^*(e_{j_1}, \ldots, e_{j_k})}_{\begin{cases}1 & \text{if } I^\tau = J, \\ 0 & \text{if } I^\tau \neq J.\end{cases}} \tag{4.64}$$

But $I^\tau = J$ only if τ is the identity permutation (because both I^τ and J are strictly increasing). The only non-zero term in the sum is when τ is the identity permutation, so

$$\psi_I(e_{j_1}, \ldots, e_{j_k}) = \begin{cases}1 & \text{if } I = J, \\ 0 & \text{if } I \neq J.\end{cases} \tag{4.65}$$

□

Corollary 5. *The alternating k-tensors ψ_I, where I is strictly increasing, are a basis of $\mathcal{A}^k(V)$.*

Proof. Take $T \in \mathcal{A}^k(V)$. The tensor T can be expanded as $T = \sum c_I e_I^*$. So

$$\begin{aligned}\operatorname{Alt}(T) &= k! \sum c_I \operatorname{Alt}(e_I^*) \\ &= k! \sum c_I \psi_I.\end{aligned} \tag{4.66}$$

If I is repeating, then $\psi_I = 0$. If I is non-repeating, then $I = J^\sigma$, where J is strictly increasing. Then $\psi_I = (-1)^\sigma \psi_J$.

So, we can replace all multi-indices in the sum by strictly increasing multi-indices,

$$T = \sum a_I \psi_I, \ \ I\text{'s strictly increasing.} \tag{4.67}$$

Therefore, the ψ_I's span $\mathcal{A}^k(V)$. Moreover, the ψ_I's are a basis if and only if the a_i's are unique. We show that the a_I's are unique.

Let J be any strictly increasing multi-index. Then

$$\begin{aligned}T(e_{j_1}, \ldots, e_{j_k}) &= \sum a_I \psi(e_{j_1}, \ldots, e_{j_k}) \\ &= a_J,\end{aligned} \tag{4.68}$$

by property (3) of the previous theorem. Therefore, the ψ_I's are a basis of $\mathcal{A}^k(V)$. □

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Lecture 19

We begin with a review of tensors and alternating tensors.

We defined $\mathcal{L}^k(V)$ to be the set of k-linear maps $T : V^k \to \mathbb{R}$. We defined $e_1, \dots, e_n$ to be a basis of V and $e_1^*, \dots, e_n^*$ to be a basis of V^*. We also defined $\{e_I^* = e_{i_1}^* \otimes \cdots \otimes e_{i_k}^*\}$ to be a basis of $\mathcal{L}^k(V)$, where $I = (i_1, \dots, i_k)$, $1 \le i_r \le n$ is a multi-index. This showed that $\dim \mathcal{L}^k = n^k$.

We defined the permutation operation on a tensor. For $\sigma \in S_n$ and $T \in \mathcal{L}^k$, we defined $T^\sigma \in \mathcal{L}^k$ by $T^\sigma(v_1, \dots, v_k) = T(v_{\sigma^{-1}(1)}, \dots, v_{\sigma^{-1}(k)})$. Then we defined that T is *alternating* if $T^\sigma = (-1)^\sigma T$. We defined $\mathcal{A}^k = \mathcal{A}^k(V)$ to be the space of all alternating k-tensors.

We defined the alternating operator $\operatorname{Alt} : \mathcal{L}^k \to \mathcal{A}^k$ by $\operatorname{Alt}(T) = \sum (-1)^\sigma T^\sigma$, and we defined $\psi_I = \operatorname{Alt}(e_I^*)$, where $I = (i_1, \dots, i_k)$ is a strictly increasing multi-index. We proved the following theorem:

Theorem 4.24. *The ψ_I's (where I is strictly increasing) are a basis for $\mathcal{A}^k(V)$.*

Corollary 6. *If $0 \le k \le n$, then*

$$\dim \mathcal{A}^k = \binom{n}{k} = \frac{n!}{k!(n-k)!}. \tag{4.69}$$

Corollary 7. *If $k > n$, then $\mathcal{A}^k = \{0\}$.*

We now ask what is the kernel of Alt ? That is, for which $T \in \mathcal{L}^k$ is $\operatorname{Alt}(T) = 0$?

Let $T \in \mathcal{L}^k$ be a decomposable k-tensor, $T = \ell_1 \otimes \cdots \otimes \ell_k$, where each $\ell_i \in V^*$.

Definition 4.25. The k-tensor T is *redundant* if $\ell_i = \ell_{i+1}$ for some $1 \le i \le k-1$.

We define

$$\mathcal{I}^k \equiv \operatorname{Span}\{ \text{ redundant } k\text{-tensors } \}. \tag{4.70}$$

Claim. *If $T \in \mathcal{I}^k$, then $\operatorname{Alt}(T) = 0$.*

Proof. It suffices to prove this for $T = \ell_1 \otimes \cdots \otimes \ell_k$, where $\ell_1 = \ell_{i+1}$ (T is redundant). Let $\tau = \tau_{i,i+1} \in S_k$. So, $T^\tau = T$. But

$$\begin{aligned} \operatorname{Alt}(T^\tau) &= (-1)^\tau \operatorname{Alt}(T) \\ &= -\operatorname{Alt}(T), \end{aligned} \tag{4.71}$$

so $\operatorname{Alt}(T) = 0$. □

Claim. *Suppose that $T \in \mathcal{I}^k$ and $T' \in \mathcal{L}^m$. Then $T' \otimes T \in \mathcal{I}^{k+n}$ and $T \otimes T' \in \mathcal{I}^{k+m}$.*

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Proof. We can assume that T and T' are both decomposable tensors.

$$T = \ell_1 \otimes \cdots \otimes \ell_k, \quad \ell_i = \ell_{i+1}, \tag{4.72}$$

$$T' = \ell'_1 \otimes \cdots \otimes \ell'_m, \tag{4.73}$$

$$T \otimes T' = \ell_1 \otimes \cdots \otimes \underbrace{\ell_i \otimes \ell_{i+1}}_{\text{a redundancy}} \otimes \cdots \otimes \ell_k \otimes \ell'_1 \otimes \cdots \otimes \ell'_m \tag{4.74}$$

$$\in \mathcal{I}^{k+m}. \tag{4.75}$$

A similar argument holds for $T' \otimes T$. □

Claim. *For each $T \in \mathcal{L}^k$ and $\sigma \in S_k$, there exists some $w \in \mathcal{I}^k$ such that*

$$T = (-1)^\sigma T^\sigma + W. \tag{4.76}$$

Proof. In proving this we can assume that T is decomposable. That is, $T = \ell_1 \otimes \cdots \otimes \ell_k$.

We first check the case $k = 2$. Let $T = \ell_1 \otimes \ell_2$. The only (non-identity) permutation is $\sigma = \tau_{1,2}$. In this case, $T = (-1)^\sigma T^\sigma + W$ becomes $W = T + T^\sigma$, so

$$\begin{aligned} W &= T + T^\sigma \\ &= \ell_1 \otimes \ell_2 + \ell_2 \otimes \ell_1 \\ &= (\ell_1 + \ell_2) \otimes (\ell_1 + \ell_2) - \ell_1 \otimes \ell_1 - \ell_2 \otimes \ell_2 \\ &\in \mathcal{I}^2. \end{aligned} \tag{4.77}$$

We now check the case k is arbitrary. Let $T = \ell_1 \otimes \cdots \otimes \ell_k$ and $\sigma = \tau_1 \tau_2 \ldots \tau_r \in S_k$, where the τ_i's are elementary transpositions. We will prove that $W \in \mathcal{I}^k$ by induction on r.

- Case $r = 1$: Then $\sigma = \tau_{i,i+1}$, and

$$\begin{aligned} W &= T + T^\sigma \\ &= (\ell_1 \otimes \cdots \otimes \ell_k) + (\ell_1 \otimes \cdots \otimes \ell_k)^\sigma \\ &= \ell_1 \otimes \cdots \otimes \ell_{i-1} \otimes (\ell_i \otimes \ell_{i+1} + \ell_{i+1} \otimes \ell_i) \otimes \ell_{i+2} \otimes \cdots \otimes \ell_k \\ &\in \mathcal{I}^k, \end{aligned} \tag{4.78}$$

  because $(\ell_i \otimes \ell_{i+1} + \ell_{i+1} \otimes \ell_i) \in \mathcal{I}^k$.

- Induction step ($(r-1) \implies r$): Let $\beta = \tau_2 \ldots \tau_r$, and let $\tau = \tau_1$ so that $\sigma = \tau_1 \tau_2 \ldots \tau_r = \tau\beta$. Then

$$T^\sigma = (T^\beta)^\tau. \tag{4.79}$$

  By induction, we know that

$$T^\beta = (-1)^\beta T + W, \tag{4.80}$$

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for some $W \in \mathcal{I}^k$. So,

$$\begin{aligned} T^\sigma &= (-1)^\beta T^\tau + W^\tau \\ &= (-1)^\beta (-1)^\tau T + W^\tau \\ &= (-1)^\sigma T + W^\tau, \end{aligned} \tag{4.81}$$

where $W^\tau = (-1)^\tau W + W' \in \mathcal{I}^k$.

□

Corollary 8. *For every $T \in \mathcal{L}^k$,*

$$\operatorname{Alt}(T) = k!T + W \tag{4.82}$$

for some $W \in \mathcal{I}^k$.

Proof.

$$\operatorname{Alt}(T) = \sum_\sigma (-1)^\sigma T^\sigma, \tag{4.83}$$

but we know that $T^\sigma = (-1)^\sigma T + W_\sigma$, for some $W_\sigma \in \mathcal{I}^k$, so

$$\begin{aligned} \operatorname{Alt}(T) &= \sum_\sigma (T + (-1)^\sigma W_\sigma) \\ &= k!T + W, \end{aligned} \tag{4.84}$$

where $W = \sum_\sigma (-1)^\sigma W_\sigma \in \mathcal{I}^k$. □

Theorem 4.26. *Every $T \in \mathcal{L}^k$ can be written uniquely as a sum*

$$T = T_1 + T_2, \tag{4.85}$$

where $T_1 \in \mathcal{A}^k$ and $T_2 \in \mathcal{I}^k$.

Proof. We know that $\operatorname{Alt}(T) = k!T + W$, for some $W \in \mathcal{I}^k$. Solving for T, we get

$$T = \underbrace{\frac{1}{k!}\operatorname{Alt}(T)}_{T_1} - \underbrace{\frac{1}{k!}W}_{T_2}. \tag{4.86}$$

We check uniqueness:

$$\operatorname{Alt}(T) = \underbrace{\operatorname{Alt}(T_1)}_{k!T_1} + \underbrace{\operatorname{Alt}(T_2)}_{0}, \tag{4.87}$$

so T_1 is unique, which implies that T_2 is also unique. □

Claim.

$$\mathcal{I}^k = \ker \operatorname{Alt}\ . \tag{4.88}$$

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Proof. If Alt $T = 0$, then

$$T = -\frac{1}{k!}W, \;\; W \in \mathcal{I}^k, \tag{4.89}$$

so $T \in \mathcal{I}^k$. □

The space $\mathcal{I}^k$ is a subspace of $\mathcal{L}^k$, so we can form the quotient space

$$\Lambda^k(V^*) \equiv \mathcal{L}^k/\mathcal{I}^k. \tag{4.90}$$

What's up with this notation $\Lambda^k(V^*)$? We motivate this notation with the case $k = 1$. There are no redundant 1-tensors, so $\mathcal{I}^1 = \{0\}$, and we already know that $\mathcal{L}^1 = V^*$. So

$$\Lambda^1(V^*) = V^*/\mathcal{I}^1 = \mathcal{L}^1 = V^*. \tag{4.91}$$

Define the map $\pi : \mathcal{L}^k \to \mathcal{L}^k/\mathcal{I}^k$. The map π is onto, and $\ker \pi = \mathcal{I}^k$.

Claim. *The map π maps $\mathcal{A}^k$ bijectively onto $\Lambda^k(V^*)$.*

Proof. Every element of Λ^k is of the form $\pi(T)$ for some $T \in \mathcal{L}^k$. We can write $T = T_1 + T_2$, where $T_1 \in \mathcal{A}^k$ and $T_2 \in \mathcal{I}^k$. So,

$$\begin{aligned} \pi(T) &= \pi(T_1) + \pi(T_2) \\ &= \pi(T_1) + 0 \\ &= \pi(T_1). \end{aligned} \tag{4.92}$$

So, π maps $\mathcal{A}^k$ onto Λ^k. Now we show that π is one-to-one. If $T \in \mathcal{A}^k$ and $\pi(T) = 0$, then $T \in \mathcal{I}^k$ as well. We know that $\mathcal{A}^k \cap \mathcal{I}^k = \{0\}$, so π is bijective. □

We have shown that

$$\mathcal{A}^k(V) \cong \Lambda^k(V^*). \tag{4.93}$$

The space $\Lambda^k(V^*)$ is not mentioned in Munkres, but sometimes it is useful to look at the same space in two different ways.

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Lecture 20

We begin with a review of last lecture.

Consider a vector space V. A tensor $T \in \mathcal{L}^k$ is *decomposable* if $T = \ell_1 \otimes \cdots \otimes \ell_k$, $\ell_i \in \mathcal{L}^1 = V^*$. A decomposable tensor T is *redundant* of $\ell_i = \ell_{i+1}$ for some i. We define

$$\mathcal{I}^k = \mathcal{I}^k(V) = \text{Span}\{ \text{ redundant } k\text{-tensors } \}. \tag{4.94}$$

Because $\mathcal{I}^k \subseteq \mathcal{L}^k$, we can take the quotient space

$$\Lambda^k = \Lambda^k(V^*) = \mathcal{L}^k / \mathcal{I}^k, \tag{4.95}$$

defining the map

$$\pi : \mathcal{L}^k \to \Lambda^k. \tag{4.96}$$

We denote by $\mathcal{A}^k(V)$ the set of all alternating k-tensors. We repeat the main theorem from last lecture:

Theorem 4.27. *The map π maps $\mathcal{A}^k$ bijectively onto Λ^k. So, $\mathcal{A}^k \cong \Lambda^k$.*

It is easier to understand the space $\mathcal{A}^k$, but many theorems are much simpler when using Λ^k. This ends the review of last lecture.

4.6 Wedge Product

Now, let $T_1 \in \mathcal{I}^{k_1}$ and $T_2 \in \mathcal{L}^{k_2}$. Then $T_1 \otimes T_2$ and $T_2 \otimes T_1$ are in $\mathcal{I}^k$, where $k = k_1 + k_2$. The following is an example of the usefulness of Λ^k.

Let $\mu_i \in \Lambda^{k_i}$, $i = 1, 2$. So, $\mu_i = \pi(T_i)$ for some $T_i \in \mathcal{L}^{k_i}$. Define $k = k_1 + k_2$, so $T_1 \otimes T_2 \in \mathcal{L}^k$. Then, we define

$$\pi(T_1 \otimes T_2) = \mu_1 \wedge \mu_2 \in \Lambda^k. \tag{4.97}$$

Claim. *The product $\mu_i \wedge \mu_2$ is well-defined.*

Proof. Take any tensors $T_i' \in \mathcal{L}^{k_i}$ with $\pi(T_i') = \mu_i$. We check that

$$\pi(T_1' \otimes T_2') = \pi(T_1 \otimes T_2). \tag{4.98}$$

We can write

$$T_1' = T_1 + W_1, \text{ where } W_1 \in \mathcal{I}^{k_1}, \tag{4.99}$$

$$T_2' = T_2 + W_2, \text{ where } W_2 \in \mathcal{I}^{k_2}. \tag{4.100}$$

Then,

$$T_1' \otimes T_2' = T_1 \otimes T_2 + \underbrace{W_1 \otimes T_2 + T_1 \otimes W_2 + W_1 \otimes W_2}_{\in \mathcal{I}^k}, \tag{4.101}$$

so

$$\mu_1 \wedge \mu_2 \equiv \pi(T_1' \otimes T_2') = \pi(T_1 \otimes T_2). \tag{4.102}$$

□

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This product ($\wedge$) is called the *wedge product*. We can define higher order wedge products. Given $\mu_i \in \Lambda^{k_i}$, $i = 1, 2, 3$, where $\mu = \pi(T_i)$, we define

$$\mu_1 \wedge \mu_2 \wedge \mu_3 = \pi(T_1 \otimes T_2 \otimes T_3). \tag{4.103}$$

We leave as an exercise to show the following claim.

Claim.

$$\begin{aligned} \mu_1 \wedge \mu_2 \wedge \mu_3 &= (\mu_1 \wedge \mu_2) \wedge \mu_3 \\ &= \mu_1 \wedge (\mu_2 \wedge \mu_3). \end{aligned} \tag{4.104}$$

Proof Hint: This triple product law also holds for the tensor product. □

We leave as an exercise to show that the two distributive laws hold:

Claim. *If $k_1 = k_2$, then*

$$(\mu_1 + \mu_2) \wedge \mu_3 = \mu_1 \wedge \mu_3 + \mu_2 \wedge \mu_3. \tag{4.105}$$

If $k_2 = k_3$, then

$$\mu_1 \wedge (\mu_2 + \mu_3) = \mu_1 \wedge \mu_2 + \mu_1 \wedge \mu_3. \tag{4.106}$$

Remember that $\mathcal{I}^1 = \{0\}$, so $\Lambda^1 = \Lambda^1/\mathcal{I}^1 = \mathcal{L}^1 = \mathcal{L}^1(V) = V^*$. That is, $\Lambda^1(V^*) = V^*$.

Definition 4.28. The element $\mu \in \Lambda^k$ is *decomposable* if it is of the form $\mu = \ell_1 \wedge \cdots \wedge \ell_k$, where each $\ell_i \in \Lambda^1 = V^*$.

That means that $\mu = \pi(\ell_1 \otimes \cdots \otimes \ell_k)$ is the projection of a decomposable k-tensor.

Take a permutation $\sigma \in S_k$ and an element $\omega \in \Lambda^k$ such that $\omega = \pi(T)$, where $T \in \mathcal{L}^k$.

Definition 4.29.

$$\omega^\sigma = \pi(T^\sigma). \tag{4.107}$$

We need to check that this definition does not depend on the choice of T.

Claim. *Define $\omega^\sigma = \pi(T^\sigma)$. Then,*

1. *The above definition does not depend on the choice of T,*
2. *$\omega^\sigma = (-1)^\sigma \omega$.*

Proof. 1. Last lecture we proved that for $T \in \mathcal{L}^k$,

$$T^\sigma = (-1)^\sigma T + W, \tag{4.108}$$

for some $W \in \mathcal{I}^k$. Hence, if $T \in \mathcal{I}^k$, then $T^\sigma \in \mathcal{I}^k$. If $\omega = \pi(T) = \pi(T')$, then $T' - T \in \mathcal{I}^k$. Thus, $(T')^\sigma - T^\sigma \in \mathcal{I}^k$, so $\omega^\sigma = \pi((T')^\sigma) = \pi(T^\sigma)$.

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2.

$$T^\sigma = (-1)^\sigma T + W, \tag{4.109}$$

for some $W \in \mathcal{I}^k$, so

$$\pi(T^\sigma) = (-1)^\sigma \pi(T). \tag{4.110}$$

That is,

$$\omega^\sigma = (-1)^\sigma \omega. \tag{4.111}$$

□

Suppose ω is decomposable, so $\omega = \ell_1 \wedge \cdots \wedge \ell_k,\ \ell_i \in V^*$. Then $\omega = \pi(\ell_1 \wedge \cdots \wedge \ell_k)$, so

$$\begin{aligned} \omega^\sigma &= \pi((\ell_1 \otimes \cdots \otimes \ell_k)^\sigma) \\ &= \pi(\ell_{\sigma(1)} \otimes \cdots \otimes \ell_{\sigma(k)}) \\ &= \ell_{\sigma(1)} \wedge \cdots \wedge \ell_{\sigma(k)}. \end{aligned} \tag{4.112}$$

Using the previous claim,

$$\ell_{\sigma(1)} \wedge \cdots \wedge \ell_{\sigma(k)} = (-1)^\sigma \ell_1 \wedge \cdots \wedge \ell_k. \tag{4.113}$$

For example, if $k = 2$, then $\sigma = \tau_{1,2}$. So, $\ell_2 \wedge \ell_1 = -\ell_1 \wedge \ell_2$. In the case $k = 3$, we find that

$$\begin{aligned} (\ell_1 \wedge \ell_2) \wedge \ell_3 &= \ell_1 \wedge (\ell_2 \wedge \ell_3) \\ &= -\ell_1 \wedge (\ell_3 \wedge \ell_2) = -(\ell_1 \wedge \ell_3) \wedge \ell_2 \\ &= \ell_3 \wedge (\ell_1 \wedge \ell_2). \end{aligned} \tag{4.114}$$

This motivates the following claim, the proof of which we leave as an exercise.

Claim. *If $\mu \in \Lambda^2$ and $\ell \in \Lambda^1$, then*

$$\mu \wedge \ell = \ell \wedge \mu. \tag{4.115}$$

Proof Hint: Write out μ as a linear combination of decomposable elements of Λ^2. □

Now, suppose $k = 4$. Moving ℓ_3 and ℓ_4 the same distance, we find that

$$(\ell_1 \wedge \ell_2) \wedge (\ell_3 \wedge \ell_4) = (\ell_3 \wedge \ell_4) \wedge (\ell_1 \wedge \ell_2). \tag{4.116}$$

The proof of the following is an exercise.

Claim. *If $\mu \in \Lambda^2$ and $\nu \in \Lambda^2$, then*

$$\mu \wedge \nu = \nu \wedge \mu. \tag{4.117}$$

We generalize the above claims in the following:

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Claim. *Left $\mu \in \Lambda^k$ and $\nu \in \Lambda^\ell$. Then*

$$\mu \wedge \nu = (-1)^{k\ell} \nu \wedge \mu. \tag{4.118}$$

Proof Hint: First assume k is even, and write out μ as a product of elements all of degree two. Second, assume that k is odd. □

Now we try to find a basis for $\Lambda^k(V^*)$. We begin with

$$e_1, \ldots, e_n \text{ a basis of } V, \tag{4.119}$$
$$e_1^*, \ldots, e_n^* \text{ a basis of } V^*, \tag{4.120}$$
$$e_I^* = e_{i_1}^* \otimes \cdots \otimes e_{i_k}^*,\ I = (i_1, \ldots, i_k), 1 \le i_r \le n, \text{ a basis of } \mathcal{L}^k, \tag{4.121}$$
$$\psi_I = \text{Alt}\,(e_I^*), \quad I\text{'s strictly increasing, a basis of } \mathcal{A}^k(V). \tag{4.122}$$

We know that π maps $\mathcal{A}^k$ bijectively onto Λ^k, so $\pi(\psi_I)$, where I is strictly increasing, are a basis of $\Lambda^k(V^*)$.

$$\psi_I = \text{Alt } e_I^* = \sum (-1)^\sigma (e_I^*)^\sigma. \tag{4.123}$$

So,

$$\begin{aligned} \pi(\psi_I) &= \sum (-1)^\sigma \pi((e_I^*)^\sigma) \\ &= \sum (-1)^\sigma (-1)^\sigma \pi(e_I^*) \\ &= k! \pi(e_I^*) \\ &\equiv k! \tilde{e}_I. \end{aligned} \tag{4.124}$$

Theorem 4.30. *The elements of $\Lambda^k(V^*)$*

$$\tilde{e}_{i_1}^* \wedge \cdots \wedge \tilde{e}_{i_k}^*,\ 1 \le i_1 < \ldots < i_k \le n \tag{4.125}$$

are a basis of $\Lambda^k(V^*)$.

Proof. The proof is above. □

Let V, W be vector spaces, and let $A : V \to W$ be a linear map. We previously defined the pullback operator $A^* : \mathcal{L}^k(W) \to \mathcal{L}^k(V)$. Also, given $T_i \in \mathcal{L}^{k_i}(W), i = 1, 2$, we showed that $A^*(T_1 \otimes T_2) = A^* T_1 \otimes A^* T_2$. So, if $T = \ell_1 \otimes \cdots \otimes \ell_k \in \mathcal{L}_k(W)$ is decomposable, then

$$A^* T = A^* \ell_1 \otimes \cdots \otimes A^* \ell_k, \quad \ell_i \in W^*. \tag{4.126}$$

If $\ell_i = \ell_{i+1}$, then $A^* \ell_1 = A^* \ell_{i+1}$. This shows that if $\ell_1 \otimes \cdots \otimes \ell_k$ is redundant, then $A^*(\ell_1 \otimes \cdots \otimes \ell_k)$ is also redundant. So,

$$A^* \mathcal{I}^k(W) \subseteq \mathcal{I}^k(V). \tag{4.127}$$

Let $\mu \in \Lambda^k(W^*)$, so $\mu = \pi(T)$ for some $T \in \mathcal{L}^k(W)$. We can pullback to get $\pi(A^* T) \in \Lambda^k(V^*)$.

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Definition 4.31. $A^*\mu = \pi(A^*T)$.

This definition makes sense. If $\mu = \pi(T) = \pi(T')$, then $T' - T \in \mathcal{I}^k(W)$. So $A^*T' - A^*T \in \mathcal{I}^k(V)$, which shows that $A^*\mu = \pi(A^*T') = \pi(A^*T)$.

We ask in the homework for you to show that the pullback operation is linear and that

$$A^*(\mu_1 \wedge \mu_2) = A^*\mu_1 \wedge A^*\mu_2. \tag{4.128}$$

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Lecture 21

Let V, W be vector spaces, and let $A : V \to W$ be a linear map. We defined the pullback operation $A^* : W^* \to V^*$. Last time we defined another pullback operator having the form $A^* : \Lambda^k(W^*) \to \Lambda^k(V^*)$. This new pullback operator has the following properties:

1. A^* is linear.
2. If $\omega_i \in \Lambda^{k_1}(W^*),\ i = 1, 2$, then $A^*\omega_1 \wedge \omega_2 = A^*\omega_1 \wedge \omega_2$.
3. If ω is decomposable, that is if $\omega = \ell_1 \wedge \cdots \wedge \ell_k$ where $\ell_i \in W^*$, then $A^*\omega = A^*\ell_1 \wedge \cdots \wedge A^*\ell_k$.
4. Suppose that U is a vector space and that $B : W \to U$ is a linear map. Then, for every $\omega \in \Lambda^k(U^*)$, $A^*B^*\omega = (BA)^*\omega$.

4.7 Determinant

Today we focus on the pullback operation in the special case where $\dim V = n$. So, we are studying $\Lambda^n(V^*)$, which is called the *nth exterior power of* V.

Note that $\dim \Lambda^n(V^*) = 1$.

Given a linear map $A : V \to V$, what is the pullback operator

$$A^* : \Lambda^n(V^*) \to \Lambda^n(V^*)? \tag{4.129}$$

Since it is a linear map from a one dimensional vector space to a one dimensional vector space, the pullback operator A^* is simply multiplication by some constant λ_A. That is, for all $\omega \in \Lambda^n(V^*)$, $A^*\omega = \lambda_A\omega$.

Definition 4.32. The *determinant of* A is

$$\det(A) = \lambda_A. \tag{4.130}$$

The determinant has the following properties:

1. If $A = I$ is the identity map, then $\det(A) = \det(I) = 1$.
2. If A, B are linear maps of V into V, then $\det(AB) = \det(A)\det(B)$.

   Proof: Let $\omega \in \Lambda^n(V^*)$. Then

$$\begin{aligned} (AB)^*\omega &= \det(AB)\omega \\ &= B^*(A^*\omega) \\ &= B^*(\det A)\omega \\ &= \det(A)\det(B)\omega. \end{aligned} \tag{4.131}$$

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3. If A is onto, then $\det(A) \neq 0$.

   Proof: Suppose that $A : V \to V$ is onto. Then there exists an inverse linear map $A^{-1} : V \to V$ such that $AA^{-1} = I$. So, $\det(A)\det(A^{-1}) = 1$.

4. If A is not onto, then $\det(A) = 0$.

   Proof: Let $W = \mathrm{Im}\,(A)$. If A is not onto, then $\dim W < \dim V$. Let $B : V \to W$ be the map A regarded as a map of V into W, and let $\iota_W : W \to V$ be inclusion. So, $A = \iota_W B$. For all $\omega \in \Lambda^n(V^*)$, $A^*\omega = B^*\iota_W^*\omega$. Note that $\iota_W^*\omega \in \Lambda^n(W^*) = \{0\}$ because $\dim W < n$. So, $A^*\omega = B^*\iota_W^* = 0$, which shows that $\det(A) = 0$.

Let W, V be n-dimensional vector spaces, and let $A : V \to W$ be a linear map. We have the bases

$$e_1, \ldots, e_n \text{ basis of } V, \tag{4.132}$$
$$e_1^*, \ldots, e_n^* \text{ dual basis of } V^*, \tag{4.133}$$
$$f_1, \ldots, f_n \text{ basis of } W, \tag{4.134}$$
$$f_1^*, \ldots, f_n^* \text{ dual basis of } W^*. \tag{4.135}$$

We can write $Ae_i = \sum a_{ij} f_j$, so that A has the associated matrix $A \sim [a_{ij}]$. Then $A^* f_j^* = \sum a_{jk} e_k^*$. Take $\omega = f_1^* \wedge \cdots \wedge f_n^* \in \Lambda^n(W^*)$, which is a basis vector of $\Lambda^n(W^*)$. Let us compute its pullback:

$$\begin{aligned} A^*(f_1^* \wedge \cdots \wedge f_n^*) &= \left(\sum_{k_1=1}^n a_{1,k_1} e_{k_1}^*\right) \wedge \cdots \wedge \left(\sum_{k_n=1}^n a_{n,k_n} e_{k_n}^*\right) \\ &= \sum_{k_1,\ldots,k_n} (a_{1,k_1} \ldots a_{m,k_n}) e_{k_1}^* \wedge \cdots \wedge e_{k_n}^*. \end{aligned} \tag{4.136}$$

Note that if $k_r = k_s$, where $r \neq s$, then $e_{k_1}^* \wedge \cdots \wedge e_{k_n}^* = 0$. If there are no repetitions, then there exists $\sigma \in S_n$ such that $k_i = \sigma(i)$. Thus,

$$\begin{aligned} A^*(f_1^* \wedge \cdots \wedge f_n^*) &= \sum_\sigma a_{1,\sigma(1)} \ldots a_{n,\sigma(n)} e_{\sigma(1)}^* \wedge \cdots \wedge e_{\sigma(n)}^* \\ &= \left(\sum_\sigma (-1)^\sigma a_{1,\sigma(1)} \ldots a_{n,\sigma(n)}\right) e_1^* \wedge \cdots \wedge e_n^*. \end{aligned} \tag{4.137}$$

Therefore,

$$\det[a_{ij}] = \sum_\sigma (-1)^\sigma a_{1,\sigma(1)} \ldots a_{n,\sigma(n)}. \tag{4.138}$$

In the case where $W = V$ and each $e_i = f_i$, we set $\omega = e_1^* \wedge \cdots \wedge e_n^*$, and we get $A^*\omega = \det[a_{ij}]\omega$. So, $\det(A) = \det[a_{ij}]$.

For basic facts about determinants, see Munkres section 2. We will use these results quite a lot in future lectures. We list some of the basic results below.

Let $A = [a_{ij}]$ be an $n \times n$ matrix.

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1. $\det(A) = \det(A^t)$. You should prove this as an exercise. You should explain the following steps:

$$\begin{aligned}\det(A) &= \sum_\sigma (-1)^\sigma a_{1,\sigma(1)} \dots a_{n,\sigma(n)} \\ &= \sum_\tau (-1)^\tau a_{\tau(1),1} \dots a_{\tau(n),n}, \text{ where } \tau = \sigma^{-1} \\ &= \det(A^t).\end{aligned} \tag{4.139}$$

2. Let

$$A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix}, \tag{4.140}$$

where B is $k \times k$, C is $k \times \ell$, D is $\ell \times \ell$, and $n = k + \ell$. Then

$$\det(A) = \det(B)\det(D) \tag{4.141}$$

4.8 Orientations of Vector Spaces

Let $\ell \subseteq \mathbb{R}^2$ be a line through the origin. Then $\ell - \{0\}$ has two connected components. An *orientation* of ℓ is a choice of one of these components.

More generally, given a one-dimensional vector space $\mathbb{L}$, the set $\mathbb{L}$ has two connected components. Choose $v \in \mathbb{L} - \{0\}$. Then the two components are

$$\{\lambda v : \lambda \in \mathbb{R}_+\} \text{ and } \{-\lambda v : \lambda \in \mathbb{R}_+\}. \tag{4.142}$$

Definition 4.33. An *orientation of* $\mathbb{L}$ is a choice of one of these components, usually labeled $\mathbb{L}_+$. We define

$$v \in \mathbb{L}_+ \iff v \text{ is } \textit{positively oriented}. \tag{4.143}$$

Let V be an n-dimensional vector space. Then $\Lambda^n(V^*)$ is a 1-dimensional vector space.

Definition 4.34. An *orientation of* V is an orientation of $\Lambda^n(V^*)$. That is, a choice of $\Lambda^n(V^*)_+$.

Suppose $e_1, \dots, e_n$ is a basis of V, so $e_1^*, \dots, e_n^*$ is the dual basis of V^*. Let $\omega = e_1^* \wedge \dots \wedge e_n^* \in \Lambda^n(V^*) - \{0\}$.

Definition 4.35. The basis $e_1, \dots, e_n$ is *positively oriented* if $\omega \in \Lambda^n(V^*)_+$.

Let $f_1, \dots, f_n$ be another basis of V and $f_1^*, \dots, f_n^*$ its dual basis. Let $w' = f_1^* \wedge \dots \wedge f_n^*$. We ask: How is ω' related to ω? The answer: If $f_j = \sum a_{ij} e_i$, then $\omega' = \det[a_{ij}]\omega$. So, if $e_1, \dots, e_n$ is positively oriented, then $f_1, \dots, f_n$ is positively oriented if and only if $\det[a_{ij}] > 0$.

Suppose V is an n-dimensional vector space and that W is a k-dimensional subspace of V.

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Claim. *If V and V/W are given orientations, then W acquires from these orientations a natural subspace orientation.*

Idea of proof: Let $\pi : V \to V/W$ be the canonical map, and choose a basis $e_1, \ldots, e_n$ of V such that $e_{\ell+1}, \ldots, e_n$ is a basis of W and such that $\pi(e_1), \ldots, \pi(e_\ell)$ is a basis of V/W, where $\ell = n - k$.

Replacing e_1 by $-e_1$ if necessary, we can assume that $\pi(e_1), \ldots, \pi(e_\ell)$ is an oriented basis of V/W. Replacing e_n by $-e_n$ if necessary, we can assume that $e_1, \ldots, e_n$ is an oriented basis of V. Now, give W the orientation for which $e_{\ell+1}, \ldots, e_n$ is an oriented basis of W. One should check that this choice of orientation for W is independent of the choice of basis (this is explained in the Multi-linear Algebra notes). □

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Lecture 22

In $\mathbb{R}^3$ we had the operators grad, div, and curl. What are the analogues in $\mathbb{R}^n$? Answering this question is the goal of today's lecture.

4.9 Tangent Spaces and k-forms

Let $p \in \mathbb{R}^n$.

Definition 4.36. The *tangent space of p in $\mathbb{R}^n$* is

$$T_p\mathbb{R}^n = \{(p, v) : v \in \mathbb{R}^n\}. \tag{4.144}$$

We identify the tangent space with $\mathbb{R}^n$ via the identification

$$T_p\mathbb{R}^n \cong \mathbb{R}^n \tag{4.145}$$

$$(p, v) \to v. \tag{4.146}$$

Via this identification, the vector space structure on $\mathbb{R}^n$ gives a vector space structure on $T_p\mathbb{R}^n$.

Let U be an open set in $\mathbb{R}^n$, and let $f : U \to \mathbb{R}^m$ be a $\mathcal{C}^1$ map. Also, let $p \in U$ and define $q = f(p)$. We define a linear map

$$df_p : T_p\mathbb{R}^n \to T_q\mathbb{R}^m \tag{4.147}$$

according to the following diagram:

$$\begin{array}{ccc} T_p\mathbb{R}^n & \xrightarrow{df_p} & T_q\mathbb{R}^m \\ \cong\downarrow & & \cong\uparrow \\ \mathbb{R}^n & \xrightarrow{Df(p)} & \mathbb{R}^m. \end{array} \tag{4.148}$$

So,

$$df_p(p, v) = (q, Df(p)v). \tag{4.149}$$

Definition 4.37. The *cotangent space of $\mathbb{R}^n$ at p* is the space

$$T_p^*\mathbb{R}^n \equiv (T_p\mathbb{R}^n)^*, \tag{4.150}$$

which is the dual of the tangent space of $\mathbb{R}^n$ at p.

Definition 4.38. Let U be an open subset of $\mathbb{R}^n$. A *k-form on U* is a function ω which assigns to every point $p \in U$ an element ω_p of $\Lambda^k(T_p^*\mathbb{R}^n)$ (the kth exterior power of $T_p^*\mathbb{R}^n$).

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Let us look at a simple example. Let $f \in \mathcal{C}^\infty(U)$, $p \in U$, and $c = f(p)$. The map

$$df_p : T_p\mathbb{R}^n \to T_c\mathbb{R} = \mathbb{R} \tag{4.151}$$

is a linear map of $T_p\mathbb{R}^n$ into $\mathbb{R}$. That is, $df_p \in T_p^*\mathbb{R}^n$. So, df is the one-form on U which assigns to every $p \in U$ the linear map

$$df_p \in T_p^*\mathbb{R}^n = \Lambda^1(T_p^*\mathbb{R}). \tag{4.152}$$

As a second example, let $f, g \in \mathcal{C}^\infty(U)$. Then gdf is the one-form that maps

$$p \in U \to g(p)df_p \in \Lambda^1(T_p^*\mathbb{R}^n). \tag{4.153}$$

As a third example, let $f, g \in \mathcal{C}^\infty(U)$. Then $\omega = df \wedge dg$ is the two-form that maps

$$p \in U \to df_p \wedge dg_p. \tag{4.154}$$

Note that $df_p, dg_p \in T_p^*\mathbb{R}$, so $df_p \wedge dg_p \in \Lambda^2(T_p^*\mathbb{R}^n)$.

As a fourth and final example, let $f_1, \ldots, f_k \in \mathcal{C}^\infty(U)$. Then $df_1 \wedge \cdots \wedge df_k$ is the k-form that maps

$$p \in U \to (df_1)_p \wedge \cdots \wedge (df_k)_p. \tag{4.155}$$

Note that each $(df_i)_p \in T_p^*\mathbb{R}^n$, so $(df_1)_p \wedge \cdots \wedge (df_k)_p \in \Lambda^k(T_p^*\mathbb{R}^n)$.

Let us now look at what k-forms look like in coordinates. Let $e_1, \ldots, d_n$ be the standard basis of $\mathbb{R}^n$. Let $p \in U$ and let $v_i = (p, e_i)$ for each i. Then, the vectors $v_1, \ldots, v_n$ are a basis of $T_p\mathbb{R}^n$.

Suppose we have a map $f \in \mathcal{C}^\infty(U)$. What is $df_p(v_i)$?

$$df_p(v_i) = De_i f(p) = \frac{\partial f}{\partial x_i}(p). \tag{4.156}$$

In particular, letting x_i be the ith coordinate function,

$$(dx_i)_p(v_j) = \frac{\partial x_i}{\partial x_j} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \tag{4.157}$$

So, $(dx_1)_p, \ldots, (dx_n)_p$ is the basis of $T_p^*\mathbb{R}^n$ dual to $v_1, \ldots, v_n$.

For any $f \in \mathcal{C}^\infty(U)$,

$$\begin{aligned} df_p(v_j) &= \frac{\partial f}{\partial x_j}(p) \\ &= \left(\sum_i \frac{\partial f}{\partial x_i}(p)(dx_i)_p \right)(v_j) \\ \implies df_p &= \sum \frac{\partial f}{\partial x_i}(p)(dx_i)_p \\ \implies df &= \sum \frac{\partial f}{\partial x_i} dx_i. \end{aligned} \tag{4.158}$$

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Since $(dx_1)_p, \ldots, (dx_n)_p$ is a basis of $T_p^*\mathbb{R}^n$, the wedge products

$$(dx_I)_p = (dx_{i_1})_p \wedge \cdots \wedge (dx_{i_k})_p, \;\; 1 \leq i_1 < \cdots < i_k \leq n, \tag{4.159}$$

(I strictly increasing) are a basis of $\Lambda^k(T_p^*\mathbb{R}^n)$.

Therefore, any element w_p of $\Lambda^k(T_p^*\mathbb{R}^n)$ can be written uniquely as a sum

$$\omega_p = \sum a_I(p)(dx_I)_p, \;\; a_I(p) \in \mathbb{R}, \tag{4.160}$$

where the I's are strictly increasing. Hence, any k-form can be written uniquely as a sum

$$\omega = \sum a_I dx_I, \;\; I \text{ strictly increasing}, \tag{4.161}$$

where each a_I is a real-valued function on U. That is, $a_I : U \to \mathbb{R}$.

Definition 4.39. The k-form ω is $\mathcal{C}^r(U)$ if each $a_I \in \mathcal{C}^r(U)$.

Just to simplify our discussion, from now on we will always take k-forms that are $\mathcal{C}^\infty$.

Definition 4.40. We define

$$\Omega^k(U) = \text{ the set of all } \mathcal{C}^\infty \text{ } k\text{-forms}. \tag{4.162}$$

So, $\omega \in \Omega^k(U)$ implies that $\omega = \sum a_I dx_I$, where $a_I \in \mathcal{C}^\infty(U)$.

Let us now study some basic operations on k-forms.

1. Let $\omega \in \Omega^k(U)$ and let $f \in \mathcal{C}^\infty(U)$. Then $f\omega \in \Omega^k(U)$ is the k-form that maps

$$p \in U \to f(p)\omega_p \in \Lambda^k(T_p^*\mathbb{R}^n). \tag{4.163}$$

2. Let $\omega_i \in \Omega^k(U), \; i = 1, 2$. Then $\omega_1 + \omega_2$ is the k-form that maps

$$p \in U \to (\omega_1)_p + (\omega_2)_p \in \Lambda^k(T_p^*\mathbb{R}^n). \tag{4.164}$$

3. Let $\omega_i \in \Omega^{k_i}(U), \; i = 1, 2$, and $k = k_1 + k_2$. Then $w_1 \wedge \omega_2 \in \Omega^k(U)$ is the k-form that maps

$$p \in U \to (\omega_1)_p \wedge (\omega_2)_p \in \Lambda^k(T_p^*\mathbb{R}^n), \tag{4.165}$$

since $(\omega_i)_p \in \Lambda^{k_i}(T_p^*\mathbb{R}^n)$.

Definition 4.41. We find it convenient to define $\Lambda^0(T_p^*\mathbb{R}^n) = \mathbb{R}$.

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A zero-form f on U is just a real-valued function, so $\Omega^0(U) = \mathcal{C}^\infty(\mathbb{R})$.

Take $f \in \mathcal{C}^\infty(U)$ and $df \in \Omega^1(U)$. This gives an operation

$$d : \Omega^0(U) \to \Omega^1(U), \tag{4.166}$$
$$f \to df. \tag{4.167}$$

Let $f, g \in \mathcal{C}^\infty(U)$ (that is, take f, g to be zero-forms). Then $d(fg) = gdf + fdg$. We can think of this operation as a slightly different notation for the gradient operation.

The maps $d : \Omega^k(U) \to \Omega^{k+1}(U),\ k = 0, \ldots, (n-1)$ give n vector calculus operations.

If $\omega \in \Omega^k(U)$, then ω can be written uniquely as the sum

$$\omega = \sum a_I dx_I,\ I \text{ strictly increasing}, \tag{4.168}$$

where each $a_I \in \mathcal{C}^\infty(U)$. We define

$$d\omega = \sum da_I \wedge dx_I. \tag{4.169}$$

This operator is the unique operator with the following properties:

1. For $k = 0$, this is the operation we already defined, $df = \sum \frac{\partial f}{\partial x_i} dx_i$.
2. If $\omega \in \Omega^k$, then $d(d\omega) = 0$.
3. If $\omega_i \in \Omega^{k_i}(U), i = 1, 2$, then $d(\omega_1 \wedge \omega_2) = d\omega_1 \wedge \omega_2 + (-1)^{k_1} \omega_1 \wedge d\omega_2$.

Let $a \in \mathcal{C}^\infty(U)$, and $adx_I \in \Omega^k(U)$, I strictly increasing. Then

$$d(adx_I) = da \wedge dx_I. \tag{4.170}$$

Suppose that I is not strictly increasing. Then

$$\begin{aligned} dx_I &= dx_{i_1} \wedge \cdots \wedge dx_{i_k} \\ &= 0 \ \text{ if } i_r = i_s. \end{aligned} \tag{4.171}$$

If there are no repetitions, then there exists $\sigma \in S_k$ such that $J = I^\sigma$ is strictly increasing. Then

$$dx_J = (-1)^\sigma dx_I, \tag{4.172}$$

so

$$\begin{aligned} d(adx_I) &= (-1)^\sigma d(adx_J) \\ &= (-1)^\sigma da \wedge dx_J \\ &= da \wedge dx_I. \end{aligned} \tag{4.173}$$

Putting this altogether, for *every* multi-index I of length k,

$$d(adx_I) = da \wedge dx_I. \tag{4.174}$$

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Lecture 23

Let U be an open set in $\mathbb{R}^n$. For each $k = 0, \ldots, n-1$, we define the differential operator

$$d : \Omega^k(U) \to \Omega^{k+1}(U). \tag{4.175}$$

These maps are the n basic vector calculus operations in n-dimensional calculus. We review how d is defined.

For $k = 0$, $\Omega^0(U) = \mathcal{C}^\infty(U)$. Let $f \in \mathcal{C}^\infty(U)$, and let $c = f(p)$, where $p \in U$. The mapping $df_p : T_p\mathbb{R}^n \to T_c\mathbb{R} = \mathbb{R}$ maps $T_p\mathbb{R}^n$ to $\mathbb{R}$, so $df_p \in T_p^*\mathbb{R}^n$. The map $df \in \Omega^1(U)$ is a one-form that maps $p \in U$ to $df_p \in T_p^*\mathbb{R}^n$. A formula for this in coordinates is

$$df = \sum \frac{\partial f}{\partial x_i} dx_i. \tag{4.176}$$

In k dimensions, d is a map

$$d : \Omega^k(U) \to \Omega^{k+1}(U). \tag{4.177}$$

Given $\omega \in \Omega^k(U)$, ω can be written uniquely as

$$\begin{aligned} \omega &= \sum_I a_I dx_I \\ &= \sum_I a_I dx_{i_1} \wedge \cdots \wedge dx_{i_k}, \end{aligned} \tag{4.178}$$

where $i_1 < \cdots < i_k$ and each $a_I \in \mathcal{C}^\infty(U)$. Then, we define

$$\begin{aligned} d\omega &= \sum da_I \wedge dx_I \\ &= \sum_{i,I} \frac{\partial a_I}{\partial x_i} dx_i \wedge dx_I, \end{aligned} \tag{4.179}$$

where each I is strictly increasing.

The following are some basic properties of the differential operator d:

1. If $\mu \in \Omega^k(U)$ and $\nu \in \Omega^\ell(U)$, then

$$d\mu \wedge \nu = d\mu \wedge \nu + (-1)^k \mu \wedge d\nu. \tag{4.180}$$

2. For and $\omega \in \Omega^k(U)$,

$$d(d\omega) = 0. \tag{4.181}$$

Remark. Let I be any multi-index, and let $a_I \in \mathcal{C}^\infty(U)$. Then

$$d(a_I dx_I) = da_I \wedge dx_I. \tag{4.182}$$

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We now prove the above two basic properties of the differential operator.

Claim. *If $\mu \in \Omega^k(U)$ and $\nu \in \Omega^\ell(U)$, then*

$$d\mu \wedge \nu = d\mu \wedge \nu + (-1)^k \mu \wedge d\nu. \tag{4.183}$$

Proof. Take $\mu = \sum a_I dx_I$ and $\nu = \sum b_J dx_J$, where I, J are strictly increasing. Then

$$\mu \wedge \nu = \sum a_I b_J \underbrace{dx_I \wedge dx_J}_{\text{no longer increasing}}. \tag{4.184}$$

Then

$$\begin{aligned} d(\mu \wedge \nu) &= \sum_{i,I,J} \frac{\partial a_I b_J}{\partial x_i} dx_i \wedge dx_I \wedge dx_J \\ &= \sum \frac{\partial a_I}{\partial x_i} b_J dx_i \wedge dx_I \wedge dx_J \qquad (I) \\ &\quad + \sum a_I \frac{\partial b_J}{\partial x_i} dx_i \wedge dx_I \wedge dx_J, \quad (II) \end{aligned} \tag{4.185}$$

We calculate sums (I) and (II) separately.

$$\begin{aligned} (I) &= \sum_{i,I,J} \frac{\partial a_I}{\partial x_i} dx_i \wedge dx_I \wedge b_J dx_J \\ &= \left(\sum_{i,I} \frac{\partial a_I}{\partial x_i} dx_i \wedge dx_I \right) \wedge \sum_J b_J dx_J \\ &= d\mu \wedge \nu. \end{aligned} \tag{4.186}$$

$$\begin{aligned} (II) &= \sum_{i,I,J} a_I \frac{\partial b_J}{\partial x_i} dx_i \wedge dx_I \wedge dx_J \\ &= (-1)^k \sum_{i,I,J} a_I dx_I \wedge \frac{\partial b_J}{\partial x_i} dx_i \wedge dx_J \\ &= (-1)^k \left(\sum_I a_I dx_I \right) \wedge \sum_{i,J} \frac{\partial b_J}{\partial x_i} dx_i \wedge dx_J \\ &= (-1)^k \mu \wedge d\nu. \end{aligned} \tag{4.187}$$

So,

$$\begin{aligned} d(\mu \wedge \nu) &= (I) + (II) \\ &= d\mu \wedge \nu + (-1)^k \mu \wedge d\nu. \end{aligned} \tag{4.188}$$

□

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Claim. *For and* $\omega \in \Omega^k(U)$,

$$d(d\omega) = 0. \tag{4.189}$$

Proof. Let $\omega = \sum a_I dx_I$, so

$$d\omega = \sum_{j,I} \frac{\partial a_I}{\partial x_j} dx_j \wedge dx_I. \tag{4.190}$$

Then,

$$d(d\omega) = \sum_{i,j,I} \frac{\partial^2 a_I}{\partial x_i \partial x_j} dx_i \wedge dx_j \wedge dx_I. \tag{4.191}$$

Note that if $i = j$, then there is a repeated term in the wedge product, so

$$d(d\omega) = \sum_{i<j} \frac{\partial^2 a_I}{\partial x_i \partial x_j} dx_i \wedge dx_j \wedge dx_I \tag{4.192}$$

$$+ \sum_{i>j} \frac{\partial^2 a_I}{\partial x_i \partial x_j} dx_i \wedge dx_j \wedge dx_I. \tag{4.193}$$

Note that $dx_i \wedge dx_j = -dx_j \wedge dx_i$. We relabel the second summand to obtain

$$\begin{aligned} d(d\omega) &= \sum_{i<j} \left(\underbrace{\frac{\partial^2 a_I}{\partial x_i \partial x_j} - \frac{\partial^2 a_I}{\partial x_j \partial x_i}}_{0} \right) dx_i \wedge dx_j \wedge dx_I \\ &= 0. \end{aligned} \tag{4.194}$$

□

Definition 4.42. A k-form $\omega \in \Omega^k(U)$ is *decomposable* if $\omega = \mu_1 \wedge \cdots \wedge \mu_k$, where each $\mu_i \in \Omega^1(U)$.

Theorem 4.43. *If* ω *is decomposable, then*

$$d\omega = \sum_{i=1}^{k} (-1)^{i-1} \mu_1 \wedge \cdots \wedge \mu_{i-1} \wedge d\mu_i \wedge \mu_{i+1} \wedge \cdots \wedge \mu_k. \tag{4.195}$$

Proof. The proof is by induction.

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The case $k = 1$ is obvious. We show that if the theorem is true for $k-1$, then the theorem is true for k.

$$\begin{aligned} d((\mu_1 \wedge \cdots \wedge \mu_{k-1}) \wedge \mu_k) &= (d(\mu_1 \wedge \cdots \wedge \mu_{k-1})) \wedge \mu_k \\ &\quad + (-1)^{k-1}(\mu_1 \wedge \cdots \wedge \mu_{k-1}) \wedge d\mu_k \\ &= \sum_{i=1}^{k-1} (-1)^{i-1} \mu_1 \wedge \cdots \wedge d\mu_i \wedge \cdots \wedge \mu_{k-1} \wedge \mu_k \\ &\quad + (-1)^{k-1}(\mu_1 \wedge \cdots \wedge \mu_{k-1} \wedge \mu_k) \\ &= \sum_{i=1}^{k} (-1)^{i-1} \mu_1 \wedge \cdots \wedge d\mu_i \wedge \cdots \wedge \mu_k. \end{aligned} \tag{4.196}$$

□

4.10 Pullback Operation on Exterior Forms

Another important operation in the theory of exterior forms is the *pullback operator*. This operation is not introduced in 18.01 or 18.02, because vector calculus in not usually taught rigorously.

Let U be open in $\mathbb{R}^n$ and V be open in $\mathbb{R}^m$, and let $f : U \to V$ be a $\mathcal{C}^\infty$ map. We can write out in components $f = (f_1, \ldots, f_n)$, where each $f_i \in \mathcal{C}^\infty(U)$. Let $p \in U$ and $q = f(p)$.

The pullback of the map $df_p : T_p\mathbb{R}^m \to T_q\mathbb{R}^n$ is

$$(df_p)^* : \Lambda^k(T_q^*\mathbb{R}^n) \to \Lambda^k(T_p^*\mathbb{R}^m). \tag{4.197}$$

Suppose you have a k-form ω on V.

$$\omega \in \Omega^k(V), \tag{4.198}$$

$$\omega_q \in \Lambda^k(T_q^*\mathbb{R}^n). \tag{4.199}$$

Then

$$(df_p)^* w_q \in \Lambda^k(T_p^*\mathbb{R}^m). \tag{4.200}$$

Definition 4.44. $f^*\omega$ is the k-from whose value at $p \in U$ is $(df_p)^*\omega_q$.

We consider two examples. Suppose $\phi \in \Omega^0(V) = \mathcal{C}^\infty(V)$. Then $f^*\phi(p) = \phi(q)$, so $f^*\phi = \phi \circ f$, where $f : U \to V$ and $\phi : V \to \mathbb{R}$.

Again, suppose that $\phi \in \Omega^0(V) = \mathcal{C}^\infty(V)$. What is $f^* d\phi$? Let $f(p) = q$. We have the map $d\phi_q : T_p\mathbb{R}^n \to T_c\mathbb{R} = \mathbb{R}$, where $c = \phi(q)$. So,

$$\begin{aligned} (df_p)^*(d\phi)_q &= d\phi_q \circ df_p \\ &= d(\phi \circ f)_p. \end{aligned} \tag{4.201}$$

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Therefore,

$$f^* d\phi = df^* \phi. \tag{4.202}$$

Suppose that $\mu \in \Omega^k(V)$ and $\nu \in \Omega^e ll(V)$. Then

$$\begin{aligned} (f^*(\mu \wedge \nu))_p &= (df_p)^*(\mu_q \wedge \nu_q) \\ &= (df_p)^* \mu_q \wedge (df_p^*)\nu_q. \end{aligned} \tag{4.203}$$

Hence,

$$f^*(\mu \wedge \nu) = f^*\mu \wedge f^*\nu. \tag{4.204}$$

We now obtain a coordinate formula for f^*.

Take $\omega \in \Omega^k(V)$. We can write $\omega = \sum a_I dx_{i_1} \wedge \cdots \wedge dx_{i_k}$, where each $a_I \in \mathcal{C}^\infty(U)$. Then

$$\begin{aligned} f^*\omega &= \sum f^* a_I f^* dx_{i_1} \wedge \cdots \wedge f^* dx_{i_k} \\ &= \sum f^* a_I df_{i_1} \wedge \cdots \wedge df_{i_k}, \end{aligned} \tag{4.205}$$

where we used the result that $f^* dx_i = dx_i \circ f = df_i$.

Note that $df_i = \sum \frac{\partial f_i}{\partial x_j} dx_j$, where $\frac{\partial f_i}{\partial x_j} \in \mathcal{C}^\infty(U)$. Also, $f^* a_I = a_I \circ f \in \mathcal{C}^\infty(U)$, which shows that

$$f^*\omega \in \Omega^k(U). \tag{4.206}$$

The following theorem states a very useful property of the pullback operator.

Theorem 4.45. *Let $\omega \in \Omega^k(V)$. Then,*

$$df^*\omega = f^* d\omega. \tag{4.207}$$

Proof. We have already checked this for $\omega = \phi \in \mathcal{C}^\infty(V)$, $k = 0$ already. We now prove the general case.

We can write $\omega = \sum a_I dx_I$. Then

$$f^*\omega = \sum f^* a_I df_{i_1} \wedge \cdots \wedge df_{i_k}. \tag{4.208}$$

So,

$$\begin{aligned} df^*\omega = &\sum df^* a_I \wedge df_{i_1} \wedge \cdots \wedge df_{i_k} \\ &+ \sum f^* a_I \wedge d(df_{i_1} \wedge \cdots \wedge df_{i_k}) \end{aligned} \tag{4.209}$$

Note that

$$d(df_{i_1} \wedge \cdots \wedge df_{i_k}) = \sum_{r=1}^{k} (-1)^{r-1} df_{i_1} \wedge \cdots \wedge d(df_{i_r}) \wedge \cdots \wedge df_{i_k}. \tag{4.210}$$

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We know that $d(df_{i_r}) = 0$, so

$$\begin{aligned} df^*\omega &= \sum_I df^* a_I \wedge df_{i_1} \wedge \cdots \wedge df_{i_k} \\ &= \sum_I f^* da_I \wedge f^*(dx_{i_1} \wedge \cdots \wedge dx_{i_k}) \\ &= f^*(\sum da_I \wedge dx_I) \\ &= f^* d\omega. \end{aligned} \tag{4.211}$$

□

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Lecture 24

We review the pullback operation from last lecture. Let U be open in $\mathbb{R}^m$ and let V be open in $\mathbb{R}^n$. Let $f : U \to V$ be a $\mathcal{C}^\infty$ map, and let $f(p) = q$. From the map

$$df_p : T_p\mathbb{R}^m \to T_q\mathbb{R}^n, \tag{4.212}$$

we obtain the pullback map

$$\begin{aligned} (df_p)^* : \Lambda^k(T_q^*) &\to \Lambda^k(T_p^*) \\ \omega \in \Omega^k(V) &\to f^*\omega \in \Omega^k(U). \end{aligned} \tag{4.213}$$

We define, $f^*\omega_p = (df_p)^*\omega_q$, when $\omega_q \in \Lambda^k(T_q^*)$.

The pullback operation has some useful properties:

1. If $\omega_i \in \Omega^{k_i}(V), i = 1, 2$, then

$$f^*(\omega_1 \wedge \omega_2) = f^*\omega_1 \wedge f^*\omega_2. \tag{4.214}$$

2. If $\omega \in \Omega^k(V)$, then

$$df^*\omega = f^*d\omega. \tag{4.215}$$

We prove some other useful properties of the pullback operation.

Claim. *For all $\omega \in \Omega^k(W)$,*

$$f^*g^*\omega = (g \circ f)^*\omega. \tag{4.216}$$

Proof. Let $f(p) = q$ and $g(q) = w$. We have the pullback maps

$$(df_p)^* : \Lambda^k(T_q^*) \to \Lambda^k(T_p^*) \tag{4.217}$$
$$(dg_q)^* : \Lambda^k(T_w^*) \to \Lambda^k(T_q^*) \tag{4.218}$$
$$(g \circ f)^* : \Lambda^k(T_w^*) \to \Lambda^k(T_p^*). \tag{4.219}$$

The chain rule says that

$$(dg \circ f)_p = (dg)_q \circ (df)_p, \tag{4.220}$$

so

$$d(g \circ f)_p^* = (df_p)^*(dg_q)^*. \tag{4.221}$$

□

Let U, V be open sets in $\mathbb{R}^n$, and let $f : U \to V$ be a $\mathcal{C}^\infty$ map. We consider the pullback operation on n-forms $\omega \in \Omega^n(V)$. Let $f(0) = q$. Then

$$(dx_i)_p, \quad i = 1, \ldots, n, \quad \text{is a basis of } T_p^*, \text{ and} \tag{4.222}$$
$$(dx_i)_q, \quad i = 1, \ldots, n, \quad \text{is a basis of } T_q^*. \tag{4.223}$$

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Using $f_i = x_i \circ f$,

$$\begin{aligned}(df_p)^*(dx_i)_q &= (df_i)_p \\ &= \sum \frac{\partial f_i}{\partial x_j}(p)(dx_j)_p.\end{aligned} \tag{4.224}$$

In the Multi-linear Algebra notes, we show that

$$(df_p)^*(dx_1)_q \wedge \cdots \wedge (dx_n)_q = \det\left[\frac{\partial f_i}{\partial x_j}(p)\right](dx_1)_p \wedge \cdots \wedge (dx_n)_p. \tag{4.225}$$

So,

$$f^* dx_1 \wedge \cdots \wedge dx_n = \det\left[\frac{\partial f_i}{\partial x_j}\right] dx_1 \wedge \cdots \wedge dx_n. \tag{4.226}$$

Given $\omega = \phi(x)dx_1 \wedge \cdots \wedge dx_n$, where $\phi \in \mathcal{C}^\infty$,

$$f^*\omega = \phi(f(x)) \det\left[\frac{\partial f_i}{\partial x_j}\right] dx_1 \wedge \cdots \wedge dx_n. \tag{4.227}$$

5 Integration with Differential Forms

Let U be an open set in $\mathbb{R}^n$, and let $\omega \in \Omega^k(U)$ be a differential k-form.

Definition 5.1. The *support* of ω is

$$\text{supp } \omega = \overline{\{p \in U : \omega_p \neq 0\}}. \tag{5.1}$$

Definition 5.2. The k-form ω is *compactly supported* if supp ω is compact. We define

$$\Omega_c^k(U) = \text{ the space of all compactly supported } k\text{-forms}. \tag{5.2}$$

Note that

$$\Omega_c^0(U) = \mathcal{C}_0^\infty(\mathbb{R}^n). \tag{5.3}$$

Given $\omega \in \Omega_c^n(U)$, we can write

$$\omega = \phi(x)dx_1 \wedge \cdots \wedge dx_n, \tag{5.4}$$

where $\phi \in \mathcal{C}_0^\infty(U)$.

Definition 5.3.

$$\int_U \omega \equiv \int_U \phi = \int_U \phi(x)dx_1 \ldots dx_n. \tag{5.5}$$

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We are going to state and prove the change of variables theorem for integrals of differential k-forms. To do so, we first need the notions of *orientation preserving* and *orientation reversing*.

Let U, V be open sets in $\mathbb{R}^n$. Let $f : U \to V$ be a $\mathcal{C}^\infty$ diffeomorphism. That is, for every $p \in U$, $Df(p) : \mathbb{R}^n \to \mathbb{R}^n$ is bijective. We associate $Df(p)$ with the matrix

$$Df(p) \cong \left[\frac{\partial f_i}{\partial x_j}(p)\right]. \tag{5.6}$$

The map f is a diffeomorphism, so

$$\det\left[\frac{\partial f_i}{\partial x_j}(p)\right] \neq 0. \tag{5.7}$$

So, if U is connected, then this determinant is either positive everywhere or negative everywhere.

Definition 5.4. The map f is *orientation preserving* if $\det > 0$ everywhere. The map f is *orientation reversing* if $\det < 0$ everywhere.

The following is the change of variables theorem:

Theorem 5.5. *If $\omega \in \Omega^n_c(V)$, then*

$$\int_U f^*\omega = \int_V \omega \tag{5.8}$$

if f is orientation preserving, and

$$\int_U f^*\omega = -\int_V \omega \tag{5.9}$$

if f is orientation reversing.

In Munkres and most texts, this formula is written in slightly uglier notation. Let $\omega = \phi(x)dx_1 \wedge \cdots \wedge dx_n$, so

$$f^*\omega = \phi(f(x)) \det\left[\frac{\partial f_i}{\partial x_j}\right] dx_1 \wedge \cdots \wedge dx_n. \tag{5.10}$$

The theorem can be written as following:

Theorem 5.6. *If f is orientation preserving, then*

$$\int_V \phi = \int_U \phi \circ f \det\left[\frac{\partial f_i}{\partial x_j}\right]. \tag{5.11}$$

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This is the coordinate version of the theorem.

We now prove a useful theorem found in the Supplementary Notes (and Spivak) called Sard's Theorem.

Let U be open in $\mathbb{R}^n$, and let $f : U \to \mathbb{R}^n$ be a $\mathcal{C}^1(U)$ map. For every $p \in U$, we have the map $Df(p) : \mathbb{R}^n \to \mathbb{R}^n$. We say that p is a *critical point of* f if $Df(p)$ is *not* bijective. Denote

$$C_f = \text{ the set of all critical points of } f. \tag{5.12}$$

Sard's Theorem. *The image $f(C_f)$ is of measure zero.*

Proof. The proof is in the Supplementary Notes. □

As an example of Sard's Theorem, let $c \in \mathbb{R}^n$ and let $f : U \to \mathbb{R}^n$ be the map defined by $f(x) = c$. Note that $Df(p) = 0$ for all $p \in U$, so $C_f = U$. The set $C_f = U$ is not a set of measure zero, but $f(C_f) = \{c\}$ is a set of measure zero.

As an exercise, you should prove the following claim:

Claim. *Sard's Theorem is true for maps $f : U \to \mathbb{R}^n$, where U is an open, connected subset of $\mathbb{R}$.*

Proof Hint: Let $f \in \mathcal{C}^\infty(U)$ and define $g = \frac{\partial f}{\partial x}$. The map g is continuous because $f \in \mathcal{C}^1(U)$. Let $I = [a, b] \subseteq U$, and define $\ell = b - a$. The continuity of g implies that g is uniformly continuous on I. That is, for every $\epsilon > 0$, there exists a number $N > 0$ such that $|g(x) - g(y)| < \epsilon$ whenever $x, y \in I$ and $|x - y| < \ell/N$.

Now, slice I into N equal subintervals. Let $I_r, r = 1, \ldots, k \leq N$ be the subintervals intersecting C_f. Prove the following lemma:

Lemma 5.7. *If $x, y \in I_r$, then $|f(x) - f(y)| < \epsilon\ell/N$.*

Proof Hint: Find $c \in I_r$ such that $f(x) - f(y) = (x - y)g(c)$. There exists $c_0 \in I_r \cap C_f$ if and only if $g(c_0) = 0$. So, we can take

$$|g(c)| = |g(c) - g(c_0)| \leq \epsilon. \tag{5.13}$$

Then $|f(x) - f(y)| \leq \epsilon\ell/N$. □

From the lemma, we can conclude that

$$f(I_r) \equiv J_r \tag{5.14}$$

is of length less than $\epsilon\ell/N$. Therefore,

$$f(C_f \cap I) \subset \bigcup_{r=1}^{k} J_r \tag{5.15}$$

is of length less than

$$\frac{\epsilon\ell}{N} k \leq \frac{\epsilon\ell N}{N} = \epsilon\ell. \tag{5.16}$$

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Letting $\epsilon \to 0$, we find that $F(C_f \cap I)$ is of measure zero.

To conclude the proof, let $I_m, m = 1, 2, 3, \ldots$, be an exhaustion of U by closed intervals $I_1 \subset I_2 \subset I_3 \subset \cdots$ such that $\bigcup I_m = U$. We have shown that $f(C_f \cap I_m)$ is measure zero. So, $f(C_f) = \bigcup f(C_f \cap I_m)$ implies that $f(C_f)$ is of measure zero. $\square$

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Lecture 25

5.1 The Poincare Lemma

Let U be an open subset of $\mathbb{R}^n$, and let $\omega \in \Omega^k(U)$ be a k-form. We can write $\omega = \sum a_I dx_I, I = (i_1, \ldots, i_k)$, where each $a_I \in \mathcal{C}^\infty(U)$. Note that

$$\omega \in \Omega_c^k \iff a_I \in \mathcal{C}_0^\infty(U) \text{ for each } I. \tag{5.17}$$

We are interested in $\omega \in \Omega_c^n(U)$, which are of the form

$$\omega = f dx_1 \wedge \cdots \wedge dx_n, \tag{5.18}$$

where $f \in \mathcal{C}_0^\infty(U)$. We define

$$\int_U \omega = \int_U f = \int_U f dx, \tag{5.19}$$

the Riemann integral of f over U.

Our goal over the next couple lectures is to prove the following fundamental theorem known as the Poincare Lemma.

Poincare Lemma. *Let U be a connected open subset of $\mathbb{R}^n$, and let $\omega \in \Omega_c^n(U)$. The following conditions are equivalent:*

1. $\int_U \omega = 0$,
2. $\omega = d\mu$, *for some* $\mu \in \Omega_c^{n-1}(U)$.

In today's lecture, we prove this for $U = \text{Int } Q$, where $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ is a rectangle.

Proof. First we show that (2) implies (1).

Notation.

$$dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n \equiv dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_n. \tag{5.20}$$

Let $\mu \in \Omega_c^{n-1}(U)$. Specifically, define

$$\mu = \sum_i f_i dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n, \tag{5.21}$$

where each $f_i \in \mathcal{C}_0^\infty(U)$. Every $\mu \in \Omega_c^{n-1}(U)$ can be written this way.

Applying d we obtain

$$d\mu = \sum_i \sum_j \frac{\partial f_i}{\partial x_j} dx_j \wedge dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n. \tag{5.22}$$

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Notice that if $i \neq j$, then the i,jth summand is zero, so

$$\begin{aligned} d\mu &= \sum_i \frac{\partial f_i}{\partial x_i} dx_i \wedge dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n \\ &= \sum (-1)^{i-1} \frac{\partial f_i}{\partial x_i} dx_1 \wedge \cdots \wedge dx_n. \end{aligned} \tag{5.23}$$

Integrate to obtain

$$\int_U d\mu = \sum (-1)^{i-1} \int_U \frac{\partial f_i}{\partial x_i}. \tag{5.24}$$

Note that

$$\int_{a_i}^{b_i} \frac{\partial f_i}{\partial x_i} dx_i = f_i(x)|_{x_i=a_i}^{x_i=b_i} = 0 - 0 = 0, \tag{5.25}$$

because f is compactly supported in U. It follows from the Fubini Theorem that

$$\int_U \frac{\partial f_i}{\partial x_i} = 0. \tag{5.26}$$

Now we prove the other direction, that (1) implies (2). Before our proof we make some remarks about functions of one variable.

Suppose $I = (a,b) \subseteq \mathbb{R}$, and let $g \in \mathcal{C}_0^\infty(I)$ such that supp $g \subseteq [c,d]$, where $a < c < d < b$. Also assume that

$$\int_a^b g(s)ds = 0. \tag{5.27}$$

Define

$$h(x) = \int_a^x g(s)ds, \tag{5.28}$$

where $a \leq x \leq b$.

Claim. *The function h is also supported on c,d.*

Proof. If $x > d$, then we can write

$$h(x) = \int_a^b g(s)ds - \int_x^b g(s)ds, \tag{5.29}$$

where the first integral is zero by assumption, and the second integral is zero because the integrand is zero. □

Now we begin our proof that (1) implies (2).

Let $\omega \in \Omega^n(U)$, where $U = Q$, and assume that

$$\int_U \omega = 0. \tag{5.30}$$

We will use the following inductive lemma:

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Lemma 5.8. *For all $0 \leq k \leq n+1$, there exists $\mu \in \Omega_c^{n-1}(U)$ and $f \in \mathcal{C}_0^\infty(U)$ such that*

$$\omega = d\mu + f dx_1 \wedge \cdots \wedge dx_n \tag{5.31}$$

and

$$\int f(x_1, \ldots, x_n) dx_k \ldots dx_n = 0. \tag{5.32}$$

Note that the hypothesis for $k = 0$ and $\mu = 0$ says that $\int \omega = 0$, which is our assumption (1). Also note that the hypothesis for $k = n+1$, $f = 0$, and $\omega = d\mu$ is the same as the statement (2). So, if we can show that (the lemma is true for k) implies (the lemma is true for $k+1$), then we will have shown that (1) implies (2) in Poincare's Lemma. We now show this.

Assume that the lemma is true for k. That is, we have

$$\omega = d\mu + f dx_1 \wedge \cdots \wedge dx_n \tag{5.33}$$

and

$$\int f(x_1, \ldots, x_n) dx_k \ldots dx_n = 0, \tag{5.34}$$

where $\mu \in \Omega_c^{n-1}(U)$, and $f \in \mathcal{C}_0^\infty(\mathbb{R})$.

We can assume that μ and f are supported on Int Q', where $Q' \subseteq$ Int Q and $Q' = [c_1, d_1] \times \cdots \times [c_n, d_n]$.

Define

$$g(x_1, \ldots, x_k) = \int f(x_1, \ldots, x_n) d_{k+1} \ldots dx_n. \tag{5.35}$$

Note that g is supported on the interior of $[c_1, d_1] \times \cdots \times [c_k, d_k]$. Also note that

$$\int_{a_k}^{b_k} g(x_1, \ldots, x_{k-1}, s) ds = \int f(x_1, \ldots, x_n) dx_k \ldots dx_n = 0 \tag{5.36}$$

by our assumption that the lemma holds true for k.

Now, define

$$h(x_1, \ldots, x_k) = \int_{a_k}^{x_k} g(x_1, \ldots, x_{k-1}, s) ds. \tag{5.37}$$

From our earlier remark about functions of one variable, h is supported on $c_k \leq x_k \leq d_k$. Also, note that h is supported on $c_i \leq x_i \leq d_i$, for $1 \leq i \leq k-1$. We conclude therefore that h is also supported on $[c_1, d_1] \times \cdots \times [c_k, d_k]$.

Both g and its "anti-derivative" are supported.

$$\frac{\partial h}{\partial x_k} = g. \tag{5.38}$$

Let $\ell = n - k$, and consider $\rho = \rho(x_{k+1}, \ldots, x_n) \in \mathcal{C}_0^\infty(\mathbb{R}^\ell)$. Assume that ρ is supported on the rectangle $[c_{k+1}, d_{k+1}] \times \cdots \times [c_n, d_n]$ and that

$$\int \rho dx_{k+1} \ldots dx_n = 1. \tag{5.39}$$

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We can always find such a function, so we just fix one such function.

Define

$$\nu = (-1)^k h(x_1, \ldots, x_k)\rho(x_{k+1}, \ldots, x_n)dx_1 \wedge \cdots \wedge \widehat{dx_k} \wedge \cdots \wedge dx_n. \tag{5.40}$$

The form ν is supported on $Q' = [c_1, d_1] \times \cdots \times [c_n, d_n]$.

Now we compute $d\nu$,

$$d\nu = (-1)^k \sum_j \frac{\partial}{\partial x_j}(h\rho)dx_j \wedge dx_1 \wedge \cdots \wedge \widehat{dx_k} \wedge \cdots \wedge dx_n. \tag{5.41}$$

Note that if $j \neq k$, then the summand is zero, so

$$\begin{aligned} d\nu &= (-1)^k \frac{\partial h}{\partial x_k}\rho dx_k \wedge dx_1 \wedge \cdots \wedge \widehat{dx_k} \wedge \cdots \wedge dx_n \\ &= (-1)\frac{\partial h}{\partial x_k}\rho dx_1 \wedge \cdots \wedge dx_n \\ &= -g\rho dx_1 \wedge \cdots \wedge dx_n. \end{aligned} \tag{5.42}$$

Now, define

$$\mu_{new} = \mu - \nu, \tag{5.43}$$

and

$$f_{new} = f(x_1, \ldots, x_n) - g(x_1, \ldots, x_k)\rho(x_{k+1}, \ldots, x_n). \tag{5.44}$$

$$\begin{aligned} \omega &= d\mu_{new} + f_{new}dx_1 \wedge \cdots \wedge dx_n \\ &= d\mu + (g(x_1, \ldots, x_k)\rho(x_{k+1}, \ldots, x_n) - f(x_1, \ldots, x_k) - g\rho)dx_1 \wedge \cdots \wedge dx_n \\ &= d\mu + f dx_1 \wedge \cdots \wedge dx_n \\ &= \omega. \end{aligned} \tag{5.45}$$

Note that

$$\begin{aligned} \int f_{new} &= \int f_{new}(x_1, \ldots, x_n)dx_{k+1} \ldots dx_n \\ &= \int f(x_1, \ldots, x_n)dx_{k+1} \ldots dx_n \\ &\qquad - g(x_1, \ldots, x_k) \int \rho(x_{k+1}, \ldots, x_n)dx_{k+1} \ldots dx_n \\ &= g(x_1, \ldots, x_k) - g(x_1, \ldots, x_k) = 0, \end{aligned} \tag{5.46}$$

which implies that the lemma is true for $k + 1$. □

Remark. In the above proof, we implicitly assumed that if $f \in \mathcal{C}_0^\infty(\mathbb{R}^n)$, then

$$g(x_1, \ldots, x_k) = \int f(x_1, \ldots, x_n)dx_{k+1} \ldots dx_m \tag{5.47}$$

is in $\mathcal{C}_0^\infty(\mathbb{R}^k)$. We checked the support, but we did not check that g is in $\mathcal{C}^\infty(\mathbb{R}^k)$. The proof of this is in the Supplementary Notes.

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Lecture 26

We continue our study of forms with compact support. Let us begin with a review. Let $U \in \mathbb{R}^n$ be open, and let

$$\omega = \sum_I f_I(x_1, \ldots, x_n) dx_I, \tag{5.48}$$

where $I = (i_1, \ldots, i_k)$ is strictly increasing and $dx_I = dx_{i_1} \wedge \cdots \wedge dx_{i_k}$. Then

$$\omega \text{ is compactly supported } \iff \text{ every } f_I \text{ is compactly supported.} \tag{5.49}$$

By definition,

$$\text{supp } f_I = \overline{\{x \in U : f_I(x) \neq 0\}}. \tag{5.50}$$

We assume that the f_I's are $\mathcal{C}^2$ maps.

Notation.

$$\Omega_c^k(U) = \text{ space of compactly supported differentiable } k\text{-forms on } U. \tag{5.51}$$

Now, let $\omega \in \Omega_c^n(U)$ defined by

$$\omega = f(x_1, \ldots, x_n) dx_1 \wedge \cdots \wedge dx_n, \tag{5.52}$$

where $f \in \Omega_c^0(U)$. Then

$$\int_{\mathbb{R}^n} \omega = \int_{\mathbb{R}^n} f(x_1, \ldots, x_n) dx_1 \wedge \cdots \wedge dx_n. \tag{5.53}$$

Last time we proved the Poincare Lemma for open rectangles R in $\mathbb{R}^n$. We assumed that $\omega \in \Omega_c^n(\text{Int } R)$. That is, we assumed that $\omega \in \Omega_c^n(\mathbb{R}^n)$ such that supp $\omega \subset$ Int R. We showed that for such ω the following two conditions are equivalent:

1. $\int_{\mathbb{R}^n} \omega = 0$,
2. There exists a $\mu \in \Omega_c^{n-1}(\text{Int } R)$ such that $d\mu = 0$.

Definition 5.9. Whenever $\omega \in \Omega^k(U)$ and $\omega = d\mu$ for some $\mu \in \Omega^{k-1}(U)$, we say that ω is *exact*.

Definition 5.10. Whenever $\omega \in \Omega^k(U)$ such that $d\omega = 0$, we say that ω is *closed*.

Observe that

$$\omega \in \Omega_c^n(U) \implies d\omega = 0. \tag{5.54}$$

Now we prove the Poincare Lemma for open connected subsets of $\mathbb{R}^n$.

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Poincare Lemma. *Let U be a connected open subset of $\mathbb{R}^n$, and let $\omega \in \Omega_c^n(U)$. The following conditions are equivalent:*

1. $\int_U \omega = 0$,
2. $\omega = d\mu$, *for some* $\mu \in \Omega_c^{n-1}(U)$.

Proof. We prove this more general case by reducing the proof to the case where U is a rectangle, which we proved in the previous lecture.

First we prove that (2) implies (1). We can choose a family of rectangles $\{R_i, i \in \mathbb{N}\}$ such that

$$U = \bigcup_{i \in \mathbb{N}} \text{Int } R_i \tag{5.55}$$

Since the support of μ is compact, the set supp μ is covered by finitely many of the rectangles.

We take a partition of unity $\{\phi_i, i \in \mathbb{N}\}$ subordinate to $\{R_i\}$, so that

$$\mu = \sum_{i=1}^{N} \underbrace{\phi_i \mu}_{\text{supported on Int } R_i} \tag{5.56}$$

Then

$$\int d\mu = \sum_i \int_{\mathbb{R}^n} d(\phi_i \mu). \tag{5.57}$$

Each term on the r.h.s is zero by the Poincare Lemma we proved last lecture.

We now prove the other direction, that (1) implies (2). It is equivalent to show that if $\omega_1, \omega_2 \in \Omega_c^n(U)$ such that

$$\int \omega_1 = \int \omega_2, \tag{5.58}$$

then $\omega_1 \sim \omega_2$, meaning that there exists a form $\mu \in \Omega_c^{n-1}(U)$ such that $\omega_1 = \omega_2 + d\mu$.

Choose a partition of unity $\{\phi_i\}$ as before. Then

$$\omega = \sum_{i=1}^{M} \underbrace{\phi_i \omega}_{\text{supported on Int } R_i} \tag{5.59}$$

Let

$$\int \omega = c \in \mathbb{R}, \tag{5.60}$$

and

$$\int \phi_i \omega = c_i. \tag{5.61}$$

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Choose a form ω_0 such that

$$\int \omega_0 = 1 \tag{5.62}$$

and such that supp $\omega_0 \subseteq Q_0 = R_j$ for some j. Then

$$\int \underbrace{\phi_i \omega}_{\text{supported in } R_i} = \int \underbrace{c_i \omega_0}_{\text{supported in } Q_0} \tag{5.63}$$

We want to show that there exists $\mu_i \in \Omega_c^{n-1}(U)$ such that $\phi_i \omega = c_i \omega_i + d\mu_i$.

Now we use the fact that U is connected. We use the following lemma.

Lemma 5.11. *Let U be connected. Given rectangles R_i such that* supp $\phi_i \omega \subset$ Int R_i, *and given a fixed rectangle Q_0 and any point $x \in U$, there exists a finite sequence of rectangles $R_0, \ldots, R_N$ with the following properties: $Q_0 = R_0$, $x \in$ Int R_N, and* (Int R_i) $\cap$ (Int R_{i+1}) *is non-empty.*

We omit the proof of this lemma.

Now, define $\omega_i = \phi_i \omega$, so

$$\int \omega_i = \int c_i \omega_0. \tag{5.64}$$

Note that

$$\text{supp}\,(c_i \omega_0) \subseteq \text{Int}\,(Q_0) \tag{5.65}$$

$$\text{supp}\,(\omega_i) \subseteq \text{Int}\,(R_i). \tag{5.66}$$

Choose forms ν_i such that supp $\nu_i \subseteq$ Int $R_i \cap$ Int R_{i+1} and such that

$$\int \nu_i = 1. \tag{5.67}$$

This implies that

$$\text{supp}\,(\nu_i - \nu_{i+1}) \subseteq \text{Int } R_{i+1} \tag{5.68}$$

By definition,

$$\int (\nu_i - \nu_{i+1}) = 0. \tag{5.69}$$

By the Poincare Lemma we proved last time, $\nu_i \sim \nu_{i+1}$, so there exists $\mu_i \in \Omega_c^{n-1}(U)$ such that $\nu_i = \nu_{i+1} + d\mu_i$.

So,

$$c_i \omega_0 \sim c_i \nu_0 \sim c_i \nu_1 \sim \ldots \sim c_i \nu_N \sim \phi_i \omega. \tag{5.70}$$

□

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5.2 Proper Maps and Degree

We introduce a class of functions that remain compactly supported under the pullback operation.

Definition 5.12. Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^k$, and let $f : U \to V$ be a continuous map. The map f is *proper* if for all compact subsets $K \subseteq V$, the set $f^{-1}(K)$ is compact.

Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^k$, and let $f : U \to V$ be a continuous map. Also let $\omega \in \Omega^k(V)$. The map

$$f^* : \Omega^k(V) \to \Omega^k(U) \tag{5.71}$$

is defined such that

$$\omega = g(y_1, \ldots, y_n) dy_{i_1} \wedge \ldots \wedge dy_{i_k} \to f^*\omega = g(f(x)) df_{i_1} \wedge \cdots df_{i_k}. \tag{5.72}$$

So,

$$f^{-1}(\text{supp } \omega) \supseteq \text{supp}\,(f^*\omega). \tag{5.73}$$

If f is proper and $\omega \in \Omega^n_c(V)$, then $\text{supp}\,(f^*\omega)$ is compact, in which case the map f^* is actually of the form

$$f^* : \Omega^k_c(V) \to \Omega^k_c(U). \tag{5.74}$$

That is, $\omega \in \Omega^n_c(V) \to f^*\omega \in \Omega^n_c(U)$. So, it makes sense to take the integral

$$\int_U f^*\omega = (\deg f) \int_V \omega. \tag{5.75}$$

Theorem 5.13. *Let U, V be connected open subsets of $\mathbb{R}^n$, and let $f : U \to V$ be a $\mathcal{C}^\infty$ map. For all $\omega \in \Omega^n_c(V)$,*

$$\int_U f^*\omega = (\deg f) \int_V \omega. \tag{5.76}$$

Proof. Take $\omega_0 \in \Omega^n_c(V)$ such that

$$\int \omega_0 = 1. \tag{5.77}$$

Define

$$\deg f \equiv \int f^*\omega_0, \tag{5.78}$$

and suppose that

$$\int \omega = c. \tag{5.79}$$

Then

$$\int \omega = \int c\omega_0. \tag{5.80}$$

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By the Poincare Lemma, $\omega \sim c\omega_0$. That is, there exists $\mu \in \Omega_c^{n-1}(V)$ such that $\omega = c\omega_0 + d\mu$. Then

$$\begin{aligned} f^*\omega &= f^*(c\omega_0) + f^*(d\mu) \\ &= f^*(c\omega_0) + d(f^*\mu), \end{aligned} \tag{5.81}$$

which shows that $f^*\omega \sim f^*(c\omega_0)$. Putting this altogether,

$$\begin{aligned} \int f^*\omega &= \int f^*(c\omega_0) \\ &= c\int f^*\omega_0 \\ &= c\deg f \\ &= \left(\int \omega\right)\deg f. \end{aligned} \tag{5.82}$$

□

We had $\omega = g(y_1, \ldots, y_n)dy_1 \wedge \cdots \wedge dy_n$, so

$$\begin{aligned} f^*\omega &= g(f(x))df_1 \wedge \cdots \wedge df_m \\ &= g(f(x))\det\left[\frac{\partial f_i}{\partial x_j}\right] dx_1 \wedge \cdots \wedge dx_n, \end{aligned} \tag{5.83}$$

where we used the fact that

$$df_i = \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j} dx_j \tag{5.84}$$

Restated in coordinates, the above theorem says that

$$\begin{aligned} &\int_U g(f(x))\det(Df)dx_1 \wedge \cdots \wedge dx_n \\ &\qquad = (\deg f)\int_V g(y_1, \ldots, y_n)dy_1 \wedge \cdots \wedge dy_n. \end{aligned} \tag{5.85}$$

Claim. *Given proper maps $f : V \to W$ and $g : U \to V$, where U, V, W are connected open subsets of $\mathbb{R}^n$,*

$$\deg(fg) = (\deg g)(\deg f). \tag{5.86}$$

Proof. Note that $(f \circ g)^* = g^* \circ f^*$, so

$$\begin{aligned} \deg(f \circ g)\int_W \omega &= \int_U (f \circ g)^*\omega \\ &= \int_V g^*(f^*\omega) \\ &= (\deg g)\int_V f^*\omega \\ &= (\deg g)(\deg f)\int_W \omega. \end{aligned} \tag{5.87}$$

□

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Lecture 27

We proved the following Poincare Lemma:

Poincare Lemma. *Let U be a connected open subset of $\mathbb{R}^n$, and let $\omega \in \Omega_c^n(U)$. The following conditions are equivalent:*

1. $\int_U \omega = 0$,
2. $\omega = d\mu$, *for some* $\mu \in \Omega_c^{n-1}(U)$.

We first proved this for the case $U = \text{Int } Q$, where Q was a rectangle. Then we used this result to generalize to arbitrary open connected sets. We discussed a nice application: proper maps and degree.

Let U, V be open subsets of $\mathbb{R}^n$, and let $f : U \to V$ be a $\mathcal{C}^\infty$ map. The map f is *proper* if for every compact set $C \subseteq V$, the pre-image $f^{-1}(C)$ is also compact. Hence, if f is proper, then

$$f^*\Omega_c^k(V) \subseteq \Omega_c^k(U). \tag{5.88}$$

That is, if $\omega \in \Omega_c^k(V)$, then $f^*\omega \in \Omega_c^k(U)$, for all k.

When $k = n$,

$$\omega \in \Omega_c^n(V). \tag{5.89}$$

In which case, we compare

$$\int_v \omega \quad \text{and} \quad \int_U f^*\omega. \tag{5.90}$$

Using the Poincare Lemma, we obtain the following theorem.

Theorem 5.14. *There exists a constant γ_f with the property that for all $\omega \in \Omega_c^n(V)$,*

$$\int_U f^*\omega = \gamma_f \int_V \omega. \tag{5.91}$$

We call this constant the *degree of f*,

Definition 5.15.

$$\deg(f) = \gamma_f. \tag{5.92}$$

Let U, V, W be open connected subsets of $\mathbb{R}^n$, and let $f : U \to V$ and $g : V \to W$ be proper $\mathcal{C}^\infty$ maps. Then the map $g \circ f : U \to W$ is proper, and

$$\deg(g \circ f) = \deg(f)\deg(g). \tag{5.93}$$

Proof Hint: For all $\omega \in \Omega_c^n(W)$, $(g \circ f)^*\omega = f^*(g^*\omega)$. □

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We give some examples of the degree of various maps. Let $f = T_a$, the transposition by a. That is, let $f(x) = x + a$. From #4 in section 4 of the Supplementary Notes, the map $T_a : \mathbb{R}^n \to \mathbb{R}^n$ is proper. One can show that $\deg(T_a) = 1$.

As another example, let $f = A : \mathbb{R}^n \to \mathbb{R}^n$ be a bijective liner map. Then

$$\deg A = \begin{cases} 1 & \text{if } \det A > 0, \\ -1 & \text{if } \det A < o. \end{cases} \tag{5.94}$$

We now study the degree as it pertains to orientation preserving and orientation reversing maps.

Let U, V be connected open sets in $\mathbb{R}^n$, and let $f : U \to V$ be a diffeomorphism. Take $p \in U$. Then $Df(p) : \mathbb{R}^n \to \mathbb{R}^n$ is one-to-one and onto. The map f is *orientation preserving* if $\det Df(p) > 0$ for all $p \in U$, and the map f is *orientation reversing* if $\deg Df(p) < 0$ for all $p \in U$.

Theorem 5.16. *If f is orientation preserving, then* $\deg(f) = 1$*; if f is orientation reversing, then* $\deg(f) = -1$.

Proof. Let $a \in U$ and $b = f(a)$. Define

$$f_{\text{old}} = f, \tag{5.95}$$

and define

$$f_{\text{new}} = T_{-b} \circ f_{\text{old}} \circ T_a, \tag{5.96}$$

where $T_{-b} : \mathbb{R}^n \to \mathbb{R}^n$ and $T_a : \mathbb{R}^n \to \mathbb{R}^n$ are transpositions by $-b$ and a, respectively.

By the formula $\deg(g \circ f) = \deg(f) \deg(g)$,

$$\begin{aligned} \deg(f_{\text{new}}) &= \deg(T_{-b}) \deg(f_{\text{old}}) \deg(T_a) \\ &= \deg(_{\text{old}}). \end{aligned} \tag{5.97}$$

By replacing f with f_{new}, we can assume that $0 \in U$ and $f(0) = 0$.

We can make yet another simplification, that $Df(0) = I$, the identity. To see this, let $Df(0) = A$, where $A : \mathbb{R}^n \to \mathbb{R}^n$. Taking our new f, we redefine $f_{\text{old}} = f$, and we redefine $f_{\text{new}} = A^{-1} \circ f_{\text{old}}$. Then,

$$\deg(f_{\text{new}}) = \deg(A) \deg \deg(f_{\text{old}}), \tag{5.98}$$

where

$$\begin{aligned} \deg A &= \deg(Df_{\text{old}}) \\ &= \begin{cases} 1 & \text{if } Df_{\text{old}} \text{ is orient. preserving,} \\ -1 & \text{if } Df_{\text{old}} \text{ is orient. reversing.} \end{cases} \end{aligned} \tag{5.99}$$

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We again replace f with f_{new}. It suffices to prove the theorem for this new f. To summarize, we can assume that

$$0 \in U, \quad f(0) = 0. \quad \text{and } Df(0) = I. \tag{5.100}$$

Consider $g(x) = x - f(x)$ (so $f(x) = x - g(x)$). Note that $(Dg)(0) = I - I = 0$. If we write $g = (g_1, \ldots, g_n)$, then

$$\left[\frac{\partial g_i}{\partial x_j}(0)\right] = 0. \tag{5.101}$$

So, each $\frac{\partial g_i}{\partial x_j}(0) = 0$.

Lemma 5.17. *There exists $\delta > 0$ such that for all $|x| < \delta$,*

$$|g(x)| \leq \frac{|x|}{2}. \tag{5.102}$$

Proof. So far, we know that $g(0) = 0 - f(0) = 0$, and $\frac{\partial g_i}{\partial x_j}(0) = 0$. By continuity, there exists $\delta > 0$ such that

$$\left|\frac{\partial g_i}{\partial x_j}(x)\right| \leq \frac{1}{2n}, \tag{5.103}$$

for all $|x| < \delta$. Using the Mean-value Theorem, for all $|x| < \delta$,

$$\begin{aligned} g_i(x) &= g_i(x) - g_i(0) \\ &= \sum \frac{\partial g_i}{\partial x_j}(c) x_j, \end{aligned} \tag{5.104}$$

where $c = t_0 x$ for some $0 < t_0 < 1$. So,

$$\begin{aligned} |g_i(x)| &\leq \sum_{i=1}^{n} \frac{1}{2n} |x_i| \\ &\leq \frac{1}{2} \max\{|x_i|\} \\ &= \frac{1}{2}|x|. \end{aligned} \tag{5.105}$$

□

Define $\tilde{f} : \mathbb{R}^n \to \mathbb{R}^n$ as follows. Let $\rho \in \mathcal{C}_0^\infty(\mathbb{R}^n)$, defined to have the follow properties

$$\rho(x) = \begin{cases} 1 & \text{if } |x| < \delta/2, \\ 0 & \text{if } |x| > \delta, \\ 0 \leq \rho(x) \leq 1 & \text{otherwise.} \end{cases} \tag{5.106}$$

Remember that $f(x) = x - g(x)$. Define

$$\tilde{f} = \begin{cases} x - \rho(x) g(x) & \text{if } |x| < \delta, \\ x & \text{if } |x| > \delta. \end{cases} \tag{5.107}$$

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Claim. *The map $\tilde{f}$ has the following properties:*

1. $\tilde{f} = f(x)$ *for all* $|x| < \frac{\delta}{2}$,
2. $\tilde{f} = x$ *for all* $|x| > \delta$,
3. $|\tilde{f}(x)| \geq \frac{|x|}{2}$,
4. $|\tilde{f}(x)| \leq 2|x|$.

Proof. We only proof properties (3) and (4). First we prove property (3). We have $\tilde{f}(x) = x - \rho(x)g(x) = x$ when $|x| \geq \delta$, so $|\tilde{f}(x)| = |x|$ when $|x| \geq \delta$. For $|x| < \delta$, we have

$$\begin{aligned} |\tilde{f}(x)| &\geq |x| - \rho(x)|g(x)| \\ &= |x| - |g(x)| \\ &\geq |x| - \frac{|x|}{2} \\ &= \frac{|x|}{2}. \end{aligned} \tag{5.108}$$

We now prove property (4). We have $\tilde{f}(x) = x - \rho(x)g(x)$, so $|\tilde{f}(x)| = |x|$ for $x \geq \delta$. For $x < \delta$, we have

$$\begin{aligned} |\tilde{f}(x)| &\leq |x| + \rho(x)|g(x)| \\ &\leq |x| + \frac{1}{2}|x| \\ &\leq 2|x|. \end{aligned} \tag{5.109}$$

□

Let $Q_r \equiv \{x \in \mathbb{R}^n : |x| \leq r\}$. The student should check that

$$\text{Property (3)} \implies \tilde{f}^{-1}(Q_r) \subseteq Q_{2r} \tag{5.110}$$

and that

$$\text{Property (4)} \implies \tilde{f}^{-1}(\mathbb{R}^n - Q_{2r}) \subseteq \mathbb{R}^n - Q_r \tag{5.111}$$

Notice that $\tilde{f}^{-1}(Q_r) \subseteq Q_{2r} \implies \tilde{f}$ is proper.

Now we turn back to the map f. Remember that $f : U \to V$ is a diffeomorphism and that $f(0) = 0$. So, the set $f(\text{Int } Q_{\delta/2})$ is an open neighborhood of 0 in $\mathbb{R}^n$. Take

$$\omega \in \Omega_c^n(f(\text{Int } Q_{\delta/2}) \cap \text{Int } Q_{\delta/4}) \tag{5.112}$$

such that

$$\int_{\mathbb{R}^n} \omega = 1. \tag{5.113}$$

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Then,

$$f^*\omega \in \Omega_c^n(Q_{\delta/2}) \tag{5.114}$$

and

$$\tilde{f}^*\omega \in \Omega_c^n(Q_{\delta/2}), \tag{5.115}$$

by Equation 5.110. This shows that $f^*\omega = \tilde{f}^*\omega$. Hence,

$$\begin{aligned} \int_U f^*\omega = \int_U \tilde{f}^*\omega &= \deg(f) \int_V \omega \\ &= \deg(\tilde{f}) \int_V \omega, \end{aligned} \tag{5.116}$$

where

$$\int_V \omega = 1. \tag{5.117}$$

Therefore,

$$\deg(f) = \deg(\tilde{f}). \tag{5.118}$$

Now, let us use Equation 5.111. Choose $\omega \in \Omega_c^n(\mathbb{R}^n - Q_{2\delta})$. So,

$$f^*\omega \in \Omega_c^n(\mathbb{R}^n - Q_\delta). \tag{5.119}$$

Again we take

$$\int_{\mathbb{R}^n} \omega = 1. \tag{5.120}$$

By property (2), $\tilde{f} = I$ on $\mathbb{R}^n - Q_\delta$, so

$$\tilde{f}^*\omega = \omega. \tag{5.121}$$

Integrating,

$$\int_{\mathbb{R}}^n \tilde{f}^*\omega = \deg(\tilde{f}) \int_{\mathbb{R}}^n \omega = \int_{\mathbb{R}}^n \omega. \tag{5.122}$$

Therefore,

$$\deg(f) = \deg(\tilde{f}) = 1. \tag{5.123}$$

□

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Lecture 28

Let U, V be connected open sets of $\mathbb{R}^n$, and let $f : U \to V$ be a diffeomorphism. Then

$$\deg(f) = \begin{cases} +1 & \text{if } f \text{ is orient. preserving,} \\ -1 & \text{if } f \text{ is orient. reversing.} \end{cases} \tag{5.124}$$

We showed that given any $\omega \in \Omega_c^n(V)$,

$$\int_U f^*\omega = \pm \int_V \omega. \tag{5.125}$$

Let $\omega = \phi(x)dx_1 \wedge \cdots \wedge dx_n$, where $\phi \in \mathcal{C}_0^\infty(V)$. Then

$$f^*\omega = \phi(f(x)) \det \left[\frac{\partial f_i}{\partial x_j}(x)\right] dx_1 \wedge \cdots \wedge dx_n, \tag{5.126}$$

so,

$$\int_U \phi(f(x)) \det \left[\frac{\partial f_i}{\partial x_j}\right] dx = \pm \int_V \phi(x)dx. \tag{5.127}$$

Notice that

$$f \text{ is orientation preserving} \iff \det \left[\frac{\partial f_i}{\partial x_j}(x)\right] > 0, \tag{5.128}$$

$$f \text{ is orientation reversing} \iff \det \left[\frac{\partial f_i}{\partial x_j}(x)\right] < 0. \tag{5.129}$$

So, in general,

$$\int_U \phi(f(x)) \left|\det \left[\frac{\partial f_i}{\partial x_j}(x)\right]\right| dx. \tag{5.130}$$

As usual, we assumed that $f \in \mathcal{C}^\infty$.

Remark. The above is true for $\phi \in \mathcal{C}_0^1$, a compactly supported continuous function. The proof of this is in section 5 of the Supplementary Notes. The theorem is true even if only $f \in \mathcal{C}^1$ (the notes prove it for $f \in \mathcal{C}^2$).

Today we show how to compute the degree in general.

Let U, V be connected open sets in $\mathbb{R}^n$, and let $f : U \to V$ be a proper $\mathcal{C}^\infty$ map.

Claim. *Let B be a compact subset of V, and let $A = f^{-1}(B)$. If U_0 is an open subset of U with $A \subseteq U_0$, then there exists an open subset V_0 of V with $B \subseteq V_0$ such that $f^{-1}(V_0) \subseteq U_0$.*

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Proof. Let $C \subseteq V$ be a compact set with $B \subseteq \text{Int } C$, and let $W = f^{-1}(C) - U_0$. The set W is compact, so the set $f(W)$ is also compact. Moreover, $f(W) \cap B = \phi$ since $f^{-1}(B) \subseteq U_0$.

Now, let $V_0 = \text{Int } C - f(W)$. This set is open, and

$$\begin{aligned} f^{-1}(V_0) &\subseteq f^{-1}(\text{Int } C) - W \\ &\subseteq U_0. \end{aligned} \tag{5.131}$$

□

Claim. *If $X \subseteq U$ is closed, then $f(X)$ is closed in V.*

Proof. Take any point $p \in V - f(x)$. Then $f^{-1}(p) \in U - X$. Apply the previous result with $B = \{p\}$, $A = f^{-1}(p)$, and $U_0 = U - X$. There exists an open set $V_0 \ni p$ such that $f^{-1} \subseteq U - X$. The set $V_0 \cap f(X) = \phi$, so $V - f(X)$ is open in V. □

We now remind you of Sard's Theorem. Let $f : U \to V$ be a proper $\mathcal{C}^\infty$ map. We define the critical set

$$C_f = \{p \in U : Df(p) \text{ is not bijective}\}. \tag{5.132}$$

The set C_f is closed. The set $f(C_f)$ in V is a set of measure zero. The set $f(C_f)$ is closed as well, since f is proper.

Definition 5.18. A point $q \in V$ is a *regular value* of f if $q \in V - f(C_f)$.

Sard's Theorem basically says that there are "lots" of regular values.

Lemma 5.19. *If q is a regular value, then $f^{-1}(q)$ is a finite set.*

Proof. First, $p \in f^{-1}(q) \implies p \notin C_f$. So, $Df(p) : \mathbb{R}^n \to \mathbb{R}^n$ is bijective. By the IFT, the map f is a diffeomorphism of a neighborhood U_p of $p \in U$ onto a neighborhood of q. In particular, since f is one-to-one and onto,

$$U_p \cap f^{-1}(q) = \{p\}. \tag{5.133}$$

Consider the collection $\{U_p : p \in f^{-1}(q)\}$, which is an open cover of $f^{-1}(q)$. The H-B Theorem tells us that there exists a finite subcover $\{U_{p_i}, i = 1, \ldots, N\}$. Hence,

$$f^{-1}(q) = \{p_1, \ldots, p_N\}. \tag{5.134}$$

□

Theorem 5.20. *The degree of f is*

$$\deg(f) = \sum_{i=1}^{N} \sigma_{p_i}, \tag{5.135}$$

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where

$$\sigma_{p_i} = \begin{cases} +1 & \text{if } Df(p_i) \text{ is orient. preserving,} \\ -1 & \text{if } Df(p_i) \text{ is orient. reversing.} \end{cases} \tag{5.136}$$

So, to calculate the degree, you just pick any regular value q and "count" the number of points in the pre-image of q, keeping track of the value of σ_{p_i}.

Proof. For each $p_i \in f^{-1}(q)$, let U_{p_i} be an open neighborhood of p_i such that f maps U_{p_i} diffeomorphically onto a neighborhood of q. We can assume that the U_{p_i}'s do not intersect.

Now, choose a neighborhood V_0 of q such that

$$f^{-1}(V_0) \subseteq \bigcup U_{p_i}. \tag{5.137}$$

Next, replace each U_{p_i} by $U_{p_i} \cap f^{-1}(V_0)$. So, we can assume the following:

1. f is a diffeomorphism of U_{p_i} onto V_0,
2. $f^{-1}(V_0) = \bigcup U_{p_i}$,
3. The U_{p_i}'s don't intersect.

Choose $\omega \in \Omega_c^n(V_0)$ such that

$$\int_V \omega = 1. \tag{5.138}$$

Then,

$$\begin{aligned} \int_U f^*\omega &= \sum_i \int_{U_{p_i}} f^*\omega \\ &= \sum_i \sigma_{p_i} \int_{V_0} \omega \\ &= \sum_i \sigma_{p_i}. \end{aligned} \tag{5.139}$$

But,

$$\int_U f^*\omega = (\deg f) \int_U \omega = \deg f, \tag{5.140}$$

so

$$\sum \sigma_{p_i} = \deg f. \tag{5.141}$$

□

This is a very nice theorem that is not often discussed in textbooks.

The following is a useful application of this theorem. Suppose $f^{-1}(q)$ is empty, so $q \notin f(U)$. Then $q \notin f(C_f)$, so q is a regular value. Therefore,

$$\deg(f) = 0. \tag{5.142}$$

This implies the following useful theorem.

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Theorem 5.21. *If* $\deg(f) \neq 0$*, then* $f : U \to V$ *is onto.*

This theorem can let us know if a system of non-linear equations has a solution, simply by calculating the degree. The way to think about this is as follows. Let $f = (f_1, \ldots, f_n)$ and let $q = (c_1, \ldots, c_n) \in V$. If $q \in f(U)$ then there exists a solution $x \in U$ to the system of non-linear equations

$$f_i(x) = c_i, \;\; i = 1, \ldots, n. \tag{5.143}$$

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Lecture 29

We have been studying the important invariant called the degree of f. Today we show that the degree is a "topological invariant."

5.3 Topological Invariance of Degree

Recall that given a subset A of $\mathbb{R}^m$ and a function $F : A \to \mathbb{R}^\ell$, we say that F is $\mathcal{C}^\infty$ if it extends to a $\mathcal{C}^\infty$ map on a neighborhood of A.

Let U be open in $\mathbb{R}^n$, let V be open in $\mathbb{R}^k$, and let $A = U \times [0, 1]$.

Definition 5.22. Let $f_0, f_1 : U \to V$ be $\mathcal{C}^\infty$ maps. The maps f_0 and f_1 are *homotopic* if there is a $\mathcal{C}^\infty$ map $F : U \times [0, 1] \to V$ such that $F(p, 0) = f_0(p)$ and $F(p, 1) = f_1(p)$ for all $p \in U$.

Let $f_t : U \to V$ be the map defined by

$$f_t(p) = F(p, t). \tag{5.144}$$

Note that $F \in \mathcal{C}^\infty \implies f_t \in \mathcal{C}^\infty$. So, $f_t : U \to V$, where $0 \le t \le 1$, gives a family of maps parameterized by t. The family of maps f_t is called a $\mathcal{C}^\infty$ *deformation* of f_0 into f_1.

Definition 5.23. The map F is a *proper homotopy* if for all compact sets $A \subseteq V$, the pre-image $F^{-1}(A)$ is compact.

Denote by π the map $\pi : U \times [0, 1] \to U$ that sends $(p, t) \to t$. Let $A \subseteq V$ be compact. Then $B = \pi(F^{-1}(A))$ is compact, and for all t, $f_t^{-1}(A) \subseteq B$. As a consequence, each f_t is proper.

We concentrate on the case where U, V are open connected subsets of $\mathbb{R}^n$ and $f_0, f_1 : U \to V$ are proper $\mathcal{C}^\infty$ maps. We now prove that the degree is a topological invariant.

Theorem 5.24. *If f_0 and f_1 are homotopic by a proper homotopy, then*

$$\deg(f_0) = \deg(f_1). \tag{5.145}$$

Proof. Let $\omega \in \Omega_c^n(V)$ and let supp $\omega = A$. Let $F : U \times I \to V$ be a proper homotopy between f_0 and f_1. Take $B = \pi(F^{-1}(A))$, which is compact. For all $t \in [0, 1]$, $f_t^{-1}(A) \subseteq B$.

Let us compute $f_t^*\omega$. We can write $\omega = \phi(x)dx_1 \wedge \cdots \wedge dx_n$, where supp $\phi \subseteq A$. So,

$$f_t^*\omega = \phi(F(x, t)) \det \left[\frac{\partial F_i}{\partial x_j}(x, t)\right] dx_1 \wedge \cdots \wedge dx_n, \tag{5.146}$$

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and

$$\begin{aligned}\int_U f_t^*\omega &= \deg(f_t)\int_V \omega. \\ &= \int_U \phi(F(x,t))\det\left[\frac{\partial F_i}{\partial x_j}(x,t)\right] dx_1\dots dx_n.\end{aligned} \tag{5.147}$$

Notice that the integrand is supported in the compact set B for all t, and it is $\mathcal{C}^\infty$ as a function of x and t. By Exercise #2 in section 2 of the Supplementary Notes, this implies that the integral is $\mathcal{C}^\infty$ in t. From Equation 5.147, we can conclude that $\deg(f_t)$ is a $\mathcal{C}^\infty$ function of t.

Now here is the trick. Last lecture we showed that $\deg(f_t)$ is an integer. Since $\deg(f_t)$ is continuous, it must be a constant $\deg(f_t) =$ constant. □

We consider a simple application of the above theorem. Let $U = V = \mathbb{R}^2$, and think of $\mathbb{R}^2 = \mathbb{C}$. We make the following associations:

$$i^2 = -1 \tag{5.148}$$
$$z = x + iy \tag{5.149}$$
$$\bar{z} = x - iy \tag{5.150}$$
$$z\bar{z} = |z|^2 = x^2 + y^2 \tag{5.151}$$
$$dz = dx + idy \tag{5.152}$$
$$d\bar{z} = dx - idy \tag{5.153}$$
$$dz \wedge d\bar{z} = -2idx \wedge dy \tag{5.154}$$
$$dx \wedge dy = \frac{1}{2}idz \wedge d\bar{z}. \tag{5.155}$$

Consider a map $f : \mathbb{R}^2 \to \mathbb{R}^2$, thinking of $\mathbb{R}^2 = \mathbb{C}$, defined by

$$f(z) = z^n + \sum_{i=0}^{n-1} c_i z^i,\ c_i \in \mathbb{C}. \tag{5.156}$$

Claim. *The map f is proper.*

Proof. Let $C = \sum |c_i|$. For $|z| > 1$,

$$\left|\sum_{i=0}^{n-1} c_i z^i\right| \le C|z|^{n-1}. \tag{5.157}$$

So,

$$\begin{aligned}|f(z)| &\ge |z|^n - \left|\sum c_i z^i\right| \\ &= |z|^n - C|z|^{n-1} \\ &= |z|^n\left(1 - \frac{C}{|z|}\right).\end{aligned} \tag{5.158}$$

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For $|z| > 2C$,

$$|f(z)| \geq \frac{|z|^n}{2}. \tag{5.159}$$

So, if $R > 1$ and $R > 2C$, then $f^{-1}(B_R) \subseteq B_{R_1}$, where $R_1^n/2 \leq R$ (and where B_r denotes the ball of radius r). So f is proper. □

Now, let us define a homotopy $F : \mathbb{C} \times [0,1] \to \mathbb{C}$ by

$$F(z,t) = z^n + t\sum_{i=0}^{n-1} c_i z^i. \tag{5.160}$$

We claim that $F^{-1}(B_R) \subseteq B_{R_1} \times [0,1]$, by exactly the same argument as above. So F is proper.

Notice that

$$F(z,1) = f_1(z) = f(z), \tag{5.161}$$
$$F(z,0) = f_0(z) = z^n. \tag{5.162}$$

So, by the above theorem, $\deg(f) = \deg(f_0)$.

Let us compute $\deg(f_0)$ by brute force. We have $f_0(z) = z^n$, so

$$f_0^* dz = dz^n = nz^{n-1}dz, \tag{5.163}$$
$$f_0^* d\bar{z} = d\bar{z}^n = n\bar{z}^{n-1}d\bar{z}. \tag{5.164}$$

Using the associations defined above,

$$\begin{aligned} f_0^*(dx \wedge dy) &= \frac{i}{2} f_0^*(dz \wedge d\bar{z}) \\ &= \frac{i}{2} f_0^* dz \wedge f_0^* d\bar{z} \\ &= \frac{i}{2} n^2 |z|^{2(n-1)} dz \wedge d\bar{z} \\ &+ n^2 |z|^{2n-2} dx \wedge dy. \end{aligned} \tag{5.165}$$

Let $\phi \in \mathcal{C}_0^\infty(\mathbb{R})$ such that

$$\int_0^\infty \phi(s)ds = 1. \tag{5.166}$$

Let $\omega = \phi(|z|^2)dx \wedge dy$. We calculate $\int_{\mathbb{R}^2} \omega$. Let us use polar coordinates, where

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$r = \sqrt{x^2+y^2} = |z|$.

$$\begin{aligned}\int_{\mathbb{R}^2} \omega &= \int_{\mathbb{R}^2} \phi(|z|^2) dxdy \\ &= \int_{\mathbb{R}^2} \phi(r^2) r dr d\theta \\ &= 2\pi \int_o^\infty \phi(r^2) r dr \\ &+ 2\pi \int_0^\infty \phi(s) \frac{ds}{2} \\ &= \pi.\end{aligned} \tag{5.167}$$

Now we calculate $\int f_0^* \omega$. First, we note that

$$f_0^* \omega = \phi(|z|^{2n}) n^2 |z|^{2n-2} dx \wedge dy. \tag{5.168}$$

So,

$$\begin{aligned}\int f_0^* \omega &= n^2 \int_0^\infty \phi(r^{2n}) r^{2n-2} r dr d\theta \\ &= n^2(2\pi) \int_0^\infty \phi(r^{2n}) r^{2n-1} dr \\ &= n^2(2\pi) \int_0^\infty \phi(s) \frac{ds}{2n} \\ &= n\pi.\end{aligned} \tag{5.169}$$

To summarize, we have calculated that

$$\int_{\mathbb{R}^2} \omega = \pi \quad \text{and} \quad \int_{\mathbb{R}^2} f_0^* \omega = n\pi. \tag{5.170}$$

Therefore,

$$\deg(f_0) = \deg(f) = n. \tag{5.171}$$

A better way to do the above calculation is in the homework: problem #6 of section 6 of the Supplementary Notes.

Last lecture we showed that if $\deg(f) \neq 0$, then the map f is onto. Applying this to the above example, we find that the algebraic equation

$$z^n + \sum_{i=0}^{n-1} c_i z^i = 0 \tag{5.172}$$

has a solution. This is known as the Fundamental Theorem of Algebra.

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Lecture 30

6 Manifolds

6.1 Canonical Submersion and Canonical Immersion Theorems

As part of today's homework, you are to prove the canonical submersion and immersion theorems for linear maps. We begin today's lecture by stating these two theorems.

Let $A : \mathbb{R}^n \to \mathbb{R}^m$ be a linear map, and let $[a_{ij}]$ be its associated matrix. We have the transpose map $A^t : \mathbb{R}^m \to \mathbb{R}^n$ with the associated matrix $[a_{ji}]$.

Definition 6.1. Let $k < n$. Define the *canonical submersion* map π and the *canonical immersion* map ι as follows:

Canonical submersion:

$$\pi : \mathbb{R}^n \to \mathbb{R}^k, \quad (x_1, \ldots, x_n) \to (x_1, \ldots, x_k). \tag{6.1}$$

Canonical immersion:

$$\iota : \mathbb{R}^k \to \mathbb{R}^n, \quad (x_1, \ldots, x_k) \to (x_1, \ldots, x_k, 0, \ldots, 0). \tag{6.2}$$

Canonical Submersion Thoerem. *Let $A : \mathbb{R}^n \to \mathbb{R}^k$ be a linear map, and suppose that A is onto. Then there exists a bijective linear map $B : \mathbb{R}^n \to \mathbb{R}^n$ such that $A \circ B = \pi$.*

Proof Hint: Show that there exists a basis $v_1, \ldots, v_n$ of $\mathbb{R}^n$ such that $Av_i = e_i,\ i = 1, \ldots, k$, (the standard basis of $\mathbb{R}^k$) and $Av_i = 0$ for all $i > k$. Then let $B : \mathbb{R}^n \to \mathbb{R}^n$ be the linear map $Be_i = v_i,\ i = 1, \ldots, n$, where $e_i, \ldots, e_n$ is the standard basis of $\mathbb{R}^n$. □

Canonical Immersion Thoerem. *As before, let $k < n$. Let $A : \mathbb{R}^k \to \mathbb{R}^n$ be a one-to-one linear map. Then there exists a bijective linear map $B : \mathbb{R}^n \to \mathbb{R}^n$ such that $B \circ A = \iota$.*

Proof Hint: Note that $B \circ A = \iota \iff A^t B^t = \pi$. Use the Canonical Submersion Theorem. □

Now we prove non-linear versions of these two theorems.

Let U be an open set in $\mathbb{R}^n$, and let $f : U \to \mathbb{R}^k$ be a $\mathcal{C}^\infty$ map. Let $p \in U$.

Definition 6.2. The map f is a *submersion at p* if $Df(p) : \mathbb{R}^n \to \mathbb{R}^k$ is onto.

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Canonical Submersion Thoerem. *Assume that f is a submersion at p and that $f(p) = 0$. Then there exists a neighborhood U_0 of p in U, a neighborhood V of 0 in $\mathbb{R}^n$, and a diffeomorphism $g : V \to U_0$ such that $f \circ g = \pi$.*

Proof. Let $T_p : \mathbb{R}^n \to \mathbb{R}^n$ be the translation defined by $x \to x + p$. Replacing f by $f \circ T_p$ we can assume that $p = 0$ and $f(0) = 0$.

Let $A = (Df)(0)$, where $A : \mathbb{R}^n \to \mathbb{R}^k$ is onto by the assumption that f is a submersion. So, there exists a bijective linear map $B : \mathbb{R}^n \to \mathbb{R}^n$ such that $A \circ B = \pi$. Replacing f by $f \circ B$ we can assume that $Df(0) = \pi$.

Define a map $h : U \to \mathbb{R}^n$ by

$$h(x_1, \ldots, x_n) = (f(x_1, \ldots, x_k); x_{k+1}, \ldots, x_n). \tag{6.3}$$

Note that (1) $Dh(0) = I$; and (2) $\pi h = f$. By (1), the function h maps a neighborhood U_0 of 0 in U diffeomorphically onto a neighborhood V of 0 in $\mathbb{R}^n$. By (2), we have $\pi = f \circ h^{-1}$. Take $g = h^{-1}$. □

There is a companion theorem having to do with immersions.

Definition 6.3. Let U be an open subset of $\mathbb{R}^k$, and let $f : U \to \mathbb{R}^n$ be a $\mathcal{C}^\infty$ map. Let $p \in U$. The map f is an *immersion at p* if $(Df)(p) : \mathbb{R}^k \to \mathbb{R}^n$ is injective (one-to-one).

Canonical Immersion Thoerem. *Let U be a neighborhood of 0 in $\mathbb{R}^k$, and let $f : U \to \mathbb{R}^n$ be a $\mathcal{C}^\infty$ map. Assume that f is an immersion at 0. Then there exists a neighborhood V of $f(0) = p$ in $\mathbb{R}^n$, a neighborhood W of 0 in $\mathbb{R}^k$, and a diffeomorphism $g : V \to W$ such that $\iota^{-1}(W) \subseteq U$ and $g \circ f = \iota$.*

Proof. Replacing f by $T_p \circ f$, we can assume that $f(0) = 0$. Let $A = Df(0)$, so $A : \mathbb{R}^k \to \mathbb{R}^n$ is injective. There exists a linear map $B : \mathbb{R}^n \to \mathbb{R}^n$ such that $BA = \iota$. Replacing f by $B \circ f$, we can assume that $Df(0) = \iota$.

Let $\ell = n - k$. Since $U \subseteq \mathbb{R}^k$, we get $U \times \mathbb{R}^\ell \subseteq \mathbb{R}^k \times \mathbb{R}^\ell = \mathbb{R}^n$. Define a map $h : U \times \mathbb{R}^\ell \to \mathbb{R}^n$ by

$$h(x_1, \ldots, x_n) = f(x_1, \ldots, x_k) + (0, \ldots, 0, x_{k+1}, \ldots, x_n). \tag{6.4}$$

One can check that (1) $Dh(0) = I$; and (2) $h \circ \iota = f$.

By (1), the function h maps a neighborhood W of 0 in $U \times \mathbb{R}^\ell$ diffeomorphically onto a neighborhood V of 0 in $\mathbb{R}^n$. Moreover, $W \subseteq U \times \mathbb{R}^\ell$, so $\iota^{-1}(W) \subseteq U$.

By (2), we obtain the canonical immersion map $\iota = h^{-1} \circ f$. Take $g = h^{-1}$. □

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6.2 Definition of Manifold

Now we move on to the study of manifolds.

Let X be a subset of $\mathbb{R}^n$, let Y be a subset of $\mathbb{R}^m$, and let $f : X \to Y$ be a continuous map. We define that the map f is a $\mathcal{C}^\infty$ map if for every point $p \in X$ there exists a neighborhood U_p of p in $\mathbb{R}^n$ and a $\mathcal{C}^\infty$ map $g_p : U_p \to \mathbb{R}^n$ such that $g_p|X \cap U_p = f$.

We showed in the homework that if $f : X \to Y$ is a $\mathcal{C}^\infty$ map, then there exists a neighborhood U of X in $\mathbb{R}^n$ and a $\mathcal{C}^\infty$ map $g : U \to \mathbb{R}^n$ extending f.

Definition 6.4. A map $f : X \to Y$ is a *diffeomorphism* if it is one-to-one, onto, a $\mathcal{C}^\infty$ map, and $f^{-1} : Y \to X$ is $\mathcal{C}^\infty$.

Let X be a subset of $\mathbb{R}^N$.

Definition 6.5. The set X is an *n-dimensional manifold* if for every point $p \in X$ there exists a neighborhood V of p in $\mathbb{R}^N$, an open set U in $\mathbb{R}^m$, and a diffeomorphism $f : U \to V \cap X$. The collection (f, U, X) is called a *parameterization of X at p.*

This definition does not illustrate how manifolds come up in nature. Usually manifolds come up in the following scenario.

Let W be open in $\mathbb{R}^N$, and let $f_i : W \to \mathbb{R},\ i = 1, \ldots, \ell$ be $\mathcal{C}^\infty$ functions. Suppose you want to study the solution space of

$$f_i(x_1, \ldots, x_N) = 0, \quad i = 1, \ldots, \ell. \tag{6.5}$$

Then you consider the mapping $f : W \to \mathbb{R}^\ell$ defined by

$$f(x) = (f_1(x), \ldots, f_\ell(x)). \tag{6.6}$$

Claim. *If for every $p \in W$ the map f is a submersion of p, then Equation 6.6 defines a k-dimensional manifold, where $k = N - \ell$.*

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Lecture 31

6.3 Examples of Manifolds

We begin with a review of the definition of a manifold.

Let X be a subset of $\mathbb{R}^n$, let Y be a subset of $\mathbb{R}^m$, and let $f : X \to Y$ be a continuous map.

Definition 6.6. The map f is $\mathcal{C}^\infty$ if for every $p \in X$, there exists a neighborhood U_p of p in $\mathbb{R}^n$ and a $\mathcal{C}^\infty$ map $g_p : U_p \to \mathbb{R}^m$ such that $g_p = f$ on $U_p \cap X$.

Claim. *If $f : X \to Y$ is continuous, then there exists a neighborhood U of X in $\mathbb{R}^n$ and a $\mathcal{C}^\infty$ map $g : U \to \mathbb{R}^m$ such that $g = f$ on $U \cap X$.*

Definition 6.7. The map $f : X \to Y$ is a *diffeomorphism* if it is one-to-one, onto, and both f and f^{-1} are $\mathcal{C}^\infty$ maps.

We define the notion of a manifold.

Definition 6.8. A subset X of $\mathbb{R}^N$ is an *n-dimensional manifold* if for every $p \in X$, there exists a neighborhood V of p in $\mathbb{R}^N$, an open set U in $\mathbb{R}^n$, and a diffeomorphism $\phi : U \to X \cap V$.

Intuitively, the set X is an n-dimensional manifold if locally near every point $p \in X$, the set X "looks like an open subset of $\mathbb{R}^n$."

Manifolds come up in practical applications as follows:

Let U be an open subset of $\mathbb{R}^N$, let $k < N$, and let $f : \mathbb{R}^N \to \mathbb{R}^k$ be a $\mathcal{C}^\infty$ map. Suppose that 0 is a regular value of f, that is, $f^{-1}(0) \cap C_f = \phi$.

Theorem 6.9. *The set $X = f^{-1}(0)$ is an n-dimensional manifold, where $n = N - k$.*

Proof. If $p \in f^{-1}(0)$, then $p \notin C_f$. So the map $Df(p) : \mathbb{R}^N \to \mathbb{R}^k$ is onto. The map f is a submersion at p.

By the canonical submersion theorem, there exists a neighborhood V of 0 in $\mathbb{R}^n$, a neighborhood U_0 of p in U, and a diffeomorphism $g : V \to U$ such that

$$f \circ g = \pi. \tag{6.7}$$

Recall that $\mathbb{R}^N = \mathbb{R}^\ell \times \mathbb{R}^n$ and $\pi : \mathbb{R}^N \to \mathbb{R}^k$ is the map that sends

$$(x, y) \in \mathbb{R}^k \times \mathbb{R}^n \to \mathbb{R}^k. \tag{6.8}$$

Hence, $\pi^{-1}(0) = \{0\} \times \mathbb{R}^n = \mathbb{R}^n$. By Equation 6.7, the function g maps $V \cap \pi^{-1}(0)$ diffeomorphically onto $U_0 \cap f^{-1}(0)$. But $V \cap \pi^{-1}(0)$ is a neighborhood of 0 in $\mathbb{R}^n$ and $U_0 \cap f^{-1}(0)$ is a neighborhood of p in X. □

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We give three examples of applications of the preceding theorem.

1. We consider the n-sphere S^n. Define a map

$$f : \mathbb{R}^{n+1} \to \mathbb{R}, \quad f(x) = x_1^2 + \ldots + x_{n+1}^2 - 1. \tag{6.9}$$

The derivative is $(Df)(x) = 2[x_1, \ldots, x_{n+1}]$, so $C_f = \{0\}$. If $a \in f^{-1}(0)$, then $\sum a_i^2 = 1$, so $a \notin C_f$. Thus, the set $f^{-1}(0) = S^n$ is an n-dimensional manifold.

2. Let $g : \mathbb{R}^n \to \mathbb{R}^k$ be a $\mathcal{C}^\infty$ map. Define

$$X = \text{graph } g = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^k : y = g(x)\}. \tag{6.10}$$

Note that $X \subseteq \mathbb{R}^n \times \mathbb{R}^k = \mathbb{R}^{n+k}$.

Claim. *The set X is an n-dimensional manifold.*

Proof. Define a map $f : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^k$ by

$$f(x, y) = y - g(x). \tag{6.11}$$

Note that $Df(x, y) = [-Dg(x), I_k]$. This is always of rank k, so $C_f = \phi$. Hence, the graph g is an n-dimensional manifold. □

3. The following example comes from Munkres section 24, exercise #6. Let

$$\mathcal{M}_n = \text{ the set of all } n \times n \text{ matrices}, \tag{6.12}$$

so

$$\mathcal{M}_n \cong \mathbb{R}^{n^2}. \tag{6.13}$$

With any element $[a_{ij}]$ in $\mathcal{M}_n$ we associate a vector

$$(a_{11}, \ldots, a_{1n}, a_{21}, \ldots, a_{2n}, \ldots). \tag{6.14}$$

Now, let

$$\mathcal{S}_n = \{A \in \mathcal{M}_n : A = A^t\}, \tag{6.15}$$

so

$$\mathcal{S}_n \cong \mathbb{R}^{\frac{n(n+1)}{2}}. \tag{6.16}$$

With any element $[a_{ij}]$ in $\mathcal{S}_n$ we associate a vector

$$(a_{11}, \ldots, a_{1n}, a_{22}, a_{23}, \ldots, a_{2n}, a_{33}, a_{34}, \ldots). \tag{6.17}$$

The above association avoids the "redundancies" $a_{12} = a_{21}, a_{31} = a_{13}, a_{32} = a_{23}$, etc.

Define

$$O(n) = \{A \in \mathcal{M}_n : A^t A = I\}, \tag{6.18}$$

which is the set of orthogonal $n \times n$ matrices.

As an exercise, the student should prove the following claim.

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Claim. *The set $O(n) \subseteq \mathcal{M}_n$ is an $\frac{n(n-1)}{2}$-dimensional manifold.*

Proof Hint: First hint: Let $f : \mathcal{M}_n \to \mathcal{S}_n$ be the map defined by

$$f(A) = A^t A - I, \tag{6.19}$$

so $O(n) = f^{-1}(0)$. Show that $f^{-1}(0) \cap C_f = \phi$. The main idea is to show that if $A \notin f^{-1}(0)$, then the map $Df(A) : \mathcal{M}_n \to \mathcal{S}_n$ is onto.

Second hint: Note that $Df(A)$ is the map the sends $B \in \mathcal{M}_n$ to $A^t B + B^t A$. □

Manifolds are often defined by systems of non-linear equations:

Let $f : \mathbb{R}^N \to \mathbb{R}^k$ be a continuous map, and suppose that $C_f \cap f^{-1}(0) = \phi$. Then $X = f^{-1}(0)$ is an n-dimensional manifold. Suppose that $f = (f_1, \ldots, f_k)$. Then X is defined by the system of equations

$$f_i(x_1, \ldots, x_N) = 0, \quad i = 1, \ldots, k. \tag{6.20}$$

This system of equations is called *non-degenerate*, since for every $x \in X$ the matrix

$$\left[\frac{\partial f_i}{\partial x_j}(x)\right] \tag{6.21}$$

is of rank k.

Claim. *Every n-dimensional manifold $X \subseteq \mathbb{R}^N$ can be described locally by a system of k non-degenerate equations of the type above.*

Proof Idea: Let $X \subseteq \mathbb{R}^N$ be an n-dimensional manifold. Let $p \in X$, let U be an open subset of $\mathbb{R}^n$, and let V be an open neighborhood of p in $\mathbb{R}^N$. Let $\phi : I \to V \cap X$ be a diffeomorphism. Modifying ϕ by a translation if necessary we can assume that $0 \in U$ and $\phi(0) = p$. We can think of ϕ as a map $\phi : U \to \mathbb{R}^N$ mapping U into X.

Claim. *The linear map $(D\phi)(0) : \mathbb{R}^n \to \mathbb{R}^N$ is injective.*

Proof. The map $\phi^{-1} : V \cap X \to U$ is a $\mathcal{C}^\infty$ map, so (shrinking V if necessary) we can assume there is a $\mathcal{C}^\infty$ map $\psi : V \to U$ with $\psi = \phi^{-1}$ on $V \cap X$. Since ϕ maps U onto $V \cap X$, we have $\psi \circ \phi = \phi^{-1} \circ \phi = I =$ the identity map of U onto itself. Thus,

$$I = D(\psi \circ \phi)(0) = (D\psi)(p)(D\phi)(0). \tag{6.22}$$

That is, $D\psi(p)$ is a "left inverse" of $D\phi(0)$. So, $D\phi(0)$ is injective. □

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We can conclude that $\phi : U \to \mathbb{R}^N$ is an immersion at 0. The canonical immersion theorem tells us that there exists a neighborhood U_0 of 0 in U, a neighborhood V_p of p in V, and a $\mathcal{C}^\infty$ map $g : V_p \to \mathbb{R}^N$ mapping p onto 0 and mapping V_p diffeomorphically onto a neighborhood $\mathcal{O}$ of 0 in $\mathbb{R}^N$ such that

$$\iota^{-1}(\mathcal{O}) = U_0 \tag{6.23}$$

and

$$g \circ \phi = \iota \tag{6.24}$$

on U_0. Here, the map ι is the canonical submersion map $\iota : \mathbb{R}^n \to \mathbb{R}^N$ that maps $(x_1, \ldots, x_n) \to (x_1, \ldots, x_n, 0, \ldots, 0)$.

By Equation 6.24, the function g maps $\phi(U_0)$ onto $\iota(U_0)$. However, by Equation 6.23, the set $\iota(U_0)$ is the subset of $\mathcal{O}$ defined by the equations

$$x_i = 0, \quad i = n+1, \ldots, N. \tag{6.25}$$

So, if $g = (g_1, \ldots, g_N)$, then $\phi(U_0) = X \cap V_p$ is defined by the equations

$$g_i = 0, \quad i = n+1, \ldots, N. \tag{6.26}$$

Moreover, the $N \times N$ matrix

$$\left[\frac{\partial g_i}{\partial x_j}(x)\right] \tag{6.27}$$

is of rank N at every point $x \in V_p$, since $g : V_p \to \mathcal{O}$ is a diffeomorphism. Hence, the last $N - n$ row vectors of this matrix

$$\left(\frac{\partial g_i}{\partial x_1}, \ldots, \frac{\partial g_i}{\partial x_N}\right), \quad i = n+1, \ldots, N, \tag{6.28}$$

are linearly independent at every point $x \in V_p$.

Now let $k = N - n$ and let $f_i = g_{i+n},\ i = 1, \ldots, k$. Then $X \cap V_p$ is defined by the equations

$$f_i(x) = 0, \quad i = 1, \ldots, k, \tag{6.29}$$

and the $k \times N$ matrix

$$\left[\frac{\partial f_i}{\partial x_k}(x)\right] \tag{6.30}$$

is of rank k at all points $x \in V_p$. In other words, the system of equations 6.29 is non-degenerate. □

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Lecture 32

6.4 Tangent Spaces of Manifolds

We generalize our earlier discussion of tangent spaces to tangent spaces of manifolds. First we review our earlier treatment of tangent spaces.

Let $p \in \mathbb{R}^n$. We define

$$T_p\mathbb{R}^n = \{(p, v) : v \in \mathbb{R}^n\}. \tag{6.31}$$

Of course, we associate $T_p\mathbb{R}^n \cong \mathbb{R}^n$ by the map $(p, v) \to v$.

If U is open in $\mathbb{R}^n$, V is open in $\mathbb{R}^k$, and $f : (U, p) \to (V, q)$ (meaning that f maps $U \to V$ and $p \to p_1$) is a $\mathcal{C}^\infty$ map, then we have the map $df_p : T_p\mathbb{R}^n \to T_q\mathbb{R}^k$. Via the identifications $T_p\mathbb{R}^n \cong \mathbb{R}^n$ and $T_p\mathbb{R}^k \cong \mathbb{R}^k$, the map df_p is just the map $Df(p) : \mathbb{R}^n \to \mathbb{R}^k$. Because these two maps can be identified, we can use the chain rule for $\mathcal{C}^\infty$ maps. Specifically, if $f : (U, p) \to (V, q)$ and $g : (V, q) \to (\mathbb{R}^\ell, w)$, then

$$d(g \circ f)_p = (dg)_q \circ (df)_p, \tag{6.32}$$

because $(Dg)(q)(Df(p)) = (Dg \circ f)(p)$.

You might be wondering: Why did we make everything more complicated by using df instead of Df? The answer is because we are going to generalize from Euclidean space to manifolds.

Remember, a set $X \subseteq \mathbb{R}^N$ is an n-dimensional manifold if for every $p \in X$, there exists a neighborhood V of p in $\mathbb{R}^N$, an open set U in $\mathbb{R}^n$, and a diffeomorphism $\phi : U \to V \cap X$. The map $\phi : U \to V \cap X$ is called a parameterization of X at p.

Let us think of ϕ as a map $\phi : U \to \mathbb{R}^N$ with Im $\phi \subseteq X$.

Claim. *Let $\phi^{-1}(p) = q$. Then the map $(d\phi)_q : T_q\mathbb{R}^n \to T_p\mathbb{R}^N$ is one-to-one.*

Reminder of proof: The map $\phi^{-1} : V \cap X \to U$ is a $\mathcal{C}^\infty$ map. So, shrinking V if necessary, we can assume that this map extends to a map $\psi : V \to U$ such that $\psi = \phi^{-1}$ on $X \cap V$. Then note that for any $u \in U$, we have $\psi(\phi(u)) = \phi^{-1}(\phi(u)) = u$. So, $\psi \circ \phi = \mathrm{id}_U =$ the identity on U.

Using the chain rule, and letting $\phi(q) = p$, we get

$$\begin{aligned} d(\psi \circ \phi)_q &= (d\psi)_o \circ (d\phi)_q \\ &= (d(\mathrm{id}_U))_q. \end{aligned} \tag{6.33}$$

So, $(d\phi)_q$ is injective. □

Today we define for any $p \in X$ the tangent space T_pX, which will be a vector subspace $T_pX \subseteq T_p\mathbb{R}^N$. The tangent space will be like in elementary calculus, that is, a space tangent to some surface.

Let $\phi : U \to V \cap X$ be a parameterization of X, and let $\phi(q) = p$. The above claim tells us that $(d\phi)_q : T_q\mathbb{R}^n \to T_p\mathbb{R}^N$ is injective.

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Definition 6.10. We define the *tangent space* of a manifold X to be

$$T_pX = \operatorname{Im}(d\phi)_q. \tag{6.34}$$

Because $(d\phi)_q$ is injective, the space T_pX is n-dimensional.

We would like to show that the space T_pX does not depend on the choice of parameterization ϕ. To do so, we will make use of an equivalent definition for the tangent space T_pX.

Last time we showed that given $p \in X \subseteq \mathbb{R}^N$, and $k = N - n$, there exists a neighborhood V of p in $\mathbb{R}^N$ and a $\mathcal{C}^\infty$ map $f : V \to \mathbb{R}^k$ mapping $f(p) = 0$ such that $X \cap V = f^{-1}(0)$. Note that $f^{-1}(0) \cap C_f = \phi$ (where here ϕ is the empty set).

We motivate the second definition of the tangent space. Since $p \in f^{-1}(0)$, the point $p \notin C_f$. So, the map $df_p : T_p\mathbb{R}^N \to T_0\mathbb{R}^k$ is surjective. So, the kernel of df_p in $T_p\mathbb{R}^N$ is of dimension $N - k = n$.

Definition 6.11. An alternate definition for the *tangent space* of a manifold is

$$T_pX = \ker df_p. \tag{6.35}$$

Claim. *These two definitions for the tangent space T_pX are equivalent.*

Proof. Let $\phi : U \to V \cap X$ be a parameterization of X at p with $\phi(p) = q$. The function $f : V \to \mathbb{R}^k$ has the property that $f^{-1}(0) = X \cap V$. So, $f \circ \phi \equiv 0$. Applying the chain rule,

$$(df_p) \circ (d\phi_q) = d(0) = 0. \tag{6.36}$$

So, Im $d\phi_q = \ker df_p$. □

We can now explain why the tangent space T_pX is independent of the chosen parameterization. We have two definitions for the tangent space. The first does not depend on the choice of ϕ, and the second does not depend on choice of f. Therefore, the tangent space depends on neither.

Lemma 6.12. *Let W be an open subset of $\mathbb{R}^\ell$, and let $g : W \to \mathbb{R}^n$ be a $\mathcal{C}^\infty$ map. Suppose that $g(W) \subseteq X$ and that $g(w) = p$, where $w \in W$. Then $(dg)_W \subseteq T_pX$.*

Proof Hint: We leave the proof as an exercise. As above, we have a map $f : V \to \mathbb{R}^k$ such that $X \cap V = f^{-1}(0)$ and $T_pX = \ker df_p$. Let $W_1 = g^{-1}(V)$, and consider the map $f \circ g : W_1 \to \mathbb{R}^k$. As before, $f \circ g = 0$, so $df_p \circ dg_w = 0$. □

Suppose that $X \subseteq \mathbb{R}^N$ is an n-dimensional manifold and $Y \subseteq \mathbb{R}^\ell$ is an m-dimensional manifold. Let $f : X \to Y$ be a $\mathcal{C}^\infty$ map, and let $f(p) = q$.. We want to define a linear map

$$df_p : T_pX \to T_qY. \tag{6.37}$$

Let v be a neighbor hood of p in $\mathbb{R}^N$, and let $g : V \to \mathbb{R}^\ell$ be a map such that $g = f$ on $V \cap X$. By definition $T_pX \subseteq T_p\mathbb{R}^N$, so we have

$$dg_p : T_p\mathbb{R}^N \to T_q\mathbb{R}^k. \tag{6.38}$$

We define the map df_p to be the restriction of dg_p to the tangent space T_pX.

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Definition 6.13.

$$df_p = dg_p|T_pX. \tag{6.39}$$

There are two questions about this definition that should have us worried:

1. Is Im $dg_p(T_pX)$ a subset of T_qY?
2. Does this definition depend on the choice of g?

We address these two questions here:

1. Is Im $dg_p(T_pX)$ a subset of T_qY?

   Let U be an open subset of $\mathbb{R}^N$, let $q = f(p)$, and let $\phi : U \to X \cap V$ be a parameterization of X at p. As before, let us think of ϕ as a map $\phi : U \to \mathbb{R}^N$ with $\phi(U) \subseteq X$.

   By definition, $T_pX = \text{Im}\,(d\phi)_r$, where $\phi(r) = p$. So, given $v \in T_pX$, one can always find $w \in T_r\mathbb{R}^n$ with $v = (d\phi)_r w$.

   Now, is it true that $(dg)_p(v) \in T_qY$? We have

   $$\begin{aligned}(dg)_p v &= (dg)_p(d\phi)_r(w)\\ &= d(g\circ\phi)_r(w),\end{aligned} \tag{6.40}$$

   and the map $(g \circ \phi)$ is of the form $g \circ \phi : U \to Y$, so

   $$d(g\circ\phi)_r(w) \in T_qY. \tag{6.41}$$

2. Does the definition depend on the choice of g?

   Consider two such maps $g_1, g_2 : V \to \mathbb{R}^\ell$. The satisfy $g_1 = g_2 = f$ on $X \cap V$. Then, with v, w as above,

   $$(dg_1)_p(v) = d(g_1\circ\phi)_r(w) \tag{6.42}$$
   $$(dg_2)_p(v) = d(g_2\circ\phi)_r(w). \tag{6.43}$$

   Since $g_1 = g_2$ on $X \cap V$, we have

   $$g_1 \circ \phi = g_2 \circ \phi = f \circ \phi. \tag{6.44}$$

   Hence,

   $$d(g_1\circ\phi)_r(w) = d(g_2\circ\phi)_r(w). \tag{6.45}$$

As an exercise, show that the chain rule also generalizes to manifolds as follows: Suppose that X_1, X_2, X_3 are manifolds with $X_i \subseteq \mathbb{R}^{N_i}$, and let $f : X_1 \to X_2$ and $g : X_2 \to X_3$ be $\mathcal{C}^\infty$ maps. Let $f(p) = q$ and $g(q) = r$.

Show the following claim.

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Claim.

$$d(g \circ f)_p = (dg_q) \circ (df)_q. \tag{6.46}$$

Proof Hint: Let V_1 be a neighborhood of p in $\mathbb{R}^{N_1}$, and let V_2 be a neighborhood of q in $\mathbb{R}^{N_2}$. Let $\tilde{f} : V_1 \to V_2$ be an extension of f to V_1, and let $\tilde{g} : V_2 \to \mathbb{R}^{N_3}$ be an extension of g to V_2.

The chain rule for f, g follows from the chain rule for $\tilde{f}, \tilde{g}$.

□

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Lecture 33

6.5 Differential Forms on Manifolds

Let $U \subseteq \mathbb{R}^n$ be open. By definition, a k-form ω on U is a function which assigns to each point $p \in U$ an element $\omega_p \in \Lambda^k(T_p^*\mathbb{R}^n)$.

We now define the notion of a k-form on a manifold. Let $X \subseteq \mathbb{R}^N$ be an n-dimensional manifold. Then, for $p \in X$, the tangent space $T_pX \subseteq T_p\mathbb{R}^N$.

Definition 6.14. A k-form ω on X is a function on X which assigns to each point $p \in X$ an element $\omega_p \in \Lambda^k((T_pX)^*)$.

Suppose that $f : X \to \mathbb{R}$ is a $\mathcal{C}^\infty$ map, and let $f(p) = a$. Then df_p is of the form

$$df_p : T_pX \to T_a\mathbb{R} \cong \mathbb{R}. \tag{6.47}$$

We can think of $df_p \in (T_pX)^* = \Lambda^1((T_pX)^*)$. So, we get a one-form df on X which maps each $p \in X$ to df_p.

Now, suppose

$$\mu \text{ is a } k\text{-form on } X, \text{ and} \tag{6.48}$$
$$\nu \text{ is an } \ell\text{-form on } X. \tag{6.49}$$

For $p \in X$, we have

$$\mu_p \in \Lambda^k(T_p^*X) \text{ and} \tag{6.50}$$
$$\nu_p \in \Lambda^\ell(T_p^*X). \tag{6.51}$$

Taking the wedge product,

$$\mu_p \wedge \nu_p \in \Lambda^{k+\ell}(T_p^*X). \tag{6.52}$$

The wedge product $\mu \wedge \nu$ is the $(k+\ell)$-form mapping $p \in X$ to $\mu_p \wedge \nu_p$.

Now we consider the pullback operation. Let $X \subseteq \mathbb{R}^N$ and $Y \subseteq \mathbb{R}^\ell$ be manifolds, and let $f : X \to Y$ be a $\mathcal{C}^\infty$ map. Let $p \in X$ and $a = f(p)$. We have the map

$$df_p : T_pX \to T_aY. \tag{6.53}$$

From this we get the pullback

$$(df_p)^* : \Lambda^k(T_a^*Y) \to \Lambda^k(T_p^*X). \tag{6.54}$$

Let ω be a k-form on Y. Then $f^*\omega$ is defined by

$$(f^*\omega)_p = (df_p)^*\omega_q. \tag{6.55}$$

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Let $f : X \to Y$ and $g : Y \to Z$ be $\mathcal{C}^\infty$ maps on manifolds X, Y, Z. Let ω be a k-form. Then

$$(g \circ f)^* \omega = f^*(g^* \omega), \tag{6.56}$$

where $g \circ f : X \to Z$.

So far, the treatment of k-forms for manifolds has been basically the same as our earlier treatment of k-forms. However, the treatment for manifolds becomes more complicated when we study $\mathcal{C}^\infty$ forms.

Let U be an open subset of $\mathbb{R}^n$, and let ω be a k-form on U. We can write

$$\omega = \sum a_I(x) dx_{i_1} \wedge \cdots \wedge dx_{i_k}, \quad I = (i_1, \ldots, i_k). \tag{6.57}$$

By definition, we say that $\omega \in \Omega^k(U)$ if each $A_I \in \mathcal{C}^\infty(U)$.

Let V be an open subset of $\mathbb{R}^k$, and let $f : U \to V$ be a $\mathcal{C}^\infty$ map. Let $\omega \in \Omega^k(V)$. Then $f^*\omega \in \Omega^k(U)$. Now, we want to define what we mean by a $\mathcal{C}^\infty$ form on a manifold.

Let $X \subseteq \mathbb{R}^n$ be an n-dimensional manifold, and let $p \in X$. There exists an open set U in $\mathbb{R}^N$, a neighborhood V of p in $\mathbb{R}^N$, and a diffeomorphism $\phi : U \to V \cap X$. The diffeomorphism ϕ is a parameterization of X at p.

We can think of ϕ in the following two ways:

1. as a map of U onto $V \cap X$, or

2. as a map of U onto V, whose image is contained in X.

The second way of thinking about ϕ is actually the map $\iota_X \circ \phi$, where $\iota_X : X \to \mathbb{R}^N$ is the inclusion map. Note that $\iota_X : X \to \mathbb{R}^N$ is $\mathcal{C}^\infty$, because it extends to the identity map $I : \mathbb{R}^N \to \mathbb{R}^N$.

We give two equivalent definitions for $\mathcal{C}^\infty$ k-forms. Let ω be a k-form on X.

Definition 6.15. The k-form ω *is* $\mathcal{C}^\infty$ *at* p if there exists a k-form $\tilde{\omega} \in \Omega^k(V)$ such that $\iota_X^* \tilde{w} = \omega$.

Definition 6.16. The k-form ω *is* $\mathcal{C}^\infty$ *at* p if there exists a diffeomorphism $\phi : U \to V \cap U$ such that $\phi^* \omega \in \Omega^k(U)$.

The first definition depends only on the choice of $\tilde{\omega}$, and the second definition depends only on the choice of ϕ. So, if the definitions are equivalent, then neither definition depends on the choice of $\tilde{\omega}$ or the choice of ϕ.

We show that these two definitions are indeed equivalent.

Claim. *The above two definitions are equivalent.*

Proof. First, we show that (def 6.15) $\implies$ (def 6.16). Let $\omega = \iota_X^* \tilde{\omega}$. Then $\phi^* \omega = (\iota_X \circ \phi)^* \tilde{\omega}$. The map $\iota \circ \phi : U \to V$ is $\mathcal{C}^\infty$, and $\tilde{\omega} \in \Omega^k(v)$, so $\phi^* \omega = (\iota_X \circ \phi)^* \tilde{\omega} \in \Omega^k(U)$.

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Second, we show that (def 6.16) $\implies$ (def 6.15). Let $\phi : U \to V \cap U$ be a diffeomorphism. Then $\phi^{-1} : V \cap X \to U$ can be extended to $\psi : V \to U$, where ψ is $\mathcal{C}^\infty$. On $V \cap X$, the map $\phi = \iota_X^* \tilde{\omega}$, where $\tilde{\omega} = \psi^*(\phi^* \omega)$. It is easy to show that $\tilde{\omega}$ is $\mathcal{C}^\infty$. □

Definition 6.17. The k-form ω *is* $\mathcal{C}^\infty$ if ω is $\mathcal{C}^\infty$ at p for every point $p \in X$.

Notation. If ω is $\mathcal{C}^\infty$, then $\omega \in \Omega^k(X)$.

Theorem 6.18. *If $\omega \in \Omega^k(X)$, then there exists a neighborhood W of X in $\mathbb{R}^N$ and a k-form $\tilde{\omega} \in \Omega^k(W)$ such that $\iota_X^* \tilde{\omega} = w$.*

Proof. Let $p \in X$. There exists a neighborhood V_p of p in $\mathbb{R}^N$ and a k-form $\omega^p \in \Omega^k(V_p)$ such that $\iota_X^* \omega^p = \omega$ on $V_p \cap X$.

Let

$$W \subseteq \bigcup_{p \in X} V_p. \tag{6.58}$$

The collection of sets $\{V_p : p \in X\}$ is an open cover of W. Let $\rho_1,\ i = 1, 2, 3, \ldots,$ be a partition of unity subordinate to this cover. So, $\rho_i \in \mathcal{C}_0^\infty(W)$ and supp $\rho_i \subset V_p$ for some p. Let

$$\tilde{\omega}_i = \begin{cases} \rho_i \omega^p & \text{on } V_p, \\ 0 & \text{elsewhere.} \end{cases} \tag{6.59}$$

Notice that

$$\begin{aligned} \iota_X^* \tilde{\omega}_i &= \iota_X^* \rho_i \iota_X^* \omega^p \\ &= (\iota_X^* \rho_i) \omega. \end{aligned} \tag{6.60}$$

Take

$$\tilde{\omega} = \sum_{i=1}^{\infty} \tilde{\omega}_i. \tag{6.61}$$

This sum makes sense since we used a partition of unity. From the sum, we can see that $\tilde{w} \in \Omega^k(W)$. Finally,

$$\begin{aligned} \iota_X^* \tilde{w} &= (\iota_X^* \sum \rho_i) \omega \\ &= \omega. \end{aligned} \tag{6.62}$$

□

Theorem 6.19. *Let $X \subseteq \mathbb{R}^N$ and $Y \subseteq \mathbb{R}^\ell$ be manifolds, and let $f : X \to Y$ be a $\mathcal{C}^\infty$ map. If $\omega \in \Omega^k(X)$, then $f^* \omega \in \Omega^k(Y)$.*

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Proof. Take an open set W in $\mathbb{R}^\ell$ such that $W \supset Y$, and take $\tilde{\omega} \in \Omega^k(W)$ such that $\iota_X^* \tilde{\omega} = \omega$. Take any $p \in X$ and $\phi : U \to V$ a parameterization of X at p.

We show that the pullback $\phi^*(f^*\omega)$ is in $\Omega^k(U)$. We can write

$$\begin{aligned} \phi^*(f^*\omega) &= \phi^* f^* (\iota_X^* \tilde{w}) \\ &= (\iota \circ f \circ \phi)^* \tilde{\omega}, \end{aligned} \tag{6.63}$$

where in the last step we used the chain rule.

The form $\tilde{\omega} \in \Omega^k(W)$, where W is open in $\mathbb{R}^\ell$, so $\iota \circ f \circ \phi : U \to W$. The theorem that we proved on Euclidean spaces shows that the r.h.s of Equation 6.63 is in $\Omega^k(U)$. □

The student should check the following claim:

Claim. *If $\mu, \nu \in \Omega^k(Y)$, then*

$$f^*(\mu \wedge \nu) = f^*\mu \wedge f^*\nu. \tag{6.64}$$

The differential operation d is an important operator on k-forms on manifolds.

$$d : \Omega^k(X) \to \Omega^{k+1}(X). \tag{6.65}$$

Let $X \subseteq \mathbb{R}^N$ be a manifold, and let $\omega \in \Omega^k(X)$. There exists an open neighborhood W of X in $\mathbb{R}^N$ and a k-form $\tilde{\omega} \in \Omega^k(W)$ such that $\iota_X^* \tilde{\omega} = \omega$.

Definition 6.20. $d\omega = \iota_X^* d\tilde{\omega}$.

Why is this definition well-defined? It seems to depend on the choice of $\tilde{\omega}$.

Take a parameterization $\phi : U \to V \cap X$ of X at p. Then

$$\begin{aligned} \phi^* \iota_X^* d\tilde{\omega} &= (\iota_X \circ \phi)^* d\tilde{\omega} \\ &= d(\iota_X \circ \phi)^* \omega \\ &= d\phi^* (\iota_X^* \tilde{\omega}) \\ &= d\phi^* \omega. \end{aligned} \tag{6.66}$$

So,

$$\phi^* \iota_X^* d\tilde{\omega} = d\phi^* \omega. \tag{6.67}$$

Take the inverse mapping $\phi^{-1} : V \cap X \to U$ and take the pullback $(\phi^{-1})^*$ of each side of Equation 6.67, to obtain

$$\iota_X^* d\tilde{\omega} = (\phi^{-1})^* d\phi^* \omega. \tag{6.68}$$

The r.h.s does not depend on $\tilde{\omega}$, so neither does the l.h.s.

To summarize this lecture, everything we did with k-forms on Euclidean space applies to k-forms on manifolds.

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Lecture 34

6.6 Orientation of Manifolds

Let X be an n-dimensional manifold in $\mathbb{R}^N$. Assume that X is a closed subset of $\mathbb{R}^N$. Let $f : X \to \mathbb{R}$ be a $\mathcal{C}^\infty$ map.

Definition 6.21. We remind you that the support of f is defined to be

$$\text{supp } f = \overline{\{x \in X : f(x) \neq 0\}}. \tag{6.69}$$

Since X is closed, we don't have to worry about whether we are taking the closure in X or in $\mathbb{R}^n$.

Note that

$$f \in \mathcal{C}_0^\infty(X) \iff \text{supp } f \text{ is compact.} \tag{6.70}$$

Let $\omega \in \Omega^k(X)$. Then

$$\text{supp } \omega = \overline{\{p \in X : \omega_p \neq 0\}}. \tag{6.71}$$

We use the notation

$$\omega \in \Omega_c^k(X) \iff \text{supp } \omega \text{ is compact.} \tag{6.72}$$

We will be using partitions of unity, so we remind you of the definition:

Definition 6.22. A collection of functions $\{\rho_i \in \mathcal{C}_0^\infty(X) : i = 1, 2, 3, \dots\}$ is a *partition of unity* if

1. $0 \leq \rho_i$,
2. For every compact set $A \subseteq X$, there exists $N > 0$ such that supp $\rho_i \cap A = \phi$ for all $i > N$,
3. $\sum \rho_i = 1$.

Suppose the collection of sets $\mathcal{U} = \{U_\alpha : \alpha \in I\}$ is a covering of X by open subsets U_α of X.

Definition 6.23. The partition of unity $\rho_i,\ i = 1, 2, 3, \dots$, is *subordinate to* $\mathcal{U}$ if for every i, there exists $\alpha \in I$ such that supp $\rho_i \subseteq U_\alpha$.

Claim. *Given a collection of sets $\mathcal{U} = \{U_\alpha : \alpha \in I\}$, there exists a partition of unity subordinate to $\mathcal{U}$.*

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Proof. For each $\alpha \in I$, let $\tilde{U_\alpha}$ be an open set in $\mathbb{R}^N$ such that $U_\alpha = \tilde{U}_\alpha \cap X$. We define the collection of sets $\tilde{\mathcal{U}} = \{\tilde{U}_\alpha : \alpha \in I\}$. Let

$$\tilde{U} = \bigcup \tilde{U}_\alpha. \tag{6.73}$$

From our study of Euclidean space, we know that there exists a partition of unity $\tilde{\rho}_i \in \mathcal{C}_0^\infty(\tilde{U}),\ i = 1, 2, 3, \ldots$, subordinate to $\tilde{\mathcal{U}}$. Let $\iota_X : X \to \tilde{\mathcal{U}}$ be the inclusion map. Then

$$\rho_i = \tilde{\rho}_i \circ \iota_X = \iota_X^* \tilde{\rho}_i, \tag{6.74}$$

which you should check. □

We review orientations in Euclidean space before generalizing to manifolds. For a more comprehensive review, read section 7 of the Multi-linear Algebra notes.

Suppose $\mathbb{L}$ is a one-dimensional vector space and that $v \in \mathbb{L} - \{0\}$. The set $\mathbb{L} - \{0\}$ has two components:

$$\{\lambda v : \lambda > 0\} \quad \text{and} \quad \{\lambda v : \lambda < 0\}. \tag{6.75}$$

Definition 6.24. An *orientation of* $\mathbb{L}$ is a choice of one of these components.

Notation. We call the preferred component $\mathbb{L}_+$ (the positive component). We call the other component $\mathbb{L}_-$ (the negative component).

We define a vector v to be *positively oriented* if $v \in \mathbb{L}_+$.

Now, let V be an n-dimensional vector space.

Definition 6.25. An *orientation of* V is an orientation of the one-dimensional vector space $\Lambda^n(V^*)$. That is, an orientation of V is a choice of $\Lambda^n(V^*)_+$.

Suppose that V_1, V_2 are oriented n-dimensional vector spaces, and let $A : V_1 \to V_2$ be a bijective linear map.

Definition 6.26. The map A is *orientation preserving* if

$$\omega \in \Lambda^n(V_2)_+ \implies A^*\omega \in \Lambda^n(V_1)_+. \tag{6.76}$$

Suppose that V_3 is also an oriented n-dimensional vector space, and let $B : V_2 \to V_3$ be a bijective linear map. If A and B are orientation preserving, then BA is also orientation preserving.

Finally, let us generalize the notion of orientation to orientations of manifolds. Let $X \subseteq \mathbb{R}^N$ be an n-dimensional manifold.

Definition 6.27. An *orientation of* X is a function on X which assigns to each point $p \in X$ an orientation of T_pX.

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We give two examples of orientations of a manifold:

Example 1: Let $\omega \in \Lambda^n(X)$, and suppose that ω is nowhere vanishing. Orient X by assigning to $p \in X$ the orientation of T_pX for which $\omega_p \in \Lambda^n(T_p^*X)_+$.

Example 2: Take $X = U$, an open subset of $\mathbb{R}^n$, and let

$$\omega = dx_1 \wedge \cdots \wedge dx_n. \tag{6.77}$$

Define an orientation as in the first example. This orientation is called the *standard orientation of* U.

Definition 6.28. An orientation of X is a $\mathcal{C}^\infty$ *orientation* if for every point $p \in X$, there exists a neighborhood U of p in X and an n-form $\omega \in \Omega^n(U)$ such that for all points $q \in U$, $\omega_q \in \Lambda^n(T_q^*X)_+$.

From now on, we will only consider $\mathcal{C}^\infty$ orientations.

Theorem 6.29. *If X is oriented, then there exists $\omega \in \Omega^n(X)$ such that for all $p \in X$, $\omega_p \in \Lambda^n(T_p^*X)_+$.*

Proof. For every point $p \in X$, there exists a neighborhood U_p of p and an n- form $\omega^{(p)} \in \Omega^n(U_p)$ such that for all $q \in U_p$, $(\omega^{(p)})_q \in \Lambda^n(T_Q^*X)_+$.

Take $\rho_i,\ i = 1, 2, \ldots,$ a partition of unity subordinate to $\mathcal{U} = \{U_p : p \in X\}$. For every i, there exists a point p such that $\rho_i \in \mathcal{C}_0^\infty(U_p)$. Let

$$\omega_i = \begin{cases} \rho_i \omega^{(p)} & \text{on } U_p, \\ 0 & \text{on the } X - U_p. \end{cases} \tag{6.78}$$

Since the ρ_i's are compactly supported, ω_i is a $\mathcal{C}^\infty$ map. Let

$$\omega = \sum \omega_i. \tag{6.79}$$

One can check that ω is positively oriented at every point. □

Definition 6.30. An n-form $\omega \in \Omega^n(X)$ with the property hypothesized in the above theorem is called a *volume form.*

Remark. If ω_1, ω_2 are volume forms, then we can write $\omega_2 = f\omega_1$, for some $f \in \mathcal{C}^\infty(X)$ (where $f \neq 0$ everywhere). In general, $f(p) > 0$ because $(\omega_1)_p, (\omega_2)_p \in \Lambda^n(T_p^*X)_+$. So, if ω_1, ω_2 are volume forms, then $\omega_2 = f\omega_1$, for some $f \in \mathcal{C}^\infty(X)$ such that $f > 0$.

Remark. Problem #6 on the homework asks you to show that if X is orientable and connected, then there are exactly two ways to orient it. This is easily proved using the above Remark.

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Suppose that $X \subseteq \mathbb{R}^n$ is a one-dimensional manifold (a "curve"). Then T_pX is one-dimensional. We can find vectors $v, -v \in T_pX$ such that $||v|| = 1$. An orientation of X is just a choice of v or $-v$.

Now, suppose that X is an $(n-1)$-dimensional manifold in $\mathbb{R}^n$. Define

$$N_pX = \{v \in T_p\mathbb{R}^n : v \perp w \text{ for all } w \in T_pX\}. \tag{6.80}$$

Then $\dim N_pX = 1$, so you can find $v, -v \in N_pX$ such that $||v|| = 1$. By Exercise #5 in section 4 of the Multi-linear Algebra Notes, an orientation of T_pX is just a choice of v or $-v$.

Suppose X_1, X_2 are oriented n-dimensional manifolds, and let $f : X_1 \to X_2$ be a diffeomorphism.

Definition 6.31. The map f is *orientation preserving* if for every $p \in X_1$,

$$df_p : T_pX_1 \to T_qX_2 \tag{6.81}$$

is orientation preserving, where $q = f(p)$.

Remark. Let ω_2 be a volume form on X_2. Then f is orientation preserving if and only if $f^*\omega_2 = \omega_1$ is a volume form on X_1.

We look at an example of what it means for a map to be orientation preserving. Let U, V be open sets on $\mathbb{R}^n$ with the standard orientation. Let $f : U \to V$ be a diffeomorphism. So, by definition, the form

$$dx_1 \wedge \cdots \wedge dx_n \tag{6.82}$$

is a volume form of V. The form

$$f^*dx_1 \wedge \cdots \wedge dx_n = \det\left[\frac{\partial f_i}{\partial x_j}\right] dx_1 \wedge \cdots \wedge dx_n \tag{6.83}$$

is a volume form of U if and only if

$$\det\left[\frac{\partial f_i}{\partial x_j}\right] > 0, \tag{6.84}$$

that is, if and only if f is orientation preserving in our old sense.

Now that we have studied orientations of manifolds, we have all of the ingredients we need to study integration theory for manifolds.

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Lecture 35

Before moving on to integration, we make a few more remarks about orientations.

Let X, Y be oriented manifolds. A diffeomorphism $f : X \to Y$ is orientation preserving if for every $p \in X$, the map

$$df_p : T_pX \to T_qY \tag{6.85}$$

is orientation preserving, where $q = f(p)$.

Let V be open in X, let U be open in $\mathbb{R}^n$, and let $\phi : U \to V$ be a parameterization.

Definition 6.32. The map ϕ is an *oriented parameterization* if it is orientation preserving.

Suppose ϕ is orientation reversing. Let $A : \mathbb{R}^n \to \mathbb{R}^n$ be the linear map defined by

$$A(x_1, \ldots, x_n) = (-x_1, x_2, \ldots, x_n). \tag{6.86}$$

The map A is orientation reversing. Let $U' = A^{-1}(U)$, and define $\phi' = \phi \circ A : U' \to V$. Both ϕ and A are orientation reversing, so ϕ' is orientation preserving.

Thus, for every point $p \in X$, there exists an oriented parameterization of X at p.

6.7 Integration on Manifolds

Our goal for today is to take any $\omega \in \Omega^n_c(X)$ and define

$$\int_X \omega. \tag{6.87}$$

First, we consider a special case:

Let $\phi : U \to V$ be an oriented parameterization. Let U be open in $\mathbb{R}^n$, and let V be open in X. Take any $\omega \in \Omega^n_c(V)$. Then

$$\int_V \omega = \int_U \phi^*\omega, \tag{6.88}$$

where $\phi^*\omega = f(x)dx_1 \wedge \cdots \wedge dx_n$, where $f \in \mathcal{C}^\infty_0(U)$ and

$$\int_U \phi^*\omega = \int_U f. \tag{6.89}$$

Claim. *The above definition for $\int \omega$ does not depend on the choice of oriented parameterization ϕ.*

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Proof. Let $\phi_i : U_i \to V,\ i = 1, 2$, be oriented parameterizations. Let $\omega \in \Omega_c^n(V_1 \cap V_2)$. Define

$$U_{1,2} = \phi_1^{-1}(V_1 \cap V_2), \tag{6.90}$$
$$U_{2,1} = \phi_2^{-1}(V_1 \cap V_2), \tag{6.91}$$

which are open sets in $\mathbb{R}^n$.

Both ϕ_1 and ϕ_2 are diffeomorphisms, and we have the diagram

$$\begin{array}{ccc} V_1 \cap V_2 & = & V_1 \cap V_2 \\ {\scriptstyle \phi_1}\uparrow & & {\scriptstyle \phi_2}\uparrow \\ U_{1,2} & \xrightarrow{f} & U_{2,1}. \end{array} \tag{6.92}$$

Therefore, $f = \phi_2^{-1} \circ \phi_1$ is a diffeomorphism, and $\phi_1 = \phi_2 \circ f$. Integrating,

$$\begin{aligned} \int_{U_1} \phi_1^* \omega &= \int_{U_{1,2}} \phi_1^* \omega \\ &= \int_{U_{1,2}} (\phi_2 \circ f)^* \omega \\ &= \int_{U_{1,2}} f^*(\phi_2^* \omega). \end{aligned} \tag{6.93}$$

Note that f is orientation preserving, because ϕ_1 and ϕ_2 are orientation preserving.

Using the change of variables formula,

$$\begin{aligned} \int_{U_{1,2}} f^* \phi_2^* \omega &= \int_{U_{2,1}} \phi_2^* \omega \\ &= \int_{U_2} \phi_2^* \omega. \end{aligned} \tag{6.94}$$

So, for all $\omega \in \Omega_c^n(V_1 \cap V_2)$,

$$\int_{V_1} \omega = \int_{U_1} \phi_1^* \omega = \int_{U_2} \phi_2^* \omega = \int_{V_2} \omega. \tag{6.95}$$

□

Above, we showed above how to take integrals over open sets, and now we generalize.

To define the integral, we need the following two inputs:

1. a set of oriented parameterizations $\phi_i : U_i \to V_i,\ i = 1, 2, \ldots$, such that $X = \bigcup V_i$,

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2. a partition of unity $\rho_i \in \mathcal{C}_0^\infty(V_i)$ subordinate to the cover $\{V_i\}$.

Definition 6.33. Let $\omega \in \Omega_c^n(X)$. We define the integral

$$\int_X \omega = \sum_{i=1}^{\infty} \int_{V_i} \rho_i \omega. \tag{6.96}$$

One can check various standard properties of integrals, such as linearity:

$$\int_X \omega_1 + \omega_2 = \int_X \omega_1 + \int_X \omega_2. \tag{6.97}$$

We now show that this definition is independent of the choice of the two inputs (the parameterizations and the partition of unity).

Consider two different inputs:

1. oriented parameterizations $\phi_j' : U_j' \to V_j',\ j = 1, 2, \ldots$, such that $X = \bigcup V_j'$,
2. a partition of unity $\rho_i' \in \mathcal{C}_0^\infty(V_j')$ subordinate to the cover $\{V_j'\}$.

Then,

$$\begin{aligned} \int_{V_i} \rho_i \omega &= \int_{V_i} \left(\sum_{j=1}^{\infty} \rho_j' \omega \right) \\ &= \sum_{j=1}^{\infty} \int_{V_i} \rho_i \rho_j' \omega \\ &= \sum_{j=1}^{\infty} \int_{V_i \cap V_j'} \rho_i \rho_j' \omega. \end{aligned} \tag{6.98}$$

Summing over i,

$$\begin{aligned} \sum_i \int_{V_i} \rho_i \omega &= \sum_{i,j=1}^{\infty} \int_{V_i \cap V_j'} \rho_i \rho_j' \omega \\ &= \sum_j \int_{V_j'} \rho_j' \omega, \end{aligned} \tag{6.99}$$

where the first term equals the last term by symmetry. Therefore, the integral $\int \omega$ is independent of the choices of these two inputs.

Let $X \subseteq \mathbb{R}^N$ be an oriented connected n-dimensional manifold.

Theorem 6.34. *For any $\omega \in \Omega_c^n(X)$, the following are equivalent:*

1. $\int_X \omega = 0$,

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2. $\omega \in d\Omega_c^{n-1}(X)$.

Proof. This will be a five step proof:

Step 1: The following lemma is called the Connectivity Lemma.

Lemma 6.35. *Given* $p, q \in X$*, there exists open sets* W_j*,* $j = 0, \ldots, N+1$*, such that each* W_j *is diffeomorphic to an open set in* $\mathbb{R}^n$*, and such that* $p \in W_0$*,* $q \in W_{N+1}$*, and* $W_i \cap W_{i+1} \neq \phi$*.*

Proof Idea: Fix p. The points q for which this is true form an open set. The points q for which this isn't true also form an open set. Since X is connected, only one of these sets is in X. □

Step 2: Let $\omega_1, \omega_2 \in \Omega_c^n(X)$. We say that $\omega_1 \sim \omega_2$ if

$$\int_X \omega_1 = \int_X \omega_2. \tag{6.100}$$

We can restate the theorem as

$$\omega_1 \sim \omega_2 \iff \omega_1 - \omega_2 \in d\Omega_c^{n-1}(X). \tag{6.101}$$

Step 3: It suffices to prove the statement (6.101) for $\omega_1 \in \Omega_c^n(V)$ and $\omega_2 \in \Omega_c^n(V')$, where V, V' are diffeomorphic to open sets in $\mathbb{R}^n$.

Step 4: We use a partition of unity

Lemma 6.36. *The theorem is true if* $V = V'$*.*

Proof. Let $\phi : U \to V$ be an orientation preserving parameterization. If $\omega_1 \sim \omega_2$, then

$$\int \phi^* \omega_1 = \int \phi^* \omega_2, \tag{6.102}$$

which is the same as saying that

$$\phi^* \omega_1 - \phi^* \omega_2 \in d\Omega_c^{n-1}(U), \tag{6.103}$$

which is the same as saying that

$$\omega_1 - \omega_2 \in d\Omega_c^{n-1}(V). \tag{6.104}$$

□

Step 5: In general, by the Connectivity Lemma, there exists sets W_i, $i = 0, \ldots, N+1$, such that each W_i is diffeomorphic to an open set in $\mathbb{R}^n$. We can choose $W_0 = V$ and $W_{N+1} = V'$ and $W_i \cap W_{i+1} \neq \phi$ (where ϕ here is the empty set).

We can choose $\mu_i \in \Omega_c^n(W_i \cap W_{i+1})$ such that

$$c = \int_V \omega_1 = \int \mu_0 = \cdots = \int \mu_{N+1} = \int_{V'} \omega_2. \tag{6.105}$$

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So,

$$\omega_1 \sim \mu_0 \sim \cdots \sim \mu_N \sim \omega_2. \tag{6.106}$$

We know that $\mu_0 - \omega_1 \in d\Omega_c^{n-1}$ and $\omega_2 - \mu_{N+1} \in d\Omega_C^{n-1}$ Also, each difference $\omega_i - \omega_{i+1} \in d\Omega_c^{n-1}$. Therefore, $\omega_1 - \omega_2 \in d\Omega_c^{n-1}$. □

6.8 Degree on Manifolds

Suppose that X_1, X_2 are oriented n-dimensional manifolds, and let $f : X_1 \to X_2$ be a proper map (that is, for every compact set $A \subseteq X$, the set pre-image $f^{-1}(A)$ is compact). It follows that if $\omega \in \Omega_c^k(X_2)$, then $f^*\omega \in \Omega_c^k(X_1)$.

Theorem 6.37. *If X_1, X_2 are connected and $f : X_1 \to X_2'$ is a proper $\mathcal{C}^\infty$ map, then there exists a topological invariant of f (called the* degree of f*) written* $\deg(f)$ *such that for every $\omega \in \Omega_c^n(X_2)$,*

$$\int_{X_1} f^*\omega = \deg(f) \int_{X_2} \omega. \tag{6.107}$$

Proof. The proof is pretty much verbatim of the proof in Euclidean space. □

Let us look at a special case. Let $\phi_1 : U \to V$ be an oriented parameterization, and let V_1 be open in X_1. Let $f : X_1 \to X_2$ be an oriented diffeomorphism. Define $\phi_2 = f \circ \phi_1$, which is of the form $\phi_2 : U \to V_2$, where $V_2 = f(V_1)$. Notice that ϕ_2 is an oriented parameterization of V_2.

Take $\omega \in \Omega_c^n(V_2)$ and compute the integral

$$\begin{aligned} \int_{V_1} f^*\omega &= \int_U \phi_1^* f^*\omega \\ &= \int_U (f \circ \phi_1)^*\omega \\ &= \int_U \phi_2^*\omega. \end{aligned} \tag{6.108}$$

The n-form ω is compactly supported on V_2, so

$$\begin{aligned} \int_{V_1} f^*\omega &= \int_U \phi_2^*\omega \\ &= \int_{X_2} \omega. \end{aligned} \tag{6.109}$$

On the other hand,

$$\int_{X_1} f^*\omega = \int_{V_1} f^*\omega. \tag{6.110}$$

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Combining these results,

$$\int_{X_1} f^*\omega = \int_{X_2} \omega. \tag{6.111}$$

Therefore,

$$\deg(f) = 1. \tag{6.112}$$

So, we have proved the following theorem, which is the Change of Variables theorem for manifolds:

Theorem 6.38. *Let X_1, X_2 be connected oriented n-dimensional manifolds, and let $f : X_1 \to X_2$ be an orientation preserving diffeomorphism. Then, for all $\omega \in \Omega_c^n(X_2)$,*

$$\int_{X_1} f^*\omega = \int_{X_2} \omega. \tag{6.113}$$

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Lecture 36

The first problem on today's homework will be to prove the inverse function theorem for manifolds. Here we state the theorem and provide a sketch of the proof.

Let X, Y be n-dimensional manifolds, and let $f : X \to Y$ be a $\mathcal{C}^\infty$ map with $f(p) = p_1$.

Theorem 6.39. *If $df_p : T_pX \to T_{p_1}Y$ is bijective, then f maps a neighborhood V of p diffeomorphically onto a neighborhood V_1 of p_1.*

Sketch of proof: Let $\phi : U \to V$ be a parameterization of X at p, with $\phi(q) = p$. Similarly, let $\phi_1 : U_1 \to V_1$ be a parameterization of Y at p_1, with $\phi_1(q_1) = p_1$.

Show that we can assume that $f : V \to V_1$ (Hint: if not, replace V by $V \cap f^{-1}(V_1)$).

Show that we have a diagram

$$\begin{array}{ccc} V & \xrightarrow{f} & V_1 \\ {\scriptstyle\phi}\uparrow & & \uparrow{\scriptstyle\phi_1} \\ U & \xrightarrow{g} & U_1, \end{array} \tag{6.114}$$

which defines g,

$$g = \phi_1^{-1} \circ f \circ \phi, \tag{6.115}$$

$$g(q) = q_1. \tag{6.116}$$

So,

$$(dg)_q = (d\phi_1)_{q_1}^{-1} \circ df_p \circ (d\phi)_q. \tag{6.117}$$

Note that all three of the linear maps on the r.h.s. are bijective, so $(dg)_q$ is a bijection. Use the Inverse Function Theorem for open sets in $\mathbb{R}^n$. □

This ends our explanation of the first homework problem.

Last time we showed the following. Let X, Y be n-dimensional manifolds, and let $f : X \to Y$ be a proper $\mathcal{C}^\infty$ map. We can define a topological invariant $\deg(f)$ such that for every $\omega \in \Omega_c^n(Y)$,

$$\int_X f^*\omega = \deg(f) \int_Y \omega. \tag{6.118}$$

There is a recipe for calculating the degree, which we state in the following theorem. We lead into the theorem with the following lemma.

First, remember that we defined the set C_f of critical points of f by

$$p \in C_f \iff df_p : T_pX \to T_qY \text{ is not surjective,} \tag{6.119}$$

where $q = f(p)$.

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Lemma 6.40. *Suppose that $q \in Y - f(C_f)$. Then $f^{-1}(q)$ is a finite set.*

Proof. Take $p \in f^{-1}(q)$. Since $p \notin C_f$, the map df_p is bijective. The Inverse Function Theorem tells us that f maps a neighborhood U_p of p diffeomorphically onto an open neighborhood of q. So, $U_p \cap f^{-1}(q) = p$.

Next, note that $\{U_p : p \in f^{-1}(q)\}$ is an open covering of $f^{-1}(q)$. Since f is proper, $f^{-1}(q)$ is compact, so there exists a finite subcover $U_{p_1}, \ldots, U_{p_N}$. Therefore, $f^{-1}(q) = \{p_1, \ldots, p_N\}$. □

The following theorem gives a recipe for computing the degree.

Theorem 6.41.

$$\deg(f) = \sum_{i=1}^{N} \sigma_{p_i}, \tag{6.120}$$

where

$$\sigma_{p_i} = \begin{cases} +1 & \text{if } df_{p_i} : T_{p_i}X \to T_qY \text{ is orientation preserving,} \\ -1 & \text{if } df_{p_i} : T_{p_i}X \to T_qY \text{ is orientation reversing,} \end{cases} \tag{6.121}$$

Proof. The proof is basically the same as the proof in Euclidean space. □

We say that $q \in Y$ is a *regular value of* f if $q \notin f(C_f)$. Do regular values exist? We showed that in the Euclidean case, the set of non-regular values is of measure zero (Sard's Theorem). The following theorem is the analogous theorem for manifolds.

Theorem 6.42. *If $q_0 \in Y$ and W is a neighborhood of q_0 in Y, then $W - f(C_f)$ is non-empty. That is, every neighborhood of q_0 contains a regular value (this is known as the Volume Theorem).*

Proof. We reduce to Sard's Theorem.

The set $f^{-1}(q_0)$ is a compact set, so we can cover $f^{-1}(q_0)$ by open sets $V_i \subset X$, $i = 1, \ldots, N$, such that each V_i is diffeomorphic to an open set in $\mathbb{R}^n$.

Let W be a neighborhood of q_0 in Y. We can assume the following:

1. W is diffeomorphic to an open set in $\mathbb{R}^n$,
2. $f^{-1}(W) \subset \bigcup V_i$ (which is Theorem 4.3 in the Supp. Notes),
3. $f(V_i) \subseteq W$ (for, if not, we can replace V_i with $V_i \cap f^{-1}(W)$).

Let U and the sets U_i, $i = 1, \ldots, N$, be open sets in $\mathbb{R}^n$. Let $\phi : U \to W$ and the maps $\phi_i : U_i \to V_i$ be diffeomorphisms. We have the following diagram:

$$\begin{array}{ccc} V_i & \xrightarrow{f} & W \\ {\scriptstyle \phi_i,\cong}\uparrow & & {\scriptstyle \phi,\cong}\uparrow \\ U_i & \xrightarrow{g_i} & U, \end{array} \tag{6.122}$$

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which define the maps g_i,

$$g_i = \phi^{-1} \circ f \circ \phi_i. \tag{6.123}$$

By the chain rule, $x \in C_{g_i} \implies \phi_i(x) \in C_f$, so

$$\phi_i(C_{g_i} = C_f \cap V_i. \tag{6.124}$$

So,

$$\phi(g_i(C_{g_i})) = f(C_f \cap V_i). \tag{6.125}$$

Then,

$$f(C_f) \cap W = \bigcup_i \phi(g_i(C_{g_i})). \tag{6.126}$$

Sard's Theorem tells us that $g_i(C_{g_i})$ is a set of measure zero in U, so

$$U - \bigcup g_i(C_{g_i}) \text{ is non-empty, so} \tag{6.127}$$

$$W - f(C_f) \text{ is also non-empty.} \tag{6.128}$$

In fact, this set is not only non-empty, but is a very, very "full" set. □

Let $f_0, f_1 : X \to Y$ be proper $\mathcal{C}^\infty$ maps. Suppose there exists a proper $\mathcal{C}^\infty$ map $F : X \times [0,1] \to Y$ such that $F(x,0) = f_0(x)$ and $F(x,1) = f_1(x)$. Then

$$\deg(f_0) = \deg(f_1). \tag{6.129}$$

In other words, the degree is a homotopy. The proof of this is essential the same as before.

6.9 Hopf Theorem

The Hopf Theorem is a nice application of the homotopy invariance of the degree.

Define the n-sphere

$$S^n = \{v \in \mathbb{R}^{n+1} : ||v|| = 1\}. \tag{6.130}$$

Hopf Theorem. *Let n be even. Let $f : S^n \to \mathbb{R}^{n+1}$ be a $\mathcal{C}^\infty$ map. Then, for some $v \in S^n$,*

$$f(v) = \lambda v, \tag{6.131}$$

for some scalar $\lambda \in \mathbb{R}$.

Proof. We prove the contrapositive. Assume that no such v exists, and take $w = f(v)$. Consider $w - \langle v, w \rangle v \equiv w - w_1$. It follows that $w - w_1 \neq 0$.

Define a new map $\tilde{f} : S^n \to S^n$ by

$$\tilde{f}(v) = \frac{f(v) - \langle v, f(x) \rangle}{||f(v) - \langle v, f(x) \rangle||} \tag{6.132}$$

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Note that $(w - w_1) \perp v$, so $\tilde{f}(v) \perp v$.

Define a family of functions

$$f_t : S^n \to S^n, \tag{6.133}$$

$$f_t(v) = (\cos t)v + (\sin t)\tilde{w}, \tag{6.134}$$

where $\tilde{w} = \tilde{f}(v)$ has the properties $||\tilde{w}|| = 1$ and $\tilde{w} \perp v$.

We compute the degree of f_t. When $t = 0$, $f_t = \mathrm{id}$, so

$$\deg(f_t) = \deg(f_0) = 1. \tag{6.135}$$

When $t = \pi$, $f_t(v) = -v$. But, if n is even, a map from $S^n \to S^n$ mapping $v \to (-v)$ has degree -1. We have arrived at a contradiction. □

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Lecture 37

6.10 Integration on Smooth Domains

Let X be an oriented n-dimensional manifold, and let $\omega \in \Omega_c^n(X)$. We defined the integral

$$\int_X \omega, \tag{6.136}$$

but we can generalize the integral

$$\int_D \omega, \tag{6.137}$$

for some subsets $D \subseteq X$. We generalize, but only to very simple subsets called *smooth domains* (essentially manifolds-with-boundary). The prototypical smooth domain is the half plane:

$$\mathbb{H}^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1 \leq 0\}. \tag{6.138}$$

Note that the boundary of the half plane is

$$\mathrm{Bd}\,(\mathbb{H}^n) = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1 = 0\}. \tag{6.139}$$

Definition 6.43. A closed subset $D \subseteq X$ is a *smooth domain* if for every point $p \in \mathrm{Bd}\,(D)$, there exists a parameterization $\phi : U \to V$ of X at p such that $\phi(U \cap \mathbb{H}^n) = V \cap D$.

Definition 6.44. The map ϕ is a *parameterization of D at p.*

Note that $\phi : U \cap \mathbb{H}^n \to V \cap D$ is a homeomorphism, so it maps boundary points to boundary points. So, it maps $U^b = U \cap \mathrm{Bd}\,(\mathbb{H}^n)$ onto $V^b = V \cap \mathrm{Bd}\,(D)$.

Let $\psi = \phi | U^b$. Then $\psi : U^b \to V^b$ is a diffeomorphism. The set U^b is an open set in $\mathbb{R}^{n-1}$, and ψ is a parameterization of the $\mathrm{Bd}\,(D)$ at p. We conclude that

$$\mathrm{Bd}\,(D) \text{ is an } (n-1)\text{-dimensional manifold.} \tag{6.140}$$

Here are some examples of how smooth domains appear in nature:

Let $f : X \to \mathbb{R}$ be a $\mathcal{C}^\infty$ map, and assume that $f^{-1}(0) \cap C_f = \phi$ (the empty set). That is, for all $p \in f^{-1}(0)$, $df_p \neq 0$.

Claim. *The set $D = \{x \in X : f(x) \leq 0\}$ is a smooth domain.*

Proof. Take $p \in \mathrm{Bd}\,(D)$, so $p = f^{-1}(0)$. Let $\phi : U \to V$ be a parameterization of X at p. Consider the map $g = f \circ \phi : U \to \mathbb{R}$. Let $q \in U$ and $p = \phi(q)$. Then

$$(dg_q) = df_p \circ (d\phi)_q. \tag{6.141}$$

We conclude that $dg_q \neq 0$.

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By the canonical submersion theorem, there exists a diffeomorphism ψ such that $g \circ \psi = \pi$, where π is the canonical submersion mapping $(x, \ldots, x_n) \to x_1$. We can write simply $g \circ \psi = x_1$. Replacing $\phi = \phi_{\text{old}}$ by $\phi = \phi_{\text{new}} = \phi_{\text{old}} \circ \psi$, we get the new map $\phi : U \to V$ which is a parameterization of X at p with the property that $f \circ \phi(x_1, \ldots, x_n) = x_1$. Thus, ϕ maps $\mathbb{H}^n \cap U$ onto $D \cap V$. □

We give an example of using the above claim to construct a smooth domain. Let $X = \mathbb{R}^n$, and define

$$f(x) = 1 - (x_1^2 + \cdots + x_n^2). \tag{6.142}$$

By definition,

$$f(x) \leq 0 \iff x \in B^n, \tag{6.143}$$

where $B^n = \{x \in \mathbb{R}^n : ||x|| \leq 1\}$ is the "unit ball." So, the unit ball B^n is a smooth domain.

We now define orientations of smooth domains. Assume that X is oriented, and let D be a smooth domain. Let $\phi : U \to V$ be a parameterization of D at p.

Definition 6.45. The map ϕ is an *oriented parameterization of D* if it is an oriented parameterization of X.

Assume that $\dim X = n > 1$. We show that you can always find an oriented parameterization.

Let $\phi : U \to V$ be a parameterization of D at p. Suppose that ϕ is *not* oriented. That is, as a diffeomorphism ϕ is orientation reversing. Let $A : \mathbb{R}^n \to \mathbb{R}^n$ be the map

$$A(x_1, \ldots, x_n) = (x_1, \ldots, x_{n-1}, -x_n). \tag{6.144}$$

Then A maps $\mathbb{H}^n \to \mathbb{H}^n$, and $\phi \circ A$ is orientation preserving. So, $\phi \circ A$ is an oriented parameterization of D at p.

Now, let $\phi : U \to V$ be an oriented parameterization of D at p. We define

$$U^b = U \cap \text{Bd}\,(\mathbb{H}^n), \tag{6.145}$$

$$V^b = V \cap \text{Bd}\,(D), \tag{6.146}$$

$$\psi = \phi | U^b, \tag{6.147}$$

where ψ is a parameterization of $\text{Bd}\,(D)$ at p.

We oriented $\text{Bd}\,(D)$ at p by requiring ψ to be an oriented parameterization. We need to check the following claim.

Claim. *The definition of* oriented *does not depend on the choice of parameterization.*

Proof. Let $\phi_i : U_i \to V_i$, $i = 1, 2$, be oriented parameterizations of D at p. Define

$$U_{1,2} = \phi_1^{-1}(V_1 \cap V_2), \tag{6.148}$$

$$U_{2,1} = \phi_2^{-1}(V_1 \cap V_2), \tag{6.149}$$

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from which we obtain the following diagram:

$$\begin{array}{ccc} V_1 \cap V_2 & = & V_1 \cap V_2 \\ {\scriptstyle \phi_1}\uparrow & & {\scriptstyle \phi_2}\uparrow \\ U_{1,2} & \xrightarrow{\ g\ } & U_{2,1}, \end{array} \tag{6.150}$$

which defines a map g. By the properties of the other maps ϕ_1, ϕ_2, the map g is an orientation preserving diffeomorphism of $U_{1,2}$ onto $U_{2,1}$. Moreover, g maps

$$U_{1,2}^b = \mathrm{Bd}\,(\mathbb{H}^n) \cap U_{1,2} \tag{6.151}$$

onto

$$U_{2,1}^b = \mathrm{Bd}\,(\mathbb{H}^n) \cap U_{2,1}. \tag{6.152}$$

Let $h = g|U_{1,2}^b$, so $h : U_{1,2}^b \to U_{2,1}^b$. We want to show that h is orientation preserving. To show this, we write g and h in terms of coordinates.

$$g = (g_1, \ldots, g_n), \quad \text{where } g_i = g_i(x_1, \ldots, x_n). \tag{6.153}$$

So,

$$g \text{ maps } \mathbb{H}^n \text{ to } \mathbb{H}^n \iff \begin{cases} g_1(x_1, \ldots, x_n) < 0 & \text{if } x_1 < 0, \\ g_1(x_1, \ldots, x_n) > 0 & \text{if } x_1 > 0, \\ g_1(0, x_2, \ldots, x_n) = 0 \end{cases} \tag{6.154}$$

These conditions imply that

$$\begin{cases} \frac{\partial}{\partial x_1} g_1(0, x_2, \ldots, x_n) \geq 0, \\ \frac{\partial}{\partial x_i} g_1(0, x_2, \ldots, x_n) = 0, \ \text{for } i \neq 1. \end{cases} \tag{6.155}$$

The map h in coordinates is then

$$\begin{aligned} h &= h(x_2, \ldots, x_n) \\ &= \left(g(0, x_2, \ldots, x_n), \ldots, g_{n-1}(0, x_2, \ldots, x_n)\right), \end{aligned} \tag{6.156}$$

which is the statement that $h = g|\,\mathrm{Bd}\,(\mathbb{H}^n)$.

At the point $(0, x_2, \ldots, x_n) \in U_{1,2}^b$,

$$Dg = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & 0 & \cdots & 0 \\ * & & & \\ \vdots & & Dh & \\ * & & & \end{bmatrix}. \tag{6.157}$$

The matrix Dg is an $n \times n$ block matrix containing the $(n-1) \times (n-1)$ matrix Dh, because

$$\frac{\partial h_i}{\partial x_j} = \frac{\partial g_i}{\partial x_j}(0, x_2, \ldots, x_n), \ i, j > 1. \tag{6.158}$$

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Note that

$$\det(Dg) = \frac{\partial g_1}{\partial x_1} \det(Dh). \tag{6.159}$$

We know that the l.h.s > 0 and that $\frac{\partial g_1}{\partial x_1} > 0$, so $\det(Dh) > 0$. Thus, the map $h : U^b_{1,2} \to U^b_{2,1}$ is orientation preserving.

To repeat, we showed that in the following diagram, the map h is orientation preserving:

$$\begin{array}{ccc} V_1 \cap V_2 \cap \mathrm{Bd}\,(D) & = & V_1 \cap V_2 \cap \mathrm{Bd}\,(D) \\ {\scriptstyle \psi_1}\uparrow & & {\scriptstyle \psi_2}\uparrow \\ U^b_{1,2} & \xrightarrow{h} & U^b_{2,1}. \end{array} \tag{6.160}$$

We conclude that ψ_1 is orientation preserving if and only if ψ_2 is orientation preserving. □

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Lecture 38

We begin with a review from last time.

Let X be an oriented manifold, and let $D \subseteq X$ be a smooth domain. Then $\mathrm{Bd}\,(D) = Y$ is an oriented $(n-1)$-dimensional manifold.

We defined integration over D as follows. For $\omega \in \Omega_c^n(X)$ we want to make sense of the integral

$$\int_D \omega. \tag{6.161}$$

We look at some special cases:

Case 1: Let $p \in \mathrm{Int}\ D$, and let $\phi : U \to V$ be an oriented parameterization of X at p, where $V \subseteq \mathrm{Int}\ D$. For $\omega \in \Omega_c^n(X)$, we define

$$\int_D \omega = \int_V \omega = \int_U \phi^*\omega = \int_{\mathbb{R}^n} \phi^*\omega. \tag{6.162}$$

This is just our old definition for

$$\int_V \omega. \tag{6.163}$$

Case 2: Let $p \in \mathrm{Bd}\,(D)$, and let $\phi : U \to V$ be an oriented parameterization of D at p. That is, ϕ maps $U \cap \mathbb{H}^n$ onto $V \cap D$. For $\omega \in \Omega_c^n(V)$, we define

$$\int_D \omega = \int_{\mathbb{H}^n} \phi^*\omega. \tag{6.164}$$

We showed last time that this definition does not depend on the choice of parameterization.

General case: For each $p \in \mathrm{Int}\ D$, let $\phi : U_p \to V_p$ be an oriented parameterization of X at p with $V_p \subseteq \mathrm{Int}\ D$. For each $p \in \mathrm{Bd}\,(D)$, let $\phi : U_p \to V_p$ be and oriented parameterization of D at p. Let

$$U = \sum_{p \in D} U_p, \tag{6.165}$$

where the set $\mathcal{U} = \{U_p : p \in D\}$ be an an open cover of U. Let $\rho_i,\ i = 1, 2, \ldots,$ be a partition of unity subordinate to this cover.

Definition 6.46. For $\omega \in \Omega_c^n(X)$ we define the integral

$$\int_D \omega = \sum_i \int_D \rho_i \omega. \tag{6.166}$$

Claim. *The r.h.s. of this definition is well-defined.*

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Proof. Since the ρ_i's are a partition of unity, there exists an N such that

$$\text{supp } \omega \cap \text{supp } \rho_i = \phi, \tag{6.167}$$

for all $i > N$.

Hence, there are only a finite number of non-zero terms in the summand. Moreover, each summand is an integral of one of the two types above (cases 1 and 2), and is therefore well-defined. □

Claim. *The l.h.s. of the definition does not depend on the choice of the partition of unity ρ_i.*

Proof. We proved an analogous assertion about the definition of $\int_X \omega$ a few lectures ago, and the proof of the present claim is exactly the same. □

6.11 Stokes' Theorem

Stokes' Theorem. *For all $\omega \in \Omega_c^{n-1}(X)$,*

$$\int_D d\omega = \int_{\text{Bd}\,(D)} \omega. \tag{6.168}$$

Proof. Let ρ_i, $i = 1, 2 \ldots$, be a partition of unity as defined above. Replacing ω with $\sum \rho_i \omega$, it suffices to prove this for the two special cases below:

Case 1: Let $p \in \text{Int } D$, and let $\phi : U \to V$ be an oriented parameterization of X at p with $V \subseteq \text{Int } D$. If $\omega \in \Omega_c^{n-1}(V)$, then

$$\int_D d\omega = \int_{\mathbb{R}^n} \phi^* d\omega = \int_{\mathbb{R}^n} d\phi^* \omega = 0. \tag{6.169}$$

Case 2: Let $p \in \text{Bd}\,(D)$, and let $\phi : U \to V$ be an oriented parameterization of D at p. Let $U^b = U \cap \text{Bd}\,(\mathbb{H}^n)$, and let $V^b = V \cap \text{Bd}\,(D)$. Define $\psi : \phi | U^b$, so $\psi : U^b \to V^b$ is an oriented parameterization of $\text{Bd}\,(D)$ at p. If $\omega \in \Omega_c^{n-1}(V)$, then

$$\phi^* \omega = \sum f_i(x_1, \ldots, x_n) dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n. \tag{6.170}$$

What is $\psi^* \omega$? Let $\iota : \mathbb{R}^{n-1} \to \mathbb{R}^n$ be the inclusion map mapping $\text{Bd}\,(\mathbb{H}^n) \to \mathbb{R}^n$. The inclusion map ι maps $(x_2, \ldots, x_n) \to (0, x_2, \ldots, x_n)$. Then $\phi \circ \iota = \psi$, so

$$\begin{aligned} \psi^* \omega &= \iota^* \phi^* \omega \\ &= \iota^* \left(\sum_{i=1}^{n} f_i dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n \right). \end{aligned} \tag{6.171}$$

But,

$$\iota^* dx_1 = d\iota^* x_1 = 0, \quad \text{since } \iota^* x_1 = 0. \tag{6.172}$$

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So,

$$\begin{aligned}\psi^*\omega &= \iota^* f_1 dx_2 \wedge \cdots \wedge dx_n \\ &= f_1(0, x_2, \ldots, x_n) dx_2 \wedge \cdots \wedge dx_n.\end{aligned} \tag{6.173}$$

Thus,

$$\int_{\mathrm{Bd}\,(D)} \omega = \int_{\mathbb{R}^{n-1}} \psi^*\omega = \int_{\mathbb{R}^{n-1}} f_1(0, x_2, \ldots, x_n) dx_2 \ldots dx_n. \tag{6.174}$$

On the other hand,

$$\int_D d\omega = \int_{\mathbb{H}^n} \phi^* d\omega = \int_{\mathbb{H}^n} d\phi^*\omega. \tag{6.175}$$

One should check that

$$\begin{aligned}d\phi^*\omega &= d\left(\sum f_i dx_1 \wedge \cdots \wedge \widehat{x x_i} \wedge \cdots \wedge dx_n\right) \\ &= \left(\sum (-1)^{i-1} \frac{\partial f_i}{\partial x_i}\right) dx_1 \wedge \cdots \wedge dx_n.\end{aligned} \tag{6.176}$$

So, each summand

$$\int \frac{\partial f_i}{\partial x_i} dx_1 \ldots dx_n \tag{6.177}$$

can be integrated by parts, integrating first w.r.t. the ith variable. For $i > 1$, this is the integral

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{\partial f_i}{\partial x_i} dx_i &= f_i(x_1, \ldots, x_n)\big|_{x_i=-\infty}^{x_i=\infty} \\ &= 0.\end{aligned} \tag{6.178}$$

For $i = 1$, this is the integral

$$\int_{-\infty}^{\infty} \frac{\partial f_1}{\partial x_1}(x_1, \ldots, x_n) dx_1 = f_1(0, x_2, \ldots, x_n). \tag{6.179}$$

Thus, the total integral of $\phi^* d\omega$ over $\mathbb{H}^n$ is

$$\int f_1(0, x_2, \ldots, x_n) dx_2 \ldots dx_n. \tag{6.180}$$

We conclude that

$$\int_D d\omega = \int_{\mathrm{Bd}\,(D)} \omega. \tag{6.181}$$

□

We look at some applications of Stokes' Theorem.

Let D be a smooth domain. Assume that D is compact and oriented, and let $Y = \mathrm{Bd}\,(D)$. Let Z be an oriented n-manifold, and let $f : Y \to Z$ be a $\mathcal{C}^\infty$ map.

Theorem 6.47. *If f extends to a $\mathcal{C}^\infty$ map $F : D \to Z$, then*

$$\deg(f) = 0. \tag{6.182}$$

Corollary 9. *The Brouwer fixed point theorem follows from the above theorem.*

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